

CHEMISTRY EXPERTS

(JEE Main & Advanced)

IIT JEE MAIN & ADVANCED PREVIOUS YEAR'S QUESTIONS BOOKLET FOR CLASS XIIth STUDENT'S



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EXERCISE # (JEE-MAIN)

1. For the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ the true relationship is – [AIEEE-2002]
- (1) $-\frac{d[\text{H}_2]}{dt} = \frac{d[\text{I}_2]}{dt} = \frac{d[\text{HI}]}{dt}$ (2) $\frac{d[\text{H}_2]}{dt} = -\frac{d[\text{I}_2]}{dt} = \frac{1}{2} \frac{d[\text{HI}]}{dt}$
- (3) $-\frac{2d[\text{H}_2]}{dt} = -\frac{2d[\text{I}_2]}{dt} = \frac{d[\text{HI}]}{dt}$ (4) $\frac{2d[\text{H}_2]}{dt} = \frac{2d[\text{I}_2]}{dt} = \frac{1}{2} \frac{d[\text{HI}]}{dt}$
2. A chemical reaction $2\text{A} + 2\text{B} + \text{C} \longrightarrow \text{product}$ follows the rate equation : $r \propto [\text{A}] [\text{B}]^2$ then order of reaction is - [AIEEE-2002]
- (1) 0 (2) 1 (3) 2 (4) 3
3. The rate constant of first & second order reaction is respectively - [AIEEE-2002]
- (1) $\text{time}^{-1}, \text{mole}^{-1} \cdot \text{litre} \cdot \text{time}^{-1}$ (2) $\text{mole}^{-1} \text{litre}^{-1}, \text{time}^{-1}$
- (3) $\text{mole}^{-1} \cdot \text{litre} \cdot \text{time}^{-1}, \text{time}^{-1}$ (4) $\text{sec}^{-1}, \text{litre}^{-1}$
4. In the equation $kt = \ln C_0 - \ln C_t$, the curve between t and $\ln C_t$ is - [AIEEE-2002]
- (1) a straight line (2) a parabola (3) a hyperbola (4) none of these
5. Consider following two reactions –
- $\text{A} \longrightarrow \text{Product} \longrightarrow -\frac{d[\text{A}]}{dt} = k_1 [\text{A}]^0$
- $\text{B} \longrightarrow \text{Product} \longrightarrow -\frac{d[\text{B}]}{dt} = k_2 [\text{B}]$
- k_1 and k_2 are expressed in terms of molarity (mol L^{-1}) and time (sec) as - [AIEEE-2002]
- (1) $\text{sec}^{-1}, \text{M sec}^{-1}$ (2) $\text{M sec}^{-1}, \text{M sec}^{-1}$ (3) $\text{sec}^{-1}, \text{M}^{-1} \text{sec}^{-1}$ (4) $\text{M sec}^{-1}, \text{sec}^{-1}$
6. H_2 gas is adsorbed on the metal surface like tungsten. This follows order reaction - [AIEEE-2002]
- (1) Third (2) Second (3) Zero (4) First
7. In respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statement is correct ? [AIEEE-2003]
- (1) E_a is energy of activation (2) R is Rydberg's constant
- (3) k is equilibrium constant (4) A is adsorption factor
8. For the reaction system: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ volume is suddenly reduced to half its value by increasing the pressure on it. If the reaction is of first order with respect to O_2 and second order with respect to NO , the rate of reaction will – [AIEEE-2003]
- (1) Increase to eight times of its initial value
- (2) Increase to four times of its initial value
- (3) Diminish to one-fourth of its initial value
- (4) Diminish to one-eighth of its initial value

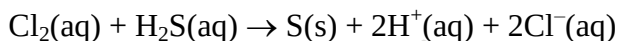
9. The rate law for a reaction between substances A and B is given by $\text{Rate} = k [A]^n [B]^m$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as : **[AIEEE-2003]**
- (1) $(n - m)$ (2) $2^{(n - m)}$ (3) $\frac{1}{2^{(m+n)}}$ (4) $(m + n)$
10. The rate equation for the reaction $2A + B \rightarrow C$ is found to be : $\text{rate} = k[A][B]$. The correct statement in relation to this reaction is that the **[AIEEE-2004]**
- (1) rate of formation of C is twice the rate of disappearance of A.
 (2) Half life is a constant
 (3) unit of k must be s^{-1}
 (4) value of k is independent of the initial concentrations of A and B
11. In a first order reaction, the concentration of the reactant, decreases from 0.8 M to 0.4 M in 15 minutes. The time taken for the concentration to change from 0.1 M to 0.025 M is **[JEE MAIN-2004]**
- (1) 7.5 minutes (2) 15 minutes (3) 30 minutes (4) 60 minutes
12. The half-life of a radioisotope is four hours. If the initial mass of the isotope was 200 g, the mass remaining after 24 hours undecayed is **[JEE MAIN-2004]**
- (1) 3.125 g (2) 2.084 g (3) 1.042 g (4) 4.167 g
13. Consider an endothermic reaction $X \rightarrow Y$ with the activation energies E_b and E_f for the backward and forward reactions, respectively. In general **[AIEEE-2005]**
- (1) $E_b > E_f$
 (2) $E_b < E_f$
 (3) there is no definite relation between E_b and E_f
 (4) $E_b = E_f$
14. A reaction involving two different reactant can never be - **[AIEEE-2005]**
- (1) first order reaction (2) unimolecular reaction
 (3) bimolecular reaction (4) second order reaction
15. $t_{1/4}$ can be taken as the time taken for the concentration of a reactant to drop to $3/4$ of its initial value. If the rate constant for a first order reaction is k, then $t_{1/4}$ can be written as— **[AIEEE-2005]**
- (1) $0.29/k$ (2) $0.10/k$ (3) $0.75/k$ (4) $0.69/k$
16. Rate of reaction can be expressed by Arrhenius equation as $k = Ae^{-E/RT}$. In this equation, E represents **[AIEEE-2005]**
- (1) the energy below which colliding molecules will not react
 (2) the total energy of the reacting molecule at a temperature, T
 (3) The fraction of molecules with energy greater than the activation energy of the reaction
 (4) None of these

17. The following mechanism has been proposed for the reaction of NO with Br₂ to form NOBr

$$\text{NO(g)} + \text{Br}_2\text{(g)} \rightleftharpoons \text{NOBr}_2\text{(g)} \quad [\text{AIEEE 2006}]$$

$$\text{NOBr}_2\text{(g)} + \text{NO(g)} \rightarrow 2\text{NOBr(g)}$$
 If the second step is the rate determining step, the order of the reaction with respect to NO(g) is :
 (1) 0 (2) 1 (3) 2 (4) 3
18. A reaction was found to be second order with respect to the concentration of carbon monoxide. If the concentration of carbon monoxide is doubled, with everything else kept the same, the rate of reaction will -
 (1) triple (2) increase by a factor of 4
 (3) double (4) remain unchanged
 [AIEEE-2006]
19. The energies of activation for forward and reverse reactions for $\text{A}_2 + \text{B}_2 \rightleftharpoons 2\text{AB}$ are 180 kJ mol⁻¹ and 200 kJ mol⁻¹ respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol⁻¹. The enthalpy change of the reaction ($\text{A}_2 + \text{B}_2 \rightarrow 2\text{AB}$) in the presence of catalyst will be (in kJ mol⁻¹) -
 (1) 300 (2) 120 (3) 280 (4) -20
 [AIEEE-2007]
20. A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial velocity is ten times the permissible value, after how many days will it be safe to enter the room ?
 (1) 100 days (2) 1000 days (3) 300 days (4) 10 days
 [JEE MAIN-2007]
21. Which of the following nuclear reactions will generate an isotope ?
 (1) β - particle emission (2) Neutron particle emission
 (3) Positron emission (4) α - Particle emission
 [JEE MAIN-2007]
22. For a reaction $\frac{1}{2} \text{A} \rightarrow 2\text{B}$, rate of disappearance of 'A' is related to the rate of appearance of 'B' by the expression.
 (1) $-\frac{d[\text{A}]}{dt} = \frac{1}{2} \frac{d[\text{B}]}{dt}$ (2) $-\frac{d[\text{A}]}{dt} = \frac{1}{4} \frac{d[\text{B}]}{dt}$
 (3) $-\frac{d[\text{A}]}{dt} = \frac{d[\text{B}]}{dt}$ (4) $-\frac{d[\text{A}]}{dt} = 4 \frac{d[\text{B}]}{dt}$
 [JEE MAIN-2008]
23. The half life period of a first order chemical reaction is 6.93 minutes. The time required for the completion of 99% of the chemical reaction will be ($\log 2 = 0.301$) :-
 (1) 46.06 minutes (2) 460.6 minutes
 (3) 230.3 minutes (4) 23.03 minutes
 [AIEEE-2009]

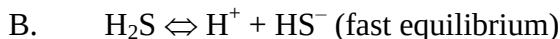
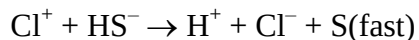
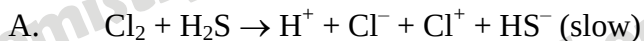
24. Consider the reaction:



The rate equation for this reaction is

$$\text{rate} = k[\text{Cl}_2][\text{H}_2\text{S}]$$

Which of these mechanisms is/are consistent with this rate equation ? **[AIEEE-2010]**



- (1) A only (2) B only (3) Both A and B (4) Neither A nor B

25. The time for half life period of a certain reaction $\text{A} \longrightarrow \text{Products}$ is 1 hour, when the initial concentration of the reactant 'A' is 2.0 mol L^{-1} . How much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction ? **[AIEEE-2010]**

- (1) 1 h (2) 4 h (3) 0.5 h (4) 0.25 h

26. The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C , the rate of the reaction increases by about :- **[AIEEE-2011]**

- (1) 32 times (2) 64 times (3) 10 times (4) 24 times

27. A reactant (A) forms two products : **[AIEEE-2011]**



If $E_{a2} = 2 E_{a1}$, then k_1 and k_2 are related as :-

- (1) $k_1 = 2k_2 e^{E_{a2}/RT}$ (2) $k_1 = k_2 e^{E_{a1}/RT}$ (3) $k_2 = k_1 e^{E_{a2}/RT}$ (4) $k_1 = Ak_2 e^{E_{a1}/RT}$

28. For a first order reaction $\text{A} \rightarrow \text{products}$ the concentration of A changes from 0.1 M to 0.025 M in 40 minutes.

The rate of reaction when the concentration of A is 0.01 M is : **[JEE MAIN-2012]**

- (1) $1.73 \times 10^{-5} \text{ M/min}$ (2) $3.47 \times 10^{-4} \text{ M/min}$
(3) $3.47 \times 10^{-5} \text{ M/min}$ (4) $1.73 \times 10^{-4} \text{ M/min}$

29. The rate of a reaction doubles when its temperature changes from 300 K to 310 K . Activation energy of such a reaction will be : ($R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ and $\log 2 = 0.301$)

[JEE MAIN-2013]

- (1) 53.6 kJ mol^{-1} (2) 48.6 kJ mol^{-1}
(3) 58.5 kJ mol^{-1} (4) 60.5 kJ mol^{-1}

30. For the non-stoichiometre reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K. **[JEE-MAIN 2014]**

Initial concentration (A)	Initial concentration (B)	Initial rate of formation of C (mol L ⁻¹ S ⁻¹)
0.1M	0.1M	1.2×10^{-3}
0.1M	0.2M	1.2×10^{-3}
0.2M	0.1M	2.4×10^{-3}

- (1) $\frac{dc}{dt} = k[A][B]^2$ (2) $\frac{dc}{dt} = k[A]$ (3) $\frac{dc}{dt} = k[A][B]$ (4) $\frac{dc}{dt} = k[A]^2[B]$

31. Higher order (>3) reactions are rare due to :- **[JEE-MAIN-(Offline)2015]**

- (1) shifting of equilibrium towards reactants due to elastic collision
 (2) loss of active species on collision
 (3) low probability of simultaneous collision of all the reacting species
 (4) increase in entropy and activation energy as more molecules are involved.

32. The reaction



follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mm Hg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant) **[JEE-MAIN-(Online)2015]**

- (1) 106.25 mm Hg (2) 116.25 mm Hg (3) 125 mm Hg (4) 150 mm Hg

33. For the equilibrium, $A(g) \rightleftharpoons B(g)$, ΔH is -40 kJ/mol. If the ratio of the activation energies

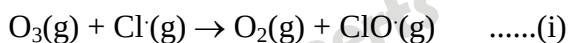
of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :- **[JEE-MAIN-(Online)2015]**

- (1) $E_f = 60$ kJ/mol; $E_b = 100$ kJ/mol (2) $E_f = 30$ kJ/mol; $E_b = 70$ kJ/mol
 (3) $E_f = 80$ kJ/mol; $E_b = 120$ kJ/mol (4) $E_f = 70$ kJ/mol; $E_b = 30$ kJ/mol

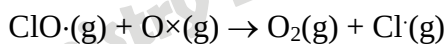
34. Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :- **[JEE-MAIN-(Offline)2016]**

- (1) 1.34×10^{-2} mol min⁻¹ (2) 6.93×10^{-2} mol min⁻¹
 (3) 6.93×10^{-4} mol min⁻¹ (4) 2.66 L min⁻¹ at STP

35. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below : [JEE-MAIN-(Online)2016]



$$k_i = 5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$$

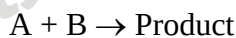


$$k_{ii} = 2.6 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1} \quad \dots\dots(\text{ii})$$

The closest rate constant for the overall reaction $\text{O}_3(\text{g}) + \text{O}(\text{g}) \rightarrow 2\text{O}_2(\text{g})$ is :

- (1) $3.1 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$ (2) $2.6 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$
 (3) $5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$ (4) $1.4 \times 10^{20} \text{ L mol}^{-1} \text{ s}^{-1}$

36. The rate law for the reaction below is given by the expression $k[\text{A}][\text{B}]$



If the concentration of B is increased from 0.1 to 0.3 mole, keeping the value of A at 0.1 mole, the rate constant will be : [JEE-MAIN-(Online)2016]

- (1) $3k$ (2) $9k$ (3) k (4) $k/3$

37. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1} . If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to : ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

[JEE-MAIN-(Offline)2017]

- (1) 6 (2) 4 (3) 8 (4) 12

38. The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A: [JEE MAIN 2017 (Online)]

- (1) 2.45 K (2) 4.92 K (3) 9.84 K (4) 19.67 K

39. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is: [JEE MAIN 2017 (Online)]

(Assume E_a and A are independent of temperature $\ln 2 = 0.693$; $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) $107.2 \text{ kJ mol}^{-1}$ (2) 53.6 kJ mol^{-1} (3) $214.4 \text{ kJ mol}^{-1}$ (4) 26.8 kJ mol^{-1}

40. N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg. The pressure of the gaseous mixture after 100 minutes at constant temperature will be: [JEE MAIN 2018 (Online)]

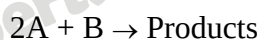
- (1) 116.25 mmHg (2) 175.0 mmHg (3) 106.25 mmHg (4) 136.25 mmHg

41. For a first order reaction, $A \rightarrow P$, $t_{1/2}$ (half-life) is 10 days. The time required for $\frac{1}{4}$ conversion of A (in days) is : ($\ln 2 = 0.693$, $\ln 3 = 1.1$) [JEE MAIN 2018 (Online)]
 (1) 5 (2) 4.1 (3) 3.2 (4) 2.5

42. If 50% of a reaction occurs in 100 second and 75% of the reaction occurs in 200 second, the order of this reaction is: [JEE MAIN 2018 (Online)]
 (1) 1 (2) 2 (3) Zero (4) 3

43. At 518°C , the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is : [JEE-MAIN-(Offline)2018]
 (1) 0 (2) 2 (3) 3 (4) 1

44. The following results were obtained during kinetic studies of the reaction :



[JEE MAIN -2019(JAN.)]

Experiment	[A] (in mol L^{-1})	[B] (in mol L^{-1})	Initial Rate of reaction (in $\text{mol L}^{-1} \text{min}^{-1}$)
(I)	0.10	0.20	6.93×10^{-3}
(II)	0.10	0.25	6.93×10^{-3}
(III)	0.20	0.30	1.386×10^{-2}

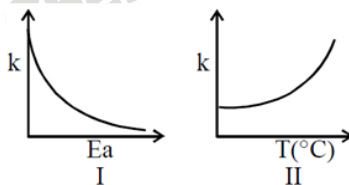
The time (in minutes) required to consume half of A is :

- (1) 10 (2) 5 (3) 100 (4) 1
45. For the reaction, $2A + B \rightarrow \text{products}$, when the concentrations of A and B both were doubled, the rate of the reaction increased from $0.3\text{ mol L}^{-1} \text{s}^{-1}$ to $2.4\text{ mol L}^{-1} \text{s}^{-1}$. When the concentration of A alone is doubled, the rate increased from $0.3\text{ mol L}^{-1} \text{s}^{-1}$ to $0.6\text{ mol L}^{-1} \text{s}^{-1}$.

[JEE MAIN -2019(JAN.)]

- (1) Total order of the reaction is 4
 (2) Order the reaction with respect to B is 1
 (3) Order of the reaction with respect to B is 2
 (4) Order of the reaction with respect to A is 2

46. Consider the given plots for a reaction obeying Arrhenius equation ($0^\circ\text{C} < T < 300^\circ\text{C}$) : (k and E_a are rate constant and activation energy, respectively) [JEE MAIN -2019(JAN.)]



Choose the correct option :

- (1) I is right but II is wrong (2) Both I and II are wrong
 (3) I is wrong but II is right (4) Both I and II are correct

47. For an elementary chemical reaction, $A_2 \xrightleftharpoons[k_{-1}]{k_1} 2A$, the expression for $\frac{d[A]}{dt}$ is:

[JEE MAIN -2019(JAN.)]

- (1) $2k_1[A_2] - 2k_{-1}[A]^2$ (2) $k_1[A_2] - k_{-1}[A]^2$ (3) $2k_1[A_2] - k_{-1}[A]^2$ (4) $k_1[A_2] + k_{-1}[A]^2$

48. If a reaction follows the Arrhenius equation, the plot $\ln k$ vs $\frac{1}{RT}$ gives straight line with a gradient ($-y$) unit. The energy required to activate the reactant is : [JEE MAIN -2019(JAN.)]

- (1) $\frac{y}{R}$ unit (2) yR unit (3) y unit (4) $-y$ unit

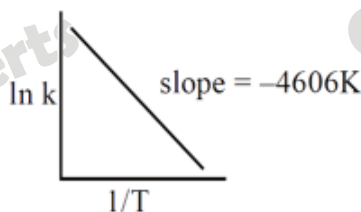
49. The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be :- [JEE MAIN -2019(JAN.)]

- (1) 18.0 h (2) 9.0 h (3) 7.2 h (4) 12.0 h

50. Decomposition of X exhibits a rate constant of $0.05 \mu\text{g}/\text{year}$. How many years are required for the decomposition of $5 \mu\text{g}$ of X into $2.5 \mu\text{g}$? [JEE MAIN -2019(JAN.)]

- (1) 25 (2) 40 (3) 50 (4) 20

51. For a reaction consider the plot of $\ln k$ versus $1/T$ given in the figure. If the rate constant of this reaction at 400 K is 10^{-5} s^{-1} , then the rate constant at 500 K is : [JEE MAIN -2019(JAN.)]



- (1) 10^{-6} s^{-1} (2) $4 \times 10^{-4} \text{ s}^{-1}$ (3) $2 \times 10^{-4} \text{ s}^{-1}$ (4) 10^{-4} s^{-1}

52. For the reaction $2A + B \rightarrow C$, the values of initial rate at different reactant concentrations are given in the table below. The rate law for the reaction is : [JEE MAIN -2019(APR.)]

[A] (mol L^{-1})	[B] (mol L^{-1})	Initial Rate ($\text{mol L}^{-1}\text{s}^{-1}$)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

- (1) $\text{Rate} = k[A][B]$ (2) $\text{Rate} = k[A]^2[B]^2$ (3) $\text{Rate} = k[A]^2[B]$ (4) $\text{Rate} = k[A][B]^2$

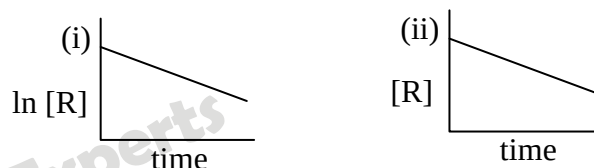
53. For a reaction scheme $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, [JEE MAIN -2019(APR.)]

If the rate of formation of B is set to be zero then the concentration of B is given by:

- (1) $k_1 k_2 [A]$ (2) $\left(\frac{k_1}{k_2}\right) [A]$ (3) $(k_1 - k_2) [A]$ (4) $(k_1 + k_2) [A]$

54. The given plots represent the variation of the concentration of a reactant R with time for two different reactions (i) and (ii). The respective orders of the reactions are :

[JEE MAIN -2019(APR.)]



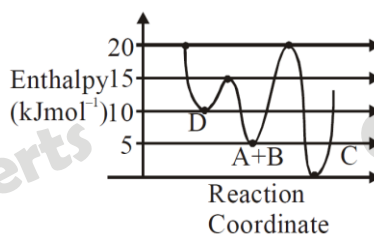
- (1) 0, 1 (2) 1, 1 (3) 0, 2 (4) 1, 0

55. Consider the given plot of enthalpy of the following reaction between A and B.



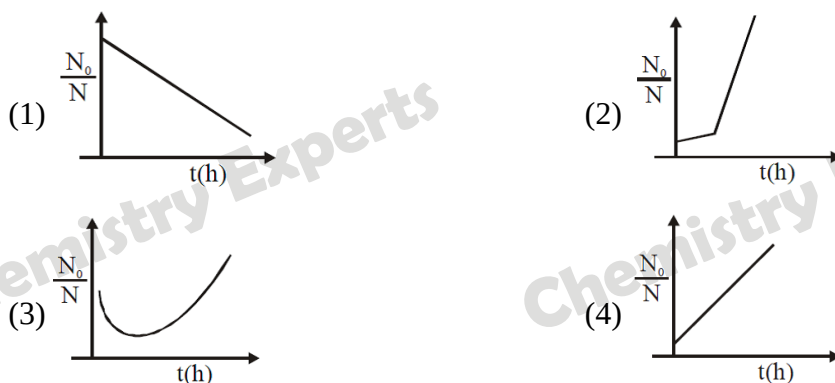
[JEE MAIN -2019(APR.)]

Identify the incorrect statement.



- (1) Formation of A and B from C has highest enthalpy of activation.
 (2) Activation enthalpy to form C is 5 kJ mol^{-1} less than that to form D.
 (3) D is kinetically stable product.
 (4) C is the thermodynamically stable product.
56. A bacterial infection in an internal wound grows as $N'(t) = N_0 \exp(lt)$, where the time t is in hours. A dose of antibiotic, taken orally, needs 1 hour to reach the wound. Once it reaches there, the bacterial population goes down as $\frac{dN}{dt} = -5N^2$. [JEE MAIN -2019(APR.)]

What will be the plot of $\frac{N_0}{N}$ vs. t after 1 hour?



57. For the reaction of H_2 with I_2 , the rate constant is $2.5 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 327°C and $1.0 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 527°C . The activation energy for the reaction, in kJ mol^{-1} is :

$$(R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1})$$

[JEE MAIN -2019(APR.)]

- (1) 150 (2) 59 (3) 166 (4) 72

58. In the following reaction; $x\text{A} \rightarrow y\text{B}$

[JEE MAIN -2019(APR.)]

$$\log_{10} \left[-\frac{d[\text{A}]}{dt} \right] = \log_{10} \left[\frac{d[\text{B}]}{dt} \right] + 0.3010$$

'A' and 'B' respectively can be:

- (1) C_2H_4 and C_4H_8 (2) C_2H_2 and C_6H_6
 (3) N_2O_4 and NO_2 (4) n-Butane and Iso-butane

59. NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per the equation,



The initial concentration of N_2O_5 is 3.00 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is:

[JEE MAIN -2019(APR.)]

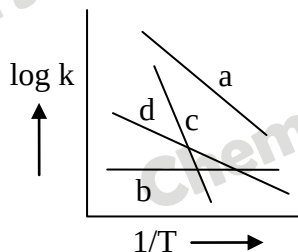
- (1) $1.667 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$ (2) $2.083 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$
 (3) $4.167 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$ (4) $8.333 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

60. The rate of a certain biochemical reaction at physiological temperature (T) occurs 10^6 times faster with enzyme than without. The change in the activation energy upon adding enzyme is :

[Jee Main (Jan. 2020)]

- (1) $+6RT$ (2) $-6(2.303)RT$ (3) $+6(2.303)RT$ (4) $-6RT$

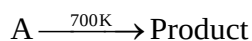
61. Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the activation energies of these reactions?



[Jee Main (Jan. 2020)]

- (1) $E_a > E_c > E_d > E_b$ (2) $E_b > E_a > E_d > E_c$
 (3) $E_c > E_a > E_d > E_b$ (4) $E_b > E_d > E_c > E_a$

62. For following reactions



It was found that the E_a is decreased by 30 kJ/mol in the presence of catalyst. If the rate remains unchanged, the activation energy for catalysed reaction is (Assume pre exponential factor is same)

[Jee Main (Jan. 2020)]

- (1) 105 kJ/mol (2) 75 kJ/mol (3) 135 kJ/mol (4) 198 kJ/mol

63. The results given in the below table were obtained during kinetic studies of the following reaction:



Experiment	[A]/molL ⁻¹	[B]/molL ⁻¹	Initial rate/ mol ⁻¹ min ⁻¹
I	0.1	0.1	6.00×10^{-3}
II	0.1	0.2	2.40×10^{-2}
III	0.2	0.1	1.20×10^{-2}
IV	X	0.2	7.20×10^{-2}
V	0.3	Y	2.88×10^{-1}

X and Y in the given table are respectively :

[Jee-Main (Sep) 2020]

- (1) 0.4, 0.3 (2) 0.3, 0.3 (3) 0.3, 0.4 (4) 0.4, 0.4

64. It is true that :

- (1) A first order reaction is always a single step reaction
 (2) A zero order reaction is a multistep reaction
 (3) A second order reaction is always a multistep reaction
 (4) A zero order reaction is a single reaction

[Jee-Main (Sep) 2020]

65. For the reaction $2A + 3B + \frac{3}{2}C \rightarrow 3P$, which statement is correct? [Jee-Main (Sep) 2020]

(1) $\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$

(2) $\frac{dn_A}{dt} = \frac{3}{2} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$

(3) $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$

(4) $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$

66. A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentration of A and B are equal initially, the time required for the initially, the time required for the concentration of A to be four times that of B (in s) is : (Use $\ln 2 = 0.693$)

[Jee-Main (Sep) 2020]

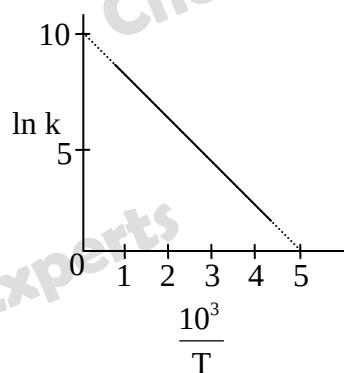
(1) 180

(2) 900

(3) 120

(4) 300

67. The rate constant (k) of a reaction is measured at different temperature (T), and the data are plotted in the given figure. The activation energy of the reaction in kJ mol^{-1} is : (R is constant)



[Jee-Main (Sep) 2020]

(1) $2R$

(2) $1/R$

(3) $2/R$

(4) R

EXERCISE # JEE ADVANCED

- Which of the following statement(s) is (are) correct [JEE 1999]
 (A) A plot of $\log K_p$ versus $1/T$ is linear
 (B) A plot of $\log [X]$ versus time is linear for a first order reaction, $X \longrightarrow P$
 (C) A plot of $\log P$ versus $1/T$ is linear at constant volume.
 (D) A plot of P versus $1/V$ is linear at constant temperature.
- The rate constant for an isomerisation reaction $A \rightarrow B$ is $4.5 \times 10^{-3} \text{ min}^{-1}$. If the initial concentration of A is 1 M. Calculate the rate of the reaction after 1 h. [JEE 1999]
- The rate constant for the reaction [JEE 2000]

$$2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$$
 is $3.0 \times 10^{-5} \text{ sec}^{-1}$ if the rate is $2.4 \times 10^{-5} \text{ mol litre}^{-1} \text{ sec}^{-1}$, then the concentration of N_2O_5 (in mol litre^{-1}) is
 (A) 1.4 (B) 1.2 (C) 0.004 (D) 0.8
- A hydrogenation reaction is carried out at 500 K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. Calculate the activation energy of the reaction if the catalyst lowers the activation barrier by 20 kJmol^{-1} . [JEE 2000]
- If I is the intensity of absorbed light and C is the concentration of AB for the photochemical process
 $\text{AB} + h\nu \longrightarrow \text{AB}$, the rate of formation of AB is directly proportional to [JEE 2001]
 (A) C (B) I (C) I^2 (D) CI
- The rate of a first order reaction is $0.04 \text{ mole litre}^{-1} \text{ s}^{-1}$ at 10 minutes and $0.03 \text{ mol litre}^{-1} \text{ s}^{-1}$ at 20 minutes after initiation. Find the half life of the reaction. [JEE 2001]
- Consider the chemical reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$. The rate of this reaction can be expressed in term of time derivative of concentration of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ or $\text{NH}_3(\text{g})$. Identify the correct relationship amongst the rate expressions. [JEE 2002]
 (A) $\text{Rate} = -d[\text{N}_2]/dt = -1/3 d[\text{H}_2]/dt = 1/2 d[\text{NH}_3]/dt$
 (B) $\text{Rate} = -d[\text{N}_2]/dt = -3 d[\text{H}_2]/dt = 2d[\text{NH}_3]/dt$
 (C) $\text{Rate} = d[\text{N}_2]/dt = 1/3 d[\text{H}_2]/dt = 1/2 d[\text{NH}_3]/dt$
 (D) $\text{Rate} = -d[\text{N}_2]/dt = -d[\text{H}_2]/dt = d[\text{NH}_3]/dt$

8. In a first order reaction the concentration of reactant decreases from 800 mol/dm^3 to 50 mol/dm^3 in $2 \times 10^4 \text{ sec}$. The rate constant of reaction in sec^{-1} is [JEE 2003]
 (A) 2×10^4 (B) 3.45×10^{-5} (C) 1.386×10^{-4} (D) 2×10^{-4}
9. The reaction, $X \longrightarrow \text{Product}$ follows first order kinetics. In 40 minutes the concentration of X changes from 0.1 M to 0.025 M. Then the rate of reaction when concentration of X is 0.01 M [JEE 2004]
 (A) $1.73 \times 10^{-4} \text{ M min}^{-1}$ (B) $3.47 \times 10^{-5} \text{ M min}^{-1}$
 (C) $3.47 \times 10^{-4} \text{ M min}^{-1}$ (D) $1.73 \times 10^{-5} \text{ M min}^{-1}$
10. For the given reactions, $A + B \rightarrow \text{Products}$, following data were obtained. [JEE-2004]
- | | $[A_0]$ | $[B_0]$ | $R_0(\text{mol L}^{-1} \text{ s}^{-1})$ |
|----|---------|---------|---|
| 1. | 0.1 | 0.2 | 0.05 |
| 2. | 0.2 | 0.2 | 0.10 |
| 3. | 0.1 | 0.1 | 0.05 |
- (a) Write the rate law expression
 (b) Find the rate constant
11. $2X(g) \longrightarrow 3Y(g) + 2Z(g)$ [JEE 2005]
- | Time (in Min) | 0 | 100 | 200 |
|-------------------------------------|-----|-----|-----|
| Partial pressure of X (in mm of Hg) | 800 | 400 | 200 |
- Assuming ideal gas condition, Calculate
 (a) Order of reaction (b) Rate constant
 (c) Time taken for 75% completion of reaction (d) Total pressure when $P_X = 700 \text{ mm}$.
12. Which of the following statement is incorrect about order of reaction? [JEE 2005]
 (A) Order of reaction is determined experimentally
 (B) It is the sum of power of concentration terms in the rate law expression
 (C) It does not necessarily depend on stoichiometric coefficients
 (D) Order of the reaction can not have fractional value.
13. Consider a reaction $aG + bH \rightarrow \text{Products}$. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is : [JEE 2007]
 (A) 0 (B) 1 (C) 2 (D) 3

14. Under the same reaction conditions, initial concentration of $1.386 \text{ mol dm}^{-3}$ of a substance becomes half in 40 seconds and 20 seconds through first order and zero order kinetics, respectively. Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constants for first order (k_1) and zero order (k_0) of the reactions is [JEE 2008]

(A) $0.5 \text{ mol}^{-1} \text{ dm}^3$ (B) 1.0 mol dm^{-3} (C) 1.5 mol dm^{-3} (D) $2.0 \text{ mol}^{-1} \text{ dm}^3$

15. For a first order reaction $A \rightarrow P$, the temperature (T) dependent rate constant (k) was found to follow the equation $\log k = - (2000) \frac{1}{T} + 6.0$. The pre-exponential factor A and the activation E_a , respectively, are [JEE-2009]

(A) $1.0 \times 10^6 \text{ s}^{-1}$ and 9.2 kJ mol^{-1} (B) 6.0 s^{-1} and 16.6 kJ mol^{-1}
 (C) $1.0 \times 10^6 \text{ s}^{-1}$ and 16.6 kJ mol^{-1} (D) $1.0 \times 10^6 \text{ s}^{-1}$ and 38.3 kJ mol^{-1}

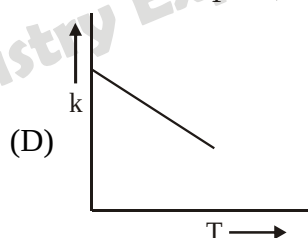
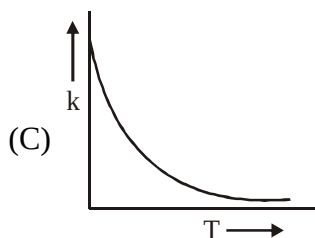
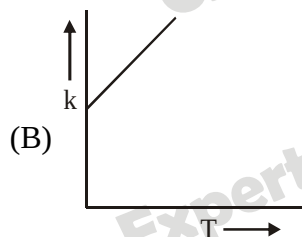
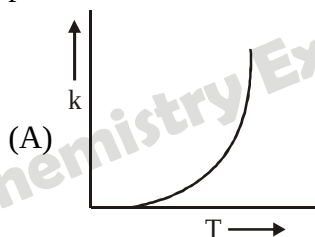
16. The concentration of R in the reaction $R \rightarrow P$ was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t(min)	0	0.05	0.12	0.18

The order of the reaction is.

[JEE 2010]

17. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is – [JEE 2010]



18. For the first order reaction



[JEE 2011]

- (A) the concentration of the reactant decreases exponentially with time
 (B) the half-life of the reaction decreases with increasing temperature.
 (C) the half-life of the reaction depends on the initial concentration of the reactant.
 (D) the reaction proceeds to 99.6% completion in eight half-life duration.

19. An organic compound undergoes first-order decomposition. The time taken for its decomposition to $1/8$ and $1/10$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of

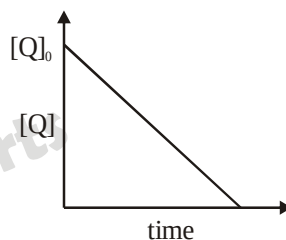
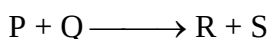
$$\frac{[t_{1/8}]}{t_{1/10}} \times 10 ?$$

(take $\log_{10} 2 = 0.3$)

[JEE 2012]

20. In the reaction :

[JEE 2013]



the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is -

- (A) 2 (B) 3 (C) 0 (D) 1

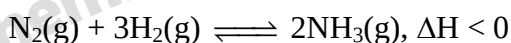
21. For the elementary reaction $M \rightarrow N$, the rate of disappearance of M increases by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect to M is

[JEE 2014]

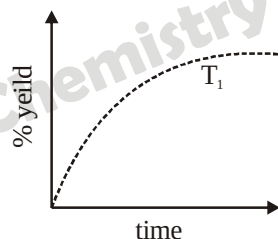
- (A) 4 (B) 3 (C) 2 (D) 1

22. The % yield of ammonia as a function of time in the reaction

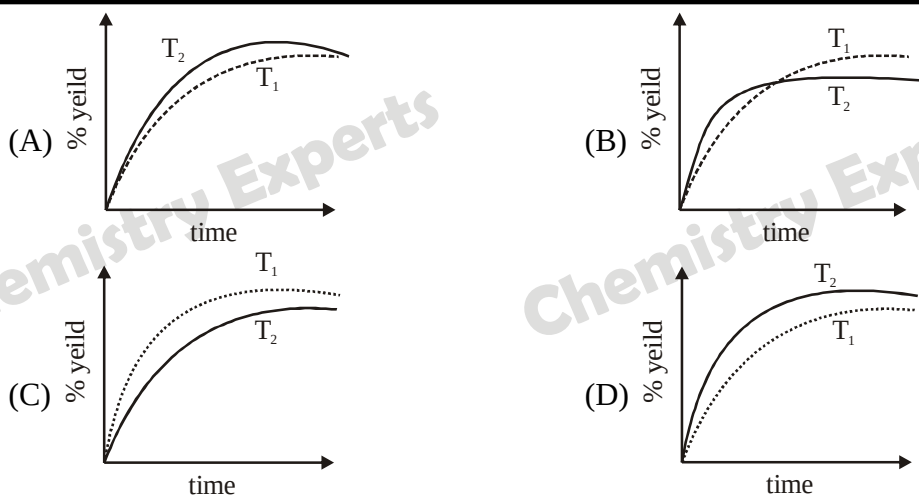
[JEE 2015]



at (P, T_1) is given below -



If this reaction is conducted at (P, T_2) , with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by -



23. In dilute aqueous H_2SO_4 , the complex diaquodioxalatoferrate(II) is oxidized by MnO_4^- . For this reaction, the ratio of the rate of change of $[\text{H}^+]$ to the rate of change of $[\text{MnO}_4^-]$ is.

[JEE 2015]

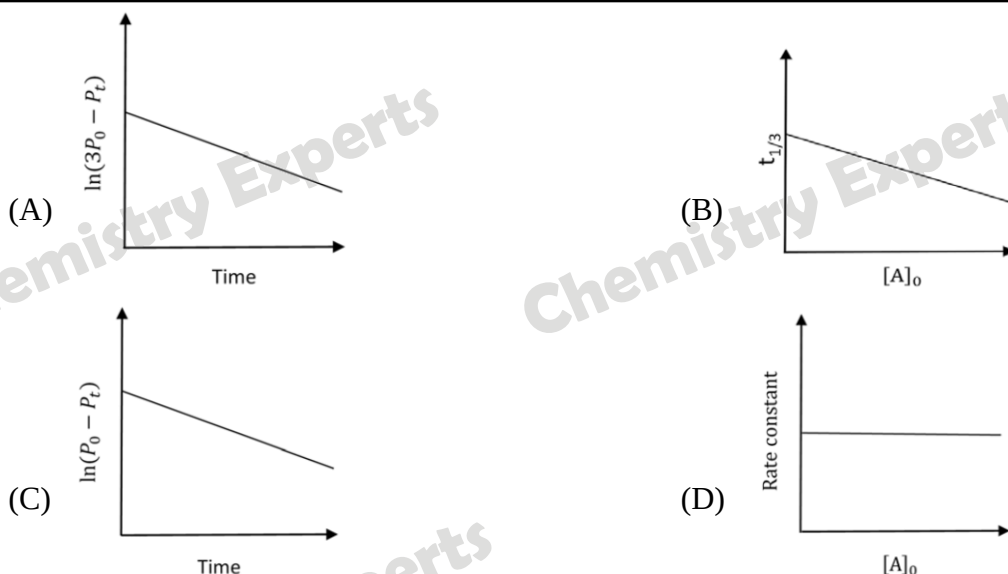
24. According to the Arrhenius equation, [JEE 2016]
- (A) A high activation energy usually implies a fast reaction
 - (B) Rate constant increase with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy
 - (C) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant
 - (D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

25. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) [JEE 2017]

- (A) The activation energy of the reaction is unaffected by the value of the steric factor
- (B) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation
- (C) Since $P = 4.5$, the reaction will not proceed unless an effective catalyst is used
- (D) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

26. For a first order reaction $\text{A(g)} \rightarrow 2\text{B(g)} + \text{C(g)}$ at constant volume and 300 K, the total pressure at the beginning ($t = 0$) and at time t are P_0 and P_t , respectively. Initially, only A is present with concentration $[\text{A}]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach $1/3^{\text{rd}}$ of its initial value. The correct option(s) is (are)
- (Assume that all these gases behave as ideal gases)

[JEE 2018]



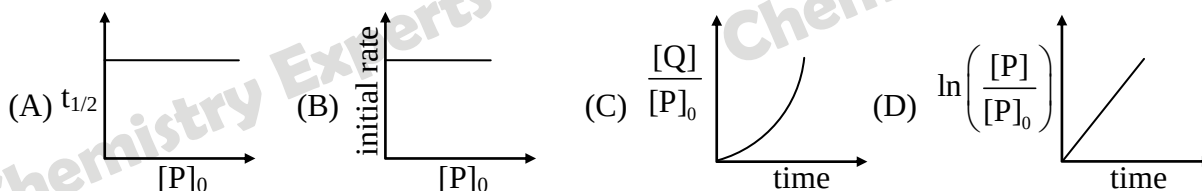
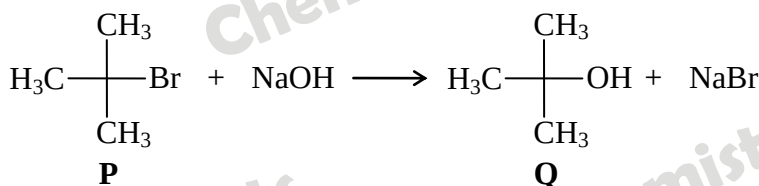
27. Consider the kinetic data given in the following table for the reaction $A + B + C \rightarrow \text{Product}$.

Experiment No.	[A] (mol dm ⁻³)	[B] (mol dm ⁻³)	[C] (mol dm ⁻³)	Rate of reaction (mol dm ³ s ⁻¹)
1	0.2	0.1	0.1	6.0×10^{-5}
2	0.2	0.2	0.1	6.0×10^{-5}
3	0.2	0.1	0.2	1.2×10^{-4}
4	0.3	0.1	0.1	9.0×10^{-5}

The rate of the reaction for $[A] = 0.15 \text{ mol dm}^{-3}$, $[B] = 0.25 \text{ mol dm}^{-3}$ and $[C] = 0.15 \text{ mol dm}^{-3}$ is found to be $Y \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$. The value of Y is _____ [JEE 2019]

28. The decomposition reaction $2\text{N}_2\text{O}_5(\text{g}) \xrightarrow{\Delta} 2\text{N}_2\text{O}_4(\text{g}) + \text{O}_2(\text{g})$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After $Y \times 10^3 \text{ s}$, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is $5 \times 10^{-4} \text{ s}^{-1}$, assuming ideal gas behavior, the value of Y is _____ [JEE 2019]

29. Which of the following plots is(are) correct for the given reaction? [JEE 2020]
 ($[P]_0$ is the initial concentration of P)



ANSWER KEY**EXERCISE # (JEE-MAIN)**

1.	3	2.	4	3.	1	4.	1	5.	4	6.	3	7.	1
8.	1	9.	2	10.	4	11.	3	12.	1	13.	2	14.	2
15.	1	16.	1	17.	3	18.	2	19.	1	20.	1	21.	2
22.	2	23.	4	24.	1	25.	4	26.	1	27.	2	28.	2
29.	1	30.	2	31.	3	32.	1	33.	3	34.	3	35.	3
36.	3	37.	2	38.	2	39.	1	40.	3	41.	2	42.	1
43.	2	44.	2	45.	3	46.	4	47.	1	48.	3	49.	1
50.	3	51.	4	52.	4	53.	2	54.	4	55.	2	56.	3
57.	3	58.	1	59.	1	60.	2	61.	3	62.	2	63.	3
64.	2	65.	3	66.	2	67.	1						

EXERCISE # JEE ADVANCED

1. ABD 2. $(3.435 \times 10^{-3} \text{ M/min})$ 3. D
4. $100 \text{ KJmol}^{-1} \Rightarrow r = k[A] = k[A]_0 e^{-kt}$ 5. B 6. 24.14 min 7. A
8. C 9. C
10. (a) $R_0 = K[A]_0$ (b) 0.5 sec^{-1}

$$k = \frac{1}{2 \times 10^4} \ln \frac{800}{50} = \frac{1}{2 \times 10^4} \ln 4 = 1.386 \times 10^{-4}$$
11. (a) 1, (b) $6.93 \times 10^{-3} \text{ min}^{-1}$, (c) 200, (d) 950 mm

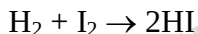
$$k = \frac{1}{40} \ln \frac{0.1}{0.025} = \frac{1}{40} \ln \frac{0.1}{0.025} = \frac{1}{40} \ln 4$$

$$r = k[X] = \frac{1}{40} \ln 4 \times 0.01 = 3.47 \times 10^{-4} \text{ M/minute}$$
12. D 13. D 14. A 15. D 16. Zero 17. A 18. ABD
19. 9 20. D 21. B 22. B 23. 8 24. BCD 25. AD
26. AD 27. 6.75 28. 2.30 29. A

HINT AND SOLUTIONS

EXERCISE # (JEE-MAIN)

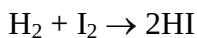
1. (3)



When 1 mole of H_2 and 1 mole of I_2 reacts, 2 moles of HI are formed in the same time interval.

Thus the rate may be expressed as
$$\frac{-\Delta[\text{H}_2]}{\Delta t} = \frac{-\Delta[\text{I}_2]}{\Delta t} = \frac{1}{2} \frac{\Delta[\text{HI}]}{\Delta t}$$

The negative sign signifies a decrease in concentration of the reactant with increase of time.



2. (4)

Order is the sum of the power of the concentrations terms in rate law expression. $R = [\text{A}][\text{B}]^2$

Thus, order of reaction = $1 + 2 = 3$.

3. (1)

$$k = \left(\frac{\text{mol}}{\text{L}} \right)^{1-n} \text{time}^{-1}, \text{ where } n = \text{order}$$

6. (3)

Surface catalysed reaction are zero order reaction

7. (1)

In Arrhenius equation, $k = Ae^{-E_a/RT}$

k = rate constant, A = frequency factor

T = temperature, R = gas constant, E_a = energy of activation.

This equation can be used for calculation of energy of activation.

8. (1)

$$r = K[\text{NO}]^2 [\text{O}_2]^1$$

$$P \propto \frac{1}{V} \Rightarrow P_f = 2P_i$$

9. (2)

$$\text{Rate}_1 = k [\text{A}]^n [\text{B}]^m$$

On doubling the concentration of A and halving the concentration of B

$$\text{Rate}_2 = k [2\text{A}]^n [\text{B}/2]^m$$

$$\text{Ratio between new and earlier rate} = \frac{k[2\text{A}]^n [\text{B}/2]^m}{k[\text{A}]^n [\text{B}]^m} = 2^n \times \left(\frac{1}{2} \right)^m = 2^{n-m}$$

10. (4)



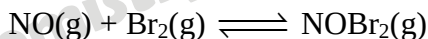
The value of k (velocity constant) is always independent of the concentration of reactant and it is a function of temperature only.

13. (2)

$$\Delta H = E_{a,f} - E_{a,b} > 0$$

$$E_{a,f} > E_{a,b}$$

17. (3)



$$\text{Rate of the reaction (r)} = k [\text{NOBr}_2] [\text{NO}]$$

$$\text{where } [\text{NOBr}_2] = K_c [\text{NO}] [\text{Br}_2]$$

$$r = k \cdot K_c [\text{NO}] [\text{Br}_2] [\text{NO}]$$

$$r = k' [\text{NO}]^2 [\text{Br}_2]$$

The order of the reaction with respect to $\text{NO(g)} = 2$.

18. (2)

Generally, molecularity of simple reactions is equal to the sum of the number of molecules of reactants involved in the balanced stoichiometric equation. Thus, a reaction involving two different reactants can never be unimolecular.

19. (1)

$$\Delta H_R = E_f - E_b = 180 - 200 = -20 \text{ kJ mol}^{-1}$$

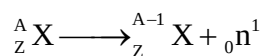
The correct answer for this question should be -20 kJ mol^{-1} . But no option given is correct. Hence we can ignore sign and select option

20. (1)

$$K = \frac{\ln 2}{30} \text{ days}^{-1}$$

$$t = \frac{1}{K} \ln 10 = \frac{30}{\ln 2} \times \ln 10 = 100 \text{ days}$$

21. (2)



22. (2)

$$-2 \frac{d[A]}{dt} = + \frac{1}{2} \frac{d[B]}{dt}$$

23. (4)

$$t_{99\%} = \frac{1}{K} \ln \frac{100}{1}$$

$$k = \frac{\ln 2}{6.93} = \frac{1}{10} \text{ min}^{-1}$$

$$t_{99\%} = 10 \ln 100 = 20 \ln 10 = 46.06 \text{ min}$$

24. (1)

From mechanism [A]

$$r = K[\text{Cl}_2][\text{H}_2\text{S}]$$

From mechanism [B]

$$r = K[\text{Cl}_2][\text{HS}]$$

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{HS}^-]}{[\text{H}_2\text{S}]}$$

$$r = K^1[\text{Cl}_2][\text{H}_2\text{S}][\text{H}^+]^{-1}$$

25. (4)

A $\longrightarrow$ product

For zero order reaction

$$t_{1/2} \propto \frac{1}{a^{n-1}} \quad a = \text{initial concentration of reactant}$$

$$t_{1/2} \propto a$$

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \frac{a_1}{a_2} ; \frac{1}{(t_{1/2})_2} = \frac{2}{0.50}$$

$$t_{1/2} = \frac{0.5}{2} = 0.25 \text{ h.}$$

26. (1)

$$r_t = r_i (\text{T.C.})^{\Delta T/10}$$

$$= r_i \times (2)^{50/10} = 32 r_i$$

27. (2)

$$K_1 = A_1 e^{-E_{a1}/RT}$$

$$K_2 = A_2 e^{-E_{a2}/RT}$$

$$\frac{K_1}{K_2} = \frac{A_1}{A_2} e^{(E_{a2} - E_{a1})/RT}$$

$$K_1 = K_2 A \times e^{E_{a1}/RT}$$

28. (2)

$$[A]_t = \frac{[A]_0}{2^n}$$

$$n = 2 \text{ so } t_{75\%} = 40 \text{ min}$$

$$t_{75\%} = 2 \times \frac{\ln 2}{K} = 40$$

$$K = \frac{\ln 2}{20} \text{ min}^{-1}$$

$$r = K[A] = \frac{\ln 2}{20} \times 0.01$$

$$= 3.47 \times 10^{-4} \text{ M/min}$$

29. (1)

$$\log \frac{K_2}{K_1} = \frac{-E_a}{2.030R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\frac{K_2}{K_1} = 2 ; T_2 = 310 \text{ K and } T_1 = 300 \text{ K}$$

$$\Rightarrow \log 2 = \frac{-E_a}{2.303 \times 8.134} \left(\frac{1}{310} - \frac{1}{300} \right)$$

$$\Rightarrow E_a = 53598.6 \text{ J/mol} = 53.6 \text{ KJ/mol}$$

Ans is (1)

30. (2)

$$1.2 \times 10^{-3} = K (0.1)^x (0.1)^y$$

$$1.2 \times 10^{-3} = K (0.1)^x (0.2)^y$$

$$2.4 \times 10^{-3} = K (0.2)^x (0.1)^y$$

$$R = K [A]^1 [B]^0$$

31. (3)

More than 3 molecules can't collide simultaneously.

32. (1)



$$t = 0 \quad 50 \quad 0 \quad 0$$

$$t = 30 \quad 50 - 2x \quad 4x \quad x$$

$$\Rightarrow 87.5 = 50 + 3x$$

$$\Rightarrow 3x = 37.5 \quad \Rightarrow \quad x = 12.5$$

$$\Rightarrow P_{\text{N}_2\text{O}_5} \text{ after 30 min} = 50 - 25 = 25$$

$$\Rightarrow t_{1/2} = 30 \text{ min.}$$

$$\text{Hence after 60 min, (two half lives), } P_{\text{N}_2\text{O}_5} \text{ remaining} = \frac{50}{4} = 12.5 \text{ torr.}$$

$$\Rightarrow \text{Hence decrease in } P_{\text{N}_2\text{O}_5} = 50 - 12.5 = 37.5 \text{ torr.}$$

$$\Rightarrow P_{\text{NO}_2} = 2 \times 37.5 = 75 \text{ torr}$$

$$P_{\text{O}_2} = \frac{37.5}{2} = 18.75 \text{ torr}$$

$$\Rightarrow P_{\text{total}} = 12.5 + 75 + 18.75 = 106.25 \text{ torr.}$$

33. (3)

$$\Delta H = E_{\text{af}} - E_{\text{ab}}$$

$$\Rightarrow -40 = 2x - 3x$$

$$\Rightarrow E_{\text{af}} = 80 \text{ kJ/mol}$$

$$E_{\text{ab}} = 120 \text{ kJ/mol}$$

34. (3)

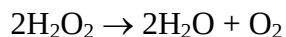
In 50 minutes, concentration of H_2O_2 becomes $\frac{1}{4}$ of initial.

$$\Rightarrow 2 \times t_{1/2} = 50 \text{ minutes}$$

$$\bullet \quad t_{1/2} = 25 \text{ minutes}$$

$$\bullet \quad k = \frac{0.693}{25} \text{ per minute}$$

$$r_{\text{H}_2\text{O}_2} = \frac{0.693}{25} \times 0.05 = 1.386 \times 10^{-3}$$



$$r_{\text{O}_2} = \frac{1}{2} \times r_{\text{H}_2\text{O}_2} ; \quad r_{\text{O}_2} = 0.693 \times 10^{-3}$$

$$r_{\text{O}_2} = 6.93 \times 10^{-4} \text{ mol/minute} \times \text{litre}$$

35. (3)

$K_i < K_{ii}$ so Step-(i) is RDS

36. (3)

$$r_1 = K[A][B] \quad \dots\dots(1)$$

$$r_2 = K[A] \times 3[B] \quad \dots\dots(2)$$

$$r_2 = 3r_1$$

37. (2)

$$K_1 = A.e^{-E_{a2}/RT}$$

$$K_1 = A.e^{-E_{a1}/RT}$$

$$E_{a1} - E_{a2} = 10 \text{ kJ/mol}$$

$$\frac{K_2}{K_1} = \frac{Ae^{-E_{a2}/RT}}{Ae^{-E_{a1}/RT}} = e^{(E_{a1} - E_{a2})}$$

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{10 \times 10^3}{8.314 \times 300} = 4$$

38. (2)

$$\ln \left(\frac{K_2}{K_1} \right)_A = \frac{(E_a)_A}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln 2 = \frac{(E_a)_A}{R} \times \left(\frac{1}{300} - \frac{1}{310} \right)$$

$$\ln 2 = \frac{(E_a)_A}{R} \times \frac{10}{300 \times 310} \quad \dots\dots(1)$$

$$\frac{(E_a)_A}{R} \times \frac{10}{300 \times 310} = \frac{2(E_a)_B}{R} \times \left(\frac{1}{300} - \frac{1}{T_2} \right)$$

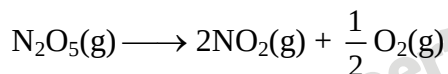
$$T_2 = 304.92$$

39. (1)

$$\ln 4 = \frac{E_a}{R} \times \left(\frac{1}{300} - \frac{1}{310} \right)$$

$$E_a = \frac{2 \times 0.693 \times 8.314 \times 300 \times 310}{10 \times 1000} \text{ kJ/mol} = 107.2 \text{ kJ/mol}$$

40. (3)



$$t = 0 \quad 50 \quad 0 \quad 0$$

$$t = t \quad 50 - x \quad 2x \quad \frac{x}{2}$$

$$t = \infty \quad 0 \quad 100 \quad 50$$

$$\Delta P = \left(P_0 + \frac{3x}{2} \right) - P_0 = 87.5 - 50 \Rightarrow 1.5x = 37.5 \Rightarrow x = 25$$

If $P_0 = 50$ and $x = 25$, $t = 50$ min is $t_{1/2}$

At $t = 100$ min

$$P_t = P_0 + \frac{3x}{2} = 50 + 1.5 \times 37.5 = 106.25 \text{ mm hg}$$

41. (2)

$$t_{1/2} = \frac{\ln 2}{K} \Rightarrow K = \frac{\ln 2}{10} \text{ days}^{-1}$$

$$t = \frac{1}{K} \ln \frac{1}{3/4} = \frac{1}{K} = \ln \frac{4}{3} = 4.1 \text{ days}$$

42. (1)

$$A_0 \xrightarrow{t} \frac{A_0}{2} \xrightarrow{t} \frac{A_0}{4}$$

Same % is reacted in same time interval. So it is 1st order

43. (2)

$$r = K[\text{acetaldehyde}]^n$$

$$1 = K \times [363 \times 0.95]^n \quad - (1)$$

$$0.5 = K \times [365 \times 0.67]^n \quad - (2)$$

$$2 = \left(\frac{0.95}{0.67} \right)^n = (2)^{n/2} \Rightarrow \frac{n}{2} = 1 \Rightarrow n = 2$$

44. (2)

$$r = k[A]^x[B]^y$$

When double the conc. of A, rate is also doubled, So order w.r.t. A is 1. But by changing the conc. of B rate does not change so order w.r.t. B is zero.

$$R = k[A]^1$$

$$6.93 \times 10^{-3} = k \times 0.1$$

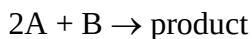
$$K = 6.93 \times 10^{-2}$$

$$\frac{k_A}{2} = k_r$$

$$k_A = 2 \times 3.93 \times 10^{-2}$$

$$t_{\frac{1}{2}} = \frac{0.693}{2 \times 6.93 \times 10^{-2}} = 5 \text{ min}$$

45. (3)



$$\text{R.O.R} = k[A]^x [B]^y$$

$$0.3 = k[A]^x [B]^y \quad \dots(1)$$

$$2.4 = k[2A]^x [2B]^y \quad \dots(2)$$

$$0.6 = k[2A]^x [B]^y \quad \dots(3)$$

Solving the above equations, we get,

$$x = 1 \text{ and } y = 2$$

i.e. order w.r.t A is 1

and order w.r.t B is 2

46. (4)

$$k = Ae^{-E_a/RT}$$

Graph(I): Higher is the value of E_a , lower will be rate constant for the reaction.

Graph(II): On increasing temperature rate constant always increases.

47. (1)

$$\text{Rate of formation of A} = 2k_1[A_2]$$

$$\text{Rate of disappearance of A} = k_{-1}[A]^2$$

$$\therefore \text{Net rate of formation of A} = 2k_1[A_2] - 2k_{-1}[A]^2$$

48. (3)

$$K = Ae^{-E_a/RT}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$y = C - mx$$

$$\text{slope} = -E_a = -y$$

$$E_a = y$$

49. (1)

For zero order :

$$k = \frac{C_0}{2 \times t_{\frac{1}{2}}} \Rightarrow k = \frac{0.2}{2 \times 6} \Rightarrow k = \frac{1}{60}$$

$$\text{Now, } \frac{C_0 - C_t}{k} = t \Rightarrow t = \frac{0.5 - 0.2}{1} \times 60 = 18 \text{ hrs}$$

50. (3)

For zero order reaction $t_{\frac{1}{2}} = \frac{[A]_0}{2K} = \frac{5}{2 \times 0.05} = 50 \text{ years}$

51. (4)

$$\therefore \ln k = \ln A - \frac{E_a}{R} \times \left(\frac{1}{T} \right)$$

$$\therefore \frac{-E_a}{R} = -4606$$

$$\therefore \ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$2.303 \log \left(\frac{k_2}{10^{-5}} \right) = 4606 \left[\frac{1}{400} - \frac{1}{500} \right]$$

$$\therefore k_2 = 10^{-4}$$

52. (4)

Let Rate = $k[A]^m [B]^n$

On putting given data

$$0.045 = k (0.5)^m (0.05)^n \quad \dots(1)$$

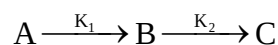
$$0.090 = k (0.10)^m (0.05)^n \quad \dots(2)$$

$$0.720 = k (0.20)^m (0.10)^n \quad \dots(3)$$

On solving

$$m = 1, n = 2$$

53. (2)



Net, rate of formation of B = $\frac{d[B]}{dt} = K_1[A] - k_2[B]$

$$\text{As, } \frac{d[B]}{dt} = 0$$

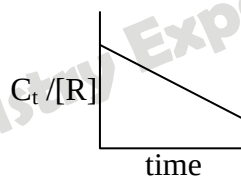
$$\Rightarrow K_1[A] - K_2[B] = 0$$

$$\Rightarrow [B] = \frac{K_1}{K_2} [A]$$

54. (4)

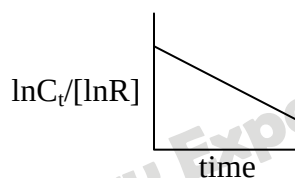
For zero order reaction

$$C_t = C_0 - kt$$

For 1st order reaction

$$C_t = C_0 e^{-kt}$$

$$\ln C_t = \ln C_0 - kt$$



55. (2)

Activation energy to from C is 5kJ/mole greater than that of to from D

$$(E_a)_C = 15 \text{ kJ/mole}$$

$$(E_a)_D = 10 \text{ kJ/mole}$$

56. (3)

$$\frac{N_0}{N} = e^{-t}$$

before 1 Hour

$$\frac{N_0}{N} \text{ Will decrease}$$

$$\int_{N_0}^N \frac{dN}{N^2} = \int -5 dt$$

$$\frac{1}{N} = \frac{1}{N_0} + 5t$$

$$\frac{N_0}{N} = 1 + 5t N_0$$

After 1Hr. $\frac{N_0}{N}$ Will increase.

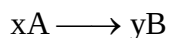
57. (3)

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \left(\frac{1}{2.5 \times 10^{-4}} \right) = \frac{E_a}{8.314} \left(\frac{1}{600} - \frac{1}{800} \right)$$

$$E_a = 166 \text{ KJ}$$

58. (1)



$$-\frac{1}{x} \frac{d[\text{A}]}{dt} = +\frac{1}{y} \frac{d[\text{B}]}{dt}$$

$$-\frac{d[\text{A}]}{dt} = \frac{x}{y} \frac{d[\text{B}]}{dt}$$

Taking log both the side,

$$\log \left(\frac{-d[\text{A}]}{dt} \right) = \log \left(\frac{d[\text{B}]}{dt} \right) + \log \left(\frac{x}{y} \right)$$

$$\Rightarrow \log \left(\frac{x}{y} \right) = 0.3010 = \log 2 \quad \Rightarrow \quad \frac{x}{y} = \frac{2}{1} \Rightarrow \text{A is } \text{C}_2\text{H}_4 \text{ \& B is } \text{C}_4\text{H}_8$$

59. (1)



$$t = 0 \quad 3\text{M}$$

$$t = 30 \quad 2.75 \text{ M}$$

$$\frac{-\Delta[\text{N}_2\text{O}_5]}{\Delta t} = \frac{0.25}{30}$$

$$\frac{1}{2} \times \frac{-\Delta(\text{N}_2\text{O}_5)}{\Delta t} = \frac{1}{4} \times \frac{\Delta[\text{NO}_2]}{\Delta t}$$

$$\frac{\Delta[\text{NO}_2]}{\Delta t} = \frac{0.25}{30} \times 2 = 1.667 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$$

60. (2)

Sol. $K_2 = A e^{-E_a/RT}$ (in presence of catalyst)

$K_1 = A e^{-E_a/RT}$ (in absence of catalyst)

$$\frac{K_2}{K_1} = e^{-\frac{(E_a' - E_a)}{RT}}$$

$$10^6 = e^{-\frac{(E_a' - E_a)}{RT}}$$

$$\frac{\Delta E_a}{RT} = -\ln 10^6$$

$$\Delta E_a = -6 \times 2.303 \times RT$$

61. (3)

Sol. For reaction E_a is more than change with temp will be more.

Magnitude of slop : $c > a > d > b$

62. (2)

Sol. $K = Ae^{-\frac{E_a}{R \times 700}}$
 $= Ae^{-\frac{[E_a - 30]}{R \times 500}}$

$$\Rightarrow \frac{E_a}{700} = \frac{E_a - 30}{500}$$

$$5E_a = 7E_a - 210$$

$$E_a = 105$$

Activation energy for catalysed reaction = $105 - 30 = 75 \text{ kJ}$

63. (3)

Sol. From rate law

$$r = -\frac{1}{2} \frac{d[A]}{dt} = \frac{-d[B]}{dt}$$

$$r = K[A]^x [B]^y$$

$$6 \times 10^{-3} = K(0.1)^x (0.1)^y \dots\dots(1)$$

$$2.4 \times 10^{-2} = K(0.1)^x (0.2)^y \dots\dots(2)$$

$$1.2 \times 10^{-2} = K(0.2)^x (0.1)^y \dots\dots(3)$$

$$(3) \div (1) \Rightarrow x = 1$$

$$(2) \div (3) \Rightarrow x = 2$$

So, order with respect to A = 1

Order with respect to B = 2

$$(4) \div (3)$$

$$\left(\frac{x}{0.2}\right) \times \left(\frac{0.2}{0.1}\right)^2 = \frac{7.2 \times 10^{-2}}{1.2 \times 10^{-2}}$$

$$x = \frac{6 \times 0.2}{4}$$

$$x = 0.3 \text{ M}$$

$$(5) \div (4)$$

$$\left(\frac{y}{0.2}\right)^2 = \frac{2.88 \times 10^{-1}}{7.2 \times 10^{-2}}$$

$$y^2 = 4 \times 0.2^2$$

$$y = 0.4 \text{ M}$$

64. (2)

Sol. Since for elementary reaction order is equal to molecularity and molecularity of reaction cannot be zero. So zero order reaction cannot be elementary reaction. It is always multistep reaction.

65. (3)

Sol. For reaction : $2A + 3B + \frac{3}{2}C \rightarrow 3P$

$$-\frac{1}{2} \frac{dn_A}{dt} = -\frac{1}{3} \frac{dn_B}{dt} = -\frac{2}{3} \frac{dn_C}{dt} = \frac{1}{3} \frac{dn_P}{dt}$$

$$\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$$

66. (2)

Sol. For 1st order reaction

$$t_{1/2} = \frac{\ln 2}{k} \propto \frac{1}{k}$$

$$k_A = \frac{\ln 2}{300}, k_B = \frac{\ln 2}{180}$$

Initially A and B are equal.

After time t : $[A]_t = [B]_t \times 4$

$$[A]_0 e^{-k_A t} = [B]_0 e^{-k_B t} \times 4$$

$$\frac{e^{-k_A t}}{e^{-k_B t}} = 4$$

$$e^{-[k_A - k_B]t} = 4$$

$$[k_B - k_A] \times t = \ln 4 = 2 \ln 2$$

$$t = \frac{2 \times \ln 2}{k_B - k_A} = \frac{2 \times \ln 2}{\frac{\ln 2}{180} - \frac{\ln 2}{300}} = 900 \text{ sec.}$$

67. (1)

Sol. $K = A e^{-E_a/RT}$

$$\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$$

$$\ln k = \ln A - \frac{E_a}{10^3 \cdot R} \cdot \frac{10^3}{T}$$

$$\text{slope} = \frac{-E_a}{1000 \cdot R} = \frac{-10}{5-0}$$

$$\frac{-E_a}{1000 \cdot R} = -2$$

$$E_a = 2R \times 10^3 \text{ (J mol)}$$

$$= 2R \text{ (kJ mol}^{-1}\text{)}$$

EXERCISE # JEE ADVANCED

2. $(3.435 \times 10^{-3} \text{ M/min})$

4. 100 KJmol^{-1}
 $r = k[A] = k[A]_0 e^{-kt}$

5. (B)

$$r = k[A] \Rightarrow 2.4 \times 10^{-5} = 3 \times 10^{-5} \times [A] \Rightarrow [A] = 0.8 \text{ M}$$

6. 24.14 min

According to Arrhenius equation $K = Ae^{-E_a/RT}$

Let E_a of the reaction in absence of catalyst = $x \text{ kJ mol}^{-1}$

Therefore E_a of the reaction in presence of catalyst = $x - 20 \text{ kJ mol}^{-1}$

The Arrhenius equations in the two conditions can thus be written as

$$K = Ae^{-\frac{x}{R \times 500}} \dots (i) \quad K = Ae^{-\frac{x-20}{R \times 400}} \dots (ii)$$

Dividing equation (i) by (ii), we get

$$e^{-\frac{x}{500R}} = e^{-\frac{x-20}{400R}} \Rightarrow \frac{x}{500} = \frac{x-20}{400} \quad \text{or} \quad x = 100 \text{ kJ mol}^{-1}$$

7. (A)

$$r = k[A] = k[A]_0 e^{-kt}$$

$$0.04 = k[A]_0 e^{-k \times 10} \quad (1)$$

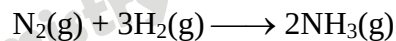
$$0.03 = k[A]_0 e^{-k \times 20} \quad (2)$$

From equation (1) and (2)

$$k = \frac{1}{10} \ln \frac{4}{3} \Rightarrow t_{1/2} = \frac{0.693}{k}$$

9. (C)

For a chemical reaction,



Concentration of N_2 and H_2 are decreased so negative sign is represented in expression, while concentration of NH_3 is increased. So +ve sign is represented in expression. Also, from stoichiometry it can be said that the time during which 1 mole of N_2 will be consumed in the same time 3 moles of H_2 is consumed and 2 moles of NH_3 is formed.

10. (a) $R_0 = K[A]_0$ (b) 0.5 sec^{-1}

$$k = \frac{1}{2 \times 10^4} \ln \frac{800}{50} = \frac{1}{2 \times 10^4} \ln 4 = 1.386 \times 10^{-4}$$

11. (a) 1, (b) $6.93 \times 10^{-3} \text{ min}^{-1}$, (c) 200, (d) 950 mm

$$k = \frac{1}{40} \ln \frac{0.1}{0.025} = \frac{1}{40} \ln \frac{0.1}{0.025} = \frac{1}{40} \ln 4$$

$$r = k[X] = \frac{1}{40} \ln 4 \times 0.01 = 3.47 \times 10^{-4} \text{ M/minute}$$

12. (D)



$$t = 0 \quad 800 \quad - \quad -$$

$$t = t \quad 800 - 2x \quad 3x \quad 2x$$

$$P_t = (800 + 3x)$$

from given data in time 100 min the partial pressure of X decreases from 800 to 400 so $t_{1/2}$ 100 min. Also in next 100 min P_x decreases from 400 to 200 so again $t_{1/2} = 100$ min. Since half life is independent of initial concentration so reaction must be 1st order with respect to X.

$$\text{Rate constant } K = \frac{\ln 2}{t_{1/2}} = 6.93 \times 10^{-3} \text{ min}^{-1}.$$

$$\text{Time taken for 75\% completion} = 2 \times t_{1/2} = 200 \text{ min.}$$

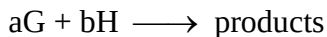
$$\text{Now when } P_x = 700 = 800 - 2x \text{ so } x = 50 \text{ mm of Hg}$$

$$\text{so total pressure} = 800 + 3x = 950 \text{ mm of Hg}$$

13. (D)

Order of reaction can have any value zero, positive, negative or any fractional value

14. (A)



$$\text{Rate} = k[G]^x [H]^y$$

$$R = K [G]_0^x [H]_0^y \quad (\text{Let initial conc. are } [G]_0 \text{ \& } [H]_0)$$

$$8R = K[2G]_0^x [2H]_0^y = K2^x \cdot 2^y R$$

$$\text{so} \quad 2^{x+y} = 8 \quad \Rightarrow x + y = 3$$

15. (D)

From Arrhenius equation

$$K = Ae^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{RT}$$

$$2.303 \log K = 2.303 \log A - \frac{E_a}{RT}$$

$$\log K = \frac{-E_a}{2.303R} \times \frac{1}{T} + \log A \quad \text{..... (A)}$$

$$\log K = - (2000) \frac{1}{T} + 6 \quad \text{.....(B)}$$

On comparing equation (A) and (B)

$$\frac{-E_a}{2.303R} = -2000$$

$$E_a = 2.303 \times 8.314 \times 2000 = 38.29 \text{ kJ and } \log A = 6 \Rightarrow A = 10^6$$

16. Zero

$$K = \frac{C_0 - C}{t} = \frac{1 - 0.75}{0.05} = \frac{0.25}{0.05} = 5$$

$$K = \frac{0.75 - 0.40}{0.07} = \frac{0.35}{0.07} = 5$$

So, reaction must be of zero order.

18. (ABD)

$$C_t = C_0 e^{-Kt}$$

$$t_{1/2} \propto \frac{1}{K}, \quad K \uparrow \text{ on increasing } T.$$

After eight half lives,

$$C = \frac{C_0}{2^8}$$

$$\Rightarrow \% \text{ completion} = \frac{C_0 - \frac{C_0}{2^8}}{C_0} \times 100 = 99.6\%$$

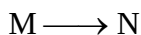
19. 9

$$Kt_{1/8} = \ln \left\{ \frac{C_0}{C_0/8} \right\} = \ln 8$$

$$Kt_{1/10} = \ln \left\{ \frac{C_0}{C_0/10} \right\} = \ln 10$$

$$\text{then } \frac{t_{1/8}}{t_{1/10}} \times 10 = \frac{\ln 8}{\ln 10} \times 10 = \frac{\log 8}{\log 10} \times 10 = 9$$

21. (B)



$$r = K [M]^x$$

as [M] is doubled, rate increases by a factor of 8.

$$\text{i.e. } 8r = K [2M]^x$$

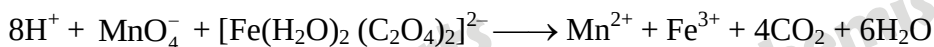
$$\Rightarrow 8 = (2)^x$$

$$x = 3$$

22. B

Initially on increasing temperature rate of reaction will increase, so % yield will also increase with time. But at equilibrium % yield at high temperature (T_2) would be less than at T_1 as reaction is exothermic.

23. 8



$$\frac{\text{rate of change of } [H^+]}{\text{rate of change of } [MnO_4^-]} = 8$$

24. (BCD)

(A) High activation energy means lower value of k and slow reaction.

(B) On increasing temperature energy of particles increases hence greater number of collisions occurs whose energy exceeds the activation energy.

$$K = Ae^{-E_a/RT}$$

$$\frac{dk}{dT} = k \frac{E_a}{RT^2}$$

$$\frac{dk}{dT} \propto \frac{E_a}{RT^2}$$

(C) Rate of increase of k with temperature is higher when E_a has a large value(D) A = Frequency factor = No of collisions per unit time per unit volume.

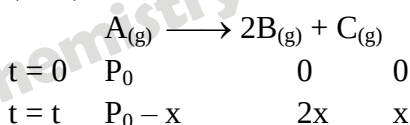
25. AD

(A) E_a is independent on stearic factor(B) $k = P.Z.e^{-E_a/RT}$

$$k = Ae^{-E_a/RT}$$

According to collision theory, P value is generally less than unity but for some reactions P is greater than one and for such reactions, observed rate is greater than rate predicted from Arrhenius equation. For a reaction with P value greater than 1, implies that the experimentally determined value of frequency factor (A) is higher than that predicted by Arrhenius equations and such reactions proceeds rapidly without the use of a catalyst. $\therefore$ (A) and (C).

26. (A,D)

Total pressure at t time $P_t = P_0 - x + 2x + x = P_0 + 2x =$

$$\therefore x = \frac{P_t - P_0}{2}$$

$$\therefore P_0 - x = P_0 - \left(\frac{P_t - P_0}{2} \right) = \frac{3P_0 - P_t}{2}$$

$$\therefore t = \frac{1}{k} \ln \left(\frac{P_0}{\frac{3P_0 - P_t}{2}} \right) \Rightarrow kt = \ln \left(\frac{2P_0}{3P_0 - P_t} \right)$$

$$\ln(3P_0 - P_t) = \ln(2P_0) - kt$$

$$\text{At } t = t_{\frac{1}{3}} ; \quad P_A = \frac{P_0}{3}$$

$$\therefore t_{\frac{1}{3}} = \frac{1}{k} \ln \left(\frac{P_0}{P_0/3} \right) = \frac{\ln 3}{k} = \text{constant}$$

Value of k doesn't depend on initial concentration of reactant.

27. 6.75

From the given data rate of reaction is obtained as :

$$r = k[A]^1[B]^0[C]^1$$

$$\text{From experiment-1 : } r = k[0.2][0.1] = 6 \times 10^{-5}$$

$$\text{So, } k = 3 \times 10^{-3} \text{ s}^{-1}$$

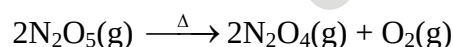
$$\therefore r = 3 \times 10^{-3} \times 0.15 \times 0.15 = 6.75 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

28. 2.30

$$\text{Rate} = -\frac{1}{2} \frac{d(\text{N}_2\text{O}_5)}{dt} = k[\text{N}_2\text{O}_5]^1$$

$$-\frac{d(\text{N}_2\text{O}_5)}{dt} = 2k[\text{N}_2\text{O}_5]^1 = k^1[\text{N}_2\text{O}_5]^1$$

$$k^1 = 2k = 2 \times 5 \times 10^{-4} = 10^{-3} \text{ sec}^{-1}$$



$$t = 0 \quad 1 \quad 0 \quad 0$$

$$t \quad 1-P \quad P \quad \frac{P}{2}$$

$$P_{\text{total}} \text{ after time 't'} = 1 + \frac{P}{2} = 1.45 \text{ atm}$$

$$P = 0.9 \text{ atm}$$

$$k' = \frac{1}{t} \ln \frac{P^\circ}{P_t}$$

$$10^{-3} = \frac{1}{y \times 10^3} \ln \frac{1}{0.1} = \frac{1}{y \times 10^3} \ln 10$$

$$y = \ln 10 = 2.303$$

29. (A)

Sol. Since given reaction follows S_N^1 mechanism

$\therefore$ reaction is 1st order

$$\therefore t_{\frac{1}{2}} = \frac{0.693}{k}$$

EXERCISE # (JEE-MAIN)

1. In a compound, atoms of element Y form ccp lattice and those of element X occupy $\frac{2}{3}$ rd of tetrahedral voids. The formula of the compound will be: **[AIEEE-08]**
 (1) X_4Y_3 (2) X_2Y_3 (3) X_2Y (4) X_3Y_4
2. Percentages of free space in cubic close packed structure and in body centered packed structure are respectively: **[AIEEE-10]**
 (1) 48% and 26% (2) 30% and 26% (3) 26% and 32% (4) 32% and 48%
3. The edge length of a face centered cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is: **[AIEEE-10]**
 (1) 144 pm (2) 288 pm (3) 398 pm (4) 618 pm
4. The number of atoms per unit cell in B.C.C. & F.C.C. is respectively: **[AIEEE-02]**
 (1) 8, 10 (2) 2, 4 (3) 1, 2 (4) 1, 3
5. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ? **[AIEEE-03]**
 (1) 1.28×10^{21} unit cells (2) 1.71×10^{21} unit cells
 (3) 2.57×10^{21} unit cells (4) 5.14×10^{21} unit cells
6. What type of crystal defect is indicated in the diagram below? **[AIEEE-04]**
- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| Na^+ | Cl^- | Na^+ | Cl^- | Na^+ | Cl^- |
| Cl^- | | Cl^- | Na^+ | | Na^+ |
| Na^+ | Cl^- | | Cl^- | Na^+ | Cl^- |
| Cl^- | Na | Cl^- | Na^+ | | Na^+ |
- (1) Frenkel defect (2) Schottky defect
 (3) Interstitial defect (4) Frenkel and Schottky defects
7. An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula of this compound would be— **[AIEEE-05]**
 (1) A_2B (2) AB (3) A_3B (4) AB_3

8. Lattice energy of an ionic compound depends upon: [AIEEE-05]
(1) Size of the ion only (2) Charge on the ion only
(3) Charge on the ion and size of the ion (4) Packing of ions only
9. Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius): [AIEEE-06]
(1) $\frac{24}{3}\pi r^3$ (2) $\frac{12}{3}\pi r^3$ (3) $\frac{16}{3}\pi r^3$ (4) $\frac{20}{3}\pi r^3$
10. Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be: [AIEEE-12]
(1) 152 pm (2) 75 pm (3) 300 pm (4) 240 pm
11. Among the following the incorrect statement is: [Jee-Main (online)-12]
(1) Density of crystals remains unaffected due to Frenkel defect
(2) In BCC unit cell the void space is 32%
(3) Electrical conductivity of semiconductors and metals increases with increase in temperature
(4) Density of crystals decreases due to Schottky defect
12. A solid has 'bcc' structure. If the distance of nearest approach between two atoms is 1.73 Å, the edge length of the cell is: [Jee-Main (online)-12]
(1) 314.20 pm (2) 216 pm (3) 200 pm (4) 1.41 pm
13. Ammonium chloride crystallizes in a body centred cubic lattice with edge length of unit cell of 390 pm. If the size of chloride ion is 180 pm, the size of ammonium ion would be: [Jee-Main (online)-12]
(1) 158 pm (2) 174 pm (3) 142 pm (4) 126 pm
14. The radius of a calcium ion is 94 pm and of the oxide ion is 146 pm. The possible crystal structure of calcium oxide will be: [Jee-Main (online)-12]
(1) Octahedral (2) Tetrahedral (3) Pyramidal (4) Trigonal
15. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be: [Jee-Main (offline)-13]
(1) 7.01% (2) 4.08% (3) 6.05% (4) 5.08

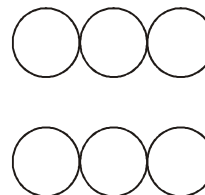
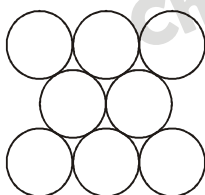
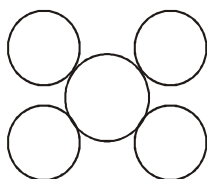
16. An element having an atomic radius of 0.14 nm crystallizes in an fcc unit cell. What is the length of a side of the cell? **[Jee-Main (online)-13]**
 (1) 0.96 nm (2) 0.4 nm (3) 0.24 nm (4) 0.56 nm
17. Which one of the following statements about packing in solids is **incorrect**? **[Jee-Main (online)-13]**
 (1) Void space in ccp mode of packing is 26%
 (2) Coordination number in hcp mode of packing is 12
 (3) Void space in hcp mode of packing is 32%
 (4) Coordination number in bcc mode of packing is 8
18. In a face centred cubic lattice, atoms of A form the corner points and atoms of B form the face centred points. If two atoms of A are missing from the corner points, the formula of the ionic compound is **[Jee-Main (online)-13]**
 (1) AB₂ (2) AB₃ (3) AB₄ (4) A₂B₅
19. Copper crystallises in fcc with a unit length of 361 pm. What is the radius of copper atom? **[AIEEE-2011, Jee-Main (online)-13]**
 (1) 181 pm (2) 128 pm (3) 157 pm (4) 108 pm
20. CsCl crystallises in body centred cubic lattice. If 'a' is its edge length then which of the following expression is correct : **[Jee-Main (offline)-14]**
 (1) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2}a$ (2) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \sqrt{3}a$ (3) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = 3a$ (4) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{3a}{2}$
21. In a face centered cubic lattice atoms A are at the corner points and atoms B at the face centered points. If atom B is missing from one of the face centered points, the formula of the ionic compound is: **[AIEEE-2011, Jee-Main (online)-14]**
 (1) AB₂ (2) A₂B₃ (3) A₅B₂ (4) A₂B₅
22. The appearance of colour in solid alkali metal halides is generally due to: **[AIEEE-2011, Jee-Main (online)-14]**
 (1) Frenkel defect (2) F-centres (3) Schottky defect (4) Interstitial position
23. In a monoclinic unit cell, the relation of sides and angles are respectively **[Jee-Main (online)-14]**
 (1) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$ (2) $a \neq b \neq c$ and $\beta = \gamma = 90^\circ \neq \alpha$
 (3) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$ (4) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$
24. The total number of octahedral void(s) per atom present in a cubic close packed structure is :- **[Jee-Main (online)-14]**
 (1) 1 (2) 2 (3) 3 (4) 4

25. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of $4.29\text{\AA}$. The radius of sodium atom is approximately: **[Jee-Main (offline)-15]**
 (1) $5.72\text{\AA}$ (2) $0.93\text{\AA}$ (3) $1.86\text{\AA}$ (4) $3.022\text{\AA}$
26. Which of the following compounds is metallic and ferromagnetic? **[Jee-Main (offline)-16]**
 (1) CrO_2 (2) VO_2 (3) MnO_2 (4) TiO_2
27. A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be **[Jee-Main (offline)-17]**
 (1) $\sqrt{2} a$ (2) $\frac{a}{\sqrt{2}}$ (3) $2a$ (4) $2\sqrt{2} a$
28. Which of the following compounds is likely to show both Frenkel and Schottky defects in its crystalline form? **[Jee Main (Jan. 2020)]**
 (1) AgBr (2) KBr (3) ZnS (4) CsCl
29. A diatomic molecule X_2 has a body-centred cubic (bcc) structure with a cell edge of 300 pm . The density of the molecule is 6.17 g cm^{-3} . The number of molecules present in 200 g of X_2 is : (Avogadro constant (N_A) = $6 \times 10^{23}\text{ mol}^{-1}$) **[Jee-Main (Sept.) 2020]**
 (1) $8 N_A$ (2) $2 N_A$ (3) $4 N_A$ (4) $40 N_A$
30. An element crystallizes in a face-centred cubic (fcc) unit cell with cell edge a. The distance between the centres of two nearest octahedral voids in the crystal lattice is : **[Jee-Main (Sept.) 2020]**
 (1) $\frac{a}{\sqrt{2}}$ (2) $\frac{a}{2}$ (3) a (4) $\sqrt{2}a$
31. A crystal is made up of metal ions ' M_1 ' and ' M_2 ' and oxide ions. Oxide ions form a ccp lattice structure. The cation ' M_1 ' occupies 50% of octahedral voids and the cation ' M_2 ' occupies 12.5% of tetrahedral voids of oxide lattice. The oxidation numbers of ' M_1 ' and ' M_2 ' are, respectively : **[Jee-Main (Sept.) 2020]**
 (1) +2, +4 (2) +3, +1 (3) +1, +3 (4) +4, +2
32. An element with molar mass $2.7 \times 10^{-2}\text{ kg mol}^{-1}$ forms a cubic unit cell with edge length 405 pm . Its density is $2.7 \times 10^3\text{ kg m}^{-3}$, the radius of the element is approximately _____ $\times 10^{-12}\text{ m}$ (to the nearest integer). **[Jee-Main (Sept.) 2020]**

EXERCISE # (JEE-ADVANCED)

1. A metal crystallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. **[JEE-1999]**

2. The figures given below show the location of atoms in three crystallographic planes in FCC lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagram. **[JEE-2000]**



3. In a solid “AB” having NaCl structure “A” atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is **[JEE-2000]**

(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4

4. In any ionic solid [MX] with schottky defects, the number of positive and negative ions are same. **[T/F]** **[JEE-2000]**

5. The coordination number of a metal crystallising in a hcp structure is **[JEE-2000]**
(A) 12 (B) 4 (C) 8 (D) 6

6. A substance A_xB_y crystallises in a FCC lattice in which atoms “A” occupy each corner of the cube and atoms “B” occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y . **[JEE-2002]**

(A) AB_3 (B) A_4B_3
(C) A_3B (D) composition cannot be specified

7. Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it. **[JEE 2003]**

8. (i) AB crystallizes in a rock salt structure with $A : B = 1 : 1$. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg m^{-3} . **[JEE-2004]**

(ii) If measured density is 20 kg m^{-3} . Identify the type of point defect.

9. Which of the following FCC structure contains cations in alternate tetrahedral voids?

[JEE 2005]

(A) NaCl (B) ZnS (C) Na_2O (D) CaF_2

10. An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstitial sites without disturbing the structure. [JEE 2005]
11. The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. [JEE 2006]
12. Match the crystal system / unit cells mentioned in Column I with their characteristic features mentioned in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2007]

Column I

- (A) simple cubic and face-centred cubic
(B) cubic and rhombohedral
(C) cubic and tetragonal
(D) hexagonal and monoclinic

Column II

- (P) have these cell parameters $a = b = c$ and $\alpha = \beta = \gamma$
(Q) are two crystal systems
(R) have only two crystallographic angles of 90°
(S) belong to same crystal system.

Paragraph for Question No. 13 to 15

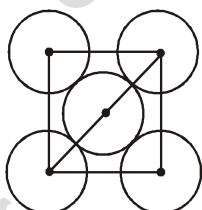
In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

13. The number of atoms in this HCP unit cells is [JEE 2008]
(A) 4 (B) 6 (C) 12 (D) 17
14. The volume of this HCP unit cell is [JEE 2008]
(A) $24\sqrt{2}r^3$ (B) $16\sqrt{2}r^3$ (C) $12\sqrt{2}r^3$ (D) $\frac{64}{3\sqrt{3}}r^3$
15. The empty space in this HCP unit cell is [JEE 2008]
(A) 74% (B) 47.6 % (C) 32% (D) 26%

16. The correct statement(s) regarding defects in solid is (are) [JEE 2009]
 (A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.
 (B) Frenkel defect is a dislocation defect
 (C) Trapping of an electron in the lattice leads to the formation of F-center.
 (D) Schottky defects have no effect on the physical properties of solids.

17. The packing efficiency of the two-dimensional square unit cell shown below is

[JEE-2010]



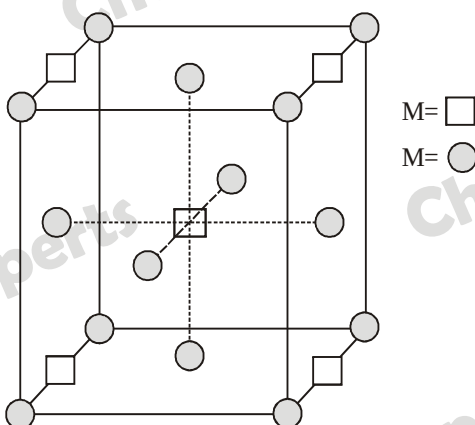
- (A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

18. The number of hexagonal faces that present in a truncated octahedron is.

[JEE-2011]

19. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is:

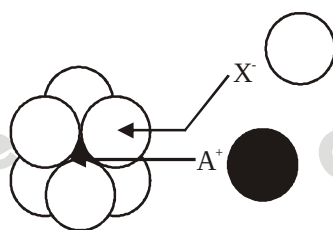
[JEE-2012]



- (A) MX (B) MX_2 (C) M_2X (D) M_5X_{14}

20. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is -

[JEE-2013]



- (A) 104 pm (B) 125 pm (C) 183 pm (D) 57 pm

21. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminum ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are – [JEE-2015]

(A) $\frac{1}{2}, \frac{1}{8}$ (B) $1, \frac{1}{4}$ (C) $\frac{1}{2}, \frac{1}{2}$ (D) $\frac{1}{4}, \frac{1}{8}$

22. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are) [JEE-2016]

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12
 (B) The efficiency of atom packing is 74%
 (C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively
 (D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

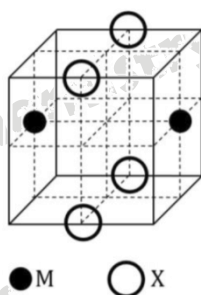
23. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8 g cm^{-3} , then the number of atoms present in 256 g of the crystal is $N \times 10^{24}$. The value of N is [JEE-2017]

24. Consider an ionic solid **MX** with NaCl structure. Construct a new structure (**Z**) whose unit cell is constructed from the unit cell of **MX** following the sequential instructions given below. Neglect the charge balance. [JEE-2018]

(i) Remove all the anions (**X**) except the central one
 (ii) Replace all the face centered cations (**M**) by anions (**X**)
 (iii) Remove all the corner cations (**M**)
 (iv) Replace the central anion (**X**) with cation (**M**)

The value of $\left(\frac{\text{number of anions}}{\text{number of cations}} \right)$ in **Z** is ____.

25. The cubic unit cell structure of a compound containing cation M and anion X is shown below. When compared to the anion, the cation has smaller ionic radius. Choose the correct statement(s). [JEE - 2020]



(A) The empirical formula of the compound is MX.
 (B) The cation M and anion X have different coordination geometries.
 (C) The ratio of M-X bond length to the cubic unit cell edge length is 0.866.
 (D) The ratio of the ionic radii of cation M to anion X is 0.414.

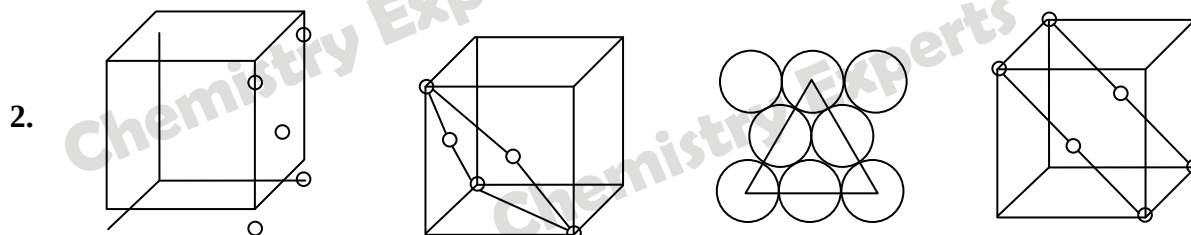
ANSWER KEY

EXERCISE # (JEE-MAIN)

1.	1	2.	1	3.	3	4.	2	5.	3	6.	2	7.	3
8.	3	9.	1	10.	2	11.	1	12.	3	13.	3	14.	1
15.	1	16.	2	17.	2	18.	3	19.	3	20.	3	21.	1
22.	4	23.	2	24.	2	25.	1	26.	2	27.	3	28.	1
29.	3	30.	1	31.	1	32.	143						

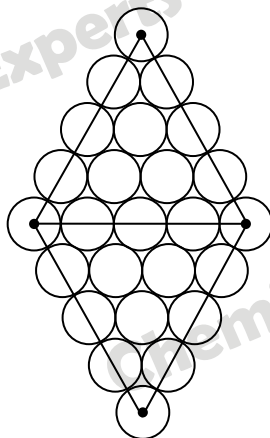
EXERCISE # (JEE-ADVANCED)

1. (1.259)



3. (D) 4. (T) 5. (A) 6. (A)

7. To have maximum number of spheres, the packing must be hcp.
Maximum number of spheres = 25.



Area of figure = 2 × Area of equilateral triangle of side = 40 mm or 4 cm.

$$= 2 \times \frac{\sqrt{3}}{4} (4)^2 = 13.856$$

$$\Rightarrow \frac{25}{13.856} = 1.804 \text{ marbles/cm}^2$$

8. (i) 5 kg m^{-3}
 (ii) There is a huge difference density & observed density. It is only possible if some foreign species occupy interstitial space i.e. substitution defect.
 (i) In Rock salt like crystal AB, there are 4 AB units per unit cell

$$\Rightarrow \text{density} = \frac{Z \times M}{a^3 \times N_A} = \frac{4 \times 6.023y}{8y \times 10^{-27} \times 6.023 \times 10^{23}}$$

$$= 5 \times 10^3 \text{ g/m}^3 \quad [\because a = 2y^{1/3} \text{ nm} \quad \text{or} \quad 2y^{1/3} \times 10^{-9} \text{ m}]$$

or 5 kg/m^3

9. (B) 10. 117.1 pm 11. 216.5 pm

12. $A \rightarrow P, S$

$B \rightarrow P, Q$

$C \rightarrow W$

$D \rightarrow Q, R$

(A) simple cubic & FCC

Have the cell parameters $a = b = c$ & $\alpha = \beta = \gamma = (p)$ & are 2 crystal systems (Q)

(C) Cubic & Tetragonal are 2 crystal system. (Q)

(D) Hexagonal & Monoclinic are 2 crystal system (Q) have 2 crystallographic angles of 90° (R).

13. B 14. A 15. D 16. BC 17. D 18. 8 19. B
 20. A 21. A 22. BCD 23. 2 24. 3 25. AC

HINT AND SOLUTIONS

EXERCISE # (JEE-MAIN)

1. Number of Atom X = $\frac{2}{3} \times 8$

Number of Atom Y = 4

$\Rightarrow$ Formula of compound is $\frac{X_{16}}{3} Y_4$ or $X_4 Y_3$

2. For C.C.P.

% Of free space = 26 %

For B.C.C.

% Of free space = 32 %

3. $(r_+ + r_-) = \frac{a}{2}$

$110 + r_- = \frac{508}{2}$

$\Rightarrow r_- = \frac{508}{2} - 110$

$\Rightarrow r_- = 144 \text{ pm}$

4. (2)

5. 58.5 gm of NaCl contains $\frac{6.022 \times 10^{23}}{4}$ unit cells

$\Rightarrow 1 \text{ gm of NaCl should contain } \frac{6.022 \times 10^{23} \times 1}{58.5 \times 4} \quad \text{i.e. } 2.57 \times 10^{21} \text{ unit cells}$

6. Schottky defect

7. Number of Atom A = $4 \times \frac{1}{8}$ i.e. 1

Number of Atom B = $6 \times \frac{1}{2}$ i.e. 3

$\Rightarrow$ Formula of compound is AB_3

8. Lattice energy of an ionic compounds depends on $\rightarrow$ (3)

i.e. charge on the ion & size of the ion.

9. Total volume of atoms present in a face. centred unit cell of metal is $4 \times \frac{4}{3} \pi r^3$

i.e. $\frac{16}{3} \pi r^3$

10. For B.C.C.

$$r = \frac{\sqrt{3}a}{4}$$

$$= \frac{\sqrt{3} \times 351}{4} \quad \text{i.e. } r = 152 \text{ pm}$$

11. The only incorrect statement is (3) i.e. electrical conductivity of semiconductors & metals increases with Increases in temperature.

12. For B.C.C

The distance of Nearest Approach between atom is equal to $\frac{\sqrt{3}a}{2}$

$$\frac{\sqrt{3}a}{2} = 1.73$$

$$\Rightarrow a = 2\text{\AA} \text{ or } 200 \text{ pm}$$

13. For NH_4Cl in B.C.C lattice:

$$(r_{\text{NH}_4^+} + r_{\text{Cl}^-}) = \frac{\sqrt{3}a}{2}$$

$$\Rightarrow r_{\text{NH}_4^+} + 180 = \frac{\sqrt{3} \times 390}{2}$$

$$\Rightarrow r_{\text{NH}_4^+} = \frac{\sqrt{3} \times 390}{2} - 180$$

$$\Rightarrow r_{\text{NH}_4^+} = 158 \text{ pm}$$

14. For CaO : $\frac{r_{\text{Ca}^{+2}}}{r_{\text{O}^{2-}}} = \frac{94}{146} = 0.643 \Rightarrow$ By radius ratio rule it should be octahedral.

15. $\text{M}_{0.98}\text{O}_1$

Consider 1 mole of oxide

mole of M = 0.98, mole of $\text{O}^{2-} = 1$

Let moles of $\text{M}^{+3} = x$

& moles of $\text{M}^{+2} = (0.98 - x)$

$\Rightarrow$ By charge balance $(0.98 - x) \cdot 2 + 3x - 2 = 0$.

$\Rightarrow x = 0.04 \Rightarrow \% \text{ of } \text{M}^{+3} = \frac{0.04}{0.98} \times 100 \quad \text{i.e. } 4.08 \%$

16. For F.C.C

$a = 2\sqrt{2} r \Rightarrow a = 2\sqrt{2} \times 0.14 \quad \text{i.e. } a = 0.4 \text{ nm}$

17. The only incorrect statement is (3) i.e., void space in hcp mode of packing is 32%

18. Number of atom A = $6 \times \frac{1}{8} \quad \text{i.e. } \frac{3}{4}$

Number of atom B = $6 \times \frac{1}{2} \quad \text{i.e. } 3 \Rightarrow$ Formula of compound is AB_4

19. For F.C.C.

$r = \frac{a}{2\sqrt{2}} = \frac{361}{2\sqrt{2}} \quad \text{i.e. } r = 128 \text{ pm}$

20. For CsCl $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2} a$

21. Number of atom A = $8 \times \frac{1}{8} \quad \text{i.e. } 1$

Number of atom B = $5 \times \frac{1}{2} \Rightarrow$ Formula compound is $(\text{AB}_{5/2}) \quad \text{i.e. } \text{A}_2\text{B}_5$

22. The appearance of colour in solid-alkali metal halide is generally due to F-centers.

23. For monoclinic unit cell :

$a \neq b \neq c \quad \& \quad \beta = \gamma = 90^\circ \neq \alpha$

24. The total number of octahedral void per atom present in cubic close packed structure is 1.

25. For B.C.C. $r = \frac{\sqrt{3}a}{4} \Rightarrow r = \frac{\sqrt{3} \times 4.29}{4}$ i.e. $r = 1.86 \text{ \AA}$

26. CrO_2 is magnetic & ferromagnetic both

27. The closest approach between 2 atoms in metallic crystal in F.C.C. structure is $\frac{a}{\sqrt{2}}$

28. AgBr can both schottky and Frenkel defect

29. $\text{Density} = \frac{Z \times M}{N_A \times a^3}$

$$6.17 = \frac{2 \times M}{6 \times 10^{23} \times (300 \times 10^{-10})^3} \Rightarrow M = 50$$

$$\therefore \text{Number of molecules in 200 gm.} = \frac{200 \times N_A}{50} = 4 N_A$$

30. Min distance between two octahedral voids in fcc = $\frac{a}{\sqrt{2}}$

(Distance between edge centre and body centre)

31. O^{2-} ions form ccp. $\text{O}_4 \rightarrow (-8 \text{ charge})$

$$M_1 = 50\% \text{ of O.V.} \Rightarrow \frac{50}{100} \times 4 = 2 : (M_1)_2$$

$$M_2 = 12.5\% \text{ of T.V.} \Rightarrow \frac{12.5}{100} \times 8 = 1 : (M_2)_1$$

So formula is : $(M_1)_2 (M_2)_1 \text{O}_4$

This must be neutral. Both metals must have +8 charge in total.

$$\text{From given options : } \begin{cases} \text{O.N. of } M_1 = +2 \\ M_2 = +4 \end{cases}$$

32. $\therefore \text{Density} = \frac{Z \times M}{N_A \times a^3}$

$$2.7 \times 10^3 = \frac{Z \times 2.7 \times 10^{-2}}{6.02 \times 10^{23} \times (405 \times 10^{-12})^3}$$

$$Z = 4$$

$\therefore$ Structure is FCC.

So, $a = 2\sqrt{2}R$

$$R = \frac{405 \times 10^{-12}}{2\sqrt{2}} \text{ m} = 143 \times 10^{-12} \text{ m}$$

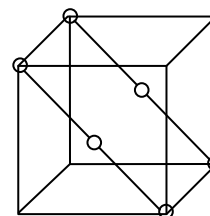
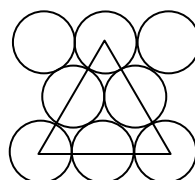
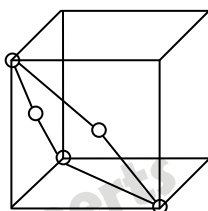
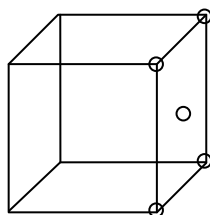
EXERCISE # (JEE-ADVANCED)

1. (1.259)

$$\text{Density} \propto \frac{Z}{a^3}$$

$$\frac{d_{\text{F.C.C}}}{d_{\text{B.C.C}}} = \frac{4}{(3.5)^3} \times \frac{(3)^3}{2} = 1.259$$

2.



3. (D)

$$\text{Number of atoms A per unit cell} = \left(\frac{1}{8} \times 8 + 4 \times \frac{1}{2} \right) = 3$$

$$\text{Number of atom B per unit cell} = 4 \cdot \left(12 \times \frac{1}{4} + 1 \right)$$

$\Rightarrow$ Formula is (A_3B_4)

4. (T)

Factual

5. (A)

3 - consecutive layers of atom in hcp are there.

Atom X is in correct of 12 like atoms, 6 from layer B and 3 from top and bottom layer A each.

$\Rightarrow$ C.N. = $(6 + 3 + 3)$ i.e. 12

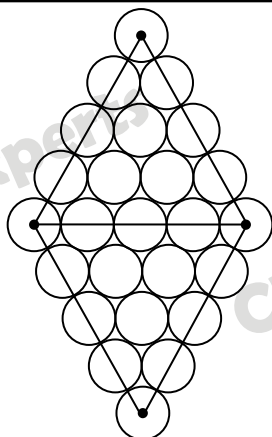
6. (A)

$$\text{Number of atom of A per unit cell} = 8 \times \frac{1}{8} \quad \text{i.e. 1}$$

$$\text{Number of atoms of B per unit cell} = 6 \times \frac{1}{2} \quad \text{i.e. 3} \quad \Rightarrow \text{formula is } AB_3$$

7. To have maximum number of spheres, the packing must be hcp.

Maximum number of spheres = 25.



Area of figure = 2 × Area of equilateral triangle of side = 40 mm or 4 cm.

$$= 2 \times \frac{\sqrt{3}}{4} (4)^2 = 13.856 \Rightarrow \frac{25}{13.856} = 1.804 \text{ marbles/cm}^2$$

8. (i) 5 kg m^{-3}
 (ii) There is a huge difference density & observed density. It is only possible if some foreign species occupy interstitial space i.e. substitution defect.

(i) In Rock salt like crystal AB, there are 4 AB units per unit cell

$$\Rightarrow \text{density} = \frac{Z \times M}{a^3 \times N_A} = \frac{4 \times 6.023y}{8y \times 10^{-27} \times 6.023 \times 10^{23}}$$

$$= 5 \times 10^3 \text{ g/m}^3 \quad [\because a = 2y^{1/3} \text{ nm} \quad \text{or} \quad 2y^{1/3} \times 10^{-9} \text{ m}]$$

or 5 kg/m^3

9. (B)

In C.C.P, number of tetrahedral voids = 2 × number of atoms.

As there are 4S^{2-} ions at lattice points & they need 4zn^{+2} , which are adjusted in alternate tetrahedral voids

10. 117.1 pm

In a cubic crystal system, there are 2 types of void known as octahedral & Tetrahedral voids.

It r_1 is the radius of void & r_2 is the radius of atom creating these voids, then

$$\left(\frac{r_1}{r_2} \right)_{\text{octahedral}} = 0.414$$

$$\left(\frac{r_1}{r_2} \right)_{\text{tetrahedral}} = 0.225$$

The above radius ratio values indicate that octahedral void has larger radius, have for maximum diameter of atom to be present in interstitial space

$$r_1 = 0.414r_2$$

Also in F.C.C., $4r_2 = \sqrt{2}a$

$$\Rightarrow \text{Diameter required } (2r_1) = 2 \times 0.414 r_2 = \frac{2 \times 0.414 \times a}{2\sqrt{2}} = \frac{0.414 \times 400}{\sqrt{2}}$$

i.e. 117.1 pm

11. 216.5 pm

$$\rho = \frac{Z \times M}{a^3 \times N_A}$$

$$\Rightarrow Z = \frac{\rho \times a^3 \times N_A}{M} = \frac{2 \times (5 \times 10^{-8})^3 \times 6 \times 10^{23}}{75} = 2$$

So, it is a B.C.C. unit cell

$$\Rightarrow R = \frac{\sqrt{3}a}{4} = \frac{\sqrt{3} \times 5}{4} \Rightarrow R = 216.5 \text{ pm}$$

12. A $\rightarrow$ P, S

B $\rightarrow$ P, Q

C $\rightarrow$ W

D $\rightarrow$ Q, R

(A) simple cubic & FCC

Have the cell parameters $a = b = c$ & $\alpha = \beta = \gamma = (p)$ & are 2 crystal systems (Q)

(C) Cubic & Tetragonal are 2 crystal system. (Q)

(D) Hexagonal & Monoclinic are 2 crystal system (Q) have 2 crystallographic angles of 90° (R).

13. (B)

$$\text{Total number of atoms in 1 unit cell} = \left(12 \times \frac{1}{6}\right) + 3 + \left(2 \times \frac{1}{2}\right) = 6$$

14. (A)

$$C = \sqrt{\frac{2}{3}} \times 4r$$

$$\text{Area of Base} = 6 \times \frac{\sqrt{3}}{4} (2r)^2$$

$$\Rightarrow \text{volume of Hexagon} = \text{Area of base} \times \text{height} = \left(6 \times \frac{\sqrt{3}}{4} \times 4r^2\right) \times \left(\sqrt{\frac{2}{3}} \times 4r\right) = 24\sqrt{2} r^3$$

15. (D)

$$\text{Packing fraction} = \frac{\text{Volume of atom in 1 unit cell}}{\text{Volume of 1 unit cell}} \times 100 = \frac{6 \times \frac{4}{3} \pi r^3}{24\sqrt{2}r^3} \times 100 = 74\%$$

 $\Rightarrow$ Empty space = 26%

16. (B), (C)

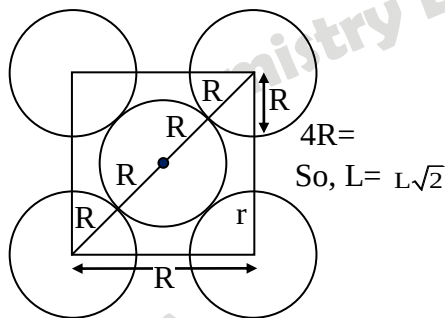
Frenkel defect is a dislocation defect. Trapping of an electron in the lattice leads to the formation of F-centres.

17. (D)

$$4R = L\sqrt{2}$$

$$\text{So, } L = 2\sqrt{2} R$$

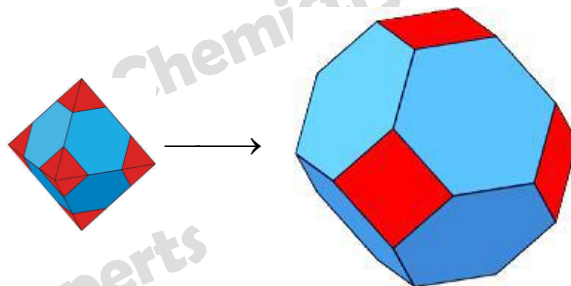
$$\Rightarrow \text{Area of square unit cell} = (2\sqrt{2} R)^2 = 8R^2$$



$$\text{Area of atoms present in 1 unit cell} = \pi R^2 + 4 \left(\frac{\pi R^2}{4} \right) = 2\pi R^2$$

$$\Rightarrow \text{Packing efficiency} = \frac{2\pi R^2}{8R^2} \times 100 = 78.54\%$$

18. (8)



Explanation : The truncated octahedron is the 14-faced Archimedean solid with 14 total faces : 6 square & 8 regular hexagons.

The truncated octahedron is formed by removing the six right square pyramids one from each point of regular octahedron.

Truncated octahedron unfolded in two dimensions.

19. (B)

$$\text{Solution} \rightarrow \text{No. of M atoms} = \frac{1}{4} \times 4 + 1 = 2$$

$$\text{Number of X atoms} = \frac{1}{2} \times 6 + \frac{1}{8} \times 8 = 3 + 1 = 4$$

So, empirical formula of compound is (MX_2) .

20. (A)

The given arrangement is octahedral void arrangement.

$$\frac{r_{A^+}}{r_X} \geq 0.414 \Rightarrow r_{A^+} \geq 0.414 \times 250$$

$$\text{i.e. } r_{A^+} \geq 103.5 \text{ pm}$$

$$\frac{r_{A^+}}{r_{X^-}} < 0.732 \Rightarrow r_{A^+} < 183 \text{ pm.}$$

So, the radius of A^+ is 104 pm.

21. (A)

In CCP, O^{2-} ions are 4

Hence total negative charge = -8

Let number of Al^{+3} ions be x

& number of Mg^{+2} ion be y

$$\Rightarrow \text{Total position charge} = 3x + 2y$$

$$\Rightarrow 3x + 2y = 8$$

This relation is satisfied only by $x = 2$ & $y = 1$

Hence number of $Al^{+3} = 2$

& number of $Mg^{+2} = 1$

$$\Rightarrow \text{m i.e. fraction of octahedral holes occupied by } Al^{+3} \text{ ions} = \frac{2}{4} \quad \text{i.e. } \frac{1}{2}$$

$$\text{and n i.e. fraction of tetrahedral holes occupied by } Mg^{+2} \text{ ions} = \frac{1}{8}$$

22. (B), (C), (D)

- (a) Nearest neighbor in the top most layer of CCP structure is 9 thus incorrect.
 (b) Packing efficiency is 74% thus correct.

$$\% \text{ of packing efficiency} = \frac{4 \times \frac{4}{3} \pi r^3}{a^3} \times 100 = \frac{4 \times \frac{4}{3} \pi r^3}{(2\sqrt{2}r)^3} \times 100 \text{ i.e. } 74\%$$

- (c) Number of tetrahedral void per atom = 2

Number of octahedral void per atom = 1

this correct

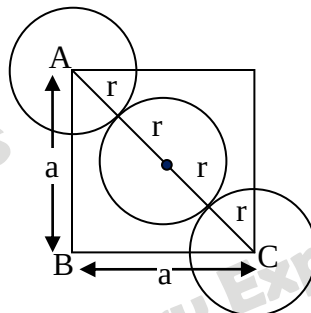
- (d) In $\triangle ABC$

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow (4r)^2 = a^2 + a^2$$

$$\Rightarrow 4r = \sqrt{2}a$$

$$\Rightarrow a = 2\sqrt{2}r$$



25. (A,C)

The given compound has CsCl type structure in which anion X^- form simple cubic unit cell and M^+ cation is present in cubic void.

$$\therefore r_{M^+} + r_{X^-} = \frac{a\sqrt{3}}{2}$$

$$\therefore \frac{r_{M^+} + r_{X^-}}{a} = \frac{\sqrt{3}}{2} = 0.866$$

Coordination no. of M^+ = 8

Coordination no. of X^- = 8

$$\text{And } 0.732 \leq \frac{r_{M^+}}{r_{X^-}} < 1$$

$\therefore$ Correct Ans. (A), (C)

EXERCISE # (JEE-MAIN)

1. If at 298 K the bond energies of C–H, C–C, C=C and H–H bonds are respectively 414, 347, 615 and 435 kJ mol⁻¹, the value of enthalpy change for the reaction : **[AIEEE-2003]**



will be :-

- (1) +125 kJ (2) -125 kJ (3) +250 kJ (4) -250 kJ

2. The enthalpies of combustion of carbon and carbon monoxide are -393.5 and -283 kJ mol⁻¹ respectively. The enthalpy of formation of carbon monoxide per mole :- **[AIEEE-2004]**

- (1) 110.5 kJ (2) 676.5 kJ (3) -676.5 kJ (4) -110.5 kJ

3. If the bond dissociation energies of XY, X₂ and Y₂ (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and Δ_fH for the formation of XY is -200 kJ mol⁻¹. The bond dissociation energy of X₂ will be :- **[AIEEE-2005]**

- (1) 200 kJ mol⁻¹ (2) 100 kJ mol⁻¹ (3) 800 kJ mol⁻¹ (4) 300 kJ mol⁻¹

4. The enthalpy changes for the following processes are listed below : **[AIEEE-2006]**



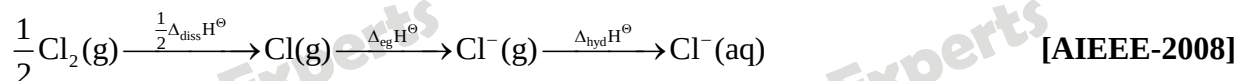
Given that the standard states for iodine and chlorine are I₂(s) and Cl₂(g), the standard enthalpy of formation for ICl(g) is :-

- (1) -16.8 kJ mol⁻¹ (2) +16.8 kJ mol⁻¹ (3) +244.8 kJ mol⁻¹ (4) -14.6 kJ mol⁻¹

5. The standard enthalpy of formation (Δ_fH°) at 298K for methane, CH₄(g), is -74.8 kJ mol⁻¹. The additional information required to determine the average energy for C–H bond formation would be:- **[AIEEE-2006]**

- (1) Latent heat of vapourization of methane
 (2) The first four ionization energies of carbon and electron gain enthalpy of hydrogen
 (3) The dissociation energy of hydrogen molecule H₂
 (4) The dissociation energy of H₂ and enthalpy of sublimation of carbon

6. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

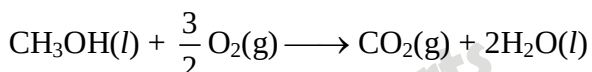


The energy involved in the conversion of $\frac{1}{2}\text{Cl}_2(\text{g})$ to $\text{Cl}^-(\text{aq})$

(using the data $\Delta_{\text{diss}}H^\ominus \text{Cl}_2 = 240 \text{ kJ mol}^{-1}$, $\Delta_{\text{eg}}H^\ominus \text{Cl} = -349 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}}H^\ominus \text{Cl}^- = -381 \text{ kJ mol}^{-1}$) will be:-

- (1) -610 kJ mol^{-1} (2) -850 kJ mol^{-1} (3) $+120 \text{ kJ mol}^{-1}$ (4) $+152 \text{ kJ mol}^{-1}$

7. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

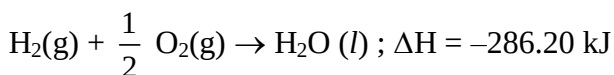


At 298 K standard Gibb's energies of formation for $\text{CH}_3\text{OH}(\text{l})$, $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$ are -166.2 , -237.2 and $-394.4 \text{ kJ mol}^{-1}$ respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1} , efficiency of the fuel cell will be

[AIEEE-2009]

- (1) 90% (2) 97% (3) 80% (4) 87%

8. On the basis of the following thermochemical data : ($\Delta G_f^\ominus \text{H}^+(\text{aq}) = 0$)



The value of enthalpy of formation of OH^- ion at 25°C is :-

[AIEEE-2009]

- (1) $+228.88 \text{ kJ}$ (2) -343.52 kJ (3) -22.88 kJ (4) -228.88 kJ

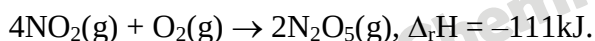
9. The standard enthalpy of formation of NH_3 is $-46.0 \text{ kJ mol}^{-1}$. If the enthalpy of formation of H_2 from its atoms is -436 kJ mol^{-1} and that of N_2 is -712 kJ mol^{-1} , the average bond enthalpy of N-H bond in NH_3 is :-

[AIEEE-2010]

- (1) $-1102 \text{ kJ mol}^{-1}$ (2) -964 kJ mol^{-1} (3) $+352 \text{ kJ mol}^{-1}$ (4) $+1056 \text{ kJ mol}^{-1}$

10. Consider the reaction :

[AIEEE-2011]



If $\text{N}_2\text{O}_5(\text{s})$ is formed instead of $\text{N}_2\text{O}_5(\text{g})$ in the above reaction, the $\Delta_r H$ value will be :-

(given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1})

- (1) -165 kJ (2) $+54 \text{ kJ}$ (3) $+219 \text{ kJ}$ (4) -219 kJ

11. The enthalpy of neutralisation of NH_4OH with HCl is $-51.46 \text{ kJ mol}^{-1}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90 \text{ kJ mol}^{-1}$. The enthalpy of ionisation of NH_4OH is:

[JEE-MAIN (online) 2012]

- (1) $+107.36 \text{ kJ mol}^{-1}$ (2) $-4.44 \text{ kJ mol}^{-1}$ (3) $-107.36 \text{ kJ mol}^{-1}$ (4) $+4.44 \text{ kJ mol}^{-1}$

12. Given [JEE-MAIN (OnLine) 2013]

Reaction	Energy Change (in kJ)
$\text{Li(s)} \longrightarrow \text{Li(g)}$	161
$\text{Li(g)} \longrightarrow \text{Li}^+(\text{g})$	520
$\frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{F(g)}$	77
$\text{F(g)} + \text{e}^- \longrightarrow \text{F}^-(\text{g})$	(Electron gain enthalpy)
$\text{Li}^+(\text{g}) + \text{F}^-(\text{g}) \longrightarrow \text{LiF(s)}$	-1047
$\text{Li(s)} + \frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{LiF(s)}$	-617

Based on data provided, the value of electron gain enthalpy of fluorine would be :

- (1) -300 kJ mol^{-1} (2) -328 kJ mol^{-1} (3) -350 kJ mol^{-1} (4) -228 kJ mol^{-1}

13. Given : [JEE-MAIN (online) 2013]



The molar enthalpy of vapourisation of water will be :-

- (1) $241.8 \text{ kJ mol}^{-1}$ (2) $527.7 \text{ kJ mol}^{-1}$ (3) 44.1 kJ mol^{-1} (4) 22.0 kJ mol^{-1}

14. The standard enthalpy of formation ($\Delta_f H_{298}^\circ$) for methane, CH_4 is $-74.9 \text{ kJ mol}^{-1}$. In order to calculate the average energy given out in the formation of a C-H bond from this it is necessary to know which one of the following? [JEE-MAIN(online) 2014]

- (1) the dissociation energy of the hydrogen molecule, H_2 .
 (2) the dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite).
 (3) the first four ionisation energies of carbon and electron affinity of hydrogen.
 (4) the first four ionisation energies of carbon.

15. For complete combustion of ethanol, $\text{C}_2\text{H}_5\text{OH(l)} + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O(l)}$, the amount of heat produced as measured in bomb calorimeter, is $1364.47 \text{ kJ mol}^{-1}$ at 25°C . Assuming ideality the Enthalpy of combustion, $\Delta_c H$, for the reaction will be :-

($R = 8.314 \text{ kJ mol}^{-1}$)

[JEE-MAIN 2014]

- (1) $-1460.50 \text{ kJ mol}^{-1}$ (2) $-1350.50 \text{ kJ mol}^{-1}$
 (3) $-1366.95 \text{ kJ mol}^{-1}$ (4) $-1361.95 \text{ kJ mol}^{-1}$

16. The heat of atomisation of methane and ethane are 360 kJ/mol and 620 kJ/mol , respectively. The longest wavelength of light capable of breaking the C-C bond is :

(Avogadro number = 6.02×10^{23} , $h = 6.62 \times 10^{-34} \text{ J s}$)

[JEE-MAIN (Online) 2015]

- (1) $2.48 \times 10^3 \text{ nm}$ (2) $1.49 \times 10^3 \text{ nm}$ (3) $2.49 \times 10^4 \text{ nm}$ (4) $2.48 \times 10^4 \text{ nm}$

17. The heats of combustion of carbon and carbon monoxide are -393.5 and $-283.5 \text{ kJ mol}^{-1}$, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :

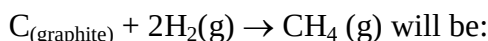
[JEE-MAIN 2016]

- (1) 676.5 (2) -676.5 (3) -110.5 (4) 110.5

18. Given



Based on the above thermochemical equations, the value of $\Delta_r H^\circ$ at 298 K for the reaction



[JEE-MAIN 2017]

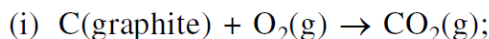
- (1) $+74.8 \text{ kJ mol}^{-1}$ (2) $+144.0 \text{ kJ mol}^{-1}$ (3) $-74.8 \text{ kJ mol}^{-1}$ (4) $-144.0 \text{ kJ mol}^{-1}$

19. The combustion of benzene (l) gives $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$. Given that heat of combustion of benzene at constant volume is $-3263.9 \text{ kJ mol}^{-1}$ at 25°C ; heat of combustion (in kJ mol^{-1}) of benzene at constant pressure will be : ($R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)

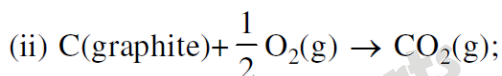
[JEE-MAIN 2018]

- (1) -3267.6 (2) 4152.6 (3) -452.6 (4) 3260

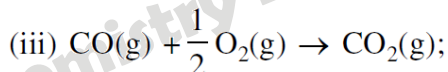
20. Given :



$$\Delta_r H^\circ = x \text{ kJ mol}^{-1}$$



$$\Delta_r H^\circ = y \text{ kJ mol}^{-1}$$



$$\Delta_r H^\circ = z \text{ kJ mol}^{-1}$$

Based on the above thermochemical equations, find out which one of the following algebraic relationships is correct ?

[JEE-MAIN 2019 (Jan.)]

- (1) $x = y - z$ (2) $y = 2z - x$ (3) $z = x + y$ (4) $x = y + z$

21. Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C . If specific heat of $\text{I}_2(\text{s})$ and $\text{I}_2(\text{vap})$ are 0.055 and $0.031 \text{ cal g}^{-1} \text{ K}^{-1}$ respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is:

[JEE-MAIN 2019 (Apr.)]

- (1) 2.85 (2) 22.8 (3) 11.4 (4) 5.7

22. If enthalpy of atomisation for $\text{Br}_{2(\text{l})}$ is $x \text{ kJ/mol}$ and bond enthalpy for Br_2 is $y \text{ kJ/mol}$, the relation between them :

[Jee Main (Jan. 2020)]

- (1) is $x = y$ (2) is $x > y$ (3) does not exist (4) is $x < y$

23. The standard heat of formation ($\Delta_f H_{298}^0$) of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are -1560 , -393.5 and -286 kJ/mol, respectively is _____.

[Jee Main (Jan. 2020)]

24. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1} , respectively. The hydration enthalpy of NaCl is :

[Jee-Main (Sep) 2020]

- (1) 784 kJ mol^{-1} (2) -780 kJ mol^{-1} (3) -784 kJ mol^{-1} (4) 780 kJ mol^{-1}

25. The heat of combustion of ethanol into carbon dioxide and water is -327 kcal at constant pressure. The heat evolved (in cal) at constant volume and 27°C (if all gases behave ideally) is ($R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$) _____.

[Jee-Main (Sep) 2020]

EXERCISE # (JEE-ADVANCED)

- Estimate the average S–F bond enthalpy in SF₆. The ΔH°_f values of SF₆ (g), S(g), and F (g) are –1100, 275 and 80 kJ/mol respectively. **[JEE 1999]**
- Which of the following is not an endothermic reaction?
 (A) Combustion of methane
 (B) Decomposition of water
 (C) Dehydrogenation of ethene to acetylene
 (D) Conversion of graphite to diamond **[JEE 1999]**
- Diborane is a potential rocket fuel which undergoes combustion according to the reaction,

$$\text{B}_2\text{H}_6(\text{g}) + 3\text{O}_2(\text{g}) \longrightarrow \text{B}_2\text{O}_3(\text{s}) + 3\text{H}_2\text{O}(\text{g})$$
 From the following data, calculate the enthalpy change for the combustion of diborane :

$$2\text{B}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g}) \longrightarrow \text{B}_2\text{O}_3(\text{s}) ; \quad \Delta H = -1273 \text{ kJ}$$

$$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{l}) ; \quad \Delta H = -286 \text{ kJ}$$

$$\text{H}_2\text{O}(\text{l}) \longrightarrow \text{H}_2\text{O}(\text{g}) ; \quad \Delta H = 44 \text{ kJ}$$

$$2\text{B}(\text{s}) + 3\text{H}_2(\text{g}) \longrightarrow \text{B}_2\text{H}_6(\text{g}) ; \quad \Delta H = 36 \text{ kJ}$$
[JEE 2000]
- ΔH°_f for CO₂(g), CO(g) and H₂O(g) are – 393.5, –110.5 and –241.8 kJ mol^{–1} respectively. The standard enthalpy change (in kJ) for the reaction

$$\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$$
 is
 (A) 524.1 (B) 41.2 (C) –262.5 (D) – 41.2 **[JEE 2000]**
- Which of the following reactions defines ΔH°_f ?
 (A) C_(diamond) + O₂(g) $\longrightarrow$ CO₂(g) (B) 1/2 H₂(g) + 1/2 F₂(g) $\longrightarrow$ HF(g)
 (C) N₂(g) + 3H₂(g) $\longrightarrow$ 2NH₃ (D) CO(g) + 1/2O₂(g) $\longrightarrow$ CO₂(g) **[JEE 2003]**
- In a constant volume calorimeter, 3.5 g of a gas with molecular weight 28 was burnt in excess oxygen at 298.0 K. The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is 2.5 kJ K^{–1}, the numerical value for the enthalpy of combustion of the gas in kJ mol^{–1} is **[JEE 2009]**

7. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a $\text{C} \equiv \text{C}$ bond in C_2H_2 . That energy is (take the bond energy of a $\text{C}-\text{H}$ bond as 350 kJ mol^{-1} .) [JEE 2012]



- (A) 1165 (B) 837 (C) 865 (D) 815

8. The standard enthalpies of formation of $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$ and glucose(s) at 25°C are -400 kJ/mol , -300 kJ/mol and -1300 kJ/mol , respectively. The standard enthalpy of combustion per gram of glucose at 25°C is [JEE (Advanced)2013]

- (A) $+2900 \text{ kJ}$ (B) -2900 kJ (C) -16.11 kJ (D) $+16.11 \text{ kJ}$

9. Choose the reaction(s) from the following options, for which the standard enthalpy of reaction is equal to the standard enthalpy of formation. [JEE (Advanced)2019]



10. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

At 298 K : $\Delta_f H^\circ(\text{SnO}_2(\text{s})) = -581.0 \text{ kJ mol}^{-1}$, $\Delta_f H^\circ(\text{CO}_2(\text{g})) = -394.0 \text{ kJ mol}^{-1}$

$S^\circ : (\text{SnO}_2(\text{s})) = 56.0 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\circ(\text{Sn}(\text{s})) = 52.0 \text{ J K}^{-1} \text{ mol}^{-1}$, [JEE (Advanced)2020]

$S^\circ : (\text{C}(\text{s})) = 6.0 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\circ(\text{CO}_2(\text{g})) = 210.0 \text{ J K}^{-1} \text{ mol}^{-1}$.

Assume that the enthalpies and the entropies are temperature independent.

ANSWER KEY

EXERCISE # (JEE - MAIN)

1. (2) 2. (3) 3. (3) 4. (1) 5. (4) 6. (1) 7. (2)
 8. (4) 9. (3) 10. (4) 11. (4) 12. (2) 13. (3) 14. (2)
 15. (3) 16. (2) 17. (3) 18. (3) 19. (1) 20. (4) 21. (2)
 22. (2) 23. -192.50 24. (3) 25. (326400) NTA Ans. (-326400.00)

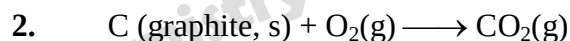
EXERCISE # (JEE-ADVANCED)

1. 309.16 kJ/mol 2. (A) 3. $-2035 \text{ kJ mol}^{-1}$ 4. (B) 5. (B)
 6. (9) 7. (D) 8. (C) 9. (CD) 10. 935.00

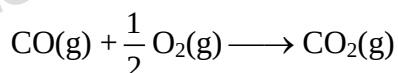
HINT AND SOLUTIONS

EXERCISE # JEE-MAIN

$$1. \quad \Delta_r H = \epsilon_{C-C} + \epsilon_{H-H} - \epsilon_{C-C} - 2 \epsilon_{C-H} \\ = 615 + 435 - 347 - 2 \times 414 = -125 \text{ kJ}$$



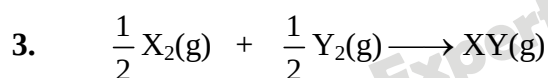
$$\Delta_r H = -393.5$$



$$\Delta_r H = -283 \text{ kJ}$$

$$-283 = -393.5 - \Delta H_f CO(g)$$

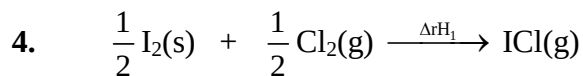
$$\Delta H_f CO(g) = -110.5 \text{ kJ}$$



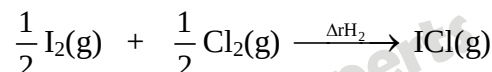
$$\Delta_r H = \frac{1}{2} \epsilon_{X-X} + \frac{1}{2} \epsilon_{Y-Y} - \epsilon_{X-Y}$$

$$-200 = \frac{1}{2} \times a + \frac{1}{2} \times 0.5a - a$$

$$A = \epsilon_{X-X} = 800$$



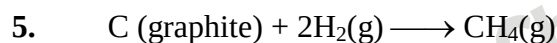
$$\downarrow 62.76/2 \quad \downarrow 0 \quad \downarrow 0$$



$$\Delta_r H_2 + \frac{62.76}{2} = \Delta_r H_1$$

$$\Delta_r H_2 = \frac{1}{2} \times 151 + \frac{1}{2} \times 242.3 - 211.3 = 196.65 - 211.3 = -14.65$$

$$\Delta_r H_1 = -14.65 + \frac{62.76}{2} = 16.73 \text{ kJ/mol}$$



$$6. \quad \Delta_r H = \frac{1}{2} \times 240 - 349 - 381 = -610 \text{ kJ/mol}$$

$$7. \quad \Delta_r G^\circ = -394.4 - 2 \times 237.2 + 166.2 \\ = -702.6 \text{ kJ/mol}$$

$$= W_{\text{non}} - PV$$

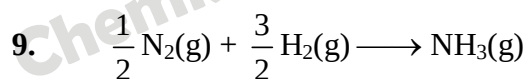
$$\Delta H = -726 \text{ kJ/mol} = q$$

$$\eta = \frac{465.4}{726} \times 100 \approx 97\%$$

8. $57.32 = \Delta H_f \text{OH}^- (\text{aq}) - \Delta H_f \text{H}_2\text{O}(\ell)$

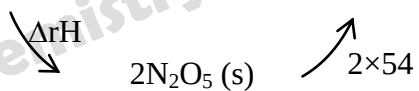
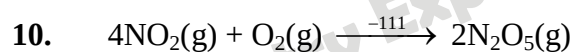
$$57.32 = \Delta H_f \text{OH}^- (\text{aq}) + 286.2$$

$$\Delta H_f \text{OH}^- (\text{aq}) = -228.88 \text{ kJ}$$



$$-46 = \frac{3}{2} \times 436 + \frac{1}{2} \times 712 - 3 \epsilon_{\text{N-H}}$$

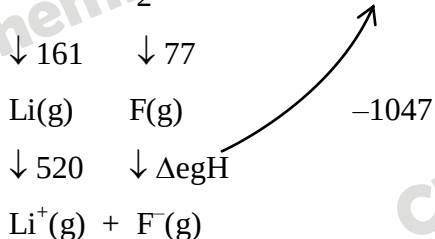
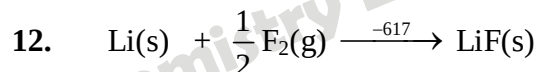
$$3 \epsilon_{\text{N-H}} = 1056 \Rightarrow \epsilon_{\text{N-H}} = 352 \text{ kJ mol}^{-1}$$



$$\Delta_r H + 2 \times 54 = -111 \Delta_r H = -219 \text{ kJ}$$

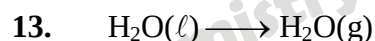
11. $-51.46 = -55.90 + \Delta H_{\text{ion}}$

$$\Delta H_{\text{ion.}} = +4.44 \text{ kJ mol}^{-1}$$

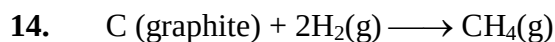


$$161 + 520 + 77 + \Delta_{\text{egH}} \text{F}(\text{g}) - 1047 = -617$$

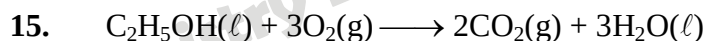
$$\Delta_{\text{egH}} \text{F}(\text{g}) = -328 \text{ kJ/mol}$$



$$\Delta_r H = b - a = -241.8 + 285.9 = +44.1 \text{ kJ}$$



$$\Delta_r H = \Delta H_{\text{sub. C(graphite)}} + 2 \epsilon_{\text{H-H}} - 4 \epsilon_{\text{C-H}}$$



$$\Delta U = -1364.47 \text{ kJ/mol}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$= -1364.47 - \frac{1 \times 8.314 \times 298}{1000} = -1366.95 \text{ kJ/mol}$$

16. $4 \epsilon_{\text{C-H}} = 360 \text{ kJ}$

$$\epsilon_{\text{C-H}} = 90 \text{ kJ/mol}$$

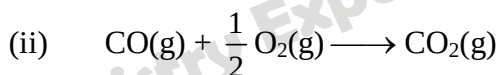
$$\epsilon_{\text{C-C}} + \epsilon_{\text{C-H}} = 620$$

$$\epsilon_{\text{C-C}} = 80 \text{ kJ/mol} = \frac{80}{96.5} \text{ eV / bond}$$

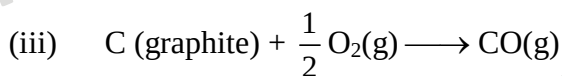
$$\lambda = \frac{1240}{80} \times 96.5 \text{ nm} = 1.493 \times 10^3 \text{ nm}$$



$$\Delta H = -393.5 \text{ kJ mol}^{-1}$$

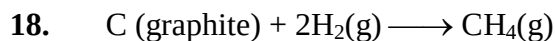


$$\Delta H = -283.5 \text{ kJ/mol}$$



$$(iii) = (i) - (ii)$$

$$\Delta_r H = -393.5 + 283.5 = -110.0 \text{ kJ/mol}$$

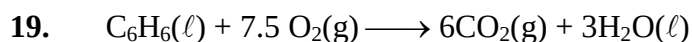


$$\Delta_r H = \Delta H_f \text{ CH}_4(\text{g})$$

$$890.3 = \Delta H_f \text{ CH}_4(\text{g}) - \Delta H_f \text{ CO}_2(\text{g}) - 2 \Delta H_f \text{ H}_2\text{O}(\ell)$$

$$890.3 = \Delta H_f \text{ CH}_4(\text{g}) + 393.5 + 2 \times 285.8$$

$$\Delta_r H = \Delta H_f \text{ CH}_4(\text{g}) = -74.8 \text{ kJ/mol}$$



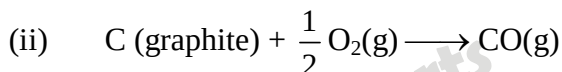
$$\Delta n_g = -1.5$$

$$\Delta H = \Delta U + \Delta n_g RT$$

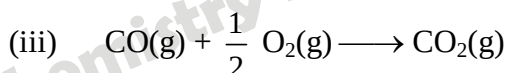
$$\Delta H = -3263.9 - \frac{1.5 \times 8.314 \times 298}{1000} = -3267.6 \text{ kJ}$$



$$\Delta_r H^\circ = x \text{ kJ/mol}$$

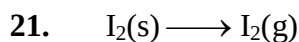


$$\Delta_r H^\circ = y \text{ kJ/mol}$$



$$\Delta_r H^\circ = z \text{ kJ/mol}$$

$$(i) = (ii) + (iii) \quad x = y + z$$

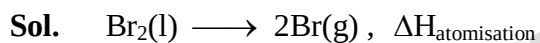


$$(\Delta C_p)_r = 0.031 - 0.055 = -0.024 \text{ cal g}^{-1} \text{ K}^{-1}$$

$$\Delta_r H_{523 \text{ K}} - \Delta_r H_{473 \text{ K}} = -0.024 \times (523 - 473)$$

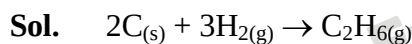
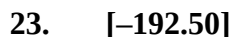
$$\Delta_r H_{523 \text{ K}} - 24 = -0.024 \times 50$$

$$\Delta_r H_{523 \text{ K}} = 22.8 \text{ cal g}^{-1}$$



$$\Delta H_{\text{atomisation}} = \Delta H_{\text{vap}} + \Delta H_{\text{BE}}$$

Therefore $x > y$

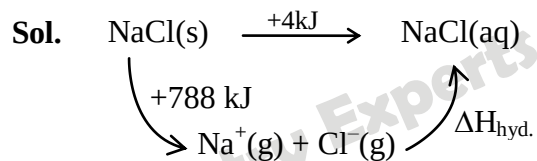


$$\Delta H_{\text{reaction}} = \sum \Delta H_{\text{C}_R} - \sum \Delta H_{\text{C}_P}$$

$$= 2 \times (-286) + 3 \times (-393.5) - (-1560)$$

$$= -192.5 \text{ kJ/mole}$$

24. (3)

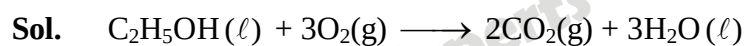


$$4 = 788 + \Delta H_{\text{hyd.}}$$

$$\Delta H_{\text{hyd}} = 4 - 788 = -784 \text{ kJ / mol}^{-1}$$

25. (326400)

NTA Ans. (-326400.00)



$$\Delta n_g = 2 - 3 = -1$$

$$\Delta_c H = \Delta_c U + (\Delta n_g) RT$$

$$\Delta_c H = \Delta_c U - RT$$

$$\Delta_c U = \Delta_c H + RT$$

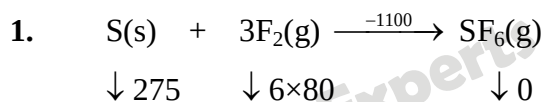
$$= -327 \times 103 + 2 \times 300$$

$$= -326400 \text{ cal.}$$

$$\therefore \text{Heat evolved}$$

$$= 326400 \text{ cal.}$$

JEE ADVANCED



$$-1100 = 275 + 6 \times 80 - 6 \epsilon_{\text{S-F}}$$

$$6 \epsilon_{\text{S-F}} = 275 - 6 \times 80 + 1100$$

$$\epsilon_{\text{S-F}} = 309.16 \text{ kJ/mol}$$

2. Combination is exothermic reaction

$$3. \quad \Delta_r H = \Delta H_f \text{B}_2\text{O}_3(\text{s}) + 3\Delta H_f \text{H}_2\text{O}(\text{g}) - \Delta H_f \text{B}_2\text{H}_6(\text{g})$$

$$\Delta H_f \text{B}_2\text{O}_3(\text{s}) = -1273 \text{ kJ}$$

$$44 = \Delta H_f \text{H}_2\text{O}(\text{g}) + 286$$

$$\Delta H_f \text{H}_2\text{O}(\text{g}) = -242 \text{ kJ}$$

$$\Delta H_f \text{B}_2\text{H}_6 = 36 \text{ kJ}$$

$$\Delta_r H = -1273 - 3 \times 242 - 36 = -2035 \text{ kJ/mol}$$

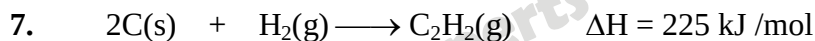
$$4. \quad \Delta_r H = \Delta H_f \text{CO}(\text{g}) + \Delta H_f \text{H}_2\text{O}(\text{g}) - \Delta H_f \text{CO}_2(\text{g})$$

$$= -110.5 - 241.8 + 393.5 = 41.2 \text{ kJ/mol}$$

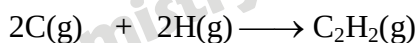
5. Refer notes

$$6. \quad 9 = 2.5 \times 0.45 \text{ kJ}$$

$$\Delta H = -\frac{2.5 \times 0.45}{3.5} \times 28 = -9 \text{ kJ/mol}$$



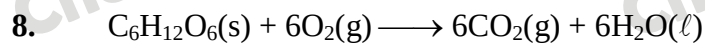
$$\downarrow 1410 \quad \downarrow 330 \quad \downarrow 0$$



$$225 = 1410 + 330 - 2 \times \epsilon_{\text{C-H}} - \epsilon_{\text{C}\equiv\text{C}}$$

$$225 = 1740 - 2 \times 350 - \epsilon_{\text{C}\equiv\text{C}}$$

$$\epsilon_{\text{C}\equiv\text{C}} = 1740 - 700 - 225 = +815 \text{ kJ}$$



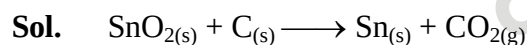
$$\Delta_r H = -6 \times 400 - 6 \times 30 + 1300$$

$$= -2700 \text{ kJ/mol}$$

$$= -\frac{2900}{180} \text{ kJ (per gram)} = -16.11 \text{ kJ}$$

9. Refer notes

10. (935.00)



$$\Delta H_{\text{reaction}}^0 = \Delta H_f^0(\text{CO}_2(\text{g})) - \Delta H_f^0(\text{SnO}_2, \text{S})$$

$$= -394.0 - [-581.0]$$

$$= 187 \text{ KJ}$$

$$\Delta S_{\text{reaction}}^0 = S^0(\text{Sn}(\text{s})) + S^0(\text{CO}_2(\text{g})) - S^0(\text{SnO}_2, (\text{s})) - S^0(\text{Cs}))$$

$$= (52 + 210) - (56 + 6)$$

$$= 200 \text{ J}$$

$$\boxed{\Delta G^0 = \Delta H^0 - T\Delta S^0}$$

For reaction to be spontaneous

$$\Delta G^0 < 0$$

$$\Delta H^0 - T\Delta S^0 < 0$$

$$187 \times 1000 - T \times 200 < 0$$

$$\boxed{T > 935\text{K}}$$

EXERCISE # (JEE-ADVANCED)

1. The nucleidic ratio ${}^3_1\text{H}$ to ${}^1_1\text{H}$ in a sample of water is $8.0 \times 10^{-18} : 1$. Tritium undergoes decay with a half-life period of 12.3 years. How many tritium atoms would 10.0 g of such a sample contain 40 years after the original sample is collected. **[JEE-1992]**

2. One of the hazards of nuclear explosion is the generation of ${}^{90}\text{Sr}$ and its subsequent incorporation in bones. This nucleide has a half-life of 28.1 years. Suppose one microgram was absorbed by a new-born child, how much ${}^{90}\text{Sr}$ will remain in his bones after 20 years. **[JEE-1995]**

3. ${}^{27}_{13}\text{Al}$ is a stable isotope. ${}^{29}_{13}\text{Al}$ is expected to disintegrate by : **[JEE-1996]**
 (A) α emission (B) β emission (C) positron emission (D) proton emission

4. ${}^{227}\text{Ac}$ has a half-life of 22.0 year with respect to radioactive decay. The decay follows two parallel paths, one leading to ${}^{227}\text{Th}$ and the other to ${}^{227}\text{Fr}$. The percentage yields of these two daughter nucleides are 2.0 and 98.0 respectively. What are the decay constants (λ) for each of the separate paths ? **[JEE-1996]**

5. Write a balanced equation for the reaction of N^{14} with α -particles. **[JEE-1997]**

6. **[JEE-1998]**
- | Assertion | Reason |
|--|---|
| Nucleide ${}^{30}_{13}\text{Al}$ is less stable than ${}^{40}_{20}\text{Ca}$ | Nucleides having odd number of protons and neutrons are generally unsatable |

- (A) if both assertion and reason are correct and reason is the correct explanation of the assertion
 (B) if both assertion and reason are correct, but reason is not the correct explanation of the assertion
 (C) if assertion is correct but reason is incorrect
 (D) if assertion is incorrect but reason is correct

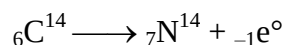
7. Decrease in atomic number is observed during **[JEE-1998]**
 (A) alpha emission (B) beta emission
 (C) positron emission (D) electron capture

8. The number of neutrons accompanying the formation of ${}_{54}\text{X}^{139}$ and ${}_{38}\text{Sr}^{94}$ from the absorption of slow neutron by ${}_{92}\text{U}^{235}$ followed by nuclear fission is [JEE 1999]
 (A) 0 (B) 2 (C) 1 (D) 3
9. ${}^{23}_{11}\text{Na}$ is the most stable isotope of Na. Find the process by which ${}^{24}_{11}\text{Na}$ can undergo radioactive decay: [JEE 2000]
 (A) β^- -emission (B) α -emission (C) β^+ -emission (D) K-electron capture
10. ${}^{64}\text{Cu}$ (half-life = 12.8 hr) decays by β^- emission (38%), β^+ emission (19%) and electron capture (43%). Write the decay product and calculate partial half-lives for each of the decay processes. [JEE 2002]
11. A radioactive sample emits n β -particles in 2 sec. In next 2 sec it emits 0.75 n β -particles, what is the mean life of the sample? [JEE 2003]
12. Fill in the blanks [JEE 2005]
 (a) ${}^{235}_{92}\text{U} + {}^1_0\text{n} \longrightarrow {}^{137}_{52}\text{A} + {}^{97}_{40}\text{B} + \text{_____}$
 (b) ${}^{82}_{34}\text{Se} \longrightarrow 2{}^0_{-1}\text{e} + \text{_____}$

Question No. 13 to 15 (3 questions) Carbon 14 is used to determine the age of organic material. The procedure is based on the formation of ${}^{14}\text{C}$ by neutron capture in the upper atmosphere.



${}^{14}\text{C}$ is absorbed by living organisms during photosynthesis. The ${}^{14}\text{C}$ content is constant in living organism once the plant or animal dies, the uptake of carbon dioxide by it ceases and the level of ${}^{14}\text{C}$ in the dead being falls due to the decay which ${}^{14}\text{C}$ undergoes.



The half life period of ${}^{14}\text{C}$ is 5770 years. The decay constant (λ) can be calculated by using the following formula $\lambda = \frac{0.693}{t_{1/2}}$

The comparison of the β^- activity of the dead matter with that of carbon still in circulation enables measurement of the period of the isolation of the material from the living cycle. The method however, ceases to be accurate over periods longer than 30,000 years. The proportion of ^{14}C to ^{12}C in living matter is $1 : 10^{12}$ [JEE 2006]

13. Which of the following option is correct ? [JEE 2006]

- (A) In living organisms, circulation of ^{14}C from atmosphere is high so the carbon content is constant in organism
- (B) Carbon dating can be used to find out the age of earth crust and rocks
- (C) Radioactive absorption due to cosmic radiation is equal to the rate of radioactive decay, hence the carbon content remains constant in living organism
- (D) Carbon dating cannot be used to determine concentration of ^{14}C in dead beings.

14. What should be the age of fossil for meaningful determination of its age? [JEE 2006]

- (A) 6 years
- (B) 6000 years
- (C) 60000 years
- (D) it can be used to calculate any age

15. A nuclear explosion has taken place leading to increase in concentration of C^{14} in nearby areas. C^{14} concentration is C_1 in nearby areas and C_2 in areas far away. If the age of the fossil is determined to be t_1 and t_2 at the places respectively, then [JEE 2006]

- (A) The age of the fossil will increase at the place where explosion has taken place and

$$t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

- (B) The age of the fossil will decrease at the place where explosion has taken place and

$$t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

- (C) The age of fossil will be determined to be the same

- (D) $\frac{t_1}{t_2} = \frac{C_1}{C_2}$

16. A positron is emitted from $^{23}_{11}\text{Na}$. The ratio of the atomic mass and atomic number of the resulting nuclide is [JEE 2006]

- (A) 22/10
- (B) 22/11
- (C) 23/10
- (D) 23/12

17. The total number of α and β particles emitted in the nuclear reaction ${}_{92}^{238}\text{U} \longrightarrow {}_{82}^{214}\text{Pb}$ is.

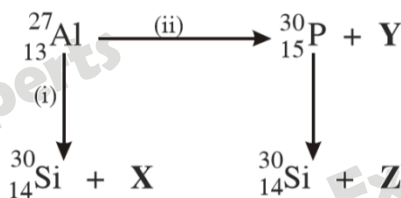
[JEE 2009]

18. The number of neutrons emitted when ${}_{92}^{235}\text{U}$ undergoes controlled nuclear fission to ${}_{54}^{142}\text{Xe}$ and ${}_{38}^{90}\text{Sr}$ is -

[JEE 2010]

19. Bombardment of aluminium by α -particle leads to its artificial disintegration in two ways, (i) and (ii) as shown. Products X, Y and Z respectively are :

[JEE 2011]



(A) proton, neutron, positron

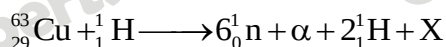
(B) neutron, positron, proton

(C) proton, positron, neutron

(D) positron, proton, neutron

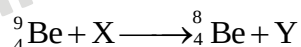
20. The periodic table consists of 18 groups. An isotope of copper, on bombardment with protons, undergoes a nuclear reaction yielding element X as shown below. To which group, element X belongs in the periodic table ?

[JEE 2012]



21. In the nuclear transmutation

[JEE-2013]



(X, Y) is(are)

(A) (γ , n)

(B) (p, D)

(C) (n, D)

(D) (γ , D)

22. A closed vessel with rigid walls contains 1 mol of ${}_{92}^{238}\text{U}$ and 1 mol of air at 298 K. Considering complete decay of ${}_{92}^{238}\text{U}$ to ${}_{82}^{206}\text{Pb}$, the ratio of the final pressure to the initial pressure of the system at 298 K is -

[JEE-2015]

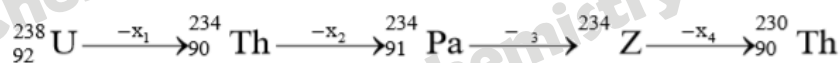
23. **STATEMENT-1** : The plot of atomic number (y-axis) versus number of neutrons (x-axis) for stable nuclei shows a curvature towards x-axis from the line of 45° slope as the atomic number is increased. and

STATEMENT-2 : Proton-proton electrostatic repulsions begin to overcome attractive forces involving protons and neutrons in heavier nuclides. **[JEE 2008]**

- (A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True

24. In the decay sequence,

[JEE-2019]



x_1 , x_2 , x_3 and x_4 are particles/radiation emitted by the respective isotopes. The correct option(s) is(are) :

- (A) x_1 will deflect towards negatively charged plate
 (B) x_2 is ${}^{-}\beta$
 (C) x_3 is γ -ray
 (D) Z is an isotope of uranium

25. ${}_{92}^{238}\text{U}$ is known to undergo radioactive decay to form ${}_{82}^{206}\text{Pb}$ by emitting alpha and beta particles.

A rock initially contained 68×10^{-6} g of ${}_{92}^{238}\text{U}$. If the number of alpha particles that it would emit during its radioactive decay of ${}_{92}^{238}\text{U}$ to ${}_{82}^{206}\text{Pb}$ in three half-lives is $Z \times 10^{18}$, then what is the value of Z ? **[JEE-2020]**

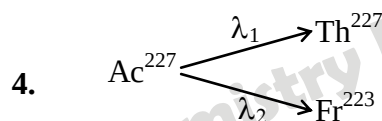
ANSWER KEY

EXERCISE # (JEE-ADVANCED)

1. $N_t = 5.6223 \times 10^6$
 moles of ${}_1\text{H}^3$ atom = $10 \times 8 \times 10^{-18}$
 number of ${}_1\text{H}^3$ atom = $10 \times 8 \times 10^{-18} \times 6 \times 10^{23} = 4.8 \times 10^7$
 $40 = \frac{12.3}{\ln 2} \ln \frac{4.8 \times 10^7}{N} = 5.6 \times 10^5$

2. $W_t = 1 \times e^{\frac{-\ln 2}{28.1} \times 20} = 0.610 \text{ m gram}$

3. (B)



$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{2}{100} \text{ and } \frac{\ln 2}{\lambda_1 + \lambda_2} = 22$$

$$\lambda_1 = \frac{2}{100} \times \frac{\ln 2}{22} = \frac{\ln 2}{11 \times 100} = 6.3 \times 10^{-4}$$

$$\lambda_2 = \frac{98}{2} \text{ and } \lambda_1 = 3.087 \times 10^{-2}$$

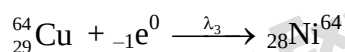
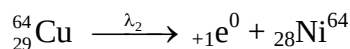
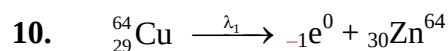


6. (B)

7. (ACD)

8. (D)

9. (A)



$$\frac{\ln 2}{\lambda_1 + \lambda_2 + \lambda_3} = 12.8 \text{ hr;}$$

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{38}{100}, \quad (t_{1/2})_1 = \frac{\ln 2}{\lambda_1} = \frac{12.8}{38} \times 100$$

$$\frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{19}{100}, \quad (t_{1/2})_2 = \frac{\ln 2}{\lambda_2} = \frac{12.8}{19} \times 100$$

$$\frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{43}{100}, \quad (t_{1/2})_3 = \frac{\ln 2}{\lambda_3} = \frac{12.8}{43} \times 100$$

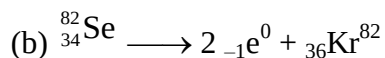
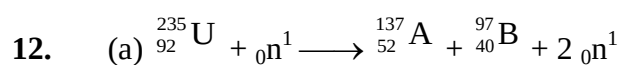
11. $n = N_0[1 - e^{-2\lambda}]$

$$1.75 n = N_0[1 - e^{-4\lambda}]$$

$$1.75 = \frac{(1 - e^{-2\lambda})(1 + e^{-2\lambda})}{1 - e^{-2\lambda}} = 1 + e^{-2\lambda}$$

$$e^{-2\lambda} = 0.75 = \frac{3}{4}$$

$$e^{2\lambda} = \frac{4}{3} \Rightarrow \lambda = \frac{1}{2} \ln \frac{4}{3}$$



13. (C) 14. (B) 15. (A) 16. (C) 17. (8) 18. (4) 19. (A)
 20. (8) 21. (AB) 22. (9) 23. (A) 24. (ABD) 25. (1.20)

HINT AND SOLUTIONS

1. $N_t = 5.6223 \times 10^6$

moles of ${}_1\text{H}^3$ atom = $10 \times 8 \times 10^{-18}$

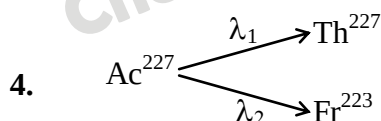
number of ${}_1\text{H}^3$ atom = $10 \times 8 \times 10^{-18} \times 6 \times 10^{23} = 4.8 \times 10^7$

$$40 = \frac{12.3}{\ln 2} \ln \frac{4.8 \times 10^7}{N} = 5.6 \times 10^5$$

2. $W_t = 1 \times e^{\frac{-\ln 2}{28.1} \times 20} = 0.610 \text{ m gram}$

3. (B)

$$\left(\frac{n}{p}\right) > \left(\frac{n}{p}\right)_{\text{stable}} \quad \beta\text{-emission}$$



$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{2}{100} \text{ and } \frac{\ln 2}{\lambda_1 + \lambda_2} = 22$$

$$\lambda_1 = \frac{2}{100} \times \frac{\ln 2}{22} = \frac{\ln 2}{11 \times 100} = 6.3 \times 10^{-4}$$

$$\lambda_2 = \frac{98}{2} \text{ and } \lambda_1 = 3.087 \times 10^{-2}$$



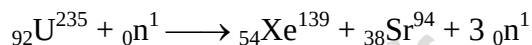
6. (B)

In ${}_{13}\text{Al}^{30}$ both neutron and proton are odd in number therefore less stable.

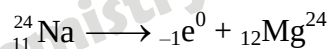
7. (ACD)

In β -emission atomic number increase in all other option it decreases

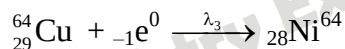
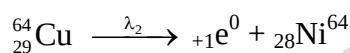
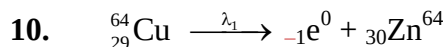
8. (D)



9. (A)



$$\frac{n}{p} > \left(\frac{n}{p} \right)_{\text{stable}} \quad \text{for Na}$$



$$\frac{\ln 2}{\lambda_1 + \lambda_2 + \lambda_3} = 12.8 \text{ hr;}$$

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{38}{100}, \quad (t_{1/2})_1 = \frac{\ln 2}{\lambda_1} = \frac{12.8}{38} \times 100$$

$$\frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{19}{100}, \quad (t_{1/2})_2 = \frac{\ln 2}{\lambda_2} = \frac{12.8}{19} \times 100$$

$$\frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{43}{100}, \quad (t_{1/2})_3 = \frac{\ln 2}{\lambda_3} = \frac{12.8}{43} \times 100$$

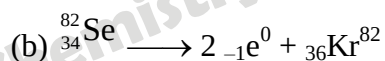
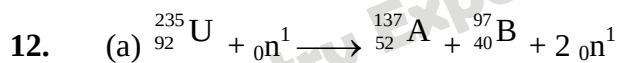
$$11. \quad n = N_0[1 - e^{-2\lambda}]$$

$$1.75 n = N_0[1 - e^{-4\lambda}]$$

$$1.75 = \frac{(1 - e^{-2\lambda})(1 + e^{-2\lambda})}{1 - e^{-2\lambda}} = 1 + e^{-2\lambda}$$

$$e^{-2\lambda} = 0.75 = \frac{3}{4}$$

$$e^{2\lambda} = \frac{4}{3} \Rightarrow \lambda = \frac{1}{2} \ln \frac{4}{3}$$



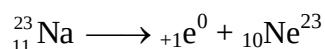
15. (A)

$$t_1 = \frac{1}{\lambda} \ln \frac{C_1}{(\%C^{14})_{\text{fossil}}}$$

$$t_2 = \frac{1}{\lambda} \ln \frac{C_2}{(\%C^{14})_{\text{fossil}}}$$

$$t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

16. (C)

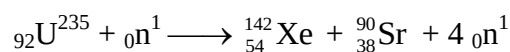


17. (8)

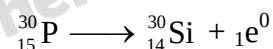
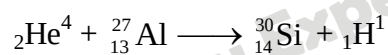
$$\text{Number of } \alpha\text{-particles} = \frac{238 - 214}{4} = 6$$

$$\text{Number of } \beta\text{-particles} = 82 - [92 - 12] = 2$$

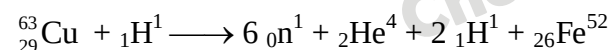
18. (4)



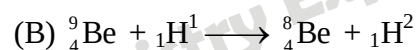
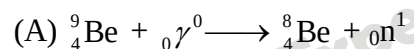
19. (A)



20. (8)



21. (AB)

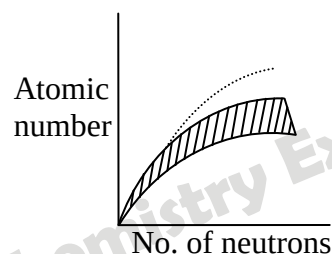


22. (9)


 $P_i \propto 1 \text{ mol of air}$
 $P_f \propto 1 \text{ mol of air} + 8 \text{ moles of } \alpha \text{ - particle}$

$$\frac{P_f}{P_i} = \frac{9}{1}$$

23. (A)



Note : Neutron increases more than proton to overcome proton-proton repulsion.

24. (ABD)

$$x_1 = \alpha, x_2 = \beta, x_3 = \beta \text{ and } x_4 = \alpha$$

25. (1.20)



$\therefore$ mole of U^{238} present in the rock initially

$$(n_0) = \frac{68 \times 10^{-6}}{238} \text{ mole}$$

After 3 half life

$$\text{moles of } \text{U}^{238} \text{ decayed} = (n_0) - \frac{n_0}{2^3} = \frac{7}{8} \times n_0$$

$$\therefore \text{ moles of } \alpha\text{-particles formed} = \frac{7}{8} \times n_0 \times 8 = 7 \times n_0 = \frac{7 \times 68 \times 10^{-6}}{238}$$

$$\therefore \text{ No. of } \alpha\text{-particles formed} = \frac{7 \times 68 \times 10^{-6}}{238} \times 6 \times 10^{23} = 1.2 \times 10^{18}$$

$$\therefore Z = 1.20$$

EXERCISE-(JEE-MAIN)

- The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy ? **[AIEEE-2003]**
 (1) < 40 kJ (2) Zero (3) 40 kJ (4) > 40 kJ
- Consider the reaction : $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ carried out at constant temperature and pressure, if ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true ? **[AIEEE-2005]**
 (1) $\Delta H = \Delta U$ (2) $\Delta H = 0$ (3) $\Delta H > \Delta U$ (4) $\Delta H < \Delta U$
- An ideal gas is allowed to expand both adiabatic reversibly and adiabatic irreversibly. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct :- **[AIEEE-2006]**
 (1) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process
 (2) $(T_f)_{\text{rev}} = (T_f)_{\text{irrev}}$
 (3) $T_f = T_i$ for both reversible and irreversible processes
 (4) $(T_f)_{\text{irrev}} > (T_f)_{\text{rev}}$
- Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and 100°C , (Given : Molar enthalpy of vapourisation of water at 1 bar and $373\text{ K} = 41\text{ kJ mol}^{-1}$ and $R = 8.3\text{ J mol}^{-1}\text{ K}^{-1}$ will be) :- **[AIEEE-2007]**
 (1) 4.100 kJ mol^{-1} (2) $3.7904\text{ kJ mol}^{-1}$
 (3) $37.904\text{ kJ mol}^{-1}$ (4) 41.00 kJ mol^{-1}
- The value of enthalpy change (ΔH) for the reaction
 $\text{C}_2\text{H}_5\text{OH}(\ell) + 3\text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)} + 3\text{H}_2\text{O}(\ell)$
 at 27°C is $-1366.5\text{ kJ mol}^{-1}$. The value of internal energy change for the above reaction at this temperature will be :- **[AIEEE-2011]**
 (1) -1371.5 kJ (2) -1369.0 kJ
 (3) -1364.0 kJ (4) -1361.5 kJ
- The difference between the reaction enthalpy change ($\Delta_r H$) and reaction internal energy change ($\Delta_r U$) for the reaction : **[JEE MAIN (online)- 2012]**
 $2\text{C}_6\text{H}_6(\ell) + 15\text{O}_2(\text{g}) \rightarrow 12\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\ell)$ at 300 K is ($R = 8.314\text{ J mol}^{-1}\text{ K}^{-1}$)
 (1) 0 J mol^{-1} (2) 2490 J mol^{-1}
 (3) -2490 J mol^{-1} (4) -7482 J mol^{-1}

7. Which of the following statements/relationships is not correct in thermodynamic changes ?

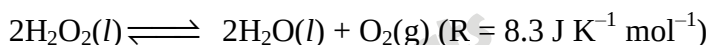
(1) $q = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas) [JEE MAIN (online)- 2014]

(2) For a system at constant volume, heat involved merely changes to internal energy.

(3) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(4) $\Delta U = 0$ (isothermal reversible expansion of a gas)

8. If 100 mole of H_2O_2 decompose at 1 bar and 300K, the work done (kJ) by one mole of $\text{O}_2(\text{g})$ as it expands against 1 bar pressure is : [JeeMain(online) 2016]



- (1) 498.00 (2) 249.00 (3) 124.50 (4) 62.25

9. ΔU is equal to : [JEE MAIN-2017]

- (1) Adiabatic work (2) Isothermal work (3) Isochoric work (4) Isobaric work

10. For a reaction, $\text{A}(\text{g}) \rightarrow \text{A}(\text{l})$; $\Delta H = -3RT$. The correct statement for the reaction is :

[Jee Main(online) 2017]

- (1) $|\Delta H| > |\Delta U|$ (2) $\Delta H = \Delta U \neq 0$ (3) $|\Delta H| < |\Delta U|$ (4) $\Delta H = \Delta U = 0$

11. A gas undergoes change from A to state B. In this process, the heat absorbed and work done by the gas is 5J and 8 J, respectively. Now gas is brought back to A by another process of during which 3 J of heat is evolve. In this reverse process of B to A :- [Jee Main(online) 2017]

- (1) 10 J of the work will be done by the surrounding on gas.
 (2) 6 J of the work will be done by the surrounding on gas.
 (3) 10 J of the work will be done by the gas.
 (4) 6 J of the work will be done by the gas.

12. An ideal gas undergoes a cyclic process as shown in figure.

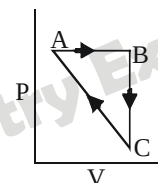
[Jee Main(online) 2018]

$$\Delta U_{BC} = -5 \text{ kJ mol}^{-1}, q_{AB} = 2 \text{ kJ mol}^{-1}$$

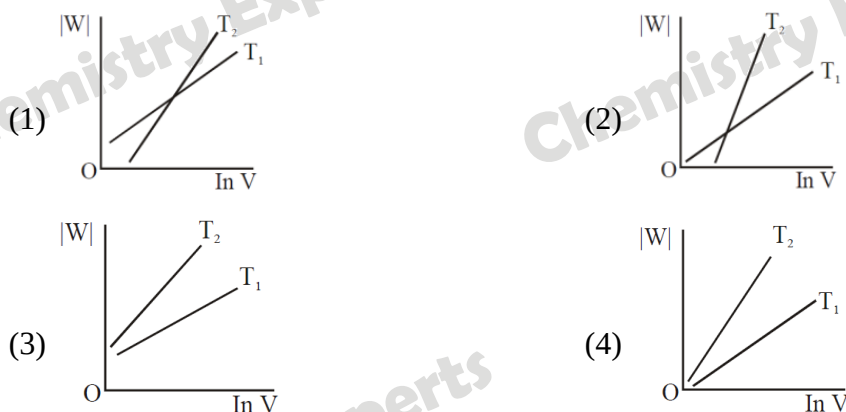
$$W_{AB} = -5 \text{ kJ mol}^{-1}, W_{CA} = 3 \text{ kJ mol}^{-1}$$

Heat absorbed by the system during process CA is :-

- (1) 18 kJ mol^{-1} (2) $+5 \text{ kJ mol}^{-1}$
 (3) -5 kJ mol^{-1} (4) -18 kJ mol^{-1}



13. Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures T_1 and T_2 ($T_1 < T_2$). The correct graphical depiction of the dependence of work done (w) on the final volume (V) is: [Jee-Main (Jan.) 2019]

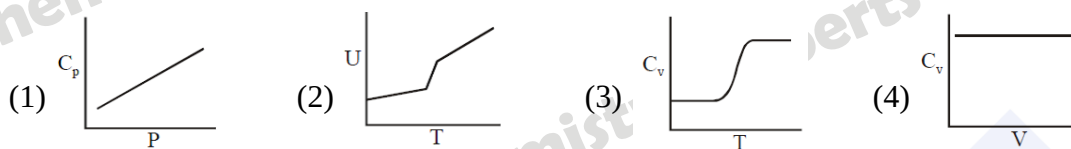


14. An ideal gas undergoes isothermal compression from 5 m^3 to 1 m^3 against a constant external pressure of 4 Nm^{-2} . Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is $24 \text{ J mol}^{-1} \text{ K}^{-1}$, the temperature of Al increases by :

[Jee-Main (Jan.) 2019]

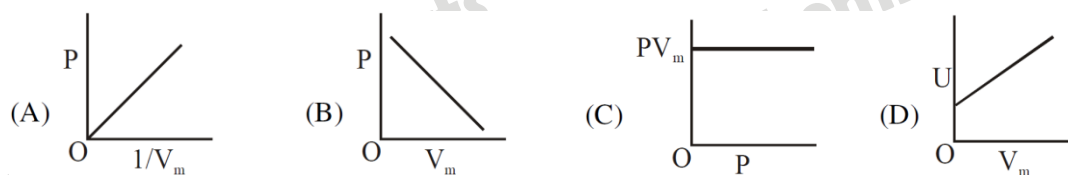
- (1) $\frac{3}{2} \text{ K}$ (2) 2 K (3) 1 K (4) $\frac{2}{3} \text{ K}$

15. For a diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities? [Jee-Main (Jan.) 2019]



16. The combination of plots which does not represent isothermal expansion of an ideal gas is:

[Jee-Main (Jan.) 2019]

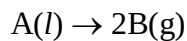


- (1) (B) and (D) (2) (A) and (D) (3) (A) and (C) (4) (B) and (C)

17. For silver, $C_p(\text{J K}^{-1} \text{mol}^{-1}) = 23 + 0.01T$. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be close to :
[Jee-Main (Apr.) 2019]
 (1) 16 kJ (2) 13 kJ (3) 62 kJ (4) 21 kJ
18. Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas ? (Assume non-expansion work is zero)
[Jee-Main (Apr.) 2019]
 (1) Cyclic process : $\Delta U = -w$ (2) Cyclic process : $q = -w$
 (3) Isochoric process : $\Delta U = q$ (4) Isothermal process : $q = -w$
19. 5 moles of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature becomes 200 K. If $C_v = 28 \text{ J K}^{-1} \text{mol}^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0 \text{ J K}^{-1} \text{mol}^{-1}$)
[Jee-Main (Apr.) 2019]
 (1) $\Delta U = 14 \text{ J}$; $\Delta(pV) = 0.8 \text{ J}$ (2) $\Delta U = 2.8 \text{ kJ}$; $\Delta(pV) = 0.8 \text{ kJ}$
 (3) $\Delta U = 14 \text{ kJ}$; $\Delta(pV) = 18 \text{ kJ}$ (4) $\Delta U = 14 \text{ kJ}$; $\Delta(pV) = 4 \text{ kJ}$
20. Among the following, the set of parameters that represents path functions, is :
 (A) $q + w$ (B) q (C) w (D) $H - TS$
 (1) (B) and (C) (2) (A) and (D) (3) (A), (B) and (C) (4) (B), (C) and (D)
[Jee-Main (Apr.) 2019]
21. During compression of a spring the work done is 10kJ and 2kJ escaped to the surroundings as heat. The change in internal energy, ΔU (in kJ) is:
[Jee-Main (Apr.) 2019]
 (1) 8 (2) -8 (3) 12 (4) -12
22. An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. The work done in kJ is:
[Jee-Main (Apr.) 2019]
 (1) -9.0 (2) -0.9 (3) -2.0 (4) + 10.0
23. The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane (I) is carried out at a temperature T, is equal to :
[Jee-Main (Apr.) 2019]
 (1) $4 RT$ (2) $-4 RT$ (3) $-3 RT$ (4) $3RT$
24. The true statement amongst the following is:
[Jee Mains (Jan. 2020)]
 (1) S is a function of temperature but ΔS is not a function of temperature.
 (2) Both S and ΔS are not functions of temperature.
 (3) Both S and ΔS are functions of temperature.
 (4) S is not a function of temperature but ΔS is a function of temperature.

25. For the reaction;

[Jee Main (Jan. 2020)]



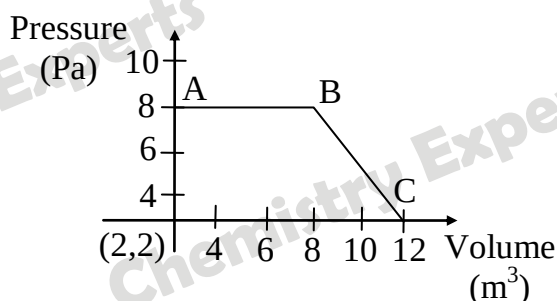
$$\Delta U = 2.1 \text{ kcal}, \Delta S = 20 \text{ cal K}^{-1} \text{ at } 300 \text{ K.}$$

Hence ΔG in kcal is _____ :

26. The magnitude of work done by a gas that undergoes a reversible expansion along the path

ABC shown in the figure is _____.

[Jee-Main (Jan.) 2020]



27. For one mole of an ideal gas, which of these statements must be true?

(a) U and H each depends only on temperature

[Jee-Main (Sep) 2020]

(b) Compressibility factor z is not equal to 1

(c) $C_{p,m} - C_{v,m} = R$

(d) $dU = C_v dT$ for any process

(1) (a) and (c)

(2) (b), (c) and (d)

(3) (a), (c) and (d)

(4) (c) and (d)

28. Five moles of an ideal gas at 1 bar and 298 K is expanded into vacuum to double the volume.

The work done is :

[Jee-Main (Sep) 2020]

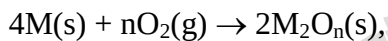
(1) $-RT \ln V_2/V_1$

(2) $-RT(V_2 - V_1)$

(3) Zero

(4) $C_v(T_2 - T_1)$

29. For a reaction, [Jee-Main (Sep) 2020]



the free energy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which :

- (1) The slope changes from positive to zero
- (2) The free energy change shows a change from negative to positive value
- (3) The slope changes from negative to positive
- (4) The slope changes from positive to negative

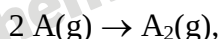
30. At constant volume, 4 mol of an ideal gas when heated from 300 K to 500 K changes its internal energy by 5000 J. The molar heat capacity at constant volume is _____.

[Jee-Main (Sep) 2020]

31. The internal energy change (in J) when 90 g of water undergoes complete evaporation at 100°C is _____. (Given : ΔH_{vap} for water at 373 K = 41 kJ/ mol, $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)

[Jee-Main (Sep) 2020]

32. For a dimerization reaction



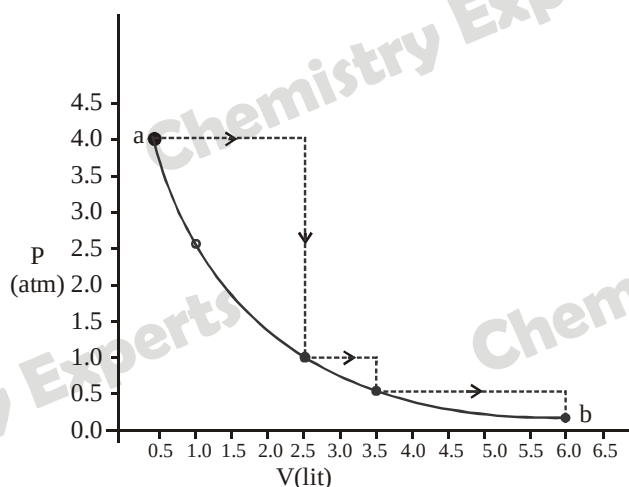
[Jee-Main (Sep) 2020]

at 298 K, $\Delta U^\ominus = -20 \text{ kJ mol}^{-1}$, $\Delta S^\ominus = -30 \text{ JK}^{-1} \text{ mol}^{-1}$, then the $\Delta G^\ominus$ will be _____ J.

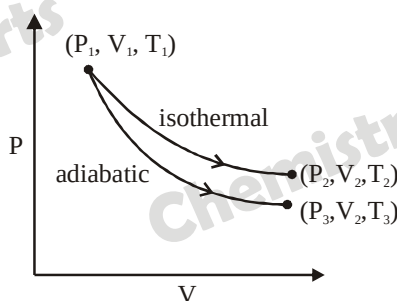
EXERCISE-(JEE-ADVANCED)

1. A sample of argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 dm³ to 2.50 dm³. Calculate the enthalpy change in this process. $C_{v,m}$ for argon is 12.48 JK⁻¹ mol⁻¹ **[JEE Adv. 2000]**
2. Which of the following statement is false? **[JEE Adv. 2001]**
 - (A) Work is a state function
 - (B) Temperature is a state function
 - (C) Change of state is completely defined when initial and final states are specified.
 - (D) Work appears at the boundary of the system
3. One mole of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy (ΔU) = 30.0 Latm. The change in enthalpy (ΔH) of the process in Latm. **[JEE Adv. 2002]**
 - (A) 40.0
 - (B) 42.3
 - (C) 44.0
 - (D) Not defined, because pressure is not constant
4. Two moles of a perfect gas undergoes the following process. **[JEE Adv. 2002]**
 - (a) A reversible isobaric expansion from (1.0 atm, 20.0 L) to (1.0 atm, 40.0 L);
 - (b) A reversible isochoric change of state from (1.0 atm, 40.0 L) to (0.5 atm, 40.0 L);
 - (c) A reversible isothermal compression from (0.5 atm, 40.0 L) to (1.0 atm, 20.0 L);
 - (i) Sketch with labels each of the processes on the same P-V diagram.
 - (ii) Calculate the total work (w) and the total heat change (q) involved in the above processes.
 - (iii) What will be the values of ΔU , ΔH and ΔS for the overall process ?
5. C_{vc} value of He is laways but C_v value of H₂ is at low temperature and moderate temperature and more than at higher temperature. Explain in two or three lines. **[JEE Adv. 2003]**
6. One mole of a liquid (1 bar, 100 ml) is taken in an adiabatic container and the pressure increases steeply to 100 bar. Then at a constant pressure of 100 bar, volume decrease by 1 ml. Find ΔU and ΔH . **[JEE Adv. 2004]**
7. Two mole of an ideal gas is expanded isothermally and reversibly from 1 litre to 10 litre at 300 K. The enthalpy change (in KJ) for the process is :- **[JEE Adv. 2004]**
 - (A) 11.4 kJ
 - (B) - 11.4 kJ
 - (C) 0 kJ
 - (D) 4.8 kJ

8. One mole of monoatomic ideal gas expands adiabatically at initial temp. T against a constant external pressure of 1 atm from one litre to two litre. Find out the final temperature.
($R = 0.0821 \text{ litre. atm K}^{-1} \text{ mol}^{-1}$) [JEE Adv. 2005]
- (A) T (B) $\frac{T}{(2)^{\frac{5}{3}-1}}$ (C) $T - \frac{2}{3 \times 0.0821}$ (D) $T + \frac{2}{3 \times 0.0821}$
9. For the reaction,
 $2\text{CO(g)} + \text{O}_2 \longrightarrow 2\text{CO}_2\text{(g)}; \quad \Delta H = -560 \text{ kJ mol}^{-1}$
 In one litre vessel at 500 K the initial pressure is 70 atm and after the reaction it becomes 40 atm at constant volume of one litre. Calculate change in internal energy. All the above gases show significant deviation from ideal behaviour. (1 L atm = 0.1 kJ) [JEE Adv. 2006]
10. The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one. [JEE Adv. 2006]
- (A) $\frac{4}{2} R$ (B) $\frac{3}{2} R$ (C) $\frac{5}{2} R$ (D) zero
11. Among the following, the state function(s) is (are) [JEE Adv. 2009]
- (A) Internal energy (B) Irreversible expansion work
 (C) Reversible expansion work (D) Molar enthalpy
12. Among the following, extensive property is (properties are) [JEE Adv.-2010]
- (A) molar conductivity (B) electromotive force
 (C) resistance (D) heat capacity
13. One mole of an ideal gas is taken from a to b along two paths denoted by the solid and the dashed lines as shown in the graph below. If the work done along the solid line path is w_s and that along the dotted line path is w_d , then the integer close to the ratio w_d / w_s is – [JEE Adv. 2010]



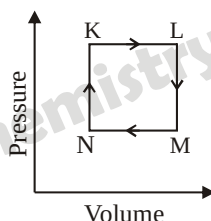
14. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) correct ? [JEE Adv. 2012]



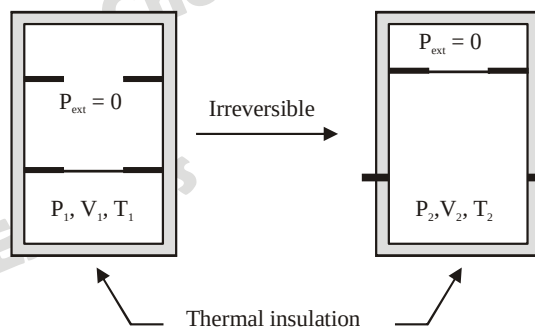
- (A) $T_1 = T_2$ (B) $T_3 > T_1$
 (C) $w_{\text{isothermal}} > w_{\text{adiabatic}}$ (D) $\Delta U_{\text{isothermal}} > \Delta U_{\text{adiabatic}}$

Paragraph for Question 4 and 5

A fixed mass 'm' of a gas is subjected to transformation of states from K to L to M to N and back to K as shown in the figure. [JEE Adv. 2013]

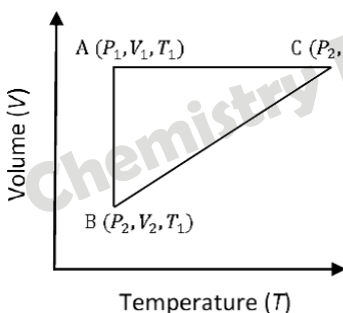


15. The pair of isochoric processes among the transformation of states is
 (A) K to L and L to M (B) L to M and N to K
 (C) L to M and M to N (D) M to N and N to K
16. The succeeding operations that enable this transformation of states are
 (A) Heating, cooling, heating, cooling (B) cooling, heating, cooling, heating
 (C) Heating, cooling, cooling, heating (D) Cooling, heating, heating, cooling
17. An ideal gas in thermally insulated vessel at internal pressure = P_1 , volume = V_1 and absolute temperature = T_1 expands irreversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2 , V_2 and T_2 , respectively. For this expansion, [JEE Adv. 2014]



- (A) $q = 0$ (B) $T_2 = T_1$ (C) $P_2 V_2 = P_1 V_1$ (D) $P_2 V_2^\gamma = P_1 V_1^\gamma$

18. An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The correct statement(s) among the following is (are) **[JEE Adv. 2017]**
- (A) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure p_1
- (B) The work done by the gas is less when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions
- (C) the change in internal energy of the gas is (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$
- (D) If the expansion is carried out freely, it is simultaneously both isothermal both isothermal as well as adiabatic.
19. A reversible cyclic process for an ideal gas is shown below. Here, P , V , and T are pressure, volume and temperature, respectively. The thermodynamic parameters q , w , H and U are heat, work, enthalpy and internal energy, respectively. **[JEE Adv. 2018]**



The correct option(s) is (are)

- (A) $q_{AC} = \Delta U_{BC}$ and $w_{AB} = P_2(V_2 - V_1)$ (B) $w_{BC} = P_2(V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$
- (C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$ (D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} > \Delta U_{CA}$
20. In thermodynamics the P-V work done is given by **[JEE Adv. 2020]**

$$w = - \int dV P_{\text{ext}}$$

For a system undergoing a particular process, the work done is ,

$$w = - \int dV \left(\frac{RT}{V-b} - \frac{a}{V^2} \right)$$

This equation is applicable to a

- (A) System that satisfies the Van der Waals equation of state.
- (B) Process that is reversible and isothermal.
- (C) Process that is reversible and adiabatic.
- (D) Process that is irreversible and at constant pressure.

ANSWER KEY

EXERCISE-(JEE-MAIN)

1. 2 2. 4 3. 4 4. 3 5. 3 6. 4 7. 1
8. 1 9. 1 10. 1 11. 2 12. 2 13. 2 14. 4
15. 1 16. 1 17. 3 18. 1 19. 4 20. 1 21. 1
22. 2 23. 2 24. 3 25. -2.70 26. 68 27. 3 28. 3
29. 2 30. 6.25, NTA Ans. (48) 31. (189494.39) NTA Ans. (189494)
32. -13537.57 Answer by NTA (-13538 J)

EXERCISE-(JEE-ADVANCED)

1. $\Delta H = -115 \text{ J}$ 2. A 3. C
4. (ii) $-w = q = 620.77 \text{ J}$, (iii) $\Delta H = 0$, $\Delta U = 0$, $\Delta S = 0$
5. Hydrogen molecule is diatomic, has three translational, two rotational and one vibrational degree of freedom. The energy spacing between adjacent levels are in the order of:
translational, rotational > vibrational
6. $\Delta U = 0.1 \text{ L atm}$, $\Delta H = 9.9 \text{ L atm}$ 7. C 8. C 9. -557 kJ/mol 10. A
11. AD 12. CD 13. 2 14. AD 15. B 16. C 17. ABC
18. ABD 19. BC 20. (ABC or ABCD)

HINT AND SOLUTIONS

JEE-MAIN

1. ΔU is state function
 $\Delta U_{AB} = +40 \text{ kJ/mol}$
 $\Delta U_{BA} = -40 \text{ kJ/mol}$
2. $\Delta n_g < 0$
 $\Delta H = \Delta U + \Delta n_g RT$
 $\Delta H < \Delta U$
3. $T_f \text{ irrev} > T_f \text{ rev}$
4. $\text{H}_2\text{O(l)} \rightleftharpoons \text{H}_2\text{O(g)}$
 $\Delta n_g = 1$
 $\Delta H = \Delta U + \Delta n_g RT$
 $41 = \Delta U + \frac{1 \times 8.3 \times 373}{1000}$
 $\Delta U = +37.9041 \text{ kJ mol}^{-1}$
5. $\text{C}_2\text{H}_5\text{OH(l)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{CO}_2\text{(g)} + 3\text{H}_2\text{O(l)}$
 $\Delta n_g = -1$
 $\Delta H = \Delta U + \Delta n_g RT$
 $-1366.5 = \Delta U - \frac{1 \times 8.314 \times 300}{1000}$
 $\Delta U = -1366.5 + 2.4942 = -1364.0058 \text{ kJ}$
6. $2\text{C}_6\text{H}_6\text{(l)} + 15\text{O}_2\text{(g)} \longrightarrow 12\text{CO}_2\text{(g)} + 6\text{H}_2\text{O(l)}$
 $\Delta n_g = -3$
 $\Delta H = \Delta U + \Delta n_g RT$
 $\Delta H - \Delta U = \Delta n_g RT$
 $= \frac{-3 \times 8.314 \times 300}{1000} = -7482.6 \text{ J mol}^{-1}$
7. For rev. isothermal process
 $W = -q = -nRT \ln \frac{V_2}{V_1}$
 $\Delta U = 0$
8. $W = -\Delta n_g RT = -1 \times 8.3 \times 300 \text{ J} = -2.49 \text{ kJ}$

9. For adiabatic process

$$q = 0, \Delta U = W$$

10. $\Delta n_g = -1$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$|\Delta U| > |\Delta H|$$

11. $q_{AB} = +5J$

$$W_{AB} = -8J$$

$$\Delta U_{AB} = q_{AB} + W_{AB} = -3J$$

$$\Delta U_{BA} = +3J$$

$$q_{BA} = -3J$$

$$W_{BA} = \Delta U_{BA} - q_{BA} = +6 J$$

12. $\Delta U = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$

$$\Delta U_{AB} = -5 + 2 = -3 \text{ kJ mol}^{-1}$$

$$-5 - 3 + \Delta U_{CA} = 0$$

$$\Delta U_{CA} = +8 \text{ KJ mol}^{-1}$$

$$q_{CA} = \Delta U_{CA} - W_{CA} = 8 - 3 = +5 \text{ kJ mol}^{-1}$$

14. $W = -4 \times (1 - 5) = +16 J$

$$q = C_m \Delta T$$

$$16 = 1 \times 24 \times \Delta T \quad \Delta T = \frac{2}{3} \text{ K}$$

15. C_p is defined at constant P only.

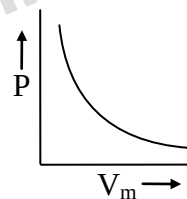
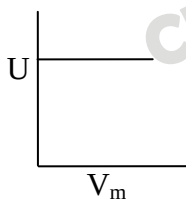
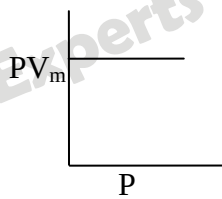
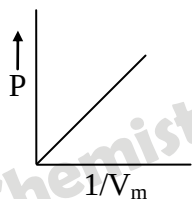
$$U = f(T)$$

C_v is defined at constant V only.

$$C_p \text{ and } C_v = f(T)$$

16. For isoT process

$$PV_m = \text{Constant}$$



$$17. \quad \Delta H = \int_{T_2}^{T_1} nC_p dT = \int_{300}^{1000} 3(23 + 0.01T) dT = 61950 \text{ J} = 62 \text{ kJ}$$

18. For adiabatic process

$$q = 0 \therefore \Delta U = W$$

19. 5 mole ideal gas is given,

$$C_{w,m} = 28 \text{ J K}^{-1} \text{ mol}^{-1}, R = 8 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$100 \text{ K} \longrightarrow 200 \text{ K}$$

$$\Delta U = nc_{v,m}dT = 5 \times 28 \times 100$$

$$= 14000 \text{ J}$$

$$= 14 \text{ kJ}$$

$$\Delta(pv) = P_2V_2 - P_1V_1$$

$$= nRT_2 - nRT_1$$

$$= 5 \times 8 \times 100 = 4 \text{ kJ}$$

20. Both work (w) and heat (q) are state function.

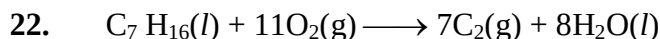
21. From 1st law of thermodynamics

$$\Delta U = q + w$$

$$w = 10 \text{ kJ} (\because \text{work is done on spring})$$

$$q = -2 \text{ kJ} (\because \text{Heat rejected into surrounding})$$

$$\Delta U = -2 + 10 = 8 \text{ kJ}$$



$$\Delta_{ng} = 7 - 11 = -4$$

$$\Delta H = \Delta U + \Delta_{ng}RT$$

$$\Delta H = \Delta U = -4RT$$

23. As process is Irreversible

$$\Rightarrow w = -P_{\text{ext}}(dv)$$

$$= -1 \times 9$$

$$= -9 \text{ lit. bar}$$

24. (3)

25. [-2.70]

$$\text{Sol. } \Delta H = \Delta U + \Delta_{ng}RT$$

$$= 2.1 \times 1000 + 2 \times 2 \times 300$$

$$= 3300 \text{ cal}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= 3.3 - \frac{300 \times 20}{1000} = -2.7 \text{ kcal}$$

26. [68]

Sol. Work done = $-\int P dv$

i.e. area under P-V curve.

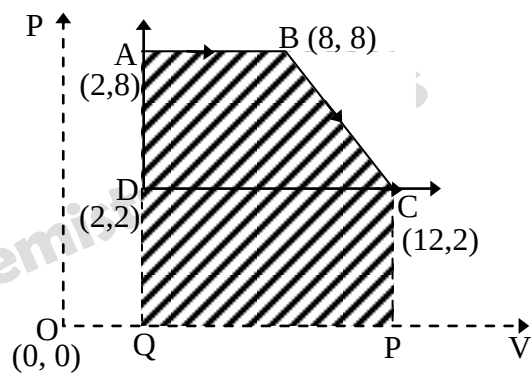
Total area under curve is equal to

(Area of trapezium (ABCD) + Area of rectangle (PCDQ))

$$= \left[\frac{1}{2} (6 + 10) \times 6 \right] + [10 \times 2]$$

$$= 48 + 20$$

$$= 68 \text{ J}$$

Hence, **answer should be [68 J]** instead of [48 J] because, origin is changed.

27. NTA Ans. (3)

Sol. For ideal Gas

$U = f(T)$, $H = f(T)$

$Z = 1$

$C_p - C_v = R$

$dU = C_v dT$

28. (3)

Sol. As the expansion is done in vacuum that is in absence of p_{ext} so

$W = \text{zero}$

29. (2)

Sol. If ΔG is negative than Oxide is stable

30. [6.25]

Sol NTA Ans. (48)

$$\Delta U = nC_v \Delta T$$

$$5000 = 4 \times C_v \times 200$$

$$C_v = 6.25 \text{ J mole}^{-1} \text{ K}^{-1}$$

31. (189494.39)

NTA Ans. (189494)

Sol. $\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(g)$

For vapourisation of $\frac{90}{18} = 5$ moles of H_2O

$$\Delta H = \Delta U + \Delta n_g RT$$

$$5 \times 41000 = \Delta U + 5 \times 8.314 \times 373$$

$$\Delta U = 189494.39 \text{ J}$$

32. -13537.57 Answer by NTA (-13538 J)

Sol. $2\text{A}(g) \longrightarrow \text{A}_2(\Delta)$

$$\begin{aligned} \Delta H^\circ &= \Delta U^\circ + \Delta n_g \cdot RT \\ &= -20 \times 1000 + (-1) \times 8.314 \times 298 \\ &= -22477.572 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta H^\circ &= \Delta H^\circ - T\Delta S^\circ \\ &= -22477.572 - 298 \times (-30) \\ &= -13537.572 \text{ J} \\ &= -13538 \text{ J} \end{aligned}$$

JEE ADVANCED

$$1. \quad T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

$$300 \times (1.25)^{\gamma-1} = T_f \times (2.5)^{\gamma-1}$$

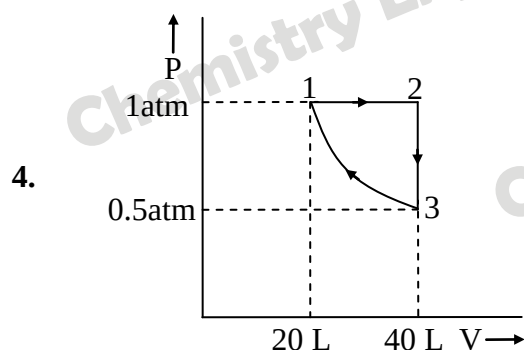
$$T_f = 300 \times \left(\frac{1.25}{2.5} \right)^{\gamma-1}$$

$$= \frac{300}{2^{2/3}} = 188.5 \text{ K}$$

$$\Delta H = n C_p \Delta T = 0.05 \times 20.8 \times (188.5 - 300) = -115.41 \text{ J}$$

2. Work is path function.

$$3. \quad \Delta H = \Delta U + P_2 V_2 - V_1 P_1 = 30 + (5 \times 4 - 3 \times 2) = 30 + 14 = +44 \text{ L-atm}$$



$$W_{\text{net}} = w_1 + w_2 + w_3$$

$$= -1 \times 20 + 0 + 20 \ln 2 = 20(\ln 2 - 1) = -20 \times 0.307 \text{ L-atm} = -6.14 \text{ L-atm}$$

$$q = -w = +6.14 \text{ L-atm}$$

for cyclic process $\Delta H = 0$, $\Delta U = 0$, $\Delta S = 0$

5. He molecule is monoatomic so it just has three degree of freedom at all temperature but H_2 is diatomic molecule at higher temperature vibrational degree of freedom is also considered.

6. For adiabatic

$$\Delta U = W$$

$$W = -P \Delta V$$

$$= 100 \times 10^{-3}$$

$$= +0.1 \text{ bar-L}$$

$$\Delta H = \Delta U + P_2 V_2 - P_1 V_1$$

$$= +0.1 + (100 \times 99 - 1 \times 100) \times 10^{-3}$$

$$= +9.9 \text{ bar-L}$$

7. $\Delta H = nC_p\Delta T$

For isoT $\Delta T = 0$

8. For irrev. adiabatic

$$nC_v\Delta T = -P_{\text{ext}}(V_2 - V_1)$$

$$1 \times \frac{3}{2} R(T_f - T) = -1 \times (2 - 1)$$

$$T_f = T - \frac{2}{3 \times 0.0821}$$

9. $\Delta H = \Delta U + V\Delta P$

$$-560 = \Delta U + 1 \times (40 - 70) \times 0.1$$

$$-560 = \Delta U - 3$$

$$\Delta U = -557 \text{ kJ mol}^{-1}$$

10. $PV^{-1} = K$

$$C = \frac{3}{2}R - \frac{R}{-2} = \frac{3}{2}R + \frac{R}{2} = 2R$$

11. State function - $\Delta U, \Delta H$

path function - W

12. Intensive properties : molar conductivity, EMF

Extensive properties : Resistance , Heat capacity

13. $W_s = -2.303RT \log \frac{V_2}{V_1} = -2.303 \times \log \frac{5.5}{0.5}$

$$= -4.606 \log 1.1 = -4.8 \text{ L-atm}$$

$$W_d = -4 \times \frac{3}{2} + (-1 \times 1) = + \left(-\frac{2}{3} \times \frac{5}{2} \right) = -\frac{26}{3} \text{ L-atm}$$

$$\frac{W_d}{W_s} = 1.8 \approx 2.0$$

14. $T_1 = T_2 > T_3$

$$T_1 > T_3$$

$W_{\text{isoT}} < W_{\text{adia}}$ (with sign)

$\Delta U_{\text{isoT}} > \Delta U_{\text{adia}}$

15. Isochoric process $\Rightarrow$ vol. constant

16. $KL \rightarrow P$ constant

$$V \uparrow, T \uparrow$$

$LM - V$ constant

$$P \downarrow, T \downarrow$$

$MN - P$ constant

$$V \downarrow, T \downarrow$$

$NK - V$ constant

$$P \uparrow, T \uparrow$$

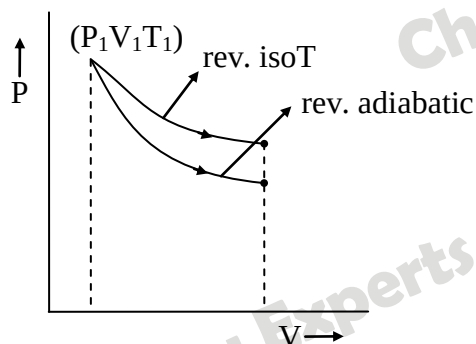
17. For free expansion

$$w = 0, q = 0 \quad \Delta T = 0$$

$$T_1 = T_2$$

$$P_1 V_1 = P_2 V_2$$

18.



(A) During irrev. compression maximum work is done on the gas.

(B) For expansion of ideal gas from same V_i to same $V_f \Rightarrow |W_{\text{isoT}}| > |W_{\text{adia}}|$

(C) If $T_1 = T_2 \Rightarrow \Delta T = 0, \Delta U = 0$

For rev. adiabatic expansion $T_2 < T_1 \Rightarrow \Delta U = -ve$

(D) For free expansion

$$P_{\text{ext}} = 0, w = 0$$

If it is carried out isothermally $\Delta U = 0, q = 0$

$\therefore$ it is adiabatic also

19. A-C isochoric process

A-B isoT process

B-C isobaric process

$$q_{AC} = \Delta U_{AC} = nC_v(T_2 - T_1) = \Delta U_{BC}$$

$$w_{AB} = -nRT_1 \ln \frac{V_2}{V_1}$$

$$w_{BC} = -P_2(V_1 - V_2) = P_2(V_2 - V_1)$$

$$q_{BC} = \Delta H_{BC} = nC_p(T_2 - T_1) = \Delta H_{AC}$$

$$\Delta U_{CA} = nC_v(T_1 - T_2)$$

$$\Delta H_{CA} < \Delta U_{CA}$$

20. (A, B, C or A, B, C, D)

Option (A) : True for Van der Waals' gas undergoing reversible process if irreversible then it will be false

Option (B) : True, Since it is reversible. ($P_{\text{ext}} = P_{\text{in}}$)

Option (C) : True, Since it is reversible ($P_{\text{ext}} = P_{\text{in}}$)

Option (D) : Given that process is irreversible at constant pressure. If internal pressure is constant then it is also true. And if only external pressure is constant then it will be false.

EXERCISE # (JEE-MAIN)

1. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels [AIEEE-2003]
 - (1) In a straight line path
 - (2) with an accelerated velocity
 - (3) In a circular path
 - (4) In a wavy path

2. What volume of hydrogen gas, at 273K and 1 atm, pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ? [AIEEE-2003]
 - (1) 44.8 L
 - (2) 22.4 L
 - (3) 89.6 L
 - (4) 67.2 L

3. As the temperature is raised from 20°C to 40°C, the average kinetic energy of neon atoms changes by factor of which of the following ? [AIEEE-2004]
 - (1) 1/2
 - (2) $\sqrt{(313/293)}$
 - (3) 313/298
 - (4) 2

4. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is - [AIEEE-2008]
 - (1) 2/3
 - (2) $\frac{1}{3} \times \frac{273}{298}$
 - (3) $\frac{1}{3}$
 - (4) $\frac{1}{2}$

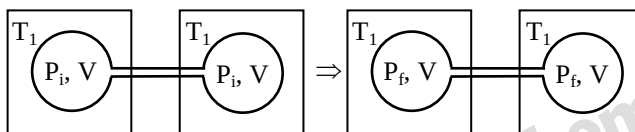
5. The molecular velocity of any gas is :- [AIEEE-2011]
 - (1) inversely proportional to the square root of temperature
 - (2) inversely proportional to absolute temperature
 - (3) directly proportional to square of temperature
 - (4) directly proportional to square root of temperature

6. a, v and u represent most probable velocity, average velocity and root mean square velocity respectively of a gas at a particular temperature. The correct order among the following is [JEE(Main)-2012]
 - (1) $a > u > v$
 - (2) $v > u > a$
 - (3) $u > v > a$
 - (4) $u > a > v$

7. An open vessel at 300 K is heated till $\frac{2}{5}$ th of the air in it is expelled. Assuming that the volume of the vessel remains constant, the temperature to which the vessel is heated is: [JEE(Main-online)-2012]
 - (1) 750 K
 - (2) 400 K
 - (3) 500 K
 - (4) 1500K

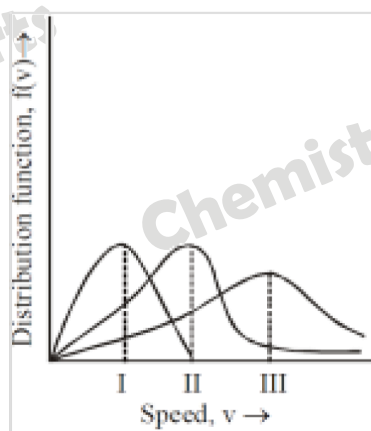
8. For 1 mol of an ideal gas at constant temperature T, the plot of (log P) against (log V) is a (P : Pressure, V : Volume) : **[JEE(Main-online)-2012]**
 (1) Straight line parallel to x-axis (2) Curve starting at origin
 (3) Straight line with a negative slope (4) Straight line passing through origin
9. The relationship among most probable velocity, average velocity and root mean square velocity is respectively :- **[JEE(Main-online)-2012]**
 (1) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (2) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$ (3) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (4) $\sqrt{8/\pi} : \sqrt{3} : \sqrt{2}$
10. Which one of the following is the wrong assumption of kinetic theory of gases ? **[JEE(Main-online)-2013]**
 (1) All the molecules move in straight line between collision and with same velocity.
 (2) Molecules are separated by great distances compared to their sizes.
 (3) Pressure is the result of elastic collision of molecules with the container's wall.
 (4) Momentum and energy always remain conserved.
11. By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from 5×10^4 cm/s to 10×10^4 cm/s ? **[JEE(Main-online)-2013]**
 (1) Four (2) three (3) Two (4) Six
12. For gaseous state, if most probable speed is denoted by C, average speed by $\bar{C}$ and mean square speed by C^2 , then for a large number of molecules the ratios of these speeds are :- **[JEE(Main-offline)-2013]**
 (1) $C : \bar{C} : C^2 = 1.225 : 1.128 : 1$ (2) $C : \bar{C} : C^2 = 1.128 : 1.225 : 1$
 (3) $C : \bar{C} : C^2 = 1 : 1.128 : 1.225$ (4) $C : \bar{C} : C^2 = 1 : 1.225 : 1.128$
13. A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is: **[JEE(Main-online)-2014]**
 (1) NH_2 (2) NH_3 (3) N_3H (4) N_2H_4
14. The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, the final volume the gas will be **[JEE(Main-online)-2014]**
 (1) 1.750 L (2) 7.50 L (3) 3.60 L (4) 4.032 L

15. The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is : (Atomic masses : He = 4 u, O = 16 u) **[JEE(Main-online)-2014]**
 (1) 1200 K (2) 600 K (3) 300 K (4) 2400 K
16. Which of the following is not an assumption of the kinetic theory of gases ? **[JEE-Mains (online)-2015]**
 (1) Gas particles have negligible volume.
 (2) A gas consists of many identical particles which are in continual motion.
 (3) At high pressure, gas particles are difficult to compress.
 (4) Collisions of gas particles are perfectly elastic.
17. Initially the root mean square (rms) velocity of N_2 molecules at certain temperature is u . If this temperature is doubled and all the nitrogen molecules dissociate into nitrogen atoms, then the new rms velocity will be : **[JEE(Main-online)-2016]**
 (1) $u/2$ (2) $4u$ (3) $14u$ (4) $2u$
18. Two closed bulbs of equal volume (V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to T_2 . The final pressure p_f is :- **[JEE-Main-2016]**

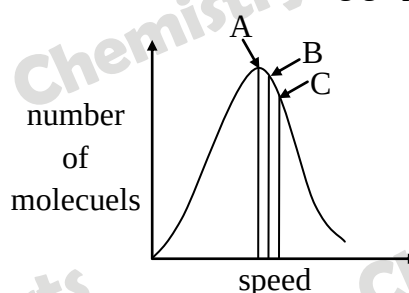


- (1) $2p_i \left(\frac{T_1 T_2}{T_1 + T_2} \right)$ (2) $p_i \left(\frac{T_1 T_2}{T_1 + T_2} \right)$ (3) $2p_i \left(\frac{T_1}{T_1 + T_2} \right)$ (4) $2p_i \left(\frac{T_2}{T_1 + T_2} \right)$
19. An open vessel at 27°C is heated until two fifth of the air (assumed as an ideal gas) in it has escaped from the vessel. Assuming that the volume of the vessel remains constant, the temperature at which the vessel has been heated is : **[JEE-Main-2019 (Jan)]**
 (1) 500°C (2) 500 K (3) 750°C (4) 750 K
20. The volume of gas A is twice than that of gas B. The compressibility factor of gas A is thrice than that of gas B at same temperature. The pressures of the gases for equal number of moles are : **[JEE-Main-2019 (Jan.)]**
 (1) $P_A = 3P_B$ (2) $3P_A = 2P_B$ (3) $2P_A = 3P_B$ (4) $P_A = 2P_B$
21. 0.5 moles of gas A and x moles of gas B exert a pressure of 200 Pa in a container of volume 10 m^3 at 1000 K. given R is the gas constant in $\text{JK}^{-1}\text{ mol}^{-1}$, x is : **[JEE-Main-2019 (Jan.)]**
 (1) $\frac{2R}{4+12}$ (2) $\frac{2R}{4-12}$ (3) $\frac{4-R}{2R}$ (4) $\frac{4+R}{2R}$

22. Points I, II and III in the following plot respectively correspond to (V_{mp} :most probable velocity)
[JEE-Main-2019 (Apr.)]



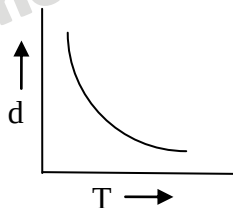
- (1) V_{mp} of O_2 (400 K) ; V_{mp} of N_2 (300 K); V_{mp} of H_2 (300 K)
 (2) V_{mp} of N_2 (300 K) ; V_{mp} of H_2 (300 K); V_{mp} of O_2 (400 K)
 (3) V_{mp} of H_2 (300 K) ; V_{mp} of N_2 (400 K); V_{mp} of O_2 (300 K)
 (4) V_{mp} of N_2 (300 K) ; V_{mp} of O_2 (400 K); V_{mp} of H_2 (300 K)
23. Identify the correct labels of A, B and C in the following graph from the options given below :



[IIT MAIN 2020(Jan.)]

Root mean square speed (V_{rms}); most probable speed (V_{mp}); Average speed (V_{av})

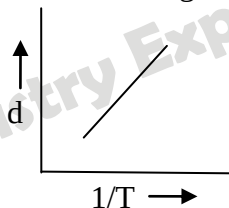
- (1) A - V_{av} ; B - V_{rms} ; C - V_{mp} (2) A - V_{mp} ; B - V_{av} ; C - V_{rms}
 (3) A - V_{rms} ; B - V_{mp} ; C - V_{av} (4) A - V_{mp} ; B - V_{rms} ; C - V_{av}
24. Which one of the following graphs is not correct for ideal gas ? [IIT MAIN 2020(Sep.)]



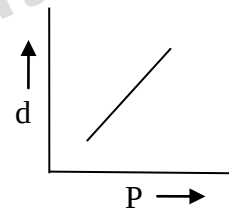
I



II



III



IV

d = Density, P = Pressure, T = Temperature

- (1) IV (2) II (3) III (4) I
25. A mixture of one mole each of H_2 , He and O_2 each are enclosed in a cylinder of volume V at temperature T . If the partial pressure of H_2 is 2 atm, the total pressure of the gases in the cylinder is: [IIT MAIN 2020(Sep.)]
- (1) 14 atm (2) 38 atm (3) 22 atm (4) 6 atm

EXERCISE # (JEE-ADVANCED)

1. Calculate the total pressure in a 10 litre cylinder which contains 0.4 g He, 1.6 g oxygen and 1.4 g of nitrogen at 27°C. Also calculate the partial pressure of He gas in the cylinder. Assume ideal behaviour for gases. **[JEE 1997]**

2. According to Graham's law, at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ of gases A and B is given by : **[JEE 1998]**

$$(A) \frac{P_A}{P_B} \left(\frac{M_A}{M_B} \right)^{1/2} \quad (B) \left(\frac{M_A}{M_B} \right) \left(\frac{P_A}{P_B} \right)^{1/2} \quad (C) \frac{P_A}{P_B} \left(\frac{M_B}{M_A} \right)^{1/2} \quad (D) \frac{M_A}{M_B} \left(\frac{P_B}{P_A} \right)^{1/2}$$

3. An evacuated glass vessel weighs 50.0 g when empty, 148.0 gm when filled with a liquid of density 0.98 g /mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 k . Determine the molecular weight of the gas. **[JEE 1998]**

4. The pressure exerted by 12 g of an ideal gas at temperature t °C in a vessel of volume V is one atmp. When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10%. Calculate the temperature 't' and volume 'V'. [molecular weight of gas = 120] **[JEE 1999]**

5. One mole of N₂ gas at 0.8 atm takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atm takes 57 sec to diffuse through the same hole. Calculate the molecular formula of the compound. (At. wt. Xe = 138, F = 19) **[JEE 1999]**

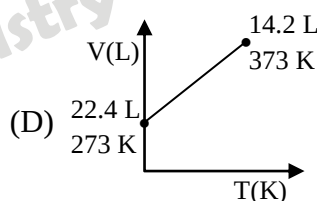
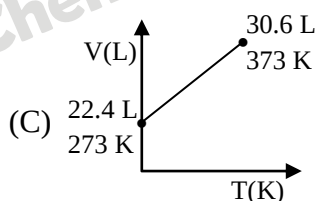
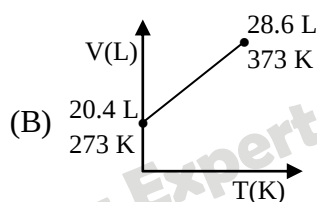
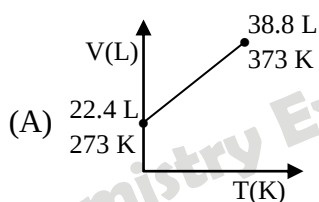
6. The r. m. s. velocity of hydrogen is $\sqrt{7}$ times the r. m. s. velocity of nitrogen. If T is the temperature of the gas : **[JEE 2000]**

$$(A) T(H_2) = T(N_2) \quad (B) T(H_2) > T(N_2) \quad (C) T(H_2) < T(N_2) \quad (D) T(H_2) = \sqrt{7} T(N_2)$$

7. **Statement-1** : The pressure of a fixed amount of an ideal gas is proportional to its temperature.
Statement-2 : Frequency of collision and their impact both increase in proportion to the square root of temperature. True / False. [JEE 2000]

8. The root mean square velocity of an ideal gas at constant pressure varies with density as [JEE 2001]
 (A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$

9. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atmp? [JEE 2002]



10. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature.
11. The root mean square velocity of one mole of a monoatomic gas having molar mass M is u_{rms} . The relation between the average kinetic energy (E) of the gas and u_{rms} is [JEE-2004]

(A) $u_{\text{rms}} = \sqrt{\frac{3E}{2M}}$ (B) $u_{\text{rms}} = \sqrt{\frac{2E}{3M}}$ (C) $u_{\text{rms}} = \sqrt{\frac{2E}{M}}$ (D) $u_{\text{rms}} = \sqrt{\frac{E}{3M}}$

12. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be [JEE 2005]

(A) 4 (B) 2 (C) 1 (D) 0.5

13. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is [JEE 2009]

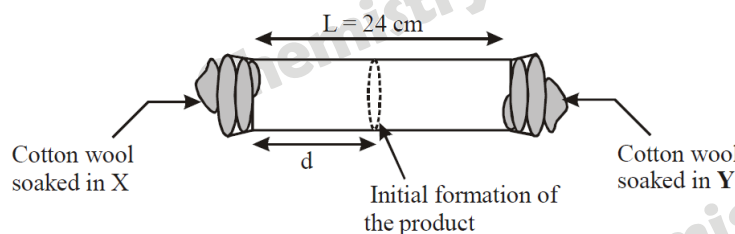
14. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to

[JEE 2011]

Paragraph for Question 15 & 16

X and Y are two volatile liquids with molar weights of 10g mol^{-1} and 40g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length $L = 24\text{ cm}$, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300K . Vapours of X and Y react to form a product which is first observed at a distance $d\text{ cm}$ from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.

[JEE 2014]



15. The value of d in cm (shown in the figure), as estimated from Graham's law, is -
 (A) 8 (B) 12 (C) 16 (D) 20
16. The experimental value of d is found to be smaller than the estimate obtained using Graham's law. This is due to -
 (A) Larger mean free path for X as compared to that of Y
 (B) Larger mean free path for Y as compared to that of X
 (C) Increased collision frequency of Y with the inert gas as compared to that of X with the inert gas
 (D) Increased collision frequency of X with the inert gas as compared to that of Y with the inert gas
17. The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. If the absolute temperature of an ideal gas increased 4 times and its pressure is increased 2 times, as a result the diffusion coefficient of a gas increased x times. The value of x is? [JEE-2016]

18. A closed tank has two compartments **A** and **B**, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does **NOT** allow the gas to leak across (Figure 2), the volume (in m^3) of the compartment **A** after the system attains equilibrium is _____. [JEE Adv. 2018]

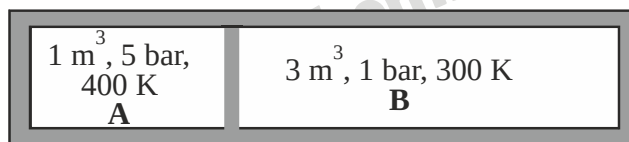


Figure 1

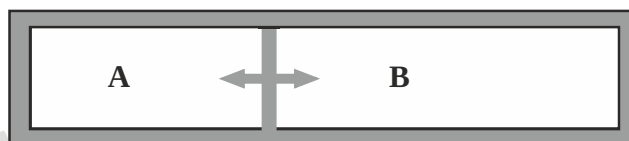
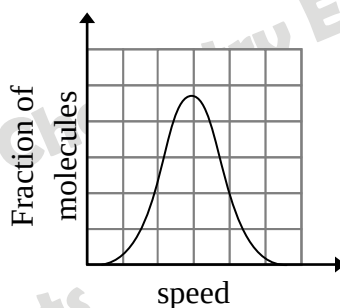


Figure 2

19. Which of the following statement(s) is (are) correct regarding the root mean square speed (U_{rms}) and average translational kinetic energy (ϵ_{av}) of a molecule in a gas at equilibrium ? [JEE Adv. 2019]
- (A) ϵ_{av} is doubled when its temperature is increased four times
- (B) U_{rms} is doubled when its temperature is increased four times
- (C) U_{rms} is inversely proportional to the square root of its molecular mass
- (D) ϵ_{av} at a given temperature does not depend on its molecular mass
20. If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is [JEE Adv. 2020]



- (A) 1 : 1 : 1 (B) 1 : 1 : 1.224 (C) 1 : 1.128 : 1.224 (D) 1 : 1.128 : 1

ANSWER KEY**EXERCISE # (JEE-MAIN)**

1.	1	2.	4	3.	3	4.	3	5.	4
6.	3	7.	3	8.	3	9.	1	10.	1
11.	1	12.	3	13.	4	14.	1	15.	4
16.	3	17.	4	18.	4	19.	2	20.	3
21.	3	22.	4	23.	2	24.	2	25.	4

EXERCISE # (JEE-ADVANCED)

1.	0.492 atmp ; 0.246 atmp	2.	C	3.	123
4.	-173°C , 0.82 L	5.	XeF ₆	6.	C
7.	Both statements are correct	8.	D	9.	C
10.	434.17 m/sec	11.	C	12.	B
13.	4	14.	7	15.	C
16.	D	17.	4	18.	2.22 or 2.23
19.	B,C,D	20.	(B or C)		

HINT & SOLUTIONS

EXERCISE # JEE-MAIN

1. (1)

Sol. Factual

2. (4)

Sol. $2\text{BCl}_3 + 3\text{H}_2 \longrightarrow 2\text{B} + 6\text{HCl}$

$$\text{mole} : \left(\frac{21.6}{10.8} \right) = 2$$

$$n_{\text{H}_2} = 3$$

$$\begin{aligned} \Rightarrow \text{Volume of H}_2 \text{ at } 273\text{K \& 1 atm} \\ &= 3 \times 22.4 \\ &= 67.2 \text{ L} \end{aligned}$$

3. (3)

Sol. $K, E \propto T$

$$\Rightarrow \frac{(K, E)_{40^\circ\text{C}}}{(L, E)_{20^\circ\text{C}}} = \frac{313}{293}$$

4. (3)

Sol. $P \propto n$

$$\Rightarrow \frac{P_{\text{CH}_4}}{P_{\text{O}_2}} = \frac{w \times 32}{16 \times w} = \frac{2}{1}$$

Fraction of total pressure exerted by O_2 is $\frac{1}{3}$

5. (4)

Sol. $u \propto \sqrt{T}$

6. (3)

Sol. $u > v > a$

7. (3)

Sol. As, V remains constant

$$n \propto \frac{1}{T}$$

$$\text{i.e. } n_1 T_1 = n_2 T_2$$

$$\Rightarrow n_1 \times 300 = \frac{3}{5} n_1 \times T_2 \Rightarrow T_2 = \frac{300 \times 5}{3} \Rightarrow T_2 = 500 \text{ K}$$

8. (3)

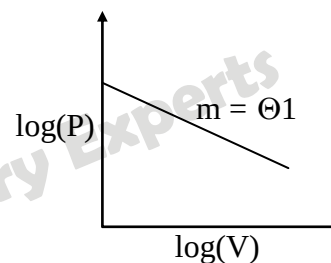
Sol. As $P = \frac{nRT}{V}$

$$\Rightarrow \log(P) = \log(nRT) + \log\left(\frac{1}{V}\right)$$

or, $\log(P) = \log(nRT) - \log(V)$

For, 1 mole ideal gas,

$$\log(P) = \log(RT) - \log(V)$$



9. (1)

Sol. $\sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$

10. (1)

Sol. Factual

11. (1)

Sol. $U_{\text{rms}} = \sqrt{\frac{3RT}{m}}$

i.e. $U_{\text{rms}} \propto \sqrt{T}$

$$\frac{5 \times 10^4}{10 \times 10^4} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{1}{4} = \frac{T_1}{T_2}$$

$$\Rightarrow (T_2 = 4T_1)$$

12. (3)

Sol. $U_{\text{mps}} = C = \sqrt{\frac{2RT}{m}}$

$$V_{\text{Avg}} = \bar{C} = \sqrt{\frac{8RT}{\pi m}} \quad \& \quad U_{\text{rms}} = C = \sqrt{\frac{3RT}{m}}$$

$$\Rightarrow C : \bar{C} : C = 1 : 1.128 : 1.225$$

13. (4)

Sol.	Element	% (by mass)	relation no. of moles	Simplest atomic ratio
	N	87.5	$\frac{87.5}{14} = 6.25$	$\frac{6.25}{6.25} = 1$
	H	12.5	$\frac{12.5}{1} = 12.5$	$\frac{12.5}{6.25} = 2$

Empirical formula = NH_2

Molar mass of compound = 32

As, molecular formula = $n \times$ empirical formula

$$\Rightarrow n = \frac{32}{16} = 2 \quad \Rightarrow (\text{m.f} = \text{N}_2\text{H}_4)$$

14. (1)

Sol. $P_1 V_1 = P_2 V_2$

$$840 \times 750 = 360 \times V_2$$

$$\Rightarrow V_2 = \frac{840 \times 750}{360}$$

$$\Rightarrow V_2 = 1750 \text{ ml} \quad \text{or} \quad 1.750 \text{ L}$$

15. (4)

Sol. $(U_{\text{rms}})_{\text{O}_2} = (U_{\text{rms}})_{\text{He}}$

$$\sqrt{\frac{3R \times T}{32}} = \sqrt{\frac{3R \times 300}{4}}$$

$$\Rightarrow \frac{T}{32} = \frac{300}{4}$$

$$\Rightarrow T = \frac{32 \times 300}{4}$$

$$\Rightarrow T = 2400 \text{ K}$$

16. (3) Factual

17. $V_{\text{rms}} \propto \sqrt{\frac{T}{M}}$

T is doubled

M is halved

$$V'_{\text{rms}} = \sqrt{4} V_{\text{rms}}$$

$$V'_{\text{rms}} = 2 V_{\text{rms}}$$

$$= 2 \mu$$

18. (4)

Sol. $(n_i)_T = (n_f)_T$

$$\frac{P_1 V}{RT_1} + \frac{P_1 V}{RT_1} = \frac{P_f V}{RT_1} + \frac{P_f V}{RT_2}$$

$$\frac{2P_1 V}{RT_1} = \frac{P_f V}{R} \left(\frac{1}{T_1} + \frac{1}{T_2} \right) \Rightarrow P_f = 2P_1 \left(\frac{T_2}{T_1 + T_2} \right)$$

19. (2)

Sol. $V_1 = V_2$, $n_2 = \frac{3}{5} n_1$

$$n_1 T_1 = n_2 T_2$$

$$n_1(300) = n_2(T_2)$$

$$300 = \frac{3}{5} T_2$$

$$T_2 = 500 \text{ K}$$

20. (3)

Sol. $V_A = 2V_B$

$$Z_A = 3Z_B$$

$$\frac{Z_A}{Z_B} = \frac{P_A}{P_B} \frac{V_A}{V_B}$$

$$3 = \frac{P_A}{P_B} \times 2 \Rightarrow 2P_A = 3P_B$$

21. (3)

Sol. $n_A = 0.5$ $n_B = x$ mole

For the container

$$P = 200 \text{ Pa}$$

$$V = 10 \text{ m}^3$$

$$T = 1000 \text{ K}$$

$$P V = n R T$$

$$200 \times 10 = (0.5 + x) R \times 1000$$

$$0.5 + x = \frac{2}{R}$$

$$x = \frac{2}{R} - 0.5 \Rightarrow \frac{4 - R}{2R}$$

22. (4)

Sol. V_{MPS} is proportional to $\sqrt{\frac{T}{M}}$

23. [2]

Sol. $V_{mp} = \sqrt{\frac{2RT}{M}}$

$$V_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

24. (2)

Sol. $d = \frac{PM}{RT}$

$$d V_s P$$

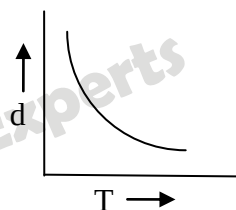
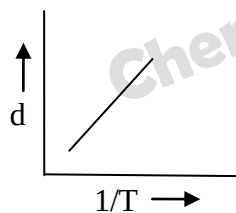
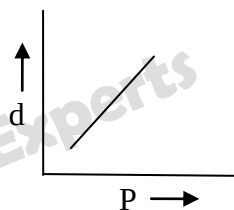
$$d = \left(\frac{M}{RT} \right) P$$

$$d V_s \frac{1}{T}$$

$$d = \left(\frac{PM}{R} \right) \cdot \frac{1}{T}$$

$$d V_s T$$

$$d \times T = \frac{PM}{R}$$



25. (4)

Sol. From $PV = nRT$

$p \propto \text{mole}$ at constant V and T .

$\therefore$ If 1 mole of H_2 exert 2 atm

$\therefore$ 1 mole of He and O_2 will also exert 2 atm each.

$\therefore$ Total pressure = $2 + 2 + 2 = 6$ atm

EXERCISE # JEE-ADVANCED

1. $n_{\text{He}} = \frac{.4}{4} = 0.1$

$$n_{\text{O}_2} = \frac{1.6}{32} = 0.05$$

$$n_{\text{N}_2} = \frac{1.4}{28} = 0.05$$

$$P_T = \frac{0.2 \times 0.0821 \times 300}{10}$$

$$P_T = 0.492 \text{ atm}$$

$$P_{\text{He}} = \frac{0.1}{0.2} \times 0.492$$

$$\Rightarrow P_{\text{He}} = 0.246 \text{ atm}$$

2. (C)

Sol. According to Graham's Law,

$$\frac{r_A}{r_B} \propto \frac{P_A}{P_B} \left(\frac{M_B}{M_A} \right)^{1/2}$$

3. Mass of liquid = $148 - 50 = 98\text{g}$

$$\text{Volume of liquid} = \frac{98}{0.98} = 100 \text{ ml} = \text{volume of flask}$$

$$\text{Mass of gas} = 50.5 - 50 = 0.50 \text{ g}$$

Using, $PV = nRT$, we get

$$M = \frac{wRT}{PV} = \frac{0.5 \times 0.082 \times 300}{1 \times 0.1}$$

$$\Rightarrow M = 123 \text{ g mol}^{-1}$$

4. For a fixed amount of gas at constant volume,

$$P \propto T$$

$$\frac{1}{1.1} = \frac{T}{T + 10}$$

$$\Rightarrow T = 100 \text{ K} \quad \text{or} \quad \ominus 173^\circ\text{C}$$

$$\text{Also, } V = \frac{nRT}{P} = \frac{12}{120} \times \frac{0.0821 \times 100}{1}$$

$$\Rightarrow V = 0.82 \text{ L}$$

5. For the same amount of gas being diffused,

$$\frac{r_1}{r_2} = \frac{t_2}{t_1} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$$

$$\frac{57}{38} = \frac{0.8}{1.6} \sqrt{\frac{M_2}{28}}$$

$$\Rightarrow M_2 = 252 \text{ g mol}^{-1}$$

The formula of compound can be considered to be XeF_n

$$\Rightarrow 131 + 19n = 252$$

$$\Rightarrow n = 6$$

$\Rightarrow$ The unknown gas is XeF_6 .

6. (C)

Sol. $(U_{\text{rms}})_{\text{H}_2} = \sqrt{7} (U_{\text{rms}})_{\text{N}_2}$

$$\frac{T_{(\text{H}_2)}}{2} = 7 \times \frac{T_{(\text{N}_2)}}{28}$$

$$\Rightarrow T_{(\text{H}_2)} = \frac{T_{(\text{N}_2)}}{2}$$

$$\Rightarrow T_{(\text{H}_2)} < T_{(\text{N}_2)}$$

7. **Statement-1:** False, Besides amount, pressure also depends on volume.

Statement-2: True as $u_{\text{rms}} \propto \sqrt{T}$

8. (D)

Sol. $U_{\text{rms}} = \sqrt{\frac{3RT}{m}} = \sqrt{\frac{3P}{d}}$

$$\Rightarrow U_{\text{rms}} \propto \left(\frac{1}{d}\right)^{1/2}$$

9. For an ideal gas, $V_m = 22.4$ lit at 273 K & 1 atm.

In option (C), at the initial point, the volume is 22.4 lit. as required by the ideal gas equation &

$\left(\frac{V}{T}\right)$ have the same value at both initial & final points.

10. $\frac{V_{\text{Avg}}}{U_{\text{rms}}} = \sqrt{\frac{8RT}{\pi m}} \div \sqrt{\frac{3RT}{m}}$

$$= \sqrt{\frac{8}{3\pi}}$$

$$\Rightarrow U_{\text{rms}} = \sqrt{\frac{3\pi}{8}} \times 400$$

$$= \sqrt{\frac{3 \times 3.14}{8}} \times 400 \Rightarrow U_{\text{rms}} = 434.17 \text{ m/s.}$$

11. $U_{\text{rms}} = \sqrt{\frac{3RT}{M}} \dots\dots(1)$

The average kinetic energy (E) of the gas is given by the following expression.

$$E = \frac{3}{2} RT$$

$$RT = \frac{2}{3} E \dots\dots(2)$$

Substitute (2) into (1)

$$U_{\text{rms}} = \sqrt{\frac{3 \times \frac{2}{3} E}{M}}$$

$$U_{\text{rms}} = \sqrt{\frac{2E}{M}}$$

12. (B)

Sol. $\frac{r_{\text{He}}}{r_{\text{CH}_4}} = \sqrt{\frac{16}{4}} = 2$

13. $(U_{\text{rms}})_x \text{ at } 400 \text{ K} = (U_{\text{rms}})_y \text{ at } 60 \text{ K}$

$$\Rightarrow \sqrt{\frac{3R \times 400}{40}} = \sqrt{\frac{2R \times 60}{m_y}}$$

$$\Rightarrow \frac{3 \times 400}{40} = \frac{2 \times 60}{m_y} \Rightarrow m_y = \frac{2 \times 60 \times 40}{3 \times 400} \Rightarrow m_y = 4$$

14. Partial pressure of He = $1 - 0.68 = 0.32$

$$V = \frac{n_{\text{He}} \cdot R \cdot T}{P_{\text{He}}} = \frac{0.1 \times 0.082 \times 273}{0.32}$$

$\Rightarrow V = 7 \text{ lit.} \Rightarrow \text{Volume of container} = \text{volume of He.}$

15. (C)

Sol. Let distance covered by X is d, then distance covered by Y is (24 – d)

If r_x & r_y are the rate of diffusion of gases X & Y

$$\frac{r_x}{r_y} = \frac{d}{24 - d} = \sqrt{\frac{40}{10}} = 2 \Rightarrow d = 16 \text{ cm}$$

16. The experimental value of d is found to be smaller than the estimated value obtained using Graham's law. This is due to increased collision frequency of Y with the inert gas as compared to that of X with the inert gas. ($\because$ As, the collision frequency increases, the molecular speed decreases much more than expected.)

17. (DC) Diffusion coefficient $\propto \lambda$ (mean free path) $\propto U_{\text{mean}}$

Thus (DC) $\propto \lambda U_{\text{mean}}$

$$\text{But } \lambda = \frac{RT}{\sqrt{2} N_0 \sigma p} \Rightarrow \lambda \propto \frac{T}{p}$$

$$\text{and } U_{\text{mean}} = \sqrt{\frac{8RT}{\pi M}}$$

$$U_{\text{mean}} \propto \sqrt{T}$$

$$\therefore \text{DC} \propto \frac{(T)^{3/2}}{p}$$

$$\frac{(\text{DC})_2}{(\text{DC})_1} = \left(\frac{p_1}{p_2} \right) \left(\frac{T_2}{T_1} \right)^{3/2} = \left(\frac{p_1}{2p_1} \right) \left(\frac{4T_1}{T_1} \right)^{3/2} = \left(\frac{1}{2} \right) (8) = 4$$

18. Finally, $P_A = P_B$ & $T_A = T_B$

$$\text{So, } \frac{n_A}{n_B} = \frac{V_A}{V_B}$$

$$\frac{5}{\frac{400R}{3}} = \frac{V_A}{300R}$$

$$\Rightarrow \frac{V_A}{V_B} = \frac{5}{4} \Rightarrow V_A = \frac{5}{9} \times 4 = \frac{20}{9} = 2.22$$

19. The root mean square speed (U_{rms}) and average translational kinetic energy (ϵ_{av}) has following relation with temperature and molecular mass

$$\epsilon_{\text{av}} = \frac{3}{2} RT, U_{\text{rms}} = \sqrt{\frac{3RT}{M}} \text{ and } U_{\text{rms}} \propto \frac{1}{\sqrt{M}}$$

$\therefore \epsilon_{\text{av}}$ doesn't depend on its molecular mass.

According to the relation given above ϵ_{av} gets doubled when the temperature is increased 2 times. Thus option B is incorrect.

20. **Explanation for option (B) :** Due to symmetrical graph given, the average value of velocities will be at the middle, symmetrical velocity which is also most probable velocity.

While calculating rms speed, we have to take average of square of speeds and its root. In square of speeds, higher speeds will give more contribution so the value will be more than the average

value. But we cannot calculate rms speed by given data as $\sqrt{\frac{3RT}{M}}$ is valid for Maxwell's

distribution.

Explanation for option (C) : According to Maxwell's distribution of speed

EXERCISE # (JEE-MAIN)

Q.1 When does a gas deviate the most from its ideal behaviour ? [JEE-MAIN(ONLINE)-2015]

- (1) At high pressure and low temperature
- (2) At high pressure and high temperature
- (3) At low pressure and low temperature
- (4) At low pressure and high temperature

Q.2 If Z is the compressibility factor, Van Der Waals' equation at low pressure can be written as :

[JEE-MAIN-2014]

- (1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$ (3) $Z = 1 + \frac{Pb}{RT}$ (4) $Z = 1 - \frac{a}{V_m RT}$

Q.3 The compressibility factor for a real gas at high pressure is :-

[AIIEEE-2012]

- (1) $1 - \frac{Pb}{RT}$ (2) $1 + \frac{PT}{Pb}$ (3) 1 (4) $1 + \frac{Pb}{RT}$

Q.4 'a' and 'b' are Van Der Waals' constants for gases. Chlorine is more easily liquefied than ethane because [AIIEEE-2011]

- (1) a for $\text{Cl}_2 < a$ for C_2H_6 but b for $\text{Cl}_2 > b$ for C_2H_6
- (2) a for $\text{Cl}_2 > a$ for C_2H_6 but b for $\text{Cl}_2 < b$ for C_2H_6
- (3) a and b for $\text{Cl}_2 > a$ and b for C_2H_6
- (4) a and b for $\text{Cl}_2 < a$ and b for C_2H_6

Q.5 In van der Waals' equation of state of the gas law, the constant 'b' is a measure of : [AIIEEE-04]

- (1) intermolecular repulsions
- (2) intermolecular attractions
- (3) volume occupied by the molecules
- (4) intermolecular collisions per unit volume

Q.6 Consider the van der Waals constants, a and b , for the following gases. [JEE-MAIN-2019]

Gas	At	Ne	Kr	Xe
$a/(\text{atm dm}^6 \text{ mol}^{-2})$	1.3	0.2	5.1	4.1
$b/(10^{-2} \text{ dm}^3 \text{ mol}^{-1})$	3.2	1.7	1.0	5.0

Which gas is expected to have the highest critical temperature ?

- (1) Xe (2) Kr (3) Ne (4) Ar

- Q.7** At a given temperature T , gases Ne, Ar, Xe and Kr are found to deviate from ideal gas behaviour. Their equation of state is given as

$$P = \frac{RT}{V - b} \text{ at } T.$$

Here, b is the Vander Waals constant. Which gas will exhibit steepest increase in the plot of Z (compression factor) vs p ? [JEE-MAIN-2019]

- (1) Kr (2) Ar (3) Xe (4) Ne

- Q.8** Consider the following table :

Gas	$a/(\text{k Pa dm}^6 \text{ mol}^{-1})$	$b/(\text{dm}^3 \text{ mol}^{-1})$
A	642.32	0.05196
B	155.21	0.04136
C	431.91	0.05196
D	155.21	0.4382

[JEE-MAIN-2019]

a and b are van der Waals constants. The correct statement about the gases is :

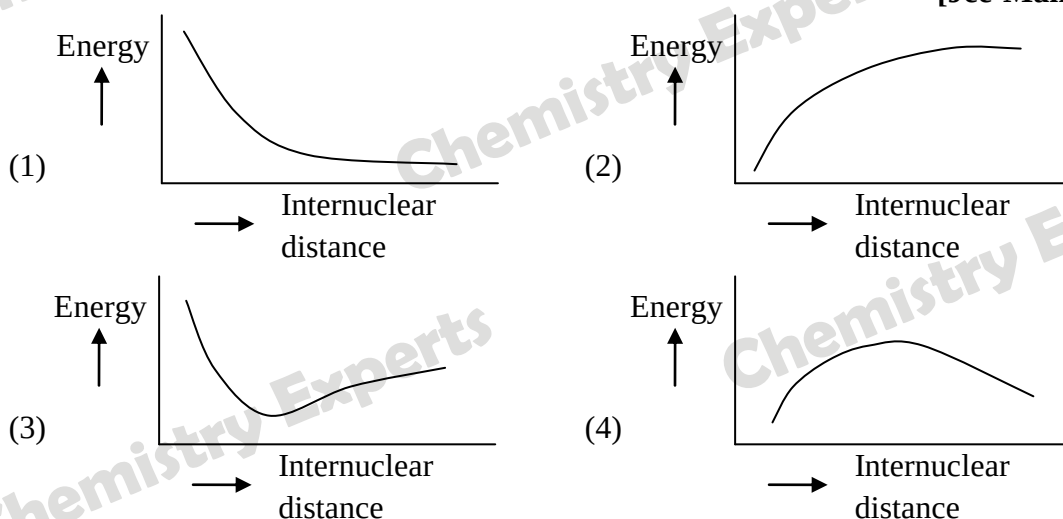
- (1) Gas C will occupy lesser volume than gas A; gas B will be more compressible than gas D
 (2) Gas C will occupy more volume than gas A; gas B will be more compressible than gas D
 (3) Gas C will occupy lesser volume than gas A; gas B will be lesser compressible than gas D
 (4) Gas C will occupy more volume than gas A; gas B will be lesser compressible than gas D

- Q.9** The volume of gas A is twice than that of gas B. The compressibility factor of gas A is thrice than that of gas B at same temperature. The pressures of the gases for equal number of moles are : [JEE-Main-2019]

- (1) $P_A = 3P_B$ (2) $3P_A = 2P_B$ (3) $2P_A = 3P_B$ (4) $P_A = 2P_B$

- Q.10** The potential energy curve for the H_2 molecule as a function of inter nuclear distance is :

[Jee-Main (Sep) 2020]



EXERCISE # (JEE-ADVANCED)

- Q.1** One way of writing the equation of state for real gases is, **[JEE 1997]**

$$P\bar{V} = RT \left[1 + \frac{B}{\bar{V}} + \dots \right] \quad \text{where } B \text{ is a constant.}$$

Derive an approximate expression for 'B' in terms of van der Waals' constants 'a' & 'b'.

- Q.2** Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atm at a temperature of 300 K. The value of "b" is 0.05 litre mol⁻¹. **[JEE 1998]**

- Q.3** A gas will approach ideal behaviour at : **[JEE 1999]**

- (A) low temperature and low pressure (B) low temperature and high pressure
(C) low pressure and high temperature (D) high temperature and high pressure .

- Q.4** The compressibility of a gas is less than unity at STP. Therefore, **[JEE 2000]**

- (A) $V_m > 22.4 \text{ L}$ (B) $V_m < 22.4 \text{ L}$ (C) $V_m = 22.4 \text{ L}$ (D) $V_m = 44.8 \text{ L}$

- Q.5** The compression factor (compressibility factor) for one mole of a van der Waals' gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the van der Waals' constant 'a'. **[JEE 2001]**

- Q.6** The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg m⁻³. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition. Determine **[JEE 2002]**

- (i) mol. wt.; (ii) molar volume; (iii) compression factor (z) of the vapour and
(iv) which forces among the gas molecules are dominating, the attractive or the repulsive

- Q.7** Positive deviation from ideal behaviour takes place because of **[JEE 2003]**

- (A) molecular attraction between atoms and $\frac{PV}{nRT} > 1$

- (B) molecular attraction between atoms and $\frac{PV}{nRT} < 1$

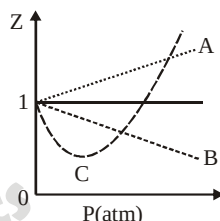
- (C) finite size of atoms and $\frac{PV}{nRT} > 1$

- (D) finite size of atoms and $\frac{PV}{nRT} < 1$

Q.8 For a real gas obeying van der Waals' equation a graph is plotted between PV_m (y-axis) and P (x-axis) where V_m is molar volume. Find y-intercept of the graph. [JEE 2004]

Q.9 The given graph represents the variation of Z

(compressibility factor = $\frac{PV}{nRT}$) versus P , for three real gases A, B and C. Identify the only INCORRECT statement. [JEE 2006]



- (A) for the gas A, $a = 0$ and its dependence on P is linear at all pressure
 (B) for the gas B, $b = 0$ and its dependence on P is linear at all pressure
 (C) for the gas C, which is typical real gas for which neither a nor $b = 0$. By knowing the minima and the point of intersection, with $Z = 1$, a and b can be calculated.
 (D) At high pressure, the slope is positive for all real gases A, B and C.

Q.10 Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2007]

Column I

- (A) Hydrogen gas ($P = 200$ atm, $T = 273$ K)
 (B) Hydrogen gas ($P \sim 0$, $T = 273$ K)
 (C) CO_2 ($P = 1$ atm, $T = 273$ K)
 (D) Real gas with very large molar volume

Column II

- (P) Compressibility factor $\neq 1$
 (Q) Attractive forces are dominant
 (R) $PV = nRT$
 (S) $P(V - nb) = nRT$

Q.11 A gas described by van der Waals' equation

[JEE 2008]

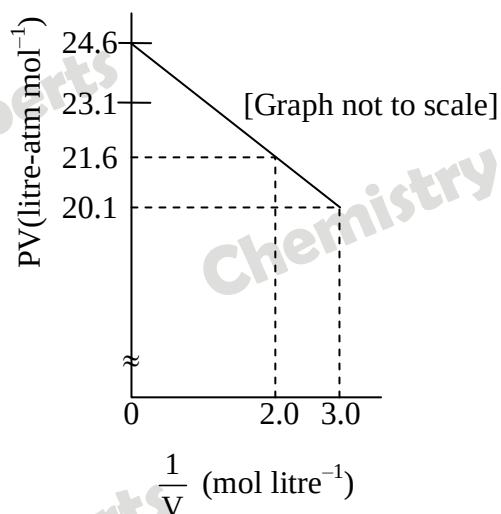
- (A) behaves similar to an ideal gas in the limit of large molar volumes
 (B) behaves similar to an ideal gas in the limit of large pressures
 (C) is characterised by van der Waals' coefficients that are dependent on the identity of the gas but are independent of the temperature
 (D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

Q.12 The term that corrects for the attractive forces present in a real gas in the Vander Waals' equation is [JEE 2009]

- (A) nb (B) $\frac{an^2}{V^2}$ (C) $-\frac{an^2}{V^2}$ (D) $-nb$

Q.13 For one mole of a van der Waals' gas when $b = 0$ and $T = 300$ K, the PV vs. $1/V$ plot is shown below. The value of the van der Waals' constant a (atm. litre² mol⁻²) is

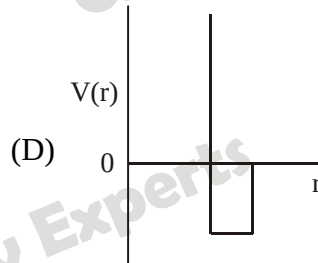
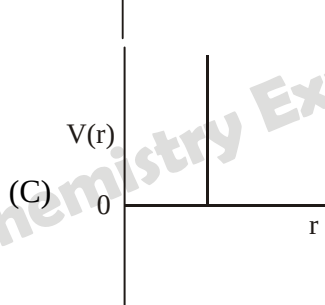
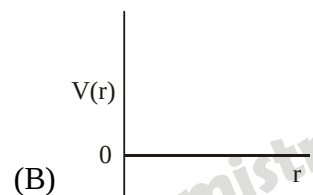
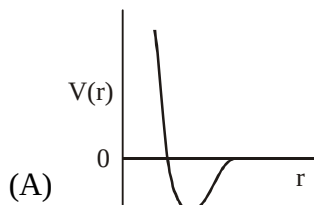
[JEE 2012]



- (A) 1.0 (B) 4.5 (C) 1.5 (D) 3.0

Q.14 One mole of a monoatomic real gas satisfies the equation $P(V - b) = RT$ where b is a constant. The relationship of interatomic potential $V(r)$ and interatomic distance r for the gas is given by-

[JEE 2015]



ANSWER KEY**EXERCISE # (JEE-MAIN)**

- | | | | | | | | | | |
|----|---|----|---|----|---|----|---|-----|---|
| 1. | 1 | 2. | 4 | 3. | 4 | 4. | 2 | 5. | 3 |
| 6. | 2 | 7. | 3 | 8. | 2 | 9. | 3 | 10. | 3 |

EXERCISE # (JEE-ADVANCED)

- | | | | | | | | |
|-----|--|-----|---|-----|---|----|---|
| 1. | $B = \left(b - \frac{a}{RT}\right)$ | 2. | 6.52 atm L ² mol ⁻² | 3. | C | 4. | B |
| 5. | 1.256 atm L ² mol ⁻² | 6. | (i) 18 g/mol , (ii) 50 L mol ⁻¹ , (iii) 1.218 , (iv) repulsive | | | | |
| 7. | C | 8. | RT | 9. | D | | |
| 10. | (A)→P,S; (B)→R; (C)→P,Q; (D)→R | 11. | A,C,D | 12. | B | | |
| 13. | C | 14. | C | | | | |

HINT AND SOLUTIONS

EXERCISE # JEE-MAIN

4. Ease of liquification $\propto \frac{a}{b}$

6. $T_c \propto \frac{a}{b}$

7. $P = \frac{RT}{V - b}$

$$PV - Pb = RT$$

$$Z = \frac{Pb}{RT}$$

$$Z = \left(\frac{b}{RT} \right) \cdot P$$

b is maximum for Xe

8. $\Rightarrow$ Gas A and C have same value of 'b' but different value of 'a' so gas having higher value of 'a' have more force of attraction so molecules will be more closer hence occupy less volume.

$\Rightarrow$ Gas B and D have same value of 'a' but different value of 'b' so gas having lesser value of 'b' will be more compressible.

10. For H_2 gas when molecules come closer to each other then first attraction among them dominates so potential energy decreases when molecules approaches too close with each other then repulsion dominates and potential energy increases.

EXERCISE # JEE-ADVANCED

1. According to the van der Waals equation,

$$\frac{P + (a/V^2)}{V - b} = RT$$

$$\text{or } P = \frac{RT}{V - b} - \frac{a}{V^2}$$

$$\text{or } PV = \frac{RTV}{V - b} - \frac{a}{V}$$

$$\text{or } PV = RT \left(1 - \frac{b}{V} \right)^{-1} - \frac{a}{V} = RT \left(1 + \frac{b}{V} \right) - \frac{a}{V}$$

Neglecting higher powers of b/V

$$\text{or } PV = RT \left(1 + \frac{b}{V} - \frac{a}{VRT} \right) = RT \left[1 + \frac{1}{V} \left(b - \frac{a}{RT} \right) \right]$$

Comparing with the given form of the equation. We get

$$B = b - \frac{a}{RT}$$

2. Vander Waal's equation for n moles of gas is,

$$\left[P + a \frac{n^2}{V^2} \right] [V - nb] = nRT.$$

Given, $v = 4L$, $P = 11 \text{ atm}$, $T = 300 \text{ K}$, $b = 0.05 \text{ litre/mol}$
 $n = 2$

$$\text{Thus, } \left[11 + a \frac{2^2}{4^2} \right] [4 - 2(0.05)] = 2 \times 0.082 \times 300$$

$$a = 6.46 \text{ atm litre}^2 \text{ mol}^{-2}$$

5. We know that, compressibility factor, $Z = PV / RT$

$$0.5 = 100 * V / 0.082 * 273$$

$$\therefore V = 0.1117 \text{ L}$$

Note :- Further when volume of a gas molecule is negligible, van der Waal's equation becomes

$$(P + a / V^2) (V - 0) = RT$$

$$\text{Or } PV = RT - a/V \text{ or } a = RTV - PV^2$$

Substituting the values

$$A = (0.082 * 0.1119 * 173) - (100 * 0.1119 * 0.1119)$$

$$= 1.253 \text{ atm L}^2 \text{ mol}^{-2}$$

$$8. \quad \left(P - \frac{a}{V_m^2} \right) (V_m - b) = RT$$

$$PV_m - Pb - \frac{a}{V_m} + \frac{ab}{V_m^2} = RT$$

at y intercept, $x = 0$

$$\text{i.e. } P = 0$$

$$\Rightarrow V_m \longrightarrow \infty$$

So the equation becomes

$$PV_m - Pb = RT$$

$$PV_m = Pb + RT$$

$$y = mx + c$$

c is the y intercept

$$\therefore c = RT$$

14. Since $a = 0$

Hence, only repulsive forces are present which are contributive only at very close distance.

Thus, the potential energy will increase abruptly.

EXERCISE # (JEE-MAIN)

- An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/s then uncertainty in position is ($h = 6.626 \times 10^{-34}$ Js) [AIEEE 2002]
 (1) 2.86×10^{-28} m (2) 2.86×10^{-32} cm (3) 1.5×10^{-27} m (4) 3.9×10^{-10} m
- Energy of H- atom in the ground state is -13.6 eV, Hence energy in the second excited state is- [AIEEE 2002]
 (1) -6.8 eV (2) -3.4 eV (3) -1.51 eV (4) -4.3 eV
- Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence uncertainty in velocity (ms^{-1}) is (Planck's constant $h = 6.6 \times 10^{-34}$ Js) [AIEEE-2002]
 (1) 2.1×10^{-28} (2) 2.1×10^{-34} (3) 0.5×10^{-34} (4) 5.0×10^{-24}
- The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$. This momentum for an s-electron will be given by [AIEEE-2003]
 (1) $\sqrt{2} \cdot \frac{h}{2\pi}$ (2) $+\frac{1}{2} \cdot \frac{2}{2\pi}$ (3) zero (4) $\frac{h}{2\pi}$
- The number of d-electrons retained in Fe^{2+} (At. number of Fe = 26) ion is : [AIEEE-2003]
 (1) 6 (2) 3 (3) 4 (4) 5
- The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 meters per second is approximately : [AIEEE 2003]
 (1) 10^{-25} meters (2) 10^{-33} meters (3) 10^{-31} meters (4) 10^{-16} meters
- In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen ? [AIEEE-2003]
 (1) $2 \rightarrow 5$ (2) $3 \rightarrow 2$ (3) $5 \rightarrow 2$ (4) $4 \rightarrow 1$
- Which of the following sets of quantum number is correct for an electron in 4f orbital ? [AIEEE-2004]
 (1) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$ (2) $n = 4, l = 4, m = -4, s = -\frac{1}{2}$
 (3) $n = 4, l = 3, m = +1, s = +\frac{1}{2}$ (4) $n = 4, l = 3, m = +4, s = +\frac{1}{2}$
- The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$) : [AIEEE-2004]
 (1) 9.1×10^{-8} nm (2) 192 nm (3) 406 nm (4) 91 nm

10. Which of the following statements in relation to the hydrogen atom is correct ?
 (1) 3s, 3p and 3d orbitals all have the same energy [AIEEE-2005]
 (2) 3s and 3p orbitals are of lower energy than 3d orbitals
 (3) 3p orbital is lower in energy than 3d orbital
 (4) 3s orbitals is lower in energy than 3p orbital
11. According to Bohr's theory angular momentum of electron in 5th shell is : [AIEEE-2006]
 (1) $1.0 \hbar/\pi$ (2) $10 \hbar/\pi$ (3) $2.5 \hbar/\pi$ (4) $25 \hbar/\pi$
12. Uncertainty in the position of an electron (mass = 9.1×10^{-31} Kg) moving with a velocity 300 ms^{-1} , accurate upto 0.001%, will be :- ($\hbar = 6.63 \times 10^{-34}$ Js) [AIEEE-2006]
 (1) $5.76 \times 10^{-2} \text{ m}$ (2) $1.92 \times 10^{-2} \text{ m}$ (3) $3.84 \times 10^{-2} \text{ m}$ (4) $19.2 \times 10^{-2} \text{ m}$
13. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite the electron in the atom from $n=1$ to $n=2$ is [AIEEE-2008]
 (1) $8.51 \times 10^5 \text{ J mol}^{-1}$ (2) $6.56 \times 10^5 \text{ J mol}^{-1}$ (3) $7.56 \times 10^5 \text{ J mol}^{-1}$ (4) $9.84 \times 10^5 \text{ J mol}^{-1}$
14. In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ($\hbar = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$, mass of electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$): [AIEEE-2009]
 (1) $1.92 \times 10^{-3} \text{ m}$ (2) $3.84 \times 10^{-3} \text{ m}$ (3) $1.52 \times 10^{-4} \text{ m}$ (4) $5.10 \times 10^{-3} \text{ m}$
15. Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 \text{ ms}^{-1}$ (Mass of proton = $1.67 \times 10^{-27} \text{ kg}$ and $\hbar = 6.63 \times 10^{-34} \text{ Js}$) : [AIEEE-2009]
 (1) 2.5 nm (2) 14.0 nm (3) 0.032 nm (4) 0.40 nm
16. The energy required to break one mole of Cl-Cl bonds in Cl_2 is 242 kJ mol^{-1} . The longest wavelength of light capable of breaking a single Cl-Cl bond is ($C = 3 \times 10^8 \text{ ms}^{-1}$ and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) [AIEEE-2010]
 (1) 494 nm (2) 594 nm (3) 640 nm (4) 700 nm
17. Ionisation energy of He^+ is $19.6 \times 10^{-18} \text{ J atom}^{-1}$. The energy of the first stationary state ($n=1$) of Li^{2+} is: [AIEEE-2010]
 (1) $8.82 \times 10^{-17} \text{ J atom}^{-1}$ (2) $4.41 \times 10^{-16} \text{ J atom}^{-1}$
 (3) $-4.41 \times 10^{-17} \text{ J atom}^{-1}$ (4) $-2.2 \times 10^{-15} \text{ J atom}^{-1}$
18. A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at : [AIEEE-2011]
 (1) 743 nm (2) 518 nm (3) 1035 nm (4) 325 nm
19. The frequency of light emitted for the transition $n=4$ to $n=2$ of He^+ is equal to the transition in H atom corresponding to which of the following [AIEEE-2011]
 (1) $n=3$ to $n=1$ (2) $n=2$ to $n=1$ (3) $n=3$ to $n=2$ (4) $n=4$ to $n=3$

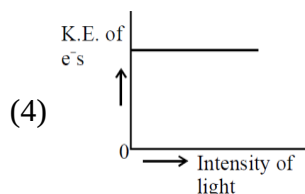
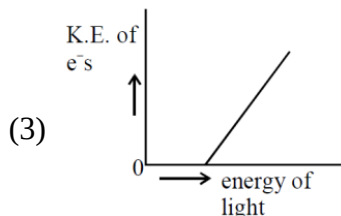
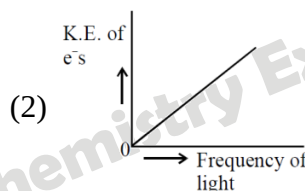
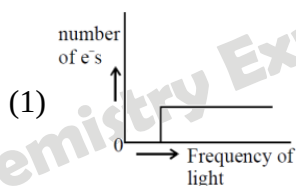
20. The electrons identified by quantum numbers n and l : **[AIEEE-2012]**
 (a) $n = 4, l = 1$ (b) $n = 4, l = 0$ (c) $n = 3, l = 2$ (d) $n = 3, l = 1$
 Can be placed in order of increasing energy as
 (1) $(a) < (c) < (b) < (d)$ (2) $(c) < (d) < (b) < (a)$
 (3) $(d) < (b) < (c) < (a)$ (4) $(b) < (d) < (a) < (c)$
21. If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become : **[JEE-Main(online) 2012]**
 (1) Two times (2) Half (3) One fourth (4) Four times
22. If the radius of first orbit of H atom is a_0 , the de-Broglie wavelength of an electron in the third orbit is : **[JEE-Main(online) 2012]**
 (1) $6\pi a_0$ (2) $8\pi a_0$ (3) $2\pi a_0$ (4) $4\pi a_0$
23. The wave number of the first emission line in the Balmer series of H-Spectrum is : **[JEE-Main(online) 2012]**
 (R = Rydberg constant) :
 (1) $\frac{3}{4} R$ (2) $\frac{9}{400} R$ (3) $\frac{5}{36} R$ (4) $\frac{7}{6} R$
24. The de Broglie wavelength of a car of mass 1000 kg and velocity 36 km/hr is : **[JEE-Main(online) 2013]**
 ($h = 6.63 \times 10^{-34}$ Js)
 (1) 6.626×10^{-31} m (2) 6.626×10^{-34} m (3) 6.626×10^{-38} m (4) 6.626×10^{-30} m
25. For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship? **[JEE-Main(online) 2014]**
 (1) a dust particle (2) an electron (3) a proton (4) an α -particle
26. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is : **[JEE-Main(online) 2014]**
 (1) 13.6 eV (2) 30.6 eV (3) 122.4 eV (4) 3.4 eV
27. Based on the equation **[JEE-Main(online) 2014]**

$$\Delta E = -2.0 \times 10^{-18} \text{ J} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$
 the wavelength of the light that must be absorbed to excite hydrogen electron from level $n = 1$ to level $n = 2$ will be ($h = 6.625 \times 10^{-34}$ Js, $C = 3 \times 10^8 \text{ ms}^{-1}$)
 (1) 2.650×10^{-7} m (2) 1.325×10^{-7} m (3) 1.325×10^{-10} m (4) 5.300×10^{-10} m
28. If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is **[JEE-Main(online) 2014]**

(1) $\sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right)}$ (2) $\sqrt{\frac{2h}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right)}$ (3) $\sqrt{\frac{2h}{m} (\lambda_0 - \lambda)}$ (4) $\sqrt{\frac{2hc}{m} (\lambda_0 - \lambda)}$

29. Ionization energy of gaseous Na atoms is $495.5 \text{ kJ mol}^{-1}$. The lowest possible frequency of light that ionizes a sodium atom is
 ($h = 6.626 \times 10^{-34} \text{ Js}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$) **[JEE-Main(online) 2014]**
 (1) $3.15 \times 10^{15} \text{ s}^{-1}$ (2) $4.76 \times 10^{14} \text{ s}^{-1}$ (3) $1.24 \times 10^{15} \text{ s}^{-1}$ (4) $7.50 \times 10^4 \text{ s}^{-1}$
30. Which of the following is the energy of a possible excited state of hydrogen?
[JEE-Main(offline) 2015]
 (1) -3.4 eV (2) $+6.8 \text{ eV}$ (3) $+13.6 \text{ eV}$ (4) -6.8 eV
31. At temperature T, the average kinetic energy of any particle is kT - The de Broglie wavelength follows the order : **[JEE-Main(online) 2015]**
 (1) Visible photon > Thermal electron > Thermal neutron
 (2) Thermal proton > Thermal electron > Visible photon
 (3) Visible photon > Thermal neutron > Thermal electron
 (4) Thermal proton > Visible photon > Thermal electron
32. The radius of the second Bohr orbit for hydrogen atom is : **[JEE Main 2017]**
 (Planck's const. $h = 6.6262 \times 10^{-34} \text{ Js}$
 mass of electron $= 9.1091 \times 10^{-31} \text{ kg}$
 charge of electron $e = 1.60210 \times 10^{-19} \text{ C}$
 permittivity of vacuum $\epsilon_0 = 8.854185 \times 10^{-12} \text{ kg}^{-1} \text{ m}^{-3} \text{ A}^2$)
 (1) 1.65A (2) 4.76A (3) 0.529A (4) 2.12A
33. For emission line of atomic hydrogen from $n_i = 8$ to $n_f =$ the plot of wave number $\left(\bar{\nu}\right)$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant, R_H is in wave number unit). **[JEE Main (Jan.) 2019]**
 (1) Linear- with slope - R_H (2) Linear with intercept - R_H
 (3) Non linear (4) Linear with slope R_H
34. Which of the following combination of statements is true regarding the interpretation of the atomic orbitals ? **[JEE Main (Jan.) 2019]**
 (a) An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbital of lower angular momentum.
 (b) For a given value of the principal quantum number, the size of the orbit is inversely proportional to the azimuthal quantum number.
 (c) According to wave mechanics, the ground state angular momentum is equal to $\frac{h}{2\pi}$.
 (d) The plot of ψ Vs r for various azimuthal quantum numbers, shows peak shifting towards higher r value.
 (1) (a), (c) (2) (a), (d) (3) (b), (c) (4) (a), (b)

35. Which of the graphs shown below does not represent the relationship between incident light and the electron ejected from metal surface? [JEE Main (Jan.) 2019]



36. The ground state energy of hydrogen atom is -13.6 eV. The energy of second excited state He^+ ion in eV is : [JEE Main (Jan.) 2019]

(1) -54.4 (2) -6.04 (3) -3.4 (4) -27.2

37. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H-atom is suitable for this purpose ? [JEE Main (Jan.) 2019]

[$R_H = 1 \times 10^5 \text{ cm}^{-1}$, $h = 6.6 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ ms}^{-1}$]

(1) Paschen, $5 \rightarrow 3$ (2) Balmer, $\infty \rightarrow 2$ (3) Lyman, $\infty \rightarrow 1$ (4) Paschen, $\infty \rightarrow 3$

38. The de Broglie wavelength (λ) associated with a photoelectron varies with the frequency (ν) of the incident radiation as, [ν_0 is threshold frequency] : [JEE Main (Jan.) 2019]

(1) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{1}{2}}}$ (2) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{1}{4}}}$ (3) $\lambda \propto \frac{1}{(\nu - \nu_0)}$ (4) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{3}{2}}}$

39. What is the work function of the metal if the light of wavelength $4000 \text{ \AA}$ generates photoelectrons of velocity $6 \times 10^5 \text{ ms}^{-1}$ from it ?

(Mass of electron = $9 \times 10^{-31} \text{ kg}$

Velocity of light = $3 \times 10^8 \text{ ms}^{-1}$

Planck's constant = $6.626 \times 10^{-34} \text{ Js}$

Charge of electron = $1.6 \times 10^{-19} \text{ JeV}^{-1}$)

[JEE Main (Jan.) 2019]

(1) 2.1 eV (2) 3.1 eV (3) 0.9 eV (4) 4.0 eV

40. If the de Broglie wavelength of the electron in n th Bohr orbit in a hydrogenic atom is equal to $1.5 \pi a_0$ (a_0 is Bohr radius), then the value of n/z is : [JEE Main (Jan.) 2019]

(1) 0.40 (2) 0.75 (3) 1.0 (4) 1.50

41. The quantum number of four electrons are given below : [JEE Main (April) 2019]

I. $n = 4, l = 2, m_l = -2, m_s = -\frac{1}{2}$ II. $n = 3, l = 2, m_l = 1, m_s = +\frac{1}{2}$

III. $n = 4, l = 1, m_l = 0, m_s = +\frac{1}{2}$ IV. $n = 3, l = 1, m_l = 1, m_s = -\frac{1}{2}$

(1) $I < III < II < IV$ (2) $IV < III < II < I$ (3) $I < II < III < I$ (4) $IV < II < III < I$

42. If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ , then for $1.5 p$ momentum of the photoelectron, the wavelength of the light should be : (Assume kinetic energy of ejected photoelectron to be very high in comparison to work function) : [JEE Main (April) 2019]

(1) $\frac{3}{4}\lambda$ (2) $\frac{4}{9}\lambda$ (3) $\frac{1}{2}\lambda$ (4) $\frac{2}{3}\lambda$

43. For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{\nu} = \bar{\nu}_{\max} - \bar{\nu}_{\min}$ be the difference in maximum and minimum frequencies in cm^{-1} . The ratio $\Delta\bar{\nu}_{\text{Lyman}} / \Delta\bar{\nu}_{\text{Balmer}}$ is:

[JEE Main (April) 2019]

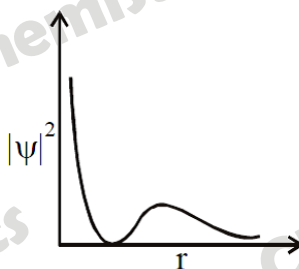
(1) 4 : 1 (2) 5 : 4 (3) 9 : 4 (4) 27 : 4

44. Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect ? (The Bohr radius is represented by a_0) [JEE Main (April) 2019]

- (1) The electron can be found at a distance $2a_0$ from the nucleus
 (2) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.
 (3) The magnitude of potential energy is double that of its kinetic energy on an average.
 (4) The probability density of finding the electron is maximum at the nucleus.

45. The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents:

[JEE Main (April) 2019]



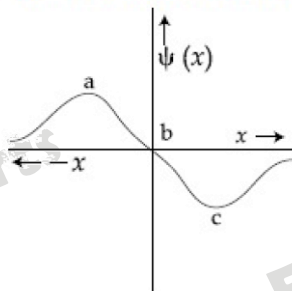
- (1) 1s orbital (2) 3s orbital (3) 2s orbital (4) 2p orbital

46. The ratio of the shortest wavelength of two special series of hydrogen spectrum is found to be about 9. The spectral series are : [JEE Main (April) 2019]

- (1) Paschen and Pfund (2) Balmer and Brackett
(3) Lyman and Paschen (4) Brackett and Pfund

47. The electrons are more likely to be found:

[JEE Main (April) 2019]



- (1) only in the region c (2) in the region a and c
(3) only in the region a (4) in the region a and b

48. The radius of the second Bohr orbit, in terms of the Bohr radius, a_0 , in Li^{2+} is :

- (1) $\frac{2a_0}{3}$ (2) $\frac{4a_0}{3}$ (3) $\frac{2a_0}{9}$ (4) $\frac{4a_0}{9}$

[Jee Main (Jan. 2020)]

49. The de Broglie wavelength of an electron in the 4th Bohr orbit is: [Jee Main (Jan. 2020)]

- (1) $6\pi a_0$ (2) $8\pi a_0$ (3) $2\pi a_0$ (4) $4\pi a_0$

50. For the Balmer series in the spectrum of H atom, $\bar{\nu} = R_H \left\{ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right\}$, the correct statements

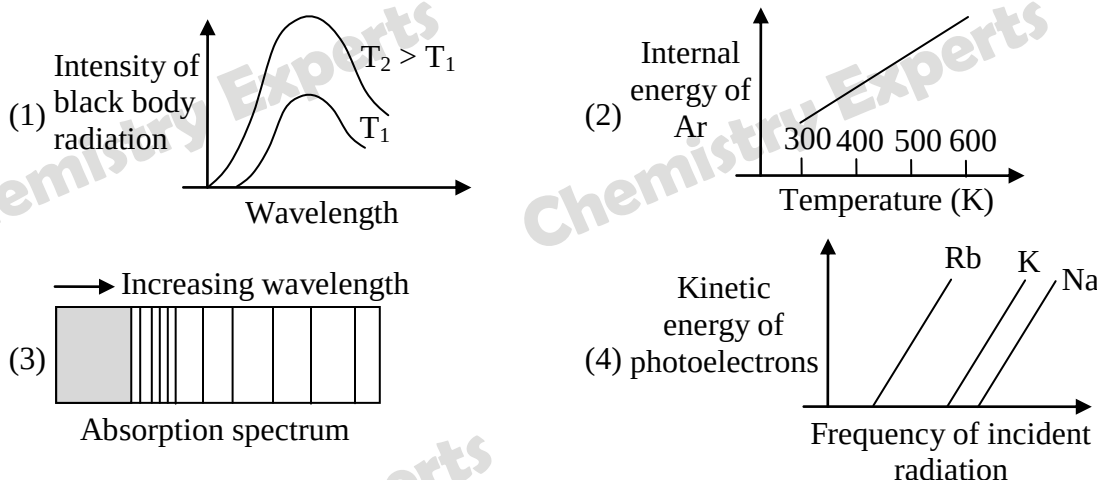
among (I) to (IV) are :

[Jee Main (Jan. 2020)]

- (I) As wavelength decreases, the lines in the series converge
(II) The integer n_1 is equal to 2
(III) The lines of longest wavelength corresponds to $n_2 = 3$
(IV) The ionization energy of hydrogen can be calculated from wave number of these lines
(1) (I), (III), (IV) (2) (I), (II), (III) (3) (II), (III), (IV) (4) (I), (II), (IV)

51. The figure that is not a direct manifestation of the quantum nature of atoms is :

[Jee-Main (Sep) 2020]



52. The work function of sodium metal is 4.41×10^{-19} J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be ($h = 6.63 \times 10^{-34}$ J s; $c = 3 \times 10^8$ m/s) _____ $\times 10^{-21}$ J. [Jee-Main (Sep) 2020]

53. The region in the electromagnetic spectrum where the Balmer series lines appear is :

[Jee-Main (Sep) 2020]

- (1) Visible (2) Microwave (3) Ultraviolet (4) Infrared

54. The shortest wavelength of H atom in the Lyman series is λ_1 . The longest wavelength in the Balmer series of He^+ is :

[Jee-Main (Sep) 2020]

- (1) $\frac{9\lambda_1}{5}$ (2) $\frac{5\lambda_1}{9}$ (3) $\frac{36\lambda_1}{5}$ (4) $\frac{27\lambda_1}{5}$

55. The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1 . The difference between the radii of 3rd and 4th orbits of He^+ is ΔR_2 . Ratio $\Delta R_1 : \Delta R_2$ is :

[Jee-Main (Sep) 2020]

- (1) 3 : 8 (2) 8 : 3 (3) 2 : 3 (4) 3 : 2

56. The correct statement about probability density (except at infinite distance from nucleus) is :

[Jee-Main (Sep) 2020]

- (1) It can be negative for 2p orbital (2) It can be zero for 1s orbital
(3) It can never be zero for 2s orbital (4) It can be zero for 3p orbital

EXERCISE # (JEE-ADVANCED)

1. The quantum numbers $+1/2$ and $-1/2$ for the electron spin represent: **[JEE 2001]**
- (A) rotation of the electron in clockwise and anticlockwise direction respectively.
- (B) rotation of the electron in anticlockwise and clockwise direction respectively.
- (C) magnetic moment of the electron pointing up and down respectively.
- (D) two quantum mechanical spin states which have no classical analogue

2. Rutherford's experiment, which established the nuclear model of atom, used a beam of :— **[JEE 2002]**
- (A) β - particles, which impinged on a metal foil and get absorbed.
- (B) γ - rays, which impinged on a metal foil and ejected electron.
- (C) Helium atoms, which impinged on a metal foil and got scattered.
- (D) Helium nuclei, which impinged on a metal foil and got scattered.

3. The magnetic moment of cobalt of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is [Given : Co^{+2}] **[JEE 2004]**
- (A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$

4. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom? **[JEE 2004]**
- (A) He^+ ($n = 2$) (B) Li^{2+} ($n = 2$) (C) Li^{2+} ($n = 3$) (D) Be^{3+} ($n = 2$)

5. (a) The Schrodinger wave equation for hydrogen atom is **[IIT-2004]**

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r_0}{a_0} \right) e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2s be at r_0 . Then find r_0 in terms of a_0 .

- (b) A base ball having mass 100 g moves with velocity 100 m/s. find out the value of wavelength of base ball.
6. (a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given $r = a_0$)
- (b) Find de-Broglie wavelength of the electron in first Bohr orbit. **[IIT-2005]**
- (c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units.

7. Given in hydrogenic atom r_n , V_n , E , K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of U, v, x, y . [JEE 2006]

- (A) $U = \frac{V_n}{K_n}$ (P) 1
 (B) $\frac{1}{r_n} \propto E^x$ (Q) -2
 (C) $r_n \propto Z^y$ (R) -1
 (Z = Atomic number)
 (D) v = (Orbital angular momentum of electron in its lowest energy) (S) 0

8. Match the entries in **Column I** with the correctly related quantum number(s) in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2008]

Column I**Column II**

- | | |
|--|--|
| (A) Orbital angular momentum of the electron in a hydrogen-like atomic orbital | (P) Principal quantum number |
| (B) A hydrogen-like one-electron wave function obeying Pauli principle | (Q) Azimuthal quantum number |
| (C) Shape, size and orientation of hydrogen like atomic orbitals | (R) Magnetic quantum number |
| (D) Probability density of electron at the nucleus | (S) Electron spin quantum number in hydrogen-like atom |

Paragraph for questions 14 to 16

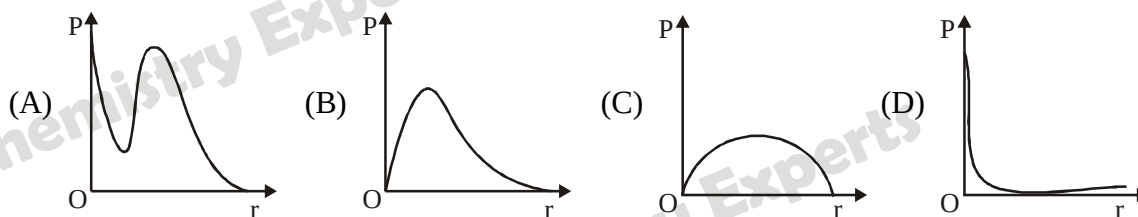
The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2 . The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom. [JEE 2010]

9. The state S_1 is :-
 (A) 1s (B) 2s (C) 2p (D) 3s
10. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-
 (A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50
11. The orbital angular momentum quantum number of the state S_2 is :-
 (A) 0 (B) 1 (C) 2 (D) 3

12. The maximum number of electrons that can have principal quantum number, $n=3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]
13. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is : – [JEE 2011]

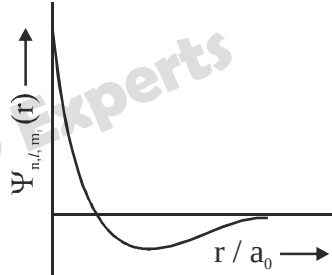
Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
$\phi(\text{eV})$	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

14. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a₀ is Bohr radius] [JEE 2012]
- (A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$ (C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{32\pi^2 m a_0^2}$
15. The atomic masses of He and Ne are 4 and 20 a.m.u. respectively. The value of the de Broglie wavelength of He gas at -73°C is "M" times that of the de Broglie wavelength of Ne at 727°C . M is. [JEE 2013]
16. In an atom, the total number of electrons having quantum numbers $n = 4$, $|m_\ell| = 1$ and $m_s = -\frac{1}{2}$ is [JEE 2014]
17. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr , at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is - [JEE 2016]



Answer 18, 19 and 20 by appropriately matching the information given in the three columns of the following table.

The wave function ψ_{n,l,m_l} is a mathematical functions whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum number n, l and m_l . Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is atomic number and a_0 is Bohr radius.

Column 1	Column 2	Column 3
(I) 1s orbital	(i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$	(P) 
(II) 2s orbital	(ii) One radial node	(Q) Probability density at nucleus $\propto \frac{1}{a_0^3}$
(III) 2p _z orbital	(iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} r e^{-\left(\frac{Zr}{2a_0}\right)} \cos \theta$	(R) Probability density is maximum at nucleus
(IV) 3d _{z²} orbital	(iv) xy-plane is a nodal plane	(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n = 6 state
[JEE Adv. 2017]		
18. For He ⁺ ion, the only INCORRECT combination is	(A) (I) (i) (S) (B) (II) (ii) (Q) (C) (I) (iii) (R) (D) (I) (i) (R)	
19. For the given orbital in Column I, the Only CORRECT combination for any hydrogen-like species is	(A) (II) (ii) (P) (B) (I) (ii) (S) (C) (IV) (iv) (R) (D) (III) (iii) (P)	
20. For hydrogen atom, the only CORRECT combination is	(A) (I) (i) (P) (B) (I) (iv) (R) (C) (II) (i) (Q) (D) (I) (i) (S)	
21. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state Ψ of He ⁺ whose energy; azimuthal quantum number and magnetic quantum number are –3.4 eV, 2 and 0, respectively. Which of the following statement(s) is(are) true for the state Ψ ? [JEE Adv. 2019]	(1) It has 2 angular nodes (2) It has 3 radial nodes (3) The nuclear charge experienced by the electron in this state is less than 2e, where e is the magnitude of the electronic charge (4) It is a 4d state.	

22. Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider the Bohr's model of a one electron atom where the electron moves around the nucleus. In the following, List-I contains some quantities for the n^{th} orbit of the atom and List-II contains showing how they depend on n . [JEE Adv. 2019]

List-I

- (I) Radius of the n^{th} orbit
 (II) Angular momentum of the electron in the n^{th} orbit
 (III) Kinetic energy of the electron in the n^{th} orbit
 (IV) Potential energy of the electron in the n^{th} orbit

List-II

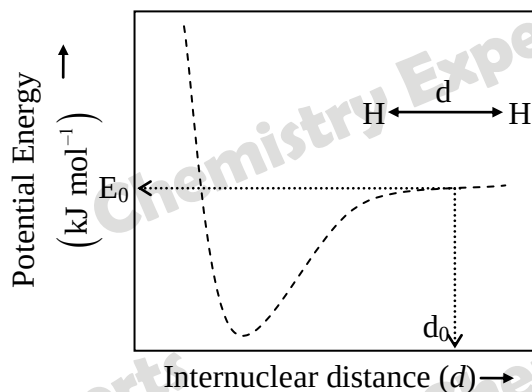
- (P) $\propto n^{-2}$
 (Q) $\propto n^{-1}$
 (R) $\propto n^0$
 (S) $\propto n^1$
 (T) $\propto n^2$
 (U) $\propto n^{1/2}$

Which of the following options has the correct combination considering List-I and List-II?

- (1) (I), (T) (2) (II), (Q) (3) (I), (P) (4) (II), (R)

23. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1} , for $d=d_0$ at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart. [JEE Adv. 2020]

Use Avogadro constant as $6.023 \times 10^{23} \text{ mol}^{-1}$.



ANSWER KEY

EXERCISE # (JEE-MAIN)

1.	1	2.	3	3.	1	4.	3	5.	1	6.	2	7.	3
8.	3	9.	4	10.	1	11.	3	12.	2	13.	4	14.	1
15.	4	16.	1	17.	3	18.	1	19.	2	20.	3	21.	2
22.	1	23.	3	24.	3	25.	1	26.	1	27.	2	28.	1
29.	3	30.	1	31.	1	32.	4	33.	4	34.	2	35.	2
36.	2	37.	4	38.	1	39.	1	40.	2	41.	2	42.	2
43.	3	44.	2	45.	3	46.	3	47.	2	48.	2	49.	2
50.	2	51.	2	52.	222.00	53.	1	54.	1	55.	3	56.	4

EXERCISE # (JEE-ADVANCED)

1. D 2. D 3. C 4. D 5. (a) $r_0 = 2a_0$ (b) $6.626 \times 10^{-25} \text{ \AA}$
6. (a) $2.197 \times 10^6 \text{ m/s}$ (b) $3.31 \text{ \AA}$ (c) $\sqrt{2} \cdot \frac{h}{2\pi}$ 7. (A) Q, (B) P, (C) R, (D) S
8. (A) $\rightarrow$ (Q,R); (B) $\rightarrow$ (P,Q,R,S) ; (C) $\rightarrow$ (P,Q,R) ; (D) $\rightarrow$ (P,Q)
9. B 10. C 11. B 12. 9 13. 4 14. C 15. 5
16. 6 17. B 18. C 19. A 20. D 21. 1 22. 2
23. (-5246.49)

HINT AND SOLUTIONS

EXERCISE # (JEE-MAIN)

1. $\Delta x \cdot \Delta v = \frac{h}{4\pi m}$ $\Delta x = \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 9.218 \times 10^{-6}} = 2.86 \times 10^{-28} \text{ m}$
2. 2nd excited state will be the 3rd energy level. $E_n = \frac{13.6}{n^2} \text{ eV}$ or $E = \frac{13.6}{9} = 1.51 \text{ eV}$.
3. $\Delta x \cdot \Delta v = \frac{h}{4\pi m}$ $\Delta v = \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 25 \times 10^{-5}} \therefore \Delta v = 2.1 \times 10^{-28} \text{ ms}^{-1}$.
4. The value of ℓ (azimuthal quantum number) for s -electron is equal to zero.
Orbital angular momentum = $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$
Substituting the value of ℓ for s-electron = $\sqrt{0(0+1)} \cdot \frac{h}{2\pi} = 0$
5. $_{26}\text{Fe} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6, 4s^2$,
 $\text{Fe}^{++} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6$
The number of d -electrons retained in $\text{Fe}^{2+} = 6$.
6. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \times 1000}{60 \times 10} = 11.05 \times 10^{-34} = 1.105 \times 10^{-33} \text{ metres}$.
7. The electron has minimum energy in the first orbit and its energy increases as n increases. Here n represents number of orbit, i.e., 1st, 2nd, 3rd The third line from the red end corresponds to yellow region i.e., 5. In order to obtain less energy electron tends to come 1st or 2nd orbit. So jump may be involved either $5 \rightarrow 1$ or $5 \rightarrow 2$
8. For 4 f orbital electrons, $n = 4$
 $\ell = 3, m = +3, +2, +1, 0, -1, -2, -3, s = +1/2$.
9. $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$; $\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) \text{ m}^{-1} \therefore \lambda = 91 \times 10^{-9} \text{ m} = 91 \text{ nm}$.
10. For hydrogen the energy order of orbital is $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d < 4f$.
11. Angular momentum of the electron, $mvr = \frac{nh}{2\pi}$ where $n = 5$ (given)
 $\therefore$ Angular momentum = $\frac{5h}{2\pi} = 2.5 \frac{h}{\pi}$
12. According to Heisenberg's uncertainty principle
 $\Delta x \times \Delta p = \frac{h}{4\pi}$
 $\Delta x \times (m \cdot \Delta v) = \frac{h}{4\pi}$; $\Delta x = \frac{h}{4\pi m \cdot \Delta v}$ $\Delta v = \frac{0.001}{100} \times 300 = 3 \times 10^{-3} \text{ ms}^{-1}$
 $\therefore \Delta x = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 3 \times 10^{-3}} = 1.92 \times 10^{-2} \text{ m}$.

13. $\text{I.E.} = 1.312 \times 10^6 \text{ J mol}^{-1}$

The energy required to excite the electron in the atom from $n_1 = 1$ to $n = 2$.

$$= 1.312 \times 10^6 \left[1 - \frac{1}{4} \right] = 9.84 \times 10^5 \text{ J mol}^{-1}$$

14. $\Delta v = \frac{0.005}{100} \times 600 = 3 \times 10^{-3} \text{ ms}^{-1}$

$$\Delta x = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 3 \times 10^{-3}} = 1.92 \times 10^{-2} \text{ m.}$$

15. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 1000} = 3.97 \times 10^{-10} = 0.397 \text{ nm.}$

16. $\text{Cl-Cl(g)} \longrightarrow 2\text{Cl(g)} ; \quad \Delta H = 242 \text{ KJ mol} = \frac{242 \times 10^3}{6.02 \times 10^{23}} \text{ J molecule}^{-1}$

$$E = \frac{hc}{\lambda} = \frac{242 \times 10^{-23} \times 10^3}{6.02} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{242 \times 10^{-23} \times 10^3} = \frac{6.6 \times 3 \times 6.02}{242} \times 10^{-6} = 0.494 \times 10^{-6} = 494 \times 10^{-9} \text{ m} = 494 \text{ nm}$$

17. $\text{I.E. of He}^+ = 19.6 \times 10^{-18} \text{ J atom}^{-1}$

$$\text{I.E.} = -E_1$$

$$E_1 \text{ for He}^+ \text{ is } = -19.6 \times 10^{-18} \text{ J atom}^{-1}$$

$$\frac{(E_1)_{\text{He}^+}}{(E_1)_{\text{Li}^{3+}}} = \frac{(Z_{\text{He}^+})^2}{(Z_{\text{Li}^{2+}})^2}$$

$$\frac{-19.6 \times 10^{-18}}{(E_1)_{\text{Li}^{2+}}} = \frac{4}{9}$$

$$E_1(\text{Li}^{2+}) = \frac{-19.6 \times 9 \times 10^{-18}}{4} = -44.1 \times 10^{-18} = -4.41 \times 10^{-17} \text{ J atom}^{-1}$$

18. $E = E_1 + E_2$

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}; \quad \frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}; \quad \frac{1}{355} = \frac{1}{680} + \frac{1}{\lambda_2}$$

$$\lambda_2 = 742.76 \text{ nm.}$$

19. $h\nu = \Delta E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$$\nu_{\text{He}^+} = \nu_{\text{H}} \times Z^2 \left(\frac{1}{\left(\frac{n_1}{2}\right)^2} - \frac{1}{\left(\frac{n_2}{2}\right)^2} \right) = \nu_{\text{H}} \left(\frac{1}{\left(\frac{2}{2}\right)^2} - \frac{1}{\left(\frac{4}{2}\right)^2} \right)$$

For H-atom

$$n_1 = 1, n_2 = 2$$

20. (a) 4 p (b) 4 s (c) 3 d (d) 3 p

Acc. to $(n + l)$ rule, increasing order of energy (d) < (b) < (c) < (a)

21. $\lambda = \frac{h}{\sqrt{2mKE}}$

22. $2\pi r = n\lambda$; $2\pi(9a_0) = 3\lambda$

$$\lambda = 6\pi a_0$$

23. $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$; $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5}{36} R$

24. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1000 \times 36 \times 1000}$

25. For macro particles (having larger mass) will be most difficult to experimentally verify the de-Broglie relationship.

26. $\Delta E = \frac{13.6 Z^2}{n^2} = \frac{13.6 \times 9}{4} = 30.6 \text{ eV}$

27. $E = -2.0 \times 10^{-18} \text{ J} \left(\frac{1}{4} - \frac{1}{1} \right) = 1.5 \times 10^{-18} \text{ J} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda}$

$$\lambda = 1.325 \times 10^{-7} \text{ m}$$

28. $KE = E_p - W$

$$\frac{1}{2}mv^2 = \frac{hc}{\lambda_0} - \frac{hc}{\lambda}; \quad v = \sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0} \right)}$$

$$(4) \sqrt{\frac{2hc}{m} (\lambda_0 - \lambda)}$$

$$29. \quad E = \frac{495.5 \times 10^3}{6.023 \times 10^{23}} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$30. \quad (E_n)_H = -13.6 \times \frac{1^2}{n^2} \text{ eV}$$

$$n = 2 \Rightarrow E_2 = -3.4 \text{ eV}$$

$$31. \quad K \cdot E = \frac{3}{2} KT = \frac{1}{2} mv^2$$

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2mK \cdot E}}$$

$$\lambda = \frac{h}{\sqrt{2m \times \frac{3}{2} KT}}$$

$$\lambda \propto \frac{1}{\sqrt{m}}$$

$$32. \quad r = \frac{n^2 h^2}{4\pi^2 m e^2 k}$$

$$r = \frac{2^2 (6.6262 \times 10^{-34} \text{ Js})^2}{4 \times (3.1416)^2 \times 9.1091 \times 10^{-31} \text{ kg} \times (1.60210 \times 10^{-19} \text{ C})^2 \times 9 \times 10^9}$$

$$r = 2.12 \times 10^{-10} \text{ m}$$

$$r = 2.12 \text{ \AA}$$

$$33. \quad \bar{v} = R_H Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = R_H \times 1^2 \times \left(\frac{1}{n_f^2} - \frac{1}{8^2} \right)$$

$$\bar{v} = R_H \times \frac{1}{n_f^2} - R_H \times \frac{1}{8^2}$$

curve of $\bar{v}$ vs $\frac{1}{n^2}$ is linear

with slope = R_H

34. An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbital of lower angular momentum since angular momentum (L) $\propto n$.

The plot of ψ Vs r for various azimuthal quantum numbers, shows peak shifting towards higher r value.

35. $\div$ Graph (2) is wrong

$$(KE)_{\max} = h\nu - h\nu$$

$$Y = mx + C$$

Because this graph have a negative intercept.

36. $E_T = -13.6 \frac{Z^2}{n^2} \frac{\text{ev}}{\text{atom}}$

For He^+

$$E_T = -13.6 \times \frac{2^2}{3^2} = -6.04 \text{ ev}$$

37. Energy of wave = $\frac{124}{\lambda(\text{nm})} \text{eV} = \frac{1240}{900} \text{eV} = 1.37 \text{ eV}$

This wave belong to paschen series of atomic spectra
for paschen series

$$\Delta E = 13.6 \times \frac{1}{9} = 1.51 \text{ eV}$$

So transition should be $\infty \rightarrow 3$

38. $h\nu = h\nu_0 + \text{K.E.} \quad \dots(1)$

$$\text{K.E.} = \frac{h^2}{2m\lambda^2} \quad \dots(2)$$

Putting (2) in (1)

$$\Rightarrow h\nu = h\nu_0 + \frac{h^2}{2m\lambda^2} \Rightarrow h(\nu - \nu_0) = \frac{h^2}{2m\lambda^2}$$

$$\Rightarrow \lambda^2 \propto \frac{1}{(\nu - \nu_0)} \Rightarrow \lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{1}{2}}}$$

39. $\frac{hc}{\lambda} = kE + W_0$

$$kE = \frac{1}{2} \times 9 \times 10^{-31} \times 36 \times 10^{10}$$

$$= 162 \times 10^{-21} \text{ J} = \frac{162 \times 10^{-21}}{1.6 \times 10^{-19}} = 1 \text{ ev}$$

$$\text{Energy of incident light} = \frac{12400}{4000} \text{ ev} = 3.1 \text{ ev}$$

$$\Rightarrow W_0 = 3.1 - 1 = 2.1 \text{ ev}$$

40. $\because 2\pi r_n = n_r \text{ \& } r_n = r_o \times \frac{n^2}{Z}$

$$\therefore 2\pi \left(r_o \times \frac{n^2}{Z} \right) = n (1.5\pi r_o)$$

$$\frac{n}{2} = \frac{1.5}{2} = 0.75$$

41. (I) $n = 4 \quad l = 2 \quad m_l = -2 \quad m_s = -1/2$ represents e^- present in 4d orbital
 (II) $n = 3 \quad l = 2 \quad m_l = 1 \quad m_s = +1/2$ represents e^- present in 3d orbital
 (IV) $n = 4 \quad l = 1 \quad m_l = 0 \quad m_s = +1/2$ represents e^- present in 4p orbital
 (V) $n = 3 \quad l = 1 \quad m_l = 1 \quad m_s = -1/2$ represents e^- present in 3p orbital

According to $(n+l)$ rule, order of energy will be

$$3p < 3d < 4p < 4d$$

42. $K \cdot E = h\nu - w_0$
 $\therefore K.E > > w_0$

$$\therefore h\nu > > w_0$$

$$\therefore K.E = h\nu = \frac{hc}{\lambda}$$

$$K \cdot E \propto \frac{1}{\lambda}$$

$$\therefore K \cdot E = \frac{p^2}{2m} \quad \therefore p^2 \propto \frac{1}{\lambda}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \left(\frac{p_1}{p_2} \right)^2 = \left(\frac{p_1}{1.5p_1} \right)^2 \Rightarrow \lambda_2 = \frac{4}{9} \lambda_1$$

43. For Lyman series maximum frequency of photon will be observed when electron makes transition from $n = \infty$ to $n = 1$ and a photon of minimum frequency will be observed when makes transition from $n = 2$ to $n = 1$

$$= R_C Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\Delta \bar{\nu}_{\text{Lyman}} = R_C Z^2 \left[\frac{1}{1} - \frac{1}{\infty} \right] - R_C Z^2 \left[\frac{1}{1} - \frac{1}{4} \right] = R_C Z^2 - \frac{3R_C Z^2}{4} = \frac{R_C Z^2}{4}$$

For Balmer series

Maximum frequency $n = \infty$ to $n = 2$

Maximum frequency $n = 3$ to $n = 2$

$$\Delta \bar{v}_{\text{Balmer}} = R_C Z^2 \left[\frac{1}{2^2} - \frac{1}{\infty} \right] - R_C Z^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{R_C Z^2}{4} - \frac{5R_C Z^2}{36} = \frac{R_C Z^2}{9}$$

$$\frac{\Delta \bar{v}_{\text{Lyman}}}{\Delta \bar{v}_{\text{Balmer}}} = \frac{\frac{R_C Z^2}{4}}{\frac{R_C Z^2}{9}} = \frac{9}{4}$$

44. Total energy of electron will be maximum at $n = \infty$

45. There is only one radial node and probability of finding the electrons is maximum near nucleus, graph belongs to 2s orbital.

46. For Lyman series

$$\frac{1}{\lambda_{\text{max}}} = R_H Z^2 \times \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right)$$

$$\frac{1}{\lambda_{\text{max}}} = R_H Z^2$$

$$\text{For Paschen series} = \frac{1}{\lambda_{\text{max}}} = R_H Z^2 \left(\frac{1}{3^2} - \frac{1}{\infty^2} \right)$$

$$\frac{1}{\lambda_{\text{max}}} = R_H Z^2 \times \frac{1}{9}$$

$$(\lambda_{\text{max}})_{\text{paschen}} = \frac{1}{9}$$

$$(\lambda_{\text{max}})_{\text{Lyman}} = 1$$

$$\frac{(\lambda_{\text{max}})_{\text{paschen}}}{(\lambda_{\text{max}})_{\text{Lyman}}} = \frac{9}{1}$$

47. We are getting maxima at point a & c

Electrons are more likely to be found in the region a & c.

48. $r = a_0 \times \frac{n^2}{Z}$

for second orbit of Li^{2+}

$$r = a_0 \times \frac{4}{3} = \frac{4a_0}{3}$$

49. $2\pi r = n\lambda$

$$2\pi \times (16a_0) = 4\lambda$$

$$8\pi a_0 = \lambda$$

50. For Balmer series $n_1 = 2$ $n_2 = 3, 4, 5, \dots$

For Balmer series, Energy of 1st line ($3 \rightarrow 2$) is minimum so λ will be max

lines in series converge, Energy is increased so λ is decreased

51. Internal energy of Ar will be equal to KE of Ar and KE is explained by KTG.

52. $E = W + KE_{\max}$

$$KE_{\max} = E - W$$

$$= \frac{hc}{\lambda} - 4.41 \times 10^{-19}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{300 \times 10^{-9}} - 4.41 \times 10^{-19}$$

$$= 2.22 \times 10^{-19} \text{ J}$$

$$= 222 \times 10^{-21} \text{ J}$$

53. Balmer series give visible lines For H-atom

54. As we know $\Delta E = \frac{hc}{\lambda}$

$$\text{So } \lambda = \frac{hc}{\Delta E} \quad \text{for } \lambda \text{ minimum i.e.}$$

shortest; $\Delta E = \text{maximum}$

for Lyman series $n = 1$ & for $\Delta E_{\max}$

Transition must be from $n = \infty$ to $n = 1$

$$\text{So } \frac{1}{\lambda_1} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda_1} = R_H Z^2 (1 - 0)$$

$$\frac{1}{\lambda_1} = R \times (1)^2 \Rightarrow \lambda_1 = \frac{1}{R}$$

For longest wavelength $\Delta E = \text{minimum}$ for Balmer series $n = 3$ to $n = 2$ will have ΔE minimum

for $\text{He}^+ Z = 2$

$$\text{So } \frac{1}{\lambda_2} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

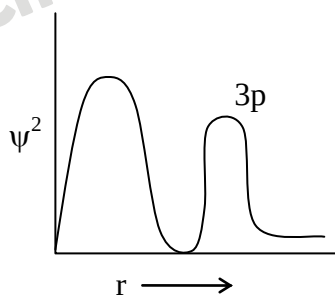
$$\frac{1}{\lambda_2} = R_H \times 4 \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\frac{1}{\lambda_2} = R_H \times \frac{5}{9}$$

$$\lambda_2 = \lambda_1 \times \frac{9}{5}$$

$$55. \quad \frac{\Delta R_1}{\Delta R_2} = \frac{(r_4 - r_3)_{4^{2+}}}{(r_4 - r_3)_{\text{He}^+}} = \frac{\frac{4^2}{2} - \frac{3^2}{2}}{\frac{4^2}{2} - \frac{3^2}{2}} = \frac{\frac{7}{2}}{\frac{7}{2}} = \frac{2}{3}$$

56. Probability density = ψ^2
 ψ^2 can be zero for 3p orbital besides of infinite distance



EXERCISE # (JEE-ADVANCED)

- The quantum numbers $+1/2$ and $-1/2$ for the electron spin represent two quantum mechanical spin states which have no classical analogue.
- α - particles are He nuclei.
- Cobalt is present as Co^{+2} which has $[\text{Ar}] 4s^0 3d^7$ configuration, which means it has 3 unpaired electrons. So the spin only magnetic moment of the compound is $\sqrt{3(3+2)} = \sqrt{15}$

$$4. \quad r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$$

$$\text{For hydrogen, } n = 1 \text{ and } Z = 1; \quad \therefore \quad r_H = 0.529 \text{ \AA}$$

$$\text{For } \text{Be}^{3+}, n = 2 \text{ and } Z = 4; \quad \therefore \quad r_{\text{Be}^{3+}} = \frac{0.529 \times 2^2}{4} = 0.529 \text{ \AA}$$

- ψ_{2s}^2 = probability of finding electron with in 2s orbital
 $\psi_{\text{at node}}^2 = 0$ (probability of finding an electron is zero at node)

$$\text{For node at } r = r_0, \quad \psi^2 = 0$$

$$\text{So, } \psi^2 = 0 = \frac{1}{4\sqrt{2}\pi} \left[\frac{1}{a_0} \right]^3 \left[2 - \frac{r_0}{a_0} \right] \times e^{r_0/2a_0}$$

$$\Rightarrow \left[2 - \frac{r_0}{a_0} \right] = 0 \text{ or } 2 = \frac{r}{a_0}$$

$$\Rightarrow r = 2a_0$$

$$(b) \quad \lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{100 \times 100 \times 10^{-3}} = \frac{6.626 \times 10^{-34}}{10,000 \times 10^{-3}} = 6.626 \times 10^{-35} \text{ m} \quad \text{or } 6.626 \times 10^{-25} \text{ \AA}$$

- For hydrogen atom, $Z = 1, n = 1$

$$v = 2.18 \times 10^6 \times \frac{Z}{n} \text{ ms}^{-1} = 2.18 \times 10^6 \text{ ms}^{-1}$$

$$(b) \quad \text{de Broglie wavelength, } \lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 2.18 \times 10^6} = 3.32 \times 10^{-10} \text{ m} = 3.3 \text{ \AA}$$

- For 2p, $l = 1$

$$\therefore \quad \text{Orbital angular momentum} = \sqrt{\ell(\ell+1)} \frac{h}{2\pi} = \sqrt{2} \frac{h}{2\pi} ..$$

$$7. \left. \begin{aligned} K_n &= \frac{KZe^2}{2r} \\ V_n &= -\frac{KZe^2}{r} \\ E_n &= -\frac{KZe^2}{2r} \end{aligned} \right\} \text{ so, } \frac{V_n}{K_n} = -2 \text{ and } E_n \propto \frac{1}{r}.$$

8. Magnetic moment of orbital angular momentum of the electron in a hydrogen-like atomic orbital depends on azimuthal quantum number and its direction depends on magnetic quantum number

9. For lower state (S_1)

$$\text{Number of radial node} = 1 = n - l - 1$$

Put $n = 2$ and $l = 0$ (as higher state S_2 has $n = 3$)

So, it would be 2s (for S_1 state)

$$\begin{aligned} 10. \text{ Energy of state } S_1 &= -13.6 \left(\frac{3^2}{2^2} \right) \text{ eV/atom} \\ &= \left(\frac{3^2}{2^2} \right) (\text{energy of H-atom in ground state}) \\ &= 2.25 (\text{energy of H-atom in ground state}). \end{aligned}$$

11. For state S_2

$$\text{Number of radial node} = 1 = n - l - 1 \quad \dots\dots (\text{eq.-1})$$

Energy of S_2 state = energy of e^- in lowest state of H-atom

$$= -13.6 \text{ eV/atom} = -13.6 \left(\frac{3^2}{n^2} \right) \text{ eV/atom}; \quad n = 3.$$

Put in equation (1) $l = 1$ so, orbital is 3p (for S_2 state).

12. Electrons with spin quantum number $= -\frac{1}{2}$ will be 1(in s) + 3(in p) + 5(in d) = 9.

$$13. E_{\text{photon}} = \frac{12400}{3000} = 4.13 \text{ eV}$$

Photoelectric effect can take place only if $E_{\text{photon}} \geq \phi$

Thus,

Li, Na, K, Mg can show photoelectric effect.

14. $mv(4a_0) = \frac{h}{\pi}$

so, $v = \frac{h}{4m\pi a_0}$

so, $KE = \frac{1}{2} mv^2 = \frac{1}{2} m \cdot \frac{h^2}{16m^2\pi^2 a_0^2} = \frac{h^2}{32m\pi^2 a_0^2}$

15. $\lambda = \frac{h}{\sqrt{2m(KE)}} \quad KE \propto T$

$$\frac{\lambda_{He}}{\lambda_{Ne}} = \sqrt{\frac{m_{Ne} KE_{Ne}}{m_{He} KE_{He}}} = \sqrt{\frac{20 \times 1000}{4 \times 200}} = 5.$$

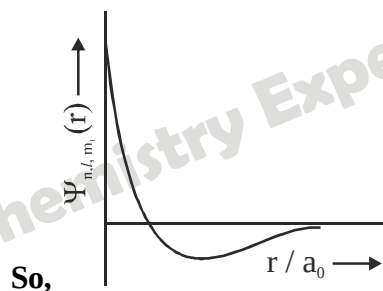
16. $n = 4, \quad m_l = 1, -1$
Hence l can be $= 3, 2, 1$

i.e. H_f ; 2 orbitals
 H_d ; 2 orbitals
 H_p ; 2 orbitals

Hence total of 6 orbitals, and we want $m_s = -\frac{1}{2}$, that is only one kind of spin. So, 6 electrons.

17. 1s orbital does not have any spherical node.

18. Number of radial nodes in 2s orbital $= n - \ell - 1 = 1$



19. 1s orbital cannot have angular part of wave function so, (D) is incorrect.

20. For H – atom $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$

$$\frac{E_4 - E_2}{E_6 - E_2} = \frac{13.6 \times \left[\frac{1}{4} - \frac{1}{9} \right]}{13.6 \times \left[\frac{1}{4} - \frac{1}{36} \right]} = \frac{27}{32}$$

21. Total energy = $-13.6 \times \frac{Z^2}{n^2}$ eV

$$-3.4\text{e} = -13.6 \times \frac{2^2}{n^2} \text{eV}$$

$$n^2 = \frac{13.6}{3.4} \times 2^2 = 16; \quad n = 4$$

Value of $\ell = 2$, So, it's a 4d orbital.

Number of angular nodes = $\ell = 2$

Number of radial nodes = $n - \ell - 1 = 4 - 2 - 1 = 1$

Effective nuclear charge at this electron will be equal to 2e since no shielding effect.

22. (1) Radius of the n^{th} orbit $\propto n^2$

(2) Angular momentum of the electron in the n^{th} orbit $\propto n$

(3) Kinetic energy of the electron in the n^{th} orbit $\propto \frac{1}{n^2}$

(4) Potential energy of the electron in the n^{th} orbit $\propto \frac{1}{n^2}$

23. For one H atom

$$\begin{aligned} \text{P.E.} &= -\frac{kq_1q_2}{r} = -\frac{9 \times 10^9 \times [1.6 \times 10^{-19}]^2}{0.529 \times 10^{-10}} \\ &= -\frac{9 \times 1.6 \times 1.6}{0.529} \times 10^{-19} \end{aligned}$$

Total potential energy for 2 H-atom (per mole)

$$\begin{aligned} &= -\frac{2 \times 9 \times 1.6 \times 1.6}{0.529} \times 10^{-19} \times 6 \times 10^{23} \\ &= -5246.5 \text{ KJ/mole} \end{aligned}$$

EXERCISE # (JEE-MAIN)

- For a cell reaction involving a two-electron change, the standard e.m.f. of the cell is found to be 0.295V at 25° C. The equilibrium constant of the reaction at 25°C will be : **[AIEEE 2003]**
 (1) 10 (2) 1×10^{10} (3) 1×10^{-10} (4) 29.5×10^{-2}
- Standard reduction electrode potentials of three metals A, B and C are respectively + 0.5V, – 3.0V and –1.2 V. The reducing powers of these metals are : **[AIEEE 2003]**
 (1) C > B > A (2) A > C > B (3) B > C > A (4) A > B > C
- For the redox reaction :
 $\text{Zn(s)} + \text{Cu}^{2+} (0.1\text{M}) \rightarrow \text{Zn}^{2+} (1\text{M}) + \text{Cu(s)}$ taking place in a cell,
 E°_{Cell} is 1.10 volt. E_{Cell} for the cell will be $\left(2.303 \frac{RT}{F} = 0.0591 \right)$ **[AIEEE 2003]**
 (1) 1.07 volt (2) 0.82 volt (3) 2.14 volt (4) 1.80 volt
- When, during electrolysis of a solution of AgNO_3 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be : **[AIEEE 2003]**
 (1) 21.6g (2) 108g (3) 1.08g (4) 10.8g
- Consider the following E° values, **[AIEEE 2004]**
 $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +0.77\text{V}$ $E^\circ_{\text{Sn}^{2+}/\text{Sn}} = -0.14\text{V}$
 Under standard conditions the potential for the reaction,
 $\text{Sn(s)} + 2\text{Fe}^{3+} (\text{aq}) \longrightarrow 2\text{Fe}^{2+} (\text{aq}) + \text{Sn}^{2+} (\text{aq})$ is:
 (1) 0.91V (2) 1.40V (3) 1.68V (4) 0.63V
- The standard e.m.f. of a cell, involving one electron change is found to be 0.591 V at 25°C. The equilibrium constant of the reaction is ($F = 96,500\text{C mol}^{-1}$; $R = 8.314\text{ JK}^{-1}\text{ mol}^{-1}$) **[AIEEE 2004]**
 (1) 1.0×10^{10} (2) 1.0×10^5 (3) 1.0×10^1 (4) 1.0×10^{30}
- The limiting molar conductivities Λ° for NaCl, KBr and KCl are 126, 152 and 150 $\text{Scm}^2\text{ mol}^{-1}$ respectively. The Λ_m° for NaBr is : **[AIEEE 2004]**
 (1) 278 $\text{S cm}^2\text{ mol}^{-1}$ (2) 176 $\text{S cm}^2\text{ mol}^{-1}$ (3) 128 $\text{S cm}^2\text{ mol}^{-1}$ (4) 302 $\text{S cm}^2\text{ mol}^{-1}$

8. In a cell that utilises the reaction, [AIEEE 2004]

$\text{Zn(s)} + 2\text{H}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ addition of H_2SO_4 to cathode compartment, will :

- (1) increase the E_{cell} and shift equilibrium to the right
- (2) lower the E_{cell} and shift equilibrium to the right
- (3) lower the E_{cell} and shift equilibrium to the left
- (4) increase the E_{cell} and shift equilibrium to the left

9. The $E_{\text{M}^{3+}/\text{M}^{2+}}^0$ values for Cr, Mn, Fe and Co are -0.41 , $+1.57$, $+0.77$ and $+1.97\text{V}$ respectively. For which one of these metals the change in oxidation state from $+2$ to $+3$ is easiest?

[AIEEE 2004]

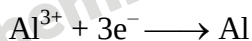
- (1) Fe (2) Mn (3) Cr (4) Co

10. For a spontaneous reaction the ΔG , equilibrium constant (K) and E_{Cell}^0 will be respectively

[AIEEE 2005]

- (1) $-\text{ve}$, < 1 , $-\text{ve}$ (2) $-\text{ve}$, > 1 , $-\text{ve}$ (3) $-\text{ve}$, > 1 , $+\text{ve}$ (4) $+\text{ve}$, > 1 , $-\text{ve}$

11. Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (1 Faraday = 96500 Coulombs). The cathode reaction is [AIEEE 2005]



To prepare 5.12 kg of aluminium metal by this method would require,

- (1) $5.49 \times 10^4 \text{ C}$ of electric charge (2) $5.49 \times 10^1 \text{ C}$ of electric charge
- (3) $5.49 \times 10^7 \text{ C}$ of electric charge (4) $1.83 \times 10^7 \text{ C}$ of electric charge

12. The highest electrical conductivity of the following aqueous solution is of [AIEEE 2005]

- (1) 0.1 M fluoroacetic acid (2) 0.1 M difluoroacetic acid
- (3) 0.1 M acetic acid (4) 0.1 M chloroacetic acid

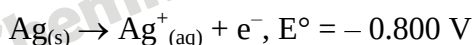
13.	Electrolyte	Λ^∞ (S cm ² mol ⁻¹)
	KCl	149.9
	KNO ₃	145.0
	HCl	426.2
	NaOAc	91.0
	NaCl	126.5

Calculate $\Lambda_{\text{HOAc}}^\infty$ using appropriate molar conductances of the electrolytes listed above at infinite dilution in H₂O at 25°C

[AIEEE 2005]

- (1) 390.7 (2) 217.5 (3) 517.2 (4) 552.7

14. Given the data at 25°C,
 $\text{Ag}_{(s)} + \text{I}^-_{(aq)} \rightarrow \text{AgI}_{(s)} + e^-$, $E^\circ = 0.152\text{V}$



What is the value of $\log K_{sp}$ for AgI?

(Where K_{sp} = solubility product)

[AIEEE 2006]

$$\left(2.303 \frac{RT}{F} = 0.59\text{V} \right)$$

- (1) -8.12 (2) +8.612 (3) -37.83 (4) -16.13

15. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω. The conductivity of this solution is 1.29 Sm⁻¹. Resistance of the same cell when filled with 0.02M of the same solution is 520Ω. The molar conductivity of 0.02M solution of the electrolyte will be.

[AIEEE 2006]

- (1) $124 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$ (2) $1240 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$
 (3) $1.24 \times 10^4 \text{ Sm}^2 \text{ mol}^{-1}$ (4) $12.4 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$

16. The molar conductivities, Λ_{NaOAc}^0 and Λ_{HCl}^0 at infinite dilution in water at 25°C are 91.0 and 426.2 S cm²/mol respectively. To calculate Λ_{HOAc}^0 the additional value required is :

[AIEEE 2006]

- (1) KCl (2) NaOH (3) NaCl (4) H₂O

17. The cell $\text{Zn} | \text{Zn}^{+2}_{(\text{aq})}(1\text{M}) || \text{Cu}^{+2}_{(\text{aq})}(1\text{M}) | \text{Cu}$ ($E^\circ_{\text{cell}} = 1.10 \text{ V}$) was allowed to be completely discharged at 298 K. The relative concentration of Zn^{+2} to Cu^{+2} , $\left\{ \frac{[\text{Zn}^{+2}]}{[\text{Cu}^{+2}]} \right\}$ is :

[AIEEE 2007]

- (1) 9.65×10^4 (2) Antilog (24.08) (3) 37.3 (4) 1037.3

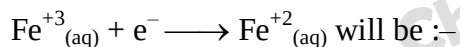
18. Given : $E^\circ_{\text{Cr}^{+3}/\text{Cr}} = -0.72 \text{ V}$, $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.42 \text{ V}$. The potential for the cell, [AIEEE 2008]



- (1) 0.26 V (2) 0.339 V (3) - 0.339 V (4) - 0.26 V

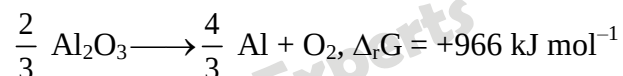
19. Given : [AIEEE-2009]

$E^\circ_{\text{Fe}^{+3}/\text{Fe}} = -0.036\text{V}$, $E^\circ_{\text{Fe}^{+2}/\text{Fe}} = -0.439\text{V}$. The value of standard electrode potential for the change.



- (1) 0.770 V (2) -0.27 V (3) -0.072 V (4) 0.385 V

20. The Gibbs energy for the decomposition of Al_2O_3 at 500°C is as follows : [AIEEE 2010]



The potential difference needed for electrolytic reduction of Al_2O_3 at 500°C is at least :-

- (1) 5.0 V (2) 4.5 V (3) 3.0 V (4) 2.5 V

21. The reduction potential of hydrogen half-cell will be negative if :- [AIEEE 2011]

- (1) $p(\text{H}_2) = 2 \text{ atm}$ $[\text{H}^+] = 1.0 \text{ M}$ (2) $p(\text{H}_2) = 2 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$
 (3) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$ (4) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 1.0 \text{ M}$

22. Resistance of 0.2 M solution of an electrolyte is 50Ω . The specific conductance of the solution is 1.3 S m^{-1} . If resistance of the 0.4M solution of the same electrolyte is 260Ω , its molar conductivity is :- [AIEEE 2011]

- (1) $6250 \text{ S m}^2 \text{ mol}^{-1}$ (2) $6.25 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 (3) $625 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$ (4) $62.5 \text{ S m}^2 \text{ mol}^{-1}$

23. The standard reduction potentials for $\text{Zn}^{2+} / \text{Zn}$, $\text{Ni}^{2+} / \text{Ni}$ and $\text{Fe}^{2+} / \text{Fe}$ are -0.76 , -0.23 and -0.44 V respectively. The reaction $\text{X} + \text{Y}^{2+} \rightarrow \text{X}^{2+} + \text{Y}$ will be spontaneous when [AIEEE 2012]
 (1) $\text{X} = \text{Zn}$, $\text{Y} = \text{Ni}$ (2) $\text{X} = \text{Ni}$, $\text{Y} = \text{Fe}$ (3) $\text{X} = \text{Ni}$, $\text{Y} = \text{Zn}$ (4) $\text{X} = \text{Fe}$, $\text{Y} = \text{Zn}$
24. Resistance of 0.2 M solution of an electrolyte is 50Ω . The specific conductance of the solution is 1.4 S m^{-1} . The resistance of 0.5 M solution of the same electrolyte is 280Ω . The molar conductivity of 0.5 M solution of the electrolyte in $\text{S m}^2 \text{ mol}^{-1}$ is : [JEE-MAIN 2014]
 (1) 5×10^3 (2) 5×10^2 (3) 5×10^{-4} (4) 5×10^{-3}
25. The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞ , respectively. The correct relationship between λ_C and λ_∞ is given as : [JEE-MAIN 2014]
 (where the constant B is positive)
 (1) $\lambda_C = \lambda_\infty - (2) \sqrt{C}$ (2) $\lambda_C = \lambda_\infty + (2) \sqrt{C}$ (3) $\lambda_C = \lambda_\infty + (2) C$ (4) $\lambda_C = \lambda_\infty - (2) (3)$
26. Given :
 $E_{\text{Cr}^{3+}/\text{Cr}}^0 = 0.74 \text{ V}$; $E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.51 \text{ V}$
 $E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}}^0 = 1.33 \text{ V}$; $E_{\text{Cl}/\text{Cl}^-}^0 = 1.36 \text{ V}$
 Based on the data given above, strongest oxidising agent will be :
 (1) Cl^- (2) Cr^{3+} (3) Mn^{2+} (4) MnO_4^-
27. Two Faraday of electricity is passed through a solution of CuSO_4 . The mass of copper deposited at the cathode is : (at. mass of $\text{Cu} = 63.5 \text{ amu}$) [JEE-MAIN 2015]
 (1) 2g (2) 127 g (3) 0 g (4) 63.5 g
28. A variable, opposite external potential (E_{ext}) is applied to the cell
 $\text{Zn}|\text{Zn}^{2+} (1 \text{ M}) \parallel \text{Cu}^{2+} (1 \text{ M}) | \text{Cu}$, of potential 1.1 V. When $E_{\text{ext}} < 1.1 \text{ V}$ and $E_{\text{ext}} > 1.1 \text{ V}$, respectively electrons flow from : [JEE-MAIN 2015]
 (1) anode to cathode in both cases (2) anode to cathode and cathode to anode
 (3) cathode to anode in both cases (4) cathode to anode and anode to cathode

29. At 298 K, the standard reduction potentials are 1.51 V for $\text{MnO}_4^-|\text{Mn}^{2+}$, 1.36 V for $\text{Cl}_2|\text{Cl}^-$, 1.07 V for $\text{Br}_2|\text{Br}^-$, and 0.54 V for $\text{I}_2|\text{I}^-$. At pH = 3, permanganate is expected to oxidize $\left(\frac{RT}{F} = 0.059\text{V}\right)$: [JEE-MAIN 2015]

(1) Cl^- and Br^- (2) Cl^- , Br^- and I^- (3) Br^- and I^- (4) I^- only

30. Oxidation of succinate ion produces ethylene and carbon dioxide gases. On passing 0.2 Faraday electricity through an aqueous solution of potassium succinate, the total volume of gases (at both cathode and anode) at STP (1 atm and 273 K) is : [JEE-MAIN 2016]

(1) 8.96 L (2) 2.24 L (3) 4.48 L (4) 6.72 L

31. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO_4 [JEE-MAIN 2016]

(1) The copper metal will dissolve and zinc metal will be deposited
 (2) No reaction will occur
 (3) The copper metal will dissolve with evolution of oxygen gas
 (4) The copper metal will dissolve with evolution of hydrogen gas

32. Given

$$E_{\text{Cl}^2/\text{Cl}^-}^0 = 1.36\text{V}, E_{\text{Cl}^{3+}/\text{Cr}}^0 = -0.74\text{V}$$

$$E_{\text{Cl}^2/\text{Cl}_2^{2-}/\text{Cr}^{3+}}^0 = 1.33\text{V}, E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.51\text{V}$$

Among the following, the strongest reducing agent is [JEE MAIN 2017]

(1) Cr (2) Mn^{2+} (3) Cr^{3+} (4) Cl^-

33. How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66 g of diborane ?

(Atomic weight of B = 10.8 u) [JEE MAIN 2018]

(1) 1.6 hours (2) 6.4 hours (3) 0.8 hours (4) 3.2 hours

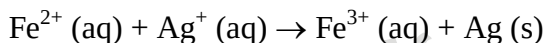
34. Given that $E_{\text{O}_2/\text{H}_2\text{O}}^0 = 1.23\text{V}$; [JEE Main 2019 (Apr.)]

$$E_{\text{S}_2\text{O}_8^{2-}/\text{SO}_4^{2-}}^0 = 2.05\text{V}; \quad E_{\text{Br}_2/\text{Br}^-}^0 = +1.09\text{V}; \quad E_{\text{Au}^{3+}/\text{Au}}^0 = +1.4\text{V}$$

The strongest oxidizing agent is :

(1) O_2 (2) $\text{S}_2\text{O}_8^{2-}$ (3) Br_2 (4) Au^{3+}

35. Calculate the standard cell potential (in V) of the cell in which following reaction takes place :



[JEE Main 2019 (Apr.)]

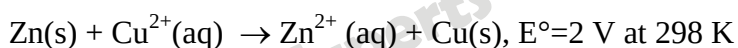
Given that

$$E_{\text{Ag}^+/\text{Ag}}^\circ = x \text{ V}$$

$$E_{\text{Fe}^{2+}/\text{Fe}}^\circ = y \text{ V}$$

$$E_{\text{Fe}^{3+}/\text{Fe}}^\circ = z \text{ V}$$

36. The standard Gibbs energy for the given cell reaction in kJ mol^{-1} at 298 K is :



(Faraday's constant, $F = 96000 \text{ C mol}^{-1}$)

[JEE MAIN 2019 (Apr.)]

37. A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode? [JEE MAIN 2019 (Apr.)]

(1) 0.20 (2) 0.15 (3) 0.10 (4) 0.05

38. Consider the statements S1 and S2 :

S1: Conductivity always increases with decrease in the concentration of electrolyte.

S2: Molar conductivity always increases with decrease in the concentration of electrolyte.

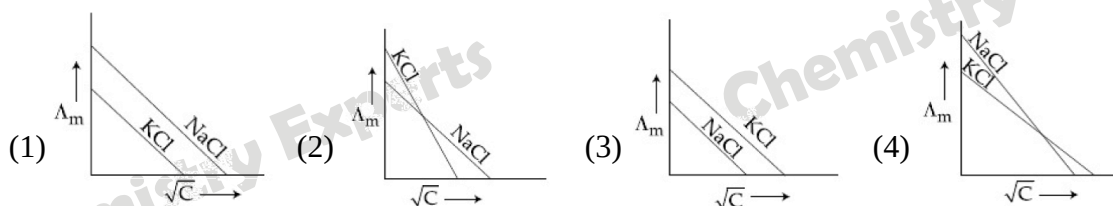
The correct option among the following is:

[JEE MAIN 2019 (Apr.)]

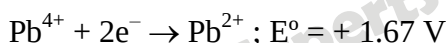
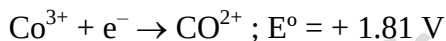
- (1) Both S1 and S2 are wrong (2) Both S1 and S2 are correct
(3) S1 is wrong and S2 is correct (4) S1 is correct and S2 is wrong

39. Which one of the following graphs between molar conductivity (Λ_m) versus $\sqrt{C}$ is correct ?

[JEE MAIN 2019 (Apr.)]



40. Given: [JEE MAIN 2019 (Apr.)]



Oxidizing power of the species will increase in the order:

- (1) $\text{Ce}^{4+} < \text{Pb}^{4+} < \text{Bi}^{3+} < \text{Co}^{3+}$ (2) $\text{Co}^{3+} < \text{Pb}^{4+} < \text{Ce}^{4+} < \text{Bi}^{3+}$
 (3) $\text{Bi}^{3+} < \text{Ce}^{4+} < \text{Pb}^{4+} < \text{Co}^{3+}$ (4) $\text{Co}^{3+} < \text{Ce}^{4+} < \text{Bi}^{3+} < \text{Pb}^{4+}$

41. The decreasing order of electrical conductivity of the following aqueous solutions is:

0.1 M Formic acid (A),

[JEE MAIN 2019 (Apr.)]

0.1 M Acetic acid (B),

0.1 M Benzoic acid (C).

- (1) $A > B > C$ (2) $C > A > B$ (3) $A > C > B$ (4) $C > B > A$

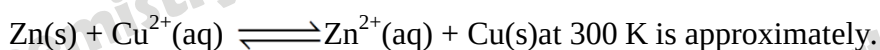
42. The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of PbSO_4 electrolyzed in g during the process is: (Molar mass of $\text{PbSO}_4 = 303 \text{ g mol}^{-1}$)

[JEE MAIN 2019 (Jan.)]

- (1) 22.8 (2) 15.2 (3) 7.6 (4) 11.4

43. If the standard electrode potential for a cell is 2 V at 300 K, the equilibrium constant (K) for the reaction

[JEE MAIN 2019 (Jan.)]

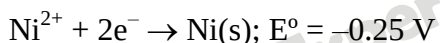
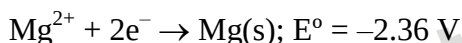
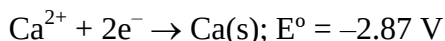
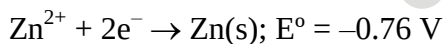


($R = 8 \text{ JK}^{-1} \text{ mol}^{-1}$, $F = 96000 \text{ C mol}^{-1}$)

- (1) e^{-80} (2) e^{-160} (3) e^{160} (4) e^{320}

44. Consider the following reduction processes :

[JEE MAIN 2019 (Jan.)]



The reducing power of the metals increases in the order :

- (1) $\text{Ca} < \text{Mg} < \text{Zn} < \text{Ni}$ (2) $\text{Zn} < \text{Mg} < \text{Ni} < \text{Ca}$
 (3) $\text{Ni} < \text{Zn} < \text{Mg} < \text{Ca}$ (4) $\text{Ca} < \text{Zn} < \text{Mg} < \text{Ni}$

45. In the cell $\text{Pt(s)}|\text{H}_2(\text{g}, 1\text{bar})|\text{HCl(aq)}|\text{AgCl(s)}|\text{Ag(s)}|\text{Pt(s)}$ the cell potential is 0.92 V when a 10^{-6} molal HCl solution is used. The standard electrode potential of $(\text{AgCl}/\text{Ag}, \text{Cl}^-)$ electrode is :

$$\left\{ \text{Given, } \frac{2.303RT}{F} = 0.06 \text{ V at } 298\text{K} \right\}$$

[JEE MAIN 2019 (Jan.)]

- (1) 0.94 V (2) 0.40 V (3) 0.20 V (4) 0.76 V

46. For the cell $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq}) || \text{M}^{x+}(\text{aq}) | \text{M(s)}$, different half cells and their standard electrode potentials are given below :

[JEE Main 2019 (Jan.)]

$\text{M}^{x+}(\text{aq})/\text{M(s)}$	$\text{Au}^{3+}(\text{aq})/\text{Au(s)}$	$\text{Ag}^+(\text{aq})/\text{Ag(s)}$	$\text{Fe}^{3+}(\text{aq})/\text{Fe}^{2+}(\text{aq})$	$\text{Fe}^{2+}(\text{aq})/\text{Fe(s)}$
$E_{\text{M}^{x+}/\text{M}}^{\circ}$	1.40	0.80	0.77	-0.44

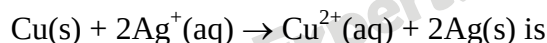
If $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$, which cathode will give a maximum value of E_{cell}° per electron transferred?

- (1) Fe^{2+}/Fe (2) Ag^{3+}/Au (3) $\text{Fe}^{3+}/\text{Fe}^{2+}$ (4) Ag^+/Ag

47. Given the equilibrium constant :

[JEE MAIN 2019 (Jan.)]

K_C of the reaction :



10×10^{15} , calculate the E_{cell}° of this reaction at 298 K

$$\left[2.303 \frac{RT}{F} \text{ at } 298\text{K is } 0.059\text{V} \right]$$

- (1) 0.4736 V (2) 0.04736 mV (3) 0.4736 mV (4) 0.04736 V

48. The standard electrode potential E° and its temperature coefficient $\left(\frac{dE^{\circ}}{dT} \right)$ for a cell are 2V and $-5 \times 10^{-4} \text{ VK}^{-1}$ at 300 K respectively. The cell reaction is $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ The standard reaction enthalpy $(\Delta_r H^{\circ})$ at 300 K in kJ mol^{-1} is,

[Use $R = 8 \text{ JK}^{-1} \text{ mol}^{-1}$ and $F = 96,000 \text{ C mol}^{-1}$]

[JEE MAIN 2019 (Jan.)]

- (1) 192.0 (2) -384.0 (3) 206.4 (4) -412.8

49. Λ_m^0 for NaCl, HCl and NaA are 126.4, 425.9 and 100.5 S cm²mol⁻¹, respectively. If the conductivity of 0.001 M HA is 5×10^{-5} S cm⁻¹, degree of dissociation of HA is:

[JEE MAIN 2019 (Jan.)]

- (1) 0.50 (2) 0.125 (3) 0.25 (4) 0.75

50. Given that the standard potentials (E°) of Cu^{2+}/Cu and Cu^+/Cu are 0.34 V and 0.522 V respectively, the E° of $\text{Cu}^{2+}/\text{Cu}^+$ is :

[Jee Main (Jan. 2020)]

- (1) -0.182 V (2) -0.158 V (3) 0.182 V (4) +0.158 V

51. The equation that is incorrect is :

(1) $(\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaCl}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{KCl}}$ [Jee Main (Jan. 2020)]

(2) $(\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaI}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}}$

(3) $(\Lambda_m^0)_{\text{H}_2\text{O}} = (\Lambda_m^0)_{\text{HCl}} + (\Lambda_m^0)_{\text{NaOH}} - (\Lambda_m^0)_{\text{NaCl}}$

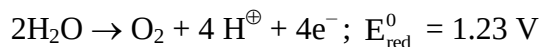
(4) $(\Lambda_m^0)_{\text{KCl}} - (\Lambda_m^0)_{\text{NaCl}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}}$

52. Amongst the following, the form of water with the lowest ionic conductance at 298 K is:

- (1) distilled water (2) saline water used for intravenous injection
(3) sea water (4) water from a well [Jee Main (Jan. 2020)]

53. What would be the electrode potential for the given half cell reaction at pH = 5 ?

[Jee Main (Jan. 2020)]



($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$; Temp = 298 K; oxygen under std. atm. Pressure of 1 bar)

54. For an electrochemical cell



the ratio $\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$ when this cell attains equilibrium is _____. [Jee Main (Jan. 2020)]

$$\left(\text{Given: } E_{\text{Sn}^{2+}|\text{Sn}}^0 = -0.14\text{V}, E_{\text{Pb}^{2+}|\text{Pb}}^0 = -0.13\text{V}, \frac{2.303RT}{F} = 0.06 \right)$$

55. For the given cell ; [Jee Main (Jan. 2020)]

$\text{Cu(s)}|\text{Cu}^{2+}(\text{C}_1\text{M})||\text{Cu}^{2+}(\text{C}_2\text{M})|\text{Cu(s)}$ change in Gibbs energy (ΔG) is negative, if :

- (1) $\text{C}_1 = 2\text{C}_2$ (2) $\text{C}_2 = \frac{\text{C}_1}{\sqrt{2}}$ (3) $\text{C}_1 = \text{C}_2$ (4) $\text{C}_2 = \sqrt{2}\text{C}_1$

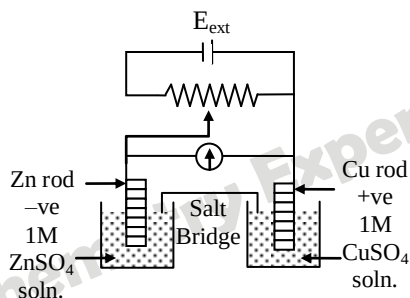
56. 108g of silver (molar mass 108g mol^{-1}) is deposited at cathode from $\text{AgNO}_3(\text{aq})$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is _____. [Jee Main (Jan. 2020)]

57. Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of NaCl and BaSO_4 , respectively, at a temperature T.

Which of the following is false ? [Jee-Main (Sep) 2020]

- (1) $C_{\text{BaSO}_4}(\text{T}_2) > C_{\text{BaSO}_4}(\text{T}_1)$ for $\text{T}_2 > \text{T}_1$
 (2) $C_{\text{NaCl}} \gg C_{\text{BaSO}_4}$ at a given T
 (3) $C_{\text{NaCl}}(\text{T}_2) > C_{\text{NaCl}}(\text{T}_1)$ for $\text{T}_2 > \text{T}_1$
 (4) Ionic mobilities of ions from both salts increase with T

58.



$$E^\circ_{\text{Cu}^{2+}|\text{Cu}} = +0.34\text{V}$$

$$E^\circ_{\text{Zn}^{2+}|\text{Zn}} = -0.76\text{V}$$

[Jee-Main (Sep) 2020]

Identify the incorrect statement from the options below for the above cell :

- (1) If $E_{\text{ext}} > 1.1\text{ V}$, e^- flows from Cu to Zn
 (2) If $E_{\text{ext}} > 1.1\text{ V}$, Zn dissolves at Zn electrode and Cu deposits at Cu electrode
 (3) If $E_{\text{ext}} < 1.1\text{ V}$, Zn dissolves at anode and Cu deposits at cathode
 (4) If $E_{\text{ext}} = 1.1\text{ V}$, no flow of e^- or current occurs

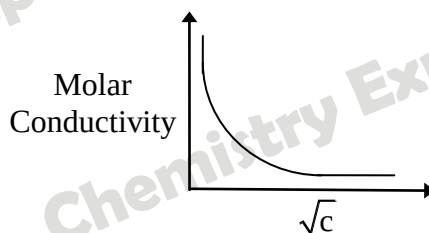
59. 250 mL of a waste solution obtained from the workshop of a goldsmith contains 0.1 M AgNO_3 and 0.1 M AuCl . The solution was electrolyzed at 2V by passing a current of 1 A for 15 minutes. The metal/metals electrodeposited will be :

$$(E_{\text{Ag}^+/\text{Ag}}^0 = 0.80\text{V}, E_{\text{Au}^+/\text{Au}}^0 = 1.69\text{V})$$

[Jee-Main (Sep) 2020]

- (1) Silver and gold in proportion to their atomic weights
- (2) Silver and gold in equal mass proportion
- (3) Only silver
- (4) Only gold

60. The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.

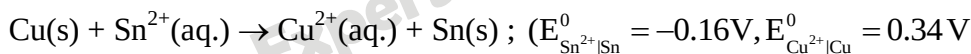


The electrolyte X is:

[Jee-Main (Sep) 2020]

- (1) NaCl (2) KNO_3 (3) CH_3COOH (4) HCl

61. The Gibbs energy change (in J) for the given reaction at $[\text{Cu}^{2+}] = [\text{Sn}^{2+}] = 1\text{ M}$ and 298 K is :

Take $F = 96500\text{ C mol}^{-1}$

[Jee-Main (Sep) 2020]

62. For the disproportionation reaction $2\text{Cu}^+(\text{aq}) \rightleftharpoons \text{Cu(s)} + \text{Cu}^{2+}(\text{aq})$ at 298 K,

In K (where k is the equilibrium constant) is $\text{_____} \times 10^{-1}$.

$$\text{Given : } (E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.16\text{V}, E_{\text{Cu}^+/\text{Cu}}^0 = 0.52\text{V}, \frac{RT}{F} = 0.025)$$

[Jee-Main (Sep) 2020]

63. The photoelectric current from Na (work function, $w_0 = 2.3\text{ eV}$) is stopped by the output voltage of the cell $\text{Pt(s)}|\text{H}_2(\text{g}, 1\text{ bar})|\text{HCl}(\text{aq.}, \text{pH} = 1)|\text{AgCl(s)}|\text{Ag(s)}$.

The pH of aq. HCl required to stop the photoelectric current from K ($w_0 = 2.25\text{ eV}$), all other conditions remaining the same, is $\text{_____} \times 10^{-2}$ (to the nearest integer).

Given,

$$2.303 \frac{RT}{F} = 0.06\text{ V}; E_{\text{AgCl}|\text{Ag}|\text{Cl}^-}^0 = 0.22\text{ V}$$

[Jee-Main (Sep) 2020]

64. An acidic solution of dichromate is electrolyzed for 8 minutes using 2A current. As per the following equation



[Jee-MAIN (Sep) 2020]

The amount of Cr^{3+} obtained was 0.104 g. The efficiency of the process (in %) is

(Take : $F = 96000 \text{ C}$, At. mass of chromium = 52) _____.

Given 60.31

65. An oxidation-reduction reaction in which 3 electrons are transferred has a ΔG° of $17.37 \text{ kJ mol}^{-1}$ at 25°C . The value of E_{Cell}° (in V) is _____ $\times 10^{-2}$.

(1 $F = 96,500 \text{ C mol}^{-1}$)

[Jee-Main (Sep) 2020]

66. Potassium chlorate is prepared by the electrolysis of KCl in basic solution



If only 60% of the current is utilized in the reaction, the time (rounded to the nearest hour) required to produce 10 g of KClO_3 using a current of 2 A is

(Given : $F = 96,500 \text{ C mol}^{-1}$; molar mass of $\text{KClO}_3 = 122 \text{ g mol}^{-1}$) [Jee-Main (Sep) 2020]

EXERCISE # (JEE-ADVANCED)

- Calculate the equilibrium constant for the reaction, $2\text{Fe}^{3+} + 3\text{I}^- \rightleftharpoons 2\text{Fe}^{2+} + \text{I}_3^-$. The standard reduction potentials in acidic conditions are 0.77 and 0.54 V respectively for $\text{Fe}^{3+} / \text{Fe}^{2+}$ and $\text{I}_3^- / \text{I}^-$ couples. **[JEE 1998]**
- Find the solubility product of a saturated solution of Ag_2CrO_4 in water at 298 K if the emf of the cell $\text{Ag}|\text{Ag}^+ (\text{satd. Ag}_2\text{CrO}_4 \text{ soln.}) || \text{Ag} + (0.1 \text{ M}) | \text{Ag}$ is 0.164 V at 298K. **[JEE 1998]**
- A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y and 1 M Z at 25°C. If the reduction potential of $Z > Y > X$, then **[JEE 1999]**
 - Y will oxidise X and not Z
 - Y will oxidise Z and X
 - Y will oxidise both X and Z
 - Y will reduce both X and Z.
- For the electrochemical cell, $\text{M} | \text{M}^+ || \text{X}^- | \text{X}$, $E^\circ (\text{M}^+/\text{M}) = 0.44 \text{ V}$ and $E^\circ (\text{X}/\text{X}^-) = 0.33 \text{ V}$. From this data, one can deduce that **[JEE 2000]**
 - $\text{M} + \text{X} \longrightarrow \text{M}^+ + \text{X}^-$ is the spontaneous reaction
 - $\text{M}^+ + \text{X}^- \longrightarrow \text{M} + \text{X}$ is the spontaneous reaction
 - $E_{\text{cell}} = 0.77 \text{ V}$
 - $E_{\text{cell}} = -0.77 \text{ V}$
- Copper sulphate solution (250 mL) was electrolysed using a platinum anode and a copper cathode. A constant current of 2 mA was passed for 16 min. It was found that after electrolysis, the absorbance (concentration) of the solution was reduced to 50% of its original value. Calculate the concentration of copper sulphate in the solution to begin with. **[JEE 2000]**
- The following electrochemical cell has been set up
 $\text{Pt}_{(\text{I})} | \text{Fe}^{3+}, \text{Fe}^{2+} (a=1) || \text{Ce}^{4+}, \text{Ce}^{3+} (a=1) | \text{Pt}_{(\text{II})}$
 $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 0.77 \text{ V}$ and $E_{\text{Ce}^{4+}/\text{Ce}^{3+}}^0 = 1.61 \text{ V}$
 If an ammeter is connected between the two platinum electrodes, predict the direction of flow of current. Will the current increase or decrease with time? **[JEE 2000]**
- The reaction, **[JEE 2001]**
 $3\text{ClO}^-(\text{aq}) \longrightarrow \text{ClO}_3^-(\text{aq}) + 2\text{Cl}^-(\text{aq})$
 is an example of
 - Oxidation reaction
 - Reduction reaction
 - Disproportionation reaction
 - Decomposition reaction

8. The correct order of equivalent conductance at infinite dilution of LiCl, NaCl and KCl is
 (A) $\text{LiCl} > \text{NaCl} > \text{KCl}$ (B) $\text{KCl} > \text{NaCl} > \text{LiCl}$ [JEE 2001]
 (C) $\text{NaCl} > \text{KCl} > \text{LiCl}$ (D) $\text{LiCl} > \text{KCl} > \text{NaCl}$
9. Saturated solution of KNO_3 is used to make salt bridge because
 (A) velocity of K^+ is greater than that of NO_3^-
 (B) velocity of NO_3^- is greater than that of K^+
 (C) velocities of both K^+ and NO_3^- are nearly the same
 (D) KNO_3 is highly soluble in water [JEE 2001]
10. The standard potential of the following cell is 0.23 V at 15°C & 0.21 V at 35°C
 $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl}(\text{aq}) | \text{AgCl}(\text{s}) | \text{Ag}(\text{s})$
 (i) Write the cell reaction.
 (ii) Calculate ΔH° , ΔS° for the cell reaction by assuming that these quantities remain unchanged in the range 15°C to 35°C .
 (iii) Calculate the solubility of AgCl in water at 25°C . Given standard reduction potential of the Ag^+/Ag couple is 0.80 V at 25°C . [JEE 2001]
11. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half cell reactions and their standard potentials are given below:
 $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \longrightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l}); E^\circ = 1.51 \text{ V}$ [JEE 2002]
 $\text{Cr}_2\text{O}_7^{2-} (\text{aq}) + 14\text{H}^+ (\text{aq}) + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} (\text{aq}) + 7\text{H}_2\text{O} (\text{l}); E^\circ = 1.38 \text{ V}$
 $\text{Fe}^{3+} (\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+} (\text{aq}); E^\circ = 0.77 \text{ V}$
 $\text{Cl}_2 (\text{g}) + 2\text{e}^- \longrightarrow 2\text{Cl}^- (\text{aq}); E^\circ = 1.40 \text{ V}$
 Identify the only incorrect statement regarding quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_3$
 (A) MnO_4^- can be used in aqueous HCl (B) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous HCl
 (C) MnO_4^- can be used in aqueous H_2SO_4 (D) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous H_2SO_4
12. In the electrolytic cell, flow of electrons is from:
 (A) Cathode to anode in solution
 (B) Cathode to anode through external supply
 (C) Cathode to anode through internal supply
 (D) Anode to cathode through internal supply.

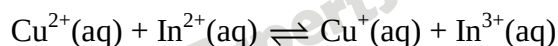
13. Two students use same stock solution of ZnSO_4 and a solution of CuSO_4 . The e.m.f of one cell is 0.03 V higher than the other. The conc. of CuSO_4 in the cell with higher e.m.f value is 0.5 M.

Find out the conc. of CuSO_4 in the other cell $\left(\frac{2.303 RT}{F} = 0.06 \right)$. [JEE 2003]

14. $\text{Zn} | \text{Zn}^{2+} (a = 0.1\text{M}) || \text{Fe}^{2+} (a = 0.01\text{M}) | \text{Fe}$. The emf of the above cell is 0.2905 V. Equilibrium constant for the cell reaction is [JEE 2004]

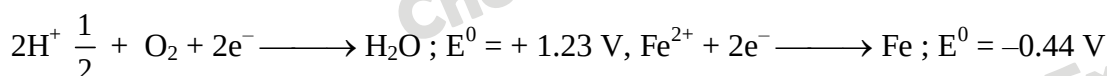
(A) $10^{0.32/0.0591}$ (B) $10^{0.32/0.0295}$ (C) $10^{0.26/0.0295}$ (D) $e^{0.32/0.295}$

15. Find the equilibrium constant at 298 K for the reaction,



Given that $E_{\text{Cu}^{2+}|\text{Cu}^{+}}^0 = 0.15\text{V}$, $E_{\text{In}^{3+}|\text{In}^{2+}}^0 = -0.42\text{V}$, $E_{\text{In}^{2+}|\text{In}^{+}}^0 = -0.40\text{V}$ [JEE 2004]

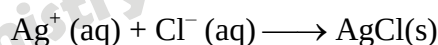
16. The half cell reactions for rusting of iron are: [JEE 2005]



ΔG^0 (in kJ) for the reaction is :

(A) -76 (B) -322 (C) -122 (D) -176

17. (a) Calculate ΔG_f^0 of the following reaction



Given : $\Delta G_f^0 (\text{AgCl}) = -109\text{ kJ/mole}$, $\Delta G_f^0 (\text{Cl}^{-}) = -129\text{ kJ/mole}$, $\Delta G_f^0 (\text{Ag}^{+}) = 77\text{ kJ/mole}$

Represent the above reaction in form of a cell

Calculate E^0 of the cell. Find $\log_{10} K_{\text{SP}}$ of AgCl

(B) (b) 6.539×10^{-2} g of metallic Zn (amu = 65.39) was added to 100 ml of saturated solution

of AgCl. Calculate $\frac{[\text{Zn}^{2+}]}{[\text{Ag}^{+}]^2} \log_{10}$, given that

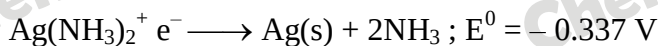


Also find how many moles of Ag will be formed ?

[JEE 2005]

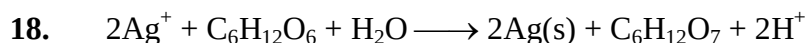
Paragraph for Question No. 18 to 20

Tollen's reagent is used for the detection of aldehyde when a solution of AgNO_3 is added to glucose with NH_4OH then gluconic acid is formed



$$\left[\text{Use } 2.303 \times \frac{RT}{F} = 0.0592 \text{ and } \frac{F}{RT} = 38.92 \text{ at } 298 \text{ K} \right]$$

[JEE 2006]



Find $\ln K$ of this reaction

- (A) 66.13 (B) 58.38 (C) 28.30 (D) 46.29

19. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

(A) E_{oxd} will increase by a factor of 0.65 from E_{oxd}^0

(B) E_{oxd} will decrease by a factor of 0.65 from E_{oxd}^0

(C) E_{red} will increase by a factor of 0.65 from E_{red}^0

(D) E_{red} will decrease by a factor of 0.65 from E_{red}^0

20. Ammonia is always added in this reaction. Which of the following must be incorrect?

(A) NH_3 combines with Ag^+ to form a complex.

(B) $\text{Ag}(\text{NH}_3)_2^+$ is a weaker oxidising reagent than Ag^+ .

(C) In absence of NH_3 silver salt of gluconic acid is formed.

(D) NH_3 has affected the standard reduction potential of glucose/gluconic acid electrode.

21. We have taken a saturated solution of AgBr . K_{sp} of AgBr is 12×10^{-14} . If 10^{-7} mole of AgNO_3 are added to 1 litre of this solution find conductivity (specific conductance) of this solution in terms of $10^{-7} \text{ S m}^{-1} \text{ mol}^{-1}$. [JEE 2006]

Given : $\lambda_{(\text{Ag}^+)}^0 = 6 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$; $\lambda_{(\text{Br}^-)}^0 = 8 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$; $\lambda_{(\text{NO}_3^-)}^0 = 7 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$

Paragraph for Question Nos. 22 & 23

Redox reactions play a pivoted role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° (V with respect to normal hydrogen electrode) values. Using this data obtain the correct explanations to Questions 14-16.

**[JEE 2007]**

22. Among the following, identify the correct statement.

- (A) Chloride ion is oxidised by O_2 (B) Fe^{2+} is oxidised by iodine
 (C) Iodine ion is oxidised by chlorine (D) Mn^{2+} is oxidised by chlorine

23. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because

- (A) O_2 oxidises Mn^{2+} to Mn^{3+}
 (B) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
 (C) Fe^{3+} oxidises H_2O to O_2
 (D) Mn^{3+} oxidises H_2O to O_2

Paragraph for Question Nos. 24 to 26

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/ electrochemical reaction, which requires a clear understanding of the mole concept. A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs) **[JEE 2007]**

24. The total number of moles of chlorine gas evolved is
 (A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0
25. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is
 (A) 200 (B) 225 (C) 400 (D) 446
26. The total charge (coulombs) required for complete electrolysis is
 (A) 24125 (B) 48250 (C) 96500 (D) 193000
27. Electrolysis of dilute aqueous NaCl solution was carried out by passing 10 milli ampere current. The time required to liberate 0.01 mol of H_2 gas at the cathode is (1 Faraday = 96500 C mol^{-1})
 (A) 9.65×10^4 sec (B) 19.3×10^4 sec (C) 28.95×10^4 sec (D) 38.6×10^4 sec [JEE 2008]
28. For the reaction of NO_3^- ion in an aqueous solution, E° is +0.96 V. Values of E° for some metal ions are given below
- | | |
|-------------------------------------|---------------------|
| $V^{2+}(aq) + 2e^- \rightarrow V$ | $E^\circ = -1.19$ V |
| $Fe^{3+}(aq) + 3e^- \rightarrow Fe$ | $E^\circ = -0.04$ V |
| $Au^{3+}(aq) + 3e^- \rightarrow Au$ | $E^\circ = +1.40$ V |
| $Hg^{2+}(aq) + 2e^- \rightarrow Hg$ | $E^\circ = +0.86$ V |
- [JEE 2009]
- The pair(s) of metal that is(are) oxidised by NO_3^- in aqueous solution is(are)
 (A) V and Hg (B) Hg and Fe (C) Fe and Au (D) Fe and V

Paragraph for Questions 29 to 30

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is :



For the above electrolytic cell the magnitude of the cell potential $|E_{\text{cell}}| = 70$ mV.

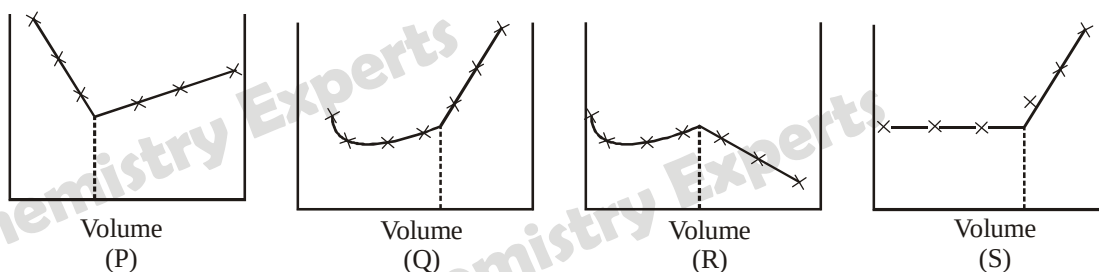
29. For the above cell :-

- (A) $E_{\text{cell}} < 0$; $\Delta G > 0$ (B) $E_{\text{cell}} > 0$; $\Delta G < 0$ (C) $E_{\text{cell}} < 0$; $\Delta G > 0$ (D) $E_{\text{cell}} > 0$; $\Delta G < 0$

30. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be :-

- (A) 35 mV (B) 70 mV (C) 140 mV (D) 700 mV

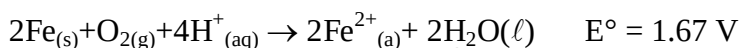
31. AgNO_3 (a) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. the plot of conductance (Λ) versus the volume of AgNO_3 is



- (A) (P) (B) (Q) (C) (R) (D) (S)

32. Consider the following cell reaction :

[JEE 2011]



At $[\text{Fe}^{2+}] = 10^{-3} \text{ M}$, $P(\text{O}_2) = 0.1 \text{ atm}$ and $\text{pH} = 3$, the cell potential at 25°C is -

- (A) 1.47 V (B) 1.77 V (C) 1.87 V (D) 1.57 V

Paragraph for Question 33 and 34

The electrochemical cell shown below is a concentration cell.

$\text{M} | \text{M}^{2+}$ (saturated solution of a sparingly soluble salt, MX_2) $|| \text{M}^{2+}$ ($0.001 \text{ mol dm}^{-3}$) $| \text{M}$ The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes.

The emf of the cell at 298 K is 0.059V.

[JEE 2012]

33. The value of ΔG (kJ mol^{-1}) for the given cell is (take $F = 96500 \text{ C mol}^{-1}$)

- (A) -5.7 (B) 5.7 (C) 11.4 (D) -11.4

34. The solubility product (K_{sp} ; $\text{mol}^3 \text{ dm}^{-9}$) of MX_2 at 298 K based on the information available for the given concentration cell is (take $2.303 \times R \times 298/F = 0.059 \text{ V}$)

- (A) 1×10^{-15} (B) 4×10^{-15} (C) 1×10^{-12} (D) 1×10^{-12}

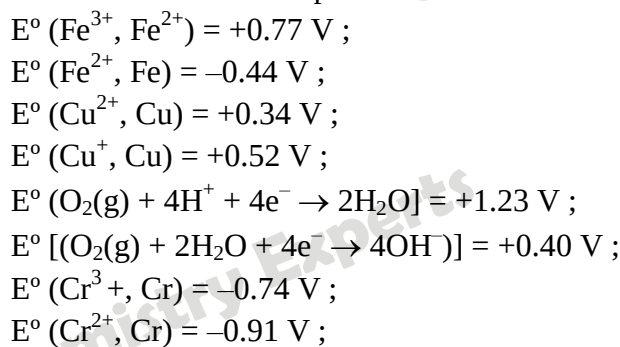
35. An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : [JEE-Adv. 2013]

List-I		List-II	
(P)	$(\text{C}_2\text{H}_5)_3\text{N} + \text{CH}_3\text{COOH}$ X Y	(1)	Conductivity decreases and then increases
(Q)	$\text{KI}(0.1\text{M}) + \text{AgNO}_3(0.01\text{M})$ X Y	(2)	Conductivity decreases and then does not change much
(R)	$\text{CH}_3\text{COOH} + \text{KOH}$ X Y	(3)	Conductivity increases and then does not change much
(S)	$\text{NaOH} + \text{HI}$ X Y	(4)	Conductivity does not change much and then increases

Codes :

	P	Q	R	S		P	Q	R	S
(A)	3	4	2	1	(B)	4	3	2	1
(C)	2	3	4	1	(D)	1	4	3	2

36. The standard reduction potential data at 25°C is given below [JEE-Adv. 2013]



Match E° of the redox pair in List-I with the values given in List-II and select the correct answer using the code given below the lists :

List-I		List-II	
(P)	$E^\circ(\text{Fe}^{3+}, \text{Fe})$	(1)	-0.18 V
(Q)	$E^\circ(4\text{H}_2\text{O} \rightleftharpoons 4\text{H}^+ + 4\text{OH}^-)$	(2)	-0.4 V
(R)	$E^\circ(\text{Cu}^{2+} + \text{Cu} \rightarrow 2\text{Cu}^+)$	(3)	-0.04 V
(S)	$E^\circ(\text{Cr}^{3+}, \text{Cr}^{2+})$	(4)	-0.83 V

Codes :

	P	Q	R	S		P	Q	R	S
(A)	4	1	2	3	(B)	2	3	4	1
(C)	1	2	3	4	(D)	3	4	1	2

37. In a galvanic cell, the salt bridge – [JEE-Adv. 2014]
- (A) Does not participate chemically in the cell reaction
 (B) Stops the diffusion of ions from one electrode to another
 (C) Is necessary for the occurrence of the cell reaction
 (D) Ensures mixing of the two electrolytic solutions
38. All the energy released from the reaction $X \rightarrow Y$, $\Delta_r G^\circ = -193 \text{ kJ mol}^{-1}$ is used for the oxidizing M^+ and $M^+ \rightarrow M^{3+} + 2e^-$, $E^\circ = -0.25 \text{ V}$.
 Under standard conditions, the number of moles of M^+ oxidized when one mole of X is converted to Y is $[F = 96500 \text{ C mol}^{-1}]$ [JEE-Adv. 2015]
39. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda_{X^-}^0 \approx \lambda_{Y^-}^0$, the difference in their pK_a values, $pK_a(\text{HX}) - pK_a(\text{HY})$, is :
 (consider degree of ionization of both acids to be $\ll 1$). [JEE-Adv. 2015]
40. For the following electrochemical cell at 298K, [JEE-Adv. 2016]
 $\text{Pt(s)} \mid \text{H}_2(\text{g}, 1\text{bar}) \mid \text{H}^+(\text{aq}, 1\text{M}) \parallel \text{M}^{4+}(\text{aq}), \text{M}^{2+}(\text{aq}) \mid \text{Pt(s)}$
 $E_{\text{cell}} = 0.092 \text{ V}$ when $\frac{[\text{M}^{2+}(\text{aq})]}{[\text{M}^{4+}(\text{aq})]} = 10^x$
 Given : $E_{\text{M}^{4+}/\text{M}^{2+}}^\circ = 0.151 \text{ V}$; $2.303 \frac{RT}{F} = 0.059 \text{ V}$
 The value of x is -
 (A) -2 (B) -1 (C) 1 (D) 2
41. The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1cm^2 . The conductance of this solution was found to be $5 \times 10^{-7} \text{ S}$. The pH of the solution is 4. The value of limiting molar conductivity (Λ_m^0) of this weak monobasic acid in aqueous solution is $Z \times 10^2 \text{ S cm}^{-1}\text{mol}^{-1}$.
 The value of Z is [JEE Adv. 2017]

42. For the following cell,

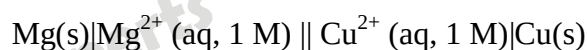


when the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+} , then expression for ΔG (in J mol^{-1}) is [F is faraday constant ; R is gas constant; T is temperature; $E^\circ(\text{cell}) = 1.1\text{V}$]

[JEE - 2017]

- (A) $2.303RT - 2.2 F$ (B) $- 2.2 F$
(C) $1.1F$ (D) $2.303RT + 1.1 F$

43. For the electrochemical cell,



the standard emf of the cell is 2.70 V at 300 K. When the concentration of Mg^{2+} is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is ____.

(given, $\frac{F}{R} = 11500 \text{ KV}^{-1}$), where F is the Faraday constant and R is the gas constant,

$\ln(10) = 2.30$

[JEE - 2018]

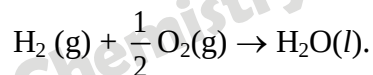
44. Consider an electrochemical cell: $\text{A(s)} \mid \text{A}^{n+}(\text{aq}, 2 \text{ M}) \parallel \text{B}^{2n+}(\text{aq}, 1 \text{ M}) \mid \text{B(s)}$. The value of ΔH° for the cell reaction is twice that of ΔG° at 300 K. If the emf of the cell is zero, the ΔS° (in $\text{J K}^{-1} \text{ mol}^{-1}$) of the cell reaction per mole of B formed at 300 K is ____.

(Given: $\ln(2) = 0.7$, $R(\text{universal gas constant}) = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$. H, S and G are enthalpy, entropy and Gibbs energy, respectively.)R

[JEE - 2018]

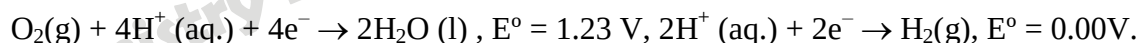
45. Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is

[JEE-2020]



The work derived from the cell on the consumption of $1.0 \times 10^{-3} \text{ mol}$ of $\text{H}_2(\text{g})$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulated container. What is the change in the temperature (in K) of the ideal gas ?

The standard reduction potentials for the two half-cells are given below.



Use $F = 96500 \text{ C mol}^{-1}$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$

ANSWER KEY**EXERCISE # (JEE-MAIN)**

1. 2 2. 3 3. 1 4. 4 5. 1 6. 1 7. 3
 8. 1 9. 3 10. 3 11. 3 12. 2 13. 1 14. 4
 15. 1 16. 3 17. 4 18. 1 19. 1 20. 4 21. 1
 22. 2 23. 1 24. 3 25. 1 26. 4 27. 4 28. 2
 29. 3 30. 1 31. 2 32. 1 33. 4 34. 2 35. 2
 36. 1 37. 4 38. 3 39. 3 40. 3 41. 3 42. 3
 43. 3 44. 3 45. 3 46. 2 47. 1 48. 4 49. 2
 50. 4 51. 2 52. 1 53. $[-0.93]$ NTA Ans. (1.52 to 1.53)
 54. $[2.15]$ NTA Ans. (2.13 to 2.16) 55. 4
 56. $[5.67 \text{ or } 5.68]$ NTA Ans. (5.66 to 5.67) 57. (None) NTA Ans. (3)
 58. 2 59. 1 60. 3 61. 96500 62. 144.00 63. 142 NTA Ans. (58)
 64. 60 65. -6 66. 11.00

EXERCISE # (JEE-ADVANCED)

1. $K_C = 6.26 \times 10^7$ 2. $K_{sp} = 2.287 \times 10^{-12}$ 3. A 4. B
 5. $7.95 \times 10^{-5} \text{ M}$ 6. decrease with time 7. C 8. B 9. C
 10. $\Delta H^0 = -49987 \text{ J mol}^{-1}$, $\Delta S^0 = -96.5 \text{ J mol}^{-1} \text{ K}^{-1}$, $s = 1.47 \times 10^{-5} \text{ M}$
 11. A 12. C 13. 0.05 14. B 15. $K_C = 10^{10}$ 16. B
 17. (a) $E^0 = 0.59 \text{ V}$, $\log_{10} K_{SP} = -10$ (b) 52.8, 10^{-6} moles 18. A 19. A
 20. D 21. 55 S m^{-1} 22. C 23. D 24. B 25. D 26. D
 27. B 28. ABD 29. B 30. C 31. D 32. D 33. D
 34. B 35. A 36. D 37. AB 38. 4 39. 3 40. D
 41. 6 42. A 43. 10 44. -11.62 45. 13.32

HINT AND SOLUTIONS

EXERCISE # (JEE-MAIN)

1. (2)

SRP $\uparrow$, Oxidising power $\uparrow$
 Reducing power $\downarrow$

2. (3)

SRP $\uparrow$, Oxidising power $\uparrow$
 Reducing power $\downarrow$

3. (1)

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \left[\frac{Zn^{2+}}{Cu^{2+}} \right]$$

$$E_{\text{cell}} = 1.10 - \frac{0.0591}{2} \log \frac{1}{0.1} = 1.07V$$

4. (4)

$$\text{number of equivalent of electrons} = \frac{9650}{96500} = 0.1 \text{ mole}$$

$$\text{number of equivalent of Ag} = 0.1 \text{ mole}$$

5. (1)

$$E^{\circ} = E^{\circ}_{\text{Sn} / \text{Sn}^{2+}} + E^{\circ}_{\text{Fe}^{3+} / \text{Fe}^{2+}} = 0.14 + 0.77 = 0.91 \text{ V}$$

6. (1)

$$E^{\circ} = \frac{0.06}{n} 10 \text{ g Keq}$$

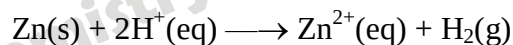
$$0.591 = \frac{0.0591}{1} 10 \text{ g Keq}$$

$$\text{Keq} = 10^{10}$$

7. (3)

$$\lambda^{\circ} \text{m NaBr} = \lambda^{\circ} \text{m KBr} + \lambda^{\circ} \text{m NaCl} - \lambda^{\circ} \text{m KCl} = 152 + 126 - 150 = 12 \text{ Scm}^2 \text{ mol}^{-1}$$

8. (1)



On adding H_2SO_4 , equilibrium shift to right

$$E = E^{\circ} - \frac{0.0591}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{H}^+]^2}$$

$$[\text{H}^+] \uparrow, E \uparrow$$

9. (3)

$$\text{Cr}^{2+} | \text{Cr}^{3+} = +0.41\text{V}$$

$$\text{Mn}^{2+} | \text{Mn}^{3+} = -1.57\text{V}$$

$$\text{Fe}^{2+} | \text{Fe}^{3+} = -0.77\text{V}$$

$$\text{Co}^{2+} | \text{Co}^{3+} = -1.97\text{V}$$

As Cr will have maximum oxidation potential value, therefore its oxidation will be easiest.

10. (3)

11. (3)

$$\frac{5.12 \times 1000}{27} \times 3 = \frac{Q}{96500}$$

$$Q = 54897 \times 10^3 \text{ C} = 5.4897 \times 10^7 \text{ C}$$

12. (2)

Difluoroacetic acid will be strongest acid due to electron withdrawing effect of two fluorine atoms so as it will show maximum electrical conductivity.

13. (1)

$$\begin{aligned} \lambda^{\circ} \text{HOAC} &= \lambda^{\circ} \text{NaOAC} + \lambda^{\circ} \text{HCl} - \lambda^{\circ} \text{NaCl} \\ &= 91 + 426.2 - 126.5 = 390.7 \end{aligned}$$

14. (4)

$$0.152 = -0.8 - \frac{0.059}{1} \log K_{\text{sp}} \quad \Rightarrow \log K_{\text{sp}} = -16.11$$

15. (1)

$$C = 0.1 \text{ M}, \quad R = 100 \, \Omega$$

$$K = 1.29 \text{ Sm}^{-1} = \frac{1}{100} \times \frac{\ell}{A}$$

$$C = 0.02 \text{ M}, \quad R = 520 \, \Omega.$$

$$K = \frac{1}{520} \times 129$$

$$\lambda_M = \frac{\frac{1}{520} \times 129}{1000 \times 0.02} = 124 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$$

16. (3)



From the reaction,

$$\Lambda_{\text{CH}_3\text{COONa}}^0 + \Lambda_{\text{HCl}}^0 = \Lambda_{\text{CH}_3\text{COOH}}^0 + \Lambda_{\text{NaCl}}^0 \quad \text{or} \quad \Lambda_{\text{CH}_3\text{COOH}}^0 = \Lambda_{\text{CH}_3\text{COONa}}^0 + \Lambda_{\text{HCl}}^0 - \Lambda_{\text{NaCl}}^0$$

Thus to calculate the value of $\Lambda_{\text{CH}_3\text{COOH}}^0$ one should know the value of Λ_{NaCl}^0 along with $\Lambda_{\text{CH}_3\text{COONa}}^0$ and Λ_{HCl}^0 .

17. (4)

$$0 = +1.1 - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

$$\log = 37.3 \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \Rightarrow \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = 10^{37.3}$$

18. (1)

$$E_{\text{cell}} = E_{\text{cell}}^0 - \log \frac{[\text{Cr}^{+3}]^2}{[\text{Fe}^{+2}]^3}$$

$$= 0.3 - \frac{0.059}{6} \log \frac{(0.1)^2}{(0.01)^3} = 0.3 - 0.04 = 0.26 \text{ V}$$

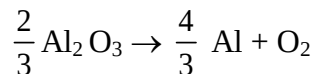
19. (1)



$$\Delta G_3 = 3 \times 0.036F - 2 \times 0.439 \times F = -1 \times E^0 (\text{Fe}^{3+}/\text{Fe}^{2+}) \times F$$

$$E^0 (\text{Fe}^{3+}/\text{Fe}^{2+}) = 2 \times 0.439 - 3 \times 0.036 = 0.878 - 0.108 = 0.770 \text{ V}$$

20. (4)



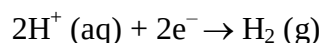
$$\Delta_r G = +966 \text{ kJ mol}^{-1} = 966 \times 10^3 \text{ J mol}^{-1}$$

$$\Delta G = -nFE_{\text{cell}}$$

$$966 \times 10^3 = -4 \times 96500 \times E_{\text{cell}}$$

$$E_{\text{cell}} = 2.5 \text{ V}$$

21. (1)



$$E_{\text{red}} = E_{\text{red}}^{\circ} - \frac{0.0591}{n} \log \frac{P_{\text{H}_2}}{(\text{H}^+)^2}; \quad E_{\text{red}} = 0 - \frac{0.0591}{2} \log \frac{2}{(1)^2}; \quad E_{\text{red}} = -\frac{0.0591}{2} \log 2$$

E_{red} is found to be negative for (3) option

22. (2)

$$\kappa = 1.3 \text{ S/m.}$$

$$R = 50 \text{ } \Omega$$

$$M = 0.2$$

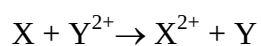
$$\kappa = \frac{1}{R} \times \frac{\ell}{A}$$

$$\Rightarrow \frac{\ell}{A} = 1.3 \times 50 \text{ m}^{-1}.$$

$$\kappa = \frac{1}{R} \times \frac{\ell}{A} = \frac{1}{260} \times 1.3 \times 50 = 0.25 \text{ Sm}^{-1}$$

$$\lambda = \frac{\kappa}{1000 \times M} = \frac{0.25}{1000 \times 0.4} = 6.25 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$$

23. (1)



For reaction to be spontaneous E° must be positive.

24. (3)

$$\kappa = 1.4 \text{ S/m.}$$

$$R = 50 \text{ } \Omega$$

$$M = 0.2$$

$$\kappa = \frac{1}{R} \times \frac{\ell}{A}$$

$$\Rightarrow \frac{\ell}{A} = 1.4 \times 50 \text{ m}^{-1}.$$

$$\kappa = \frac{1}{R} \times \frac{\ell}{A} = \frac{1}{280} \times 1.4 \times 50 = 0.25 \text{ Sm}^{-1}$$

$$\lambda = \frac{\kappa}{1000 \times M} = \frac{0.25}{1000 \times 0.5} = 5 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$$

25. (1)

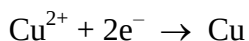
$$\lambda_c = \lambda_\infty - B\sqrt{C}$$

26. (4)

Higher the SRP, better is oxidising agent

Hence MnO_4^- is strongest oxidising agent

27. (4)

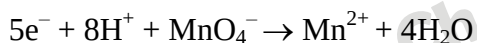


$$2F \quad 1 \text{ mole} = 63.5 \text{ g.}$$

28. (2)

If $E_{\text{ext}} > E^\circ$ Then electron flow from cathode to anode

29. (3)



$$E = 1.51 - \frac{0.06}{5} \log \frac{[\text{H}^+]^8 f}{[\text{H}^+]^8 i}$$

$$E = 1.51 - \frac{0.06}{5} \log \frac{(10^{-3})^8}{(10^{-7})^8}$$

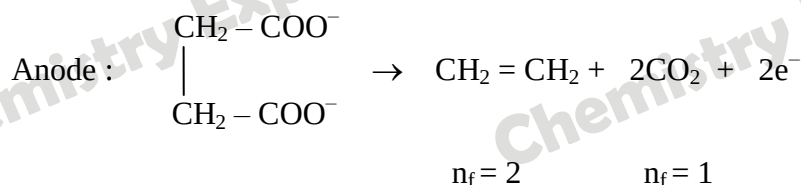
$$E = 1.51 - \frac{0.06}{5} \times 32 = 1.126 \text{ V}$$

So it will oxidize Br^- & I^-

30. (1)



$$n_f = 2$$



$$n_f = 2$$

$$n_f = 1$$

Total equivalent of gases ($CH_2 = CH_2$, CO_2 , H_2) = $0.2 + 0.2 + 0.2 = 0.6$

$$\text{Total moles of gases} = \frac{0.2}{2} + 0.2 + \frac{0.2}{2} = 0.4$$

$$\text{Total Vol. of gases} = 0.4 \times 22.4 = 8.96 \text{ L}$$

31. (2)

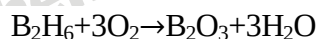


No reaction Cu lies below Zn in electro chemical series.

32. (1)

SRP $\uparrow$, Oxidising power $\uparrow$
 Reducing power $\downarrow$

33. (4)



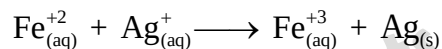
Moles of O_2 required = $3 \times$ moles of $B_2H_6 = 3$

$$\frac{I \times t}{96500} = \text{moles of } O_2 \times 4 = 12 \Rightarrow t = 3.2 \text{ hrs}$$

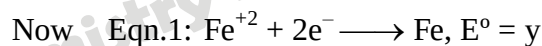
34. (2)

Since $S_2O_8^{2-}$ has highest value of reduction potential among given species therefore $S_2O_8^{2-}$ is strongest oxidizing agent

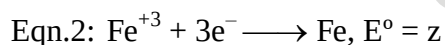
35. (2)



$$E^{\circ} = E_{\text{Ag}^{+}/\text{Ag}}^{\circ} - E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^{\circ}$$

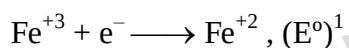


$$\Delta G^{\circ} = -2Fy$$



$$\Delta G^{\circ} = -3Fz$$

Eqn.2 – Eqn. 1, We get



$$\Delta G^{\circ} = -3Fz + 2Fy$$

$$\Rightarrow -1 \times F \times (E^{\circ})^1 = -3Fz + 2Fy$$

$$\Rightarrow (E^{\circ})^1 = E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^{\circ} = 3z - 2y$$

Now $E^{\circ} = E_{\text{Ag}^{+}/\text{Ag}}^{\circ} - E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^{\circ}$

$$= x - (3z - 2y)$$

$$= x - 3z + 2y$$

36. (1)

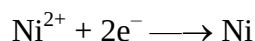
$$\Delta G^{\circ} = -nFE^{\circ}$$

$$= -2 \times 96000 \times 2$$

$$= -384000 \text{ J} = -384 \text{ kJ}$$

37. (4)

Reaction at cathode



0.1 mole

$$\text{Deposited moles of Ni} = \frac{0.1}{2} = 0.05$$

38. (3)

Conductivity increases with increasing concentration of electrolyte and molar conductivity decreases with increasing concentration of electrolyte.

39. (3)

$$\lambda_m = \lambda_m^\circ - A\sqrt{C}$$

value of A will be same for NaCl and KCl

So Slope for both curve will be same.

But λ_m° for KCl $>$ λ_m° for NaCl

So λ_m KCl $>$ λ_m NaCl

(for a given Concentration)

40. (3)

oxidising power $\propto$ S.R.P value.

$\Rightarrow$ correct order is : $\text{Bi}^{+3} < \text{Ce}^{+4} < \text{Pb}^{+4} < \text{CO}^{+3}$

$E_{\text{red}}^\circ \uparrow \Rightarrow$ oxidising power $\uparrow$

41. (3)

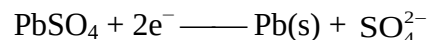
As, Acidic Strength $\propto K_a$

$\Rightarrow$ order of Acidic Strength or K_a is $A > C > B$

As, Acidic Strength ($\uparrow$) $\Rightarrow K_a(\uparrow) \Rightarrow$ electrical conductivity ($\uparrow$)

42. (3)

Anode



Equivalent of PbSO_4 electrolysed = 0.05

$$\text{Moles of PbSO}_4 = \frac{0.05}{2}$$

$$\text{Mass of PbSO}_4 = \frac{0.05}{2} \times 303 = 7.6 \text{ g}$$

43. (3)

$$E_{\text{cell}}^\circ = 2V$$

$$\text{As, } \Delta G^\circ = -nFE_{\text{cell}}^\circ \quad \dots (1)$$

$$\text{and also, } \Delta G^\circ = -RT \ln K_C \quad \dots (2)$$

Equating (1) and (2)

$$\cancel{n}FE_{\text{aq}}^\circ = \cancel{n}RT \ln K_C$$

$$2 \times 96000 = 8 \times 300 \ln K_C$$

$$\Rightarrow K_C = e^{160}$$

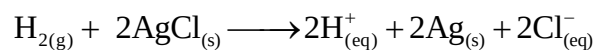
$\Rightarrow$ Correct option is 3

44. (3)

SRP $\uparrow$, Oxidising power $\uparrow$
 Reducing power $\downarrow$

45. (3)

Cell reaction :



$$Q_{\text{C}} = \frac{[\text{H}^{+}]^2 [\text{Cl}^{-}]^2}{P_{\text{H}_2}} = \frac{(10^{-6})^2 (10^{-6})^2}{1} = 10^{-24}$$

$$E = E^{\circ} - \frac{0.06}{n} \log Q$$

$$0.92 = E^{\circ} - \frac{0.06}{2} \log 10^{-24}$$

$$E^{\circ} = 0.2\text{V}$$

46. (2)

$$E_{\text{cell}}^{\circ} = (E_{\text{ox}}^{\circ})_{\text{anode}} + (E_{\text{red}}^{\circ})_{\text{cathode}}$$

$$= 0.76 + (E_{\text{red}}^{\circ})_{\text{cathode}}$$

A electrode having maximum SRP will produce maximum E_{cell}°

47. (1)

$$K_{\text{C}} \rightarrow 10 \times 10^{15}$$

$$E_{\text{Cell}}^{\circ} \rightarrow ?$$

$$\Delta G^{\circ} = -nF E_{\text{Cell}}^{\circ} \quad \dots(1)$$

$$\text{Also, } \Delta G^{\circ} = -RT \ln K_{\text{C}} \quad \dots(2)$$

Equate (1) and (2)

$$\cancel{-} nF E_{\text{Cell}}^{\circ} = \cancel{-} RT \ln K_{\text{C}}$$

$$\Rightarrow 2 \times E_{\text{Cell}}^{\circ} = \frac{2.303RT}{F} \log 10^{16}$$

48. (4)

$$\Delta S^\circ = nF \frac{dE^\circ}{dT}$$

$$\Delta G^\circ = -nFE^\circ$$

$$\Delta G = \Delta H^\circ - T \Delta S^\circ$$

$$\Delta S^\circ = -2 \times 96000 \times 5 \times 10^{-4} = -96.5 \text{ J}$$

$$\Delta G^\circ = -2 \times 96000 \times 2 = -384 \text{ kJ}$$

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$-384 = \Delta H^\circ - \frac{300 \times (-96.0)}{1000}$$

$$\Delta H^\circ = -384 - 28.8 = -412.8 \text{ kJ mol}^{-1}$$

49. (2)

$$\Lambda_{\text{mHA}}^\infty = \Lambda_{\text{mHCl}}^\infty + \Lambda_{\text{mNaA}}^\infty - \Lambda_{\text{mNaCl}}^\infty$$

$$= 425.9 + 100.5 - 126.4 = 400$$

$$\Lambda_{\text{m}} = \frac{K \times 1000}{M} = \frac{5 \times 10^{-5} \times 1000}{0.001} = 50$$

$$\alpha = \frac{\Lambda_{\text{m}}}{\Lambda_{\text{m}}^\infty} = \frac{50}{400} = \frac{1}{8} = 0.125$$

50. [4]

Sol. (1) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu} : E^\circ = 0.34\text{V}$ (2) $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu} : E^\circ = 0.522\text{V}$ (3) $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$

$$(3) = (1) - (2)$$

$$E^\circ = \frac{(2 \times 0.34) - (0.522)}{1} = 0.158\text{V}$$

51. (2)

52. (1)

53. [−0.93] NTA Ans. (1.52 to 1.53)

Sol. $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ $E_{\text{Red}}^0 = 1.23 \text{ Volt}$

but since it is an oxidation reaction

Hence, $E_{\text{rxn}}^0 = -1.23 \text{ Volt}$

$$\begin{aligned} E &= E^0 - \frac{0.0591}{4} \log [\text{H}^+]^4 \\ &= -1.23 - \frac{0.0591}{4} \log (10^{-5})^4 \\ &= -1.23 - \frac{0.0591}{4} \times (-20) \\ &= -1.23 + 0.2955 \\ &= -0.9345 \end{aligned}$$

$\therefore$ Answer should be **−0.93 V**
instead of 1.52 V or 1.53 V

54. [2.15] NTA Ans. (2.13 to 2.16)

Sol. $E^0 = -0.13 + 0.14 = 0.01 \text{ V}$

at equilibrium $E^0 = \frac{0.06}{n} \log K$

$$0.01 = \frac{0.06}{2} \log \frac{[\text{Sn}^{+2}]}{[\text{Pb}^{2+}]}$$

$$\log \frac{[\text{Sn}^{+2}]}{[\text{Pb}^{2+}]} = \frac{0.01 \times 2}{0.06}$$

$$\frac{[\text{Sn}^{+2}]}{[\text{Pb}^{2+}]} = 10^{1/3} = 2.15$$

55. (4)

Sol. $\Delta G = -n F E_{\text{cell}}$

ΔG is negative, if E_{cell} is positive

Anode : $\text{Cu(s)} \longrightarrow \text{Cu}^{+2}(\text{C}_1) + 2\text{e}^- : E^0$

Cathode : $\text{Cu}^{+2}(\text{s}) + 2\text{e}^- \longrightarrow \text{Cu(s)} :- E^0$

Cell reaction : $\text{Cu}^{+2}(\text{C}_2) \longrightarrow \text{Cu}^{+2}(\text{C}_1) \quad E_{\text{cell}}^0 = 0$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{2.303RT}{nF} \log Q$$

$$E_{\text{cell}} = 0 - \frac{2.303RT}{nF} \log \left(\frac{\text{C}_1}{\text{C}_2} \right)$$

$$E_{\text{cell}} > 0 : \text{if } \frac{\text{C}_1}{\text{C}_2} < 1 \Rightarrow \text{C}_1 < \text{C}_2$$

56. [5.67 or 5.68] NTA Ans. (5.66 to 5.67)

Sol. 108 gm Ag require 1 equivalent 7 charge

Equivalent of O_2 produced = 1

$$\text{Moles of } O_2 = \frac{1}{4}$$

$$\text{Volume of } O_2 \text{ at STP} = \frac{1}{4} \times 22.71 = 5.67 \text{ \& } 5.68$$

57. (None) NTA Ans. (3)

Sol. With increase in temperature conductance of ions increases.

58. (2)

Sol. $E^\circ_{\text{cell}} = 0.34 - (-0.76)$

$$= 1.10 \text{ volt}$$

If $E_{\text{ext}} > 1.10 \text{ volt}$

$\text{Cu} \rightarrow \text{Anode}$

$\text{Zn} \rightarrow \text{Cathode}$

If $E_{\text{ext}} = 1.10 \text{ volt}$

$\text{Zn} \rightarrow \text{Anode}$

$\text{Cu} \rightarrow \text{Cathode}$

59. (1)

Sol. As voltage is '2V' so both Ag^+ & Au^+ will reduce and their equal gm equivalent will reduce so $\text{gmeq Ag} = \text{gmeq of Au}$

$$\frac{Wt_{\text{Ag}}}{E_{\text{qwt Ag}}} = \frac{Wt_{\text{Au}}}{E_{\text{qwt Au}}}$$

$$\text{So } \frac{wt_{\text{Ag}}}{wt_{\text{qwt Au}}} = \frac{E_{\text{qwt Ag}}}{E_{\text{qwt Au}}} = \frac{At wt_{\text{Ag}}}{At wt_{\text{Au}}}$$

60. (3)

Sol. Given curve of λ_m vs $\sqrt{c}$ is for weak electrolyte, so ans. is CH_3COOH .

61. (96500)

Sol. $E^\circ_{\text{cell}} = E^\circ \text{ Cu/Cu}^{2+} + E^\circ \text{ Sn}^{2+}/\text{Sn}$

$$= -0.34 - 0.16$$

$$= -0.5 \text{ V}$$

$$\therefore [\text{Cu}^{2+}] = [\text{Sn}^{2+}] = 1 \text{ M}$$

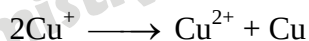
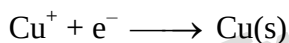
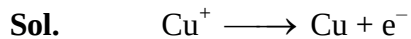
$$\therefore E^\circ_{\text{cell}} = E_{\text{cell}}$$

$$\Delta G = -nF E_{\text{cell}}$$

$$= -2 \times 96500 \times (-0.5)$$

$$= 96500 \text{ J}$$

62. (144.00)



$$E_{\text{Cell}}^{\circ} = E_{\text{Cu}^+/\text{Cu}}^{\circ} - E_{\text{Cu}^{2+}/\text{Cu}^+}^{\circ}$$

$$= 0.52 - 0.16 = 0.36 \text{ V}$$

At equilibrium $\rightarrow E_{\text{cell}} = 0$

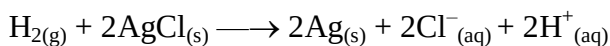
$$E_{\text{Cell}}^{\circ} = \frac{RT}{nF} \ln K$$

$$\ln K = \frac{E_{\text{Cell}}^{\circ} \times nF}{RT}$$

$$\ln K = \frac{0.36 \times 1}{0.025} = 14.4 = 144 \times 10^{-1}$$

63. (142) NTA Ans. (58)

Sol. $K.E_{\text{max}} = h\nu - W = \text{stopping potential.}$



$$Q = \frac{[\text{Cl}]^2 [\text{H}^+]^2}{P_{\text{H}_2}}$$

$$E = E^{\circ} - \frac{0.06}{n} \log Q = K.E_{\text{max}}$$

For Na :

$$0.22 - \frac{0.06}{2} \log \frac{[0.1]^2 [0.1]^2}{1} = h\nu - 2.3$$

$$\therefore h\nu = 2.64 \text{ eV}$$

For K :

$$0.22 - \frac{0.06}{2} \log (x)^2 (x)^2 = h\nu - 2.25$$

$$= 2.64 - 2.25$$

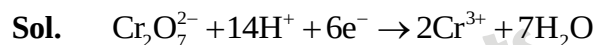
$$-0.12 \log x = 0.17$$

$$-\log x = \text{pH} = \frac{0.17}{0.12} = 1.42 = 142 \times 10^{-2}$$

Correct Ans. 142

NTA Ans. 58

64. (60)



Change flow = $2 \times 8 \times 60$ coul.

$$= \frac{2 \times 8 \times 60}{96000} \text{ F.} = 10^{-2}$$

Mole of Cr^{3+} obtained = $\frac{0.104}{52}$ mole

Equivalent of $\text{Cr}^{3+} = \frac{0.104}{52} \times 3 \text{ eq.}$

$$= 6 \times 10^{-3} \text{ eq.}$$

$$\therefore \text{Current efficiency} = \frac{6 \times 10^{-3}}{10^{-2}} \times 100 = 60$$

So correct answer is 60.

65. (-6)

Sol. $\Delta G^\circ = -nFE^\circ$

$$17.37 \times 10^3 = -3 \times 96500 \times E^\circ$$

$$E^\circ = -0.06$$

$$= -6 \times 10^{-2}$$

66. (11.00)

Sol. $\{2 \times t \times 60 \times 60 / 96500\} \times 0.60 = \frac{10}{122} \times 6$

So $t = 10.98$ hours

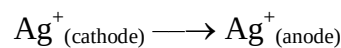
EXERCISE # (JEE-ADVANCED)

1. $K_C = 6.26 \times 10^7$

$$E^\circ = 0.77 - 0.54 = 0.23 = \frac{0.06}{2} \log K$$

$$K = 6.26 \times 10^7$$

2. $K_{sp} = 2.287 \times 10^{-12}$



$$0.1 \text{ M} \qquad 2 \text{ S}$$

$$Q = \frac{2S}{0.1}$$

$$E = 0.164 = - \frac{0.06}{2} \log \left(\frac{2S}{0.1} \right)$$

$$S = 9.23 \times 10^{-3}$$

$$K_{sp} = 4S^3 = 2.287 \times 10^{-12}$$

3. (A)

The specie having higher reduction potential oxidizes the specie having lower reduction potential.

4. (B)

$$E^\circ_{\text{cell}} = 0.33 + 0.44 = 0.77 \text{ Volt}$$

since E°_{cell} is positive therefore reaction is spontaneous.

5. $7.95 \times 10^{-5} \text{ M}$

Let initially $[Cu^{2+}] = C \text{ M}$

$$\therefore \text{moles of } Cu^{2+} \text{ in the solution} = \frac{C \times 250}{1000} = 0.25 C \text{ moles}$$

After electrolysis

$$\text{Moles of } Cu^{2+} = \frac{0.25 \times C}{2}$$

$$\text{Eq. of } Cu^{2+} = \frac{0.25 \times C}{2} \times 2 = \text{faraday of electricity.}$$

$$0.25 \times C = 2 \times 10^{-3} \times 16 \times 60 / 96500$$

$$C = \frac{2 \times 10^{-3} \times 16 \times 60}{0.25 \times 96500} = 7.95 \times 10^{-5} \text{ M}$$

6. decrease with time

Cell reaction :



$$Q = 1$$

$$E = E^\circ = 1.61 - 0.77 = 0.84 \text{ V}$$

$\therefore$ Direction of flow of current is cathode to anode which will decrease with time.

7. (C)

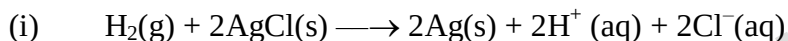
Disproportionate reaction

8. (B)

$$\text{conductance} \propto \frac{1}{\text{Hydrated radius of ion}}$$

9. (C)

10. $\Delta H^\circ = -49987 \text{ J mol}^{-1}$, $\Delta S^\circ = -96.5 \text{ J mol}^{-1} \text{ K}^{-1}$, $s = 1.47 \times 10^{-5} \text{ M}$



(ii) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -nFE^\circ$

At 15°C : $\Delta H^\circ - 288 \times \Delta S^\circ = -2 \times 96500 \times 0.23$

At 35°C : $\Delta H^\circ - 308 \times \Delta S^\circ = -2 \times 96500 \times 0.21$

On solving

$$\Delta H^\circ = -49587 \text{ J/mole}$$

$$\Delta S^\circ = -96.5 \text{ J/mole-K}$$

11. (A)

MnO_4^- ion can oxidise both Fe^{2+} to Fe^{3+} as well as Cl^- to Cl_2 . So $\text{Fe}(\text{NO}_3)_2$ cannot be estimated quantitatively with MnO_4^- ion in HCl .

E°_{Cell} for the cell $\text{Pt}, \text{Cl}_2(\text{g}) (1 \text{ atm}) | \text{Cl}^-(\text{aq}) || \text{MnO}_4^-(\text{aq}) | \text{Mn}^{2+}(\text{aq})$. is equal to $(1.51 - 1.4) = 0.11 \text{ V}$.

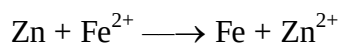
12. (C)

13. (0.05)

$$E + 0.03 = E^\circ - \frac{0.06}{2} \log \frac{[\text{Zn}^{2+}]}{0.5}.$$

$$E = E^\circ - \frac{0.06}{2} \log \frac{[\text{Zn}^{2+}]}{C}. \quad M = 0.05 \text{ M}.$$

14. (B)



$$Q = \frac{0.1}{0.01} = 10$$

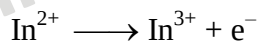
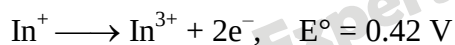
$$E = E^\circ - \frac{0.0591}{2} \log Q$$

$$0.2905 = E^\circ - \frac{0.0591}{2} \log(10)$$

$$E^\circ = 0.32$$

$$E^\circ = \frac{0.0591}{2} \log K = 0.32$$

$$K = 10^{0.32/0.0295}$$

15. $K_C = 10^{10}$ 

$$E^\circ = 0.44 \text{ V}$$

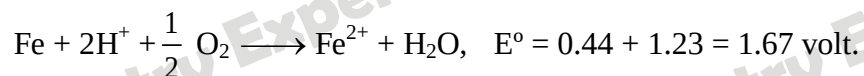
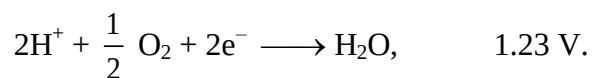
$$E^\circ_{\text{cell}} = 0.15 + 0.44 = 0.59 \text{ V}$$

$$0 = 0.59 - \frac{0.059}{1} \log K.$$

$$K = 10^{10}.$$

16. (B)





$$\Delta G^\circ = -2 \times 1.67 \times 96500 = -322.3 \text{ kJ.}$$

17. (a) $E^\circ = 0.59 \text{ V}$, $\log_{10} K_{\text{SP}} = -10$ (b) $52.8, 10^{-6}$ moles



$$\Delta G_{\text{rxn}}^\circ = -109 - (-129) - 79$$

$$= -57 \text{ kJ/mole}$$

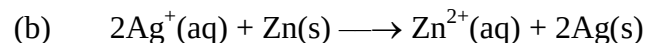
$$\Delta G^\circ = -nFE^\circ$$

$$-57 \times 100 = -1 \times 96500 \times E^\circ$$

$$E^\circ = 0.591 \text{ V}$$

$$\Delta G^\circ = -RT \ln \frac{1}{K_{\text{sp}}} = -57 \times 100$$

$$\log K_{\text{sp}} = -10$$

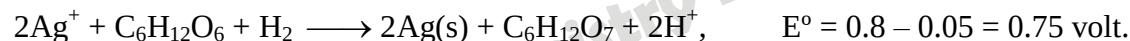


$$E^\circ = 0.8 + 0.76 = 1.56 \text{ V}$$

$$E^\circ = \frac{0.06}{2} \log K_{\text{q}} = 1.56$$

$$K_{\text{q}} = 10^{52}$$

18. (A)



$$0 = 0.75 - \frac{0.0592}{2} \log K.$$

$$\ln K = 2.303 \times \log K = 2.303 \times 25.34 = 58.38.$$

19. (A)

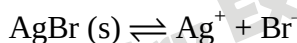
$$[\text{H}^+] = 10^{-11} \text{ M.}$$

$$E_{\text{oxide}} = -0.05 - \frac{0.0591}{2} \log(10^{-11})^2 = -0.05 + 0.65$$

$$\text{or, } \Delta H = 0.65 \text{ volt.}$$

20. (D)

Standards electrode potential does not depend upon on concentration.

21. (55 S m⁻¹)

$$(s + 10^{-7}) \times s = K_{sp} = 12 \times 10^{-14}.$$

$$s = 3 \times 10^{-7} \text{ M.}$$

$$[\text{Ag}^+] = 4 \times 10^{-7} \text{ M} ; [\text{Br}^-] = 3 \times 10^{-7} \text{ M} ; [\text{NO}_3^-] = 10^{-7} \text{ M.}$$

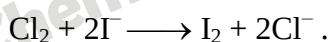
$$K_{\text{total}} = \Lambda_{\text{Ag}^+} + \Lambda_{\text{Br}^-} + \Lambda_{\text{NO}_3^-}$$

$$K_{\text{total}} = 4 \times 10^{-4} \times 6 \times 10^{-3} + 3 \times 10^{-4} \times 8 \times 10^{-3} + 1 \times 10^{-4} \times 7 \times 10^{-3}.$$

$$K_{\text{total}} = 24 + 24 + 7.$$

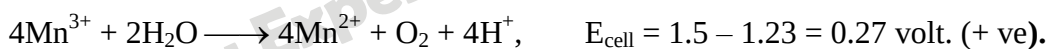
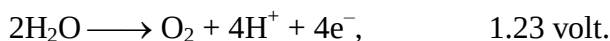
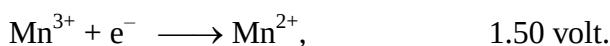
$$K_{\text{total}} = 55 \text{ Scm}^{-1}.$$

22. (C)



$$E^\circ = 1.36 + (-0.54) = 0.82 \text{ V (+ve). Spontaneous.}$$

23. (D)

So Mn³⁺ will oxidise H₂O.

24. (B)

$$\text{Mol of NaCl} = 4 \times 0.5 = 2 \text{ mol.}$$

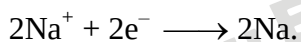
$$\text{No. of mole of Cl}_2 \text{ evolved} = \frac{1}{2} \times \text{mol of NaCl} = \frac{1}{2} \times 2 = \mathbf{1 \text{ mol.}}$$

25. (D)

Taking the 1 : 1 molar combination of Na–Hg amalgam.

$$\text{weight} = 2 \times 23 + 2 \times 200 = 446 \text{ g.}$$

26. (D)



$$\text{No. of Faraday required} = 2.$$

$$\therefore \text{total charge} = 2 \times 96500 = \mathbf{193000 \text{ coulomb.}}$$

27. (B)

$$\text{Faraday law equivalents of H}_2 \text{ produced} = \frac{I \times t \text{ (sec)}}{96500}$$

$$0.01 \times 2 = \frac{10 \times 10^{-3} \times t}{96500} = 96500 \times 2 = t$$

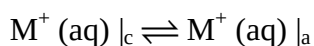
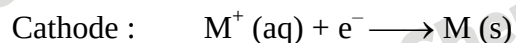
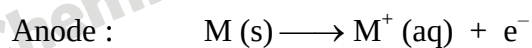
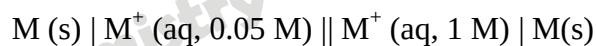
$$t = 19.3 \times 10^4 \text{ sec}$$

28. (ABD)

The species having less reduction potential with respect to NO_3^- ($E^\circ = 0.96 \text{ V}$) will be oxidised by NO_3^- . These species are V, Fe, Hg.

29. (B)

30. (C)



$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{1} \log \frac{\text{M}^+ (\text{aq}) \mid_{\text{a}}}{\text{M}^+ (\text{aq}) \mid_{\text{c}}}$$

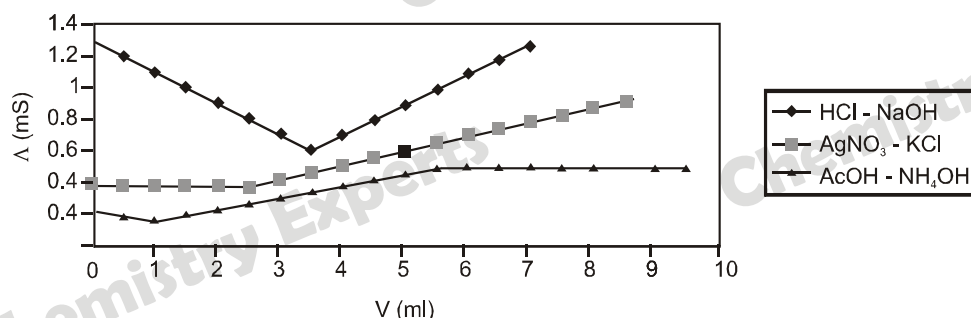
$$= 0 - \frac{0.0591}{1} \log$$

$$= + \text{ve} = 70 \text{ mV and hence } \Delta G = -nFE_{\text{cell}} = - \text{ve.}$$

$$= 70 \text{ mV} + \frac{0.0591}{1} \log 20 = 140 \text{ mV.}$$

31. (D)

Typical titration curves



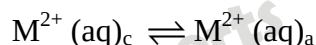
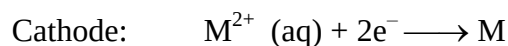
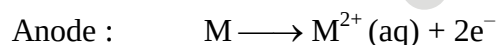
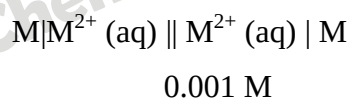
32. (D)

$$\begin{aligned}
 E &= E^\circ - \frac{0.059}{4} \log \frac{[\text{Fe}^{2+}]^2}{[\text{H}^+]^4 P_{\text{O}_2}} \\
 &= 1.67 - \frac{0.06}{4} \log \frac{(10^{-3})^2}{(10^{-3})^4 \times 0.1} = 1.67 - \frac{0.03}{2} \log 10^7 \\
 &= 1.67 - \frac{0.03}{2} \times 7 = 1.67 - 0.105 = 1.565 = 1.57 \text{ V.}
 \end{aligned}$$

33. (D)

$$\begin{aligned}
 \Delta G &= -nFE_{\text{cell}} = -2 \times 96500 \times 0.059 \times 10^{-3} \text{ kJ/mole} \\
 &= -11.4 \text{ kJ/mole.}
 \end{aligned}$$

34. (B)



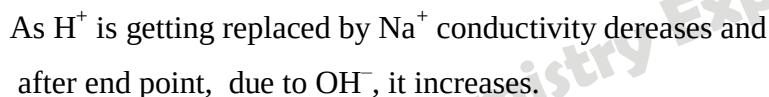
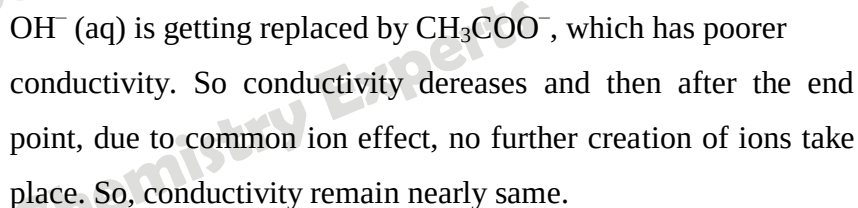
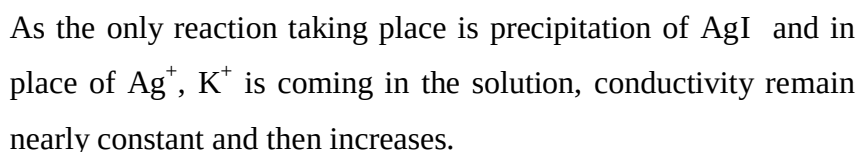
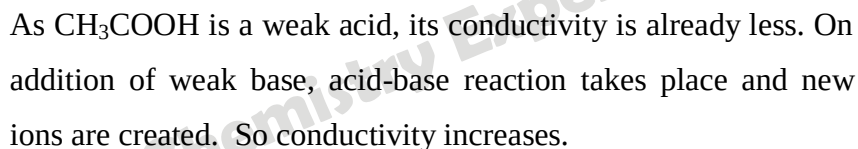
$$E_{\text{cell}} = 0 - \frac{0.059}{2} \log \left\{ \frac{\text{M}^{2+} (\text{aq})_{\text{a}}}{10^{-3}} \right\}$$

$$0.059 = -\frac{0.059}{2} \log \left\{ \frac{\text{M}^{2+} (\text{aq})_{\text{a}}}{10^{-3}} \right\}$$

$$-2 = \log \left\{ \frac{\text{M}^{2+} (\text{aq})_{\text{a}}}{10^{-3}} \right\}$$

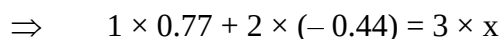
$$10^{-2} \times 10^{-3} = \text{M}^{2+} (\text{aq})_{\text{a}} = \text{solubility} = s$$

$$K_{\text{sp}} = 4s^3 = 4 \times (10^{-5})^3 = 4 \times 10^{-15}$$

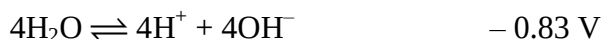


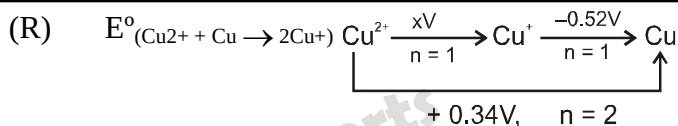
So answer of 39 is : (P) – (3) ; (Q) – (4) ; (R) – (2) ; (S) – (1). Answer is (D).

36. (D)



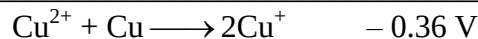
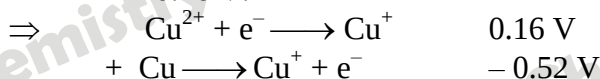
(Q) $4\text{H}_2\text{O} \rightleftharpoons 4\text{H}^+ + 4\text{OH}^-$



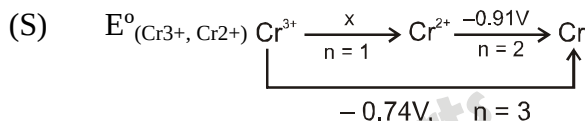


$$x \times 1 + 0.52 \times 1 = 0.34 \times 2$$

$$x = 0.16 \text{ V.}$$



However, in the given option, -0.18 V is printed.



$$x \times 1 + 2 \times (-0.91) = 3 \times (-0.74)$$

$$x - 1.82 = -2.22$$

$$\Rightarrow x = -0.4 \text{ V}$$

Hence, most appropriate is (D).

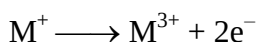
(P) – (3) ; (Q) – (4) ; (R) – (1) ; (S) – 2.

37. (AB)

Salt bridge is introduced to keep the solutions of two electrodes separate, such that the ions in electrode do not mix freely with each other. But it cannot stop the process of diffusion.

It does not participate in the chemical reaction. However, it is not necessary for occurrence of cell reaction, as we know that designs like lead accumulator, there was no salt bridge, but still reactions takes place.

38. (4)



$$\Delta G^0 = -nFE^0 \text{ For 1 mole of } \text{M}^+$$

$$\begin{aligned} \Delta G^0 &= -2 \times 96500 \times (-0.25) \text{ J} \\ &= + 48250 \text{ J/mole} = 48.25 \text{ KJ/mole} \end{aligned}$$

Energy released by conversion of 1 mole of



$$\text{Hence mole of } \text{M}^+ \text{ convert} = \frac{193}{48.25} = 4$$

39. (3)

$$\lambda_{X^-}^{\circ} \approx \lambda_{Y^-}^{\circ}$$

$$\Rightarrow \lambda_{H^+}^{\circ} + \lambda_{X^-}^{\circ} \approx \lambda_{H^+}^{\circ} + \lambda_{Y^-}^{\circ}$$

$$\Rightarrow \lambda_{HX}^{\circ} \approx \lambda_{HY}^{\circ} \quad (1)$$

Also $\frac{\lambda_m}{\lambda_m^{\circ}} = \alpha$, So $\lambda_m(HX) = \lambda_m^{\circ} \alpha_1$ and $\lambda_m(HY) = \lambda_m^{\circ} \alpha_2$

(Where α_1 and α_2 are degrees of dissociation of HX and HY respectively.)

Now, Given that

$$\lambda_m(HY) = 10 \lambda_m(HX).$$

$$\Rightarrow \lambda_m^{\circ} \alpha_2 = 10 \times \lambda_m^{\circ} \alpha_1$$

$$\alpha_2 = 10 \alpha_1 \quad (2)$$

$$K_a = \frac{C\alpha^2}{1-\alpha}, \quad \text{but } \alpha \ll 1, \quad \text{therefore } K_a = C\alpha^2.$$

$$\Rightarrow \frac{K_a(HX)}{K_a(HY)} = \frac{0.01\alpha_1^2}{0.1\alpha_2^2} = \frac{0.01}{0.1} \times \left(\frac{1}{10}\right)^2 = \frac{1}{1000}.$$

$$\Rightarrow \log(K_a(HX)) - \log(K_a(HY)) = -3.$$

$$\Rightarrow pK_a(HX) - pK_a(HY) = 3.$$

40. (D)



$$Q = \frac{[H^+]^2[M^{2+}]}{P_{H_2}[M^{4+}]} = \frac{1 \times [M^{2+}]}{1 \times [M^{4+}]} = 10^x$$

$$E = 0.092 = 0.151 - \frac{0.059}{2} \log 10^x \Rightarrow x = 2$$

41. (6)

$$\kappa = G \times \frac{\ell}{A} = 5 \times 10^{-7} \times \frac{120}{1} = 6 \times 10^{-5} \text{ Scm}^{-1}$$

$$[H^+] = c\alpha = 10^{-4} \text{ M} \Rightarrow \alpha = \frac{10^{-4}}{0.0015}$$

$$\therefore \Lambda_m = \frac{K \times 1000}{0.0015} = \frac{6 \times 10^{-5} \times 1000}{0.0015} = 40 \text{ Scm}^2\text{mol}^{-1}$$

$$\therefore \alpha = \frac{\Lambda_m}{\Lambda_m^\infty} \Rightarrow \Lambda_m^\infty = \frac{\Lambda_m}{\alpha}$$

$$\therefore \Lambda_m^\infty = \frac{40}{\frac{10^{-4}}{0.0015}} = 6 \times 10^2$$

$$\therefore Z = 6$$

42. (A)

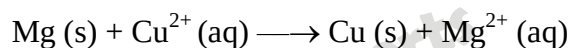


$$Q_c = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = 10$$

$$\Delta G^\circ = -2 \times F \times 1.1$$

$$\Delta G = \Delta G^\circ + RT \ln Q = -2.2 F + RT \ln 10 = 2.303 RT - 2.2 F$$

43. (10)



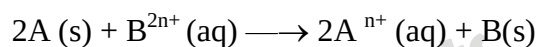
$$\text{Initially : } E = 2.7 = E^\circ - \frac{RT}{2 \times F} \ln \left(\frac{1}{1} \right) \Rightarrow E^\circ = 2.7$$

$$\text{When : } [\text{Mg}^{2+}] = x$$

$$E = 2.67 = 2.7 - \frac{300}{2 \times 11500} \times \ln \left(\frac{x}{1} \right)$$

$$\ln x = 2.3 \Rightarrow x = 10$$

44. -11.62



$$\text{Given } \Delta H^\circ = 2\Delta G^\circ$$

$$\therefore \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\therefore \Delta G^\circ = 2\Delta G^\circ - T\Delta S^\circ$$

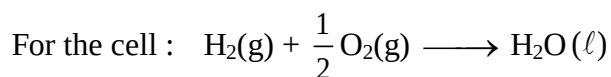
$$\Delta G^\circ = T\Delta S^\circ$$

$$\therefore \Delta G = \Delta G^\circ + RT \ln Q = 0$$

$$\Delta G^\circ = -8.3 \times 300 \times \ln \left(\frac{2^2}{1} \right) = 300 \times \Delta S^\circ$$

$$\Delta S^\circ = -8.3 \times \ln 4 = -8.3 \times 2 \times 0.7 = -11.62 \text{ J/K}$$

45. (13.32)



$$E^\circ_{\text{Cell}} = 1.23 - 0 = 1.23 \text{ V}$$

$$\therefore \Delta G^\circ_{\text{Cell}} = -n F E^\circ_{\text{Cell}}$$

$$= -2 \times 96500 \times 1.23 \text{ J/mole}$$

$$= -237390 \text{ J/mole}$$

$$= -237.39 \text{ k J/mole}$$

$\therefore$ efficiency of the cell is 70%

$$\therefore \text{Work obtained per mole reaction} = 237.39 \times \frac{70}{100} \text{ k J/mole}$$

$$\therefore \text{when } 10^{-3} \text{ mole } \text{H}_2(\text{g}) \text{ is used then work obtained} = 237.39 \times \frac{70}{100} \times 10^{-3} \text{ k J}$$

The work obtained is used to compress monoatomic gas present in insulated container

$$\therefore q = 0$$

$$\therefore \Delta U = nC_V \Delta T = W$$

$$\therefore \Delta T = \frac{W}{nC_V} = \frac{237.39 \times 0.7 \times 10^{-3} \text{ kJ}}{1 \times \frac{3}{2} \times 8.314 \times 10^{-3} \text{ kJ}}$$

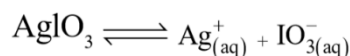
$$\Delta T = 13.327 \text{ K}$$

EXERCISE # (JEE-MAIN)

- The solubility in water of a sparingly soluble salt AB_2 is $1.0 \times 10^{-5} \text{ mol L}^{-1}$. Its solubility product will be **[AIEEE-2003]**
 (1) 1×10^{-15} (2) 1×10^{-10} (3) 4×10^{-15} (4) 4×10^{-10}
- The solubility of $Mg(OH)_2$ is x mole/lit. then its solubility product is: **[AIEEE-2002]**
 (1) x^3 (2) $5x^3$ (3) $4x^3$ (4) $2x^2$
- The molar solubility in mol L^{-1} of a sparingly soluble salt MX_4 is 's'. The corresponding solubility product is K_{SP} . 's' is given in terms of K_{SP} by relation : **[AIEEE-2004]**
 (1) $s = (K_{SP} / 128)^{1/4}$ (2) $s = (128K_{SP})^{1/4}$ (3) $s = (256K_{SP})^{1/5}$ (4) $s = (K_{SP}/256)^{1/5}$
- The conjugate base of $H_2PO_4^-$ is **[AIEEE-2004]**
 (1) H_3PO_4 (2) P_2O_5 (3) (4)
- Hydrogen ion concentration in mol/L in a solution of $pH = 5.4$ will be : **[AIEEE-2005]**
 (1) 3.88×10^6 (2) 3.98×10^8 (3) 3.98×10^{-6} (4) 3.68×10^{-6}
- The solubility product of a salt having general formula MX_2 , in water is : 4×10^{-12} . The concentration of M^{2+} ions in the aqueous solution of the salt is : **[AIEEE-2005]**
 (1) $1.0 \times 10^{-4} \text{ M}$ (2) $2.0 \times 10^{-6} \text{ M}$ (3) $4.0 \times 10^{-10} \text{ M}$ (4) $1.6 \times 10^{-4} \text{ M}$
- What is the conjugate base of OH^- ? **[AIEEE-2005]**
 (1) O^{2-} (2) O^- (3) H_2O (4) O_2
- The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. the overall dissociation constant of the acid will be **[AIEEE-2007]**
 (1) 0.2×10^5 (2) 5.0×10^{-5} (3) 5.0×10^{15} (4) 5.0×10^{-15}

9. The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffer solution of HA in which 50% of the acid is ionised is **[AIEEE-2007]**
 (1) 7.0 (2) 4.5 (3) 2.5 (4) 9.5

10. In a saturated solution of the sparingly soluble strong electrolyte $AgIO_3$ (molecular mass = 283) the equilibrium which sets in is : **[AIEEE-2007]**



If the solubility product constant K_{sp} of $AgIO_3$ at a given temperature is 1.0×10^{-8} , what is the mass of $AgIO_3$ contained in 100 ml of its saturated solution ?

- (1) 28.3×10^{-2} g (2) 2.83×10^{-3} g (3) 1.0×10^{-7} g (4) 1.0×10^{-4} g
11. The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be : **[AIEEE-2008]**
 (1) 9.58 (2) 4.79 (3) 7.01 (4) 9.22

12. Solid $Ba(NO_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At what concentration of Ba^{2+} will a precipitate begin to form? (K_{sp} for $BaCO_3 = 5.1 \times 10^{-9}$) **[AIEEE-2009]**
 (1) 8.1×10^{-8} M (2) 8.1×10^{-7} M (3) 4.1×10^{-5} M (4) 5.1×10^{-5} M

13. Solubility product of silver bromide is 5.0×10^{-13} . The quantity of potassium bromide (molar mass taken as 120 g mol^{-1}) to be added to 1 litre of 0.05 M solution of silver nitrate to start the precipitation of AgBr is : **[AIEEE-2010]**
 (1) 5.0×10^{-8} g (2) 1.2×10^{-10} g (3) 1.2×10^{-9} g (4) 6.2×10^{-5} g

14. In aqueous solution the ionization constants for carbonic acid are **[AIEEE-2010]**
 $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$

Select the correct statement for a saturated 0.034 M solution of the carbonic acid :-

- (1) The concentration of H^+ is double that of CO_3^{2-}
 (2) The concentration of CO_3^{2-} is 0.034 M
 (3) The concentration of CO_3^{2-} is greater than that of HCO_3^-
 (4) The concentrations of H^+ and HCO_3^- are approximately equal

15. At 25° C, the solubility product of $\text{Mg}(\text{OH})_2$ is 1.0×10^{-11} . At which pH, will Mg^{2+} ions start precipitating in the form of $\text{Mg}(\text{OH})_2$ from a solution of 0.001 M Mg^{2+} ions ? [AIEEE–2010]
 (1) 8 (2) 9 (3) 10 (4) 11
16. Three reactions involving H_2PO_4^- are given below:
 (i) $\text{H}_3\text{PO}_4 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{H}_2\text{PO}_4^-$ (ii) $\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \rightarrow \text{HPO}_4^{2-} + \text{H}_3\text{O}^+$
 (iii) $\text{H}_2\text{PO}_4^- + \text{OH}^- \rightarrow \text{H}_3\text{PO}_4 + \text{O}^{2-}$
 In which of the above does H_2PO_4^- act as an acid ? [AIEEE–2010]
 (1) (ii) only (2) (i) and (ii) (3) (iii) only (4) (i) only
17. The K_{sp} for $\text{Cr}(\text{OH})_3$ is 1.6×10^{-30} . The molar solubility of this compound in water is: [AIEEE–2011]
 (1) $\sqrt[3]{1.6 \times 10^{-30}}$ (2) $\sqrt[4]{1.6 \times 10^{-30}}$ (3) $\sqrt[4]{1.6 \times 10^{-30} / 27}$ (4) $1.6 \times 10^{-30} / 27$
18. An acid HA ionises as
 $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$
 The pH of 1.0 M solution is 5. Its dissociation constant would be : [AIEEE–2011]
 (1) 1×10^{-10} (2) 5 (3) 5×10^{-8} (4) 1×10^{-5}
19. If K_{sp} of CaF_2 at 25°C is 1.7×10^{-10} , the combination amongst the following which gives a precipitate of CaF_2 is : [JEE-MAIN(online)–2012]
 (1) 1×10^{-2} M Ca^{2+} and 1×10^{-5} M F^- (2) 1×10^{-4} M Ca^{2+} and 1×10^{-4} M F^-
 (3) 1×10^{-3} M Ca^{2+} and 1×10^{-5} M F^- (4) 1×10^{-2} M Ca^{2+} and 1×10^{-3} M F^-
20. The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is : [AIEEE–2012]
 (1) 1×10^{-7} (2) 3×10^{-7} (3) 1×10^{-3} (4) 1×10^{-5}
21. Solid $\text{Ba}(\text{NO}_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At which concentration of Ba^{2+} , precipitate of BaCO_3 begins to form ?
 (K_{sp} for $\text{BaCO}_3 = 5.1 \times 10^{-9}$) [JEE-MAIN(Online)–2013]
 (1) 5.1×10^{-5} M (2) 8.1×10^{-7} M (3) 4.1×10^{-5} M (4) 7.1×10^{-8} M

22. NaOH is a strong base. What will be pH of 5.0×10^{-2} M NaOH solution ? ($\log 2 = 0.3$)

[JEE-MAIN(Online)-2013]

- (1) 13.70 (2) 13.00 (3) 14.00 (4) 12.70

23. Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts Hg_2Cl_2 , $\text{Cr}_2(\text{SO}_4)_3$, BaSO_4 and CrCl_3 respectively ?

[JEE-MAIN(Online)-2013]

- (1) $\left(\frac{K_{\text{sp}}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{\text{sp}}}{108}\right)^{\frac{1}{8}}$, $(K_{\text{sp}})^{\frac{1}{2}}$, $\left(\frac{K_{\text{sp}}}{27}\right)^{\frac{1}{4}}$ (2) $(K_{\text{sp}})^{\frac{1}{2}}$, $\left(\frac{K_{\text{sp}}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{\text{sp}}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{\text{sp}}}{108}\right)^{\frac{1}{3}}$
- (3) $(K_{\text{sp}})^{\frac{1}{2}}$, $\left(\frac{K_{\text{sp}}}{108}\right)^{\frac{1}{3}}$, $\left(\frac{K_{\text{sp}}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{\text{sp}}}{4}\right)^{\frac{1}{3}}$ (4) $\left(\frac{K_{\text{sp}}}{108}\right)^{\frac{1}{3}}$, $\left(\frac{K_{\text{sp}}}{27}\right)^{\frac{1}{4}}$, $(K_{\text{sp}})^{\frac{1}{2}}$, $\left(\frac{K_{\text{sp}}}{4}\right)^{\frac{1}{3}}$

24. What would be the pH of a solution obtained by mixing 5 g of acetic acid and 7.5 g of sodium acetate and making the volume equal to 500 mL?

[JEE-MAIN(Online)-2013]

($K_a = 1.75 \times 10^{-5}$, $\text{p}K_a = 4.76$)

- (1) $4.76 < \text{pH} < 5.0$
 (2) $\text{pH} < 4.70$
 (3) pH of solution will be equal to pH of acetic acid
 (4) $\text{pH} = 4.70$

25. How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ?

[AIEEE-2013]

- (1) 0.1 L (2) 0.9 L (3) 2.0 L (4) 9.0 L

26. In some solutions, the concentration of H_3O^+ remains constant even when small amounts of strong acid or strong base are added to them. These solutions are known as:

[JEE-MAIN(Online)-2014]

- (1) Colloidal solutions (2) True solutions (3) Ideal solutions (4) Buffer solutions

27. Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge +4 and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct ? **[JEE-MAIN(Online)-2014]**

(1) $S = \{K_{\text{sp}}/144\}^{1/7}$ (2) $S = \{K_{\text{sp}}/(6912)\}^{1/7}$
 (3) $S = (K_{\text{sp}}/6912)^{1/7}$ (4) $S = \{K_{\text{sp}}/6912\}^7$

28. pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is **[JEE-MAIN-2017]**

(1) 7.2 (2) 6.9 (3) 7.0 (4) 1.0

29. An alkali is titrated against an acid with methyl orange as indicator, which of the following is a correct combination ? **[JEE-MAIN-2018]**

	Base	Acid	End Point
(1)	Strong	Strong	Pink to colourless
(2)	Weak	Strong	Colourless to pink
(3)	Strong	Strong	Pinkish red to yellow
(4)	Weak	Strong	Yellow to Pinkish red

30. An aqueous solution contains an unknown concentration of Ba^{2+} . When 50 mL of a 1 M solution of Na_2SO_4 is added, BaSO_4 just begins to precipitate. The final volume is 500 mL. The solubility product of BaSO_4 is 1×10^{-10} . What is the original concentration of Ba^{2+} ? **[JEE-MAIN-2018]**

(1) $1.0 \times 10^{-10} \text{ M}$ (2) $5 \times 10^{-9} \text{ M}$ (3) $2 \times 10^{-9} \text{ M}$ (4) $1.1 \times 10^{-9} \text{ M}$

31. An aqueous solution contains 0.10 M H_2S and 0.20 M HCl . If the equilibrium constants for the formation of HS^- from H_2S is 1.0×10^{-7} and that S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is **[JEE-MAIN-2018]**

(1) 5×10^{-19} (2) 5×10^{-8} (3) 3×10^{-20} (4) 6×10^{-21}

32. Which of the following salts is the most basic in aqueous solution? **[JEE-MAIN-2018]**

(1) $\text{Pb}(\text{CH}_3\text{COO})_2$ (2) $\text{Al}(\text{CN})_3$ (3) CH_3COOK (4) FeCl_3

33. The minimum volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution (K_{sp} of $PbCl_2 = 3.2 \times 10^{-8}$; atomic mass of Pb = 207 u) is : **[JEE Main-2018 Online]**

(1) 0.36 L (2) 0.18 L (3) 17.98 L (4) 1.798 L

34. Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1 ?

[JEE Main-2018 Online]

(1) $75\text{mL } \frac{M}{5} \text{ HCl} + 25\text{mL } \frac{M}{5} \text{ NaOH}$ (2) $100\text{mL } \frac{M}{10} \text{ HCl} + 100\text{mL } \frac{M}{10} \text{ NaOH}$

(3) $55\text{mL } \frac{M}{10} \text{ HCl} + 40\text{mL } \frac{M}{10} \text{ NaOH}$ (4) $60\text{mL } \frac{M}{10} \text{ HCl} + 40\text{mL } \frac{M}{10} \text{ NaOH}$

35. 20 mL of 0.1 M H_2SO_4 solution is added to 30 mL of 0.2 M NH_4OH solution. The pH of the resultant mixture is : [pK_b of $NH_4OH = 4.7$]. **[JEE Main-2019 (Jan)]**

(1) 9.4 (2) 5.0 (3) 9.0 (4) 5.2

36. A mixture of 100 mmol of $Ca(OH)_2$ and 2g of sodium sulphate was dissolved in water and the volume was made up to 100 mL. The mass of calcium sulphate formed and the concentration of OH^- in resulting solution, respectively, are : (Molar mass of $Ca(OH)_2$, Na_2SO_4 and $CaSO_4$ are 74, 143 and 136 g mol^{-1} , respectively; K_{sp} of $Ca(OH)_2$ is 5.5×10^{-6}) **[JEE Main-2019 (Jan)]**

(1) 13.6 g, 0.14 mol L^{-1} (2) 1.9 g, 0.14 mol L^{-1}

(3) 1.9 g, 0.28 mol L^{-1} (4) 13.6 g, 0.28 mol L^{-1}

37. If K_{sp} of Ag_2CO_3 is 8×10^{-12} , the molar solubility of Ag_2CO_3 in 0.1M $AgNO_3$ is :

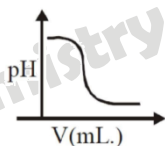
[JEE Main-2019 (Jan)]

(1) $8 \times 10^{-10} \text{ M}$ (2) $8 \times 10^{-13} \text{ M}$ (3) $8 \times 10^{-12} \text{ M}$ (4) $8 \times 10^{-11} \text{ M}$

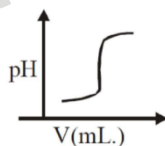
38. If solubility product of $Zr_3(PO_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S, then which of the following relation between A and K_{sp} is correct ? **[JEE Main-2019 (Apr)]**

(1) $S = \left(\frac{K_{sp}}{6912} \right)^{\frac{1}{7}}$ (2) $S = \left(\frac{K_{sp}}{144} \right)^{\frac{1}{6}}$ (3) $S = \left(\frac{K_{sp}}{929} \right)^{\frac{1}{9}}$ (4) $S = \left(\frac{K_{sp}}{216} \right)^{\frac{1}{7}}$

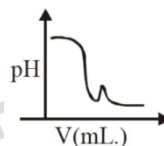
39. In an acid-base titration, 0.1 M HCl solution was added to the NaOH solution of unknown strength. Which of the following correctly shows the change of pH of the titration mixture in this experiment? **[JEE Main-2019 (Apr)]**



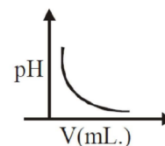
(A)



(B)



(C)



(D)

(1) D

(2) A

(3) C

(4) B

40. Consider the following statements

[JEE Main-2019 (Apr)]

- (a) The pH of a mixture containing 400 mL of 0.1 M H_2SO_4 and 400 mL of 0.1 M NaOH will be approximately 1.3.
 (b) Ionic product of water is temperature dependent.
 (c) A monobasic acid with $K_a = 10^{-5}$ has a pH = 5. The degree of dissociation of this acid is 50%.
 (d) The Le Chatelier's principle is not applicable to common-ion effect.

The correct statements are :

- (1) (a), (b) and (c) (2) (a) and (b) (3) (b) and (c) (4) (a), (b) and (d)

41. The pH of a 0.02 M NH_4Cl solution will be [given $K_b(\text{NH}_4\text{OH}) = 10^{-5}$ and $\log 2 = 0.301$]

[JEE Main-2019 (Apr)]

- (1) 5.35 (2) 4.65 (3) 4.35 (4) 2.65

42. What is the molar solubility of $\text{Al}(\text{OH})_3$ in 0.2 M NaOH solution? Given that, solubility product of $\text{Al}(\text{OH})_3 = 2.4 \times 10^{-24}$;

[JEE Main-2019 (Apr)]

- (1) 12×10^{-23} (2) 3×10^{-22} (3) 12×10^{-21} (4) 3×10^{-19}

43. The molar solubility of $\text{Cd}(\text{OH})_2$ is 1.84×10^{-5} M in water. The expected solubility of $\text{Cd}(\text{OH})_2$ in a buffer solution of pH = 12 is:

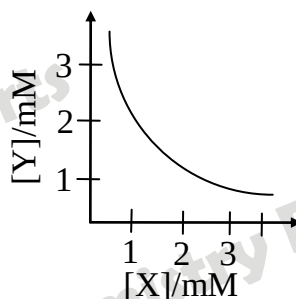
[JEE Main-2019 (Apr)]

- (1) 6.23×10^{-11} M (2) 2.49×10^{-10} M (3) 1.84×10^{-9} M (4) $\times 10^{-9}$ M

44. The strength of an aqueous NaOH solution is most accurately determined by titrating :
(Note : consider that an appropriate indicator is used) **[Jee Main (Jan. 2020)]**

- (1) Aq. NaOH in a pipette and aqueous oxalic acid in a burette
- (2) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask
- (3) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask
- (4) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask

45. The stoichiometry and solubility product of a salt with the solubility curve given below is, respectively : **[Jee Mains (Jan. 2020)]**



- (1) XY , $2 \times 10^{-6} \text{ M}^3$
 - (2) X_2Y , $2 \times 10^{-9} \text{ M}^3$
 - (3) XY_2 , $1 \times 10^{-9} \text{ M}^3$
 - (4) XY_2 , $4 \times 10^{-9} \text{ M}^3$
46. The K_{sp} for the following dissociation is 1.6×10^{-5}
 $\text{PbCl}_{2(\text{s})} \rightleftharpoons \text{Pb}_{(\text{aq})}^{2+} + 2\text{Cl}_{(\text{aq})}^{-}$
 Which of the following choices is correct for a mixture of 300 mL 0.134 M $\text{Pb}(\text{NO}_3)_2$ and 100 mL 0.4 M NaCl ? **[Jee Main (Jan. 2020)]**
- (1) $Q = K_{\text{sp}}$
 - (2) $Q < K_{\text{sp}}$
 - (3) Not enough data provided
 - (4) $Q > K_{\text{sp}}$
47. The solubility product of $\text{Cr}(\text{OH})_3$ at 298 K is 6.0×10^{-31} . The concentration of hydroxide ions in a saturated solution of $\text{Cr}(\text{OH})_3$ will be : **[Jee Main (Jan. 2020)]**
- (1) $(18 \times 10^{-31})^{1/2}$
 - (2) $(18 \times 10^{-31})^{1/4}$
 - (3) $(2.22 \times 10^{-31})^{1/4}$
 - (4) $(4.86 \times 10^{-29})^{1/4}$
48. Two solutions, A and B, each of 100 L was made by dissolving 4 g of NaOH and 9.8 g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40 L of solution A and 10 L of solution B is _____ : **[Jee Main (Jan. 2020)]**
49. 3 g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is _____.
 [Given ; pK_a of acetic acid = 4.75, molar mass of acetic acid = 60 g/mol, $\log 3 = 0.4771$]
 Neglect any changes in volume. **[Jee Main (Jan. 2020)]**

50. For the following Assertion and Reason the correct option is

Assertion (A) : When Cu (II) and sulphide ions are mixed, they react together extremely quickly to give a solid. **[Jee-Main (Sep) 2020]**

Reason (R) : The equilibrium constant of $\text{Cu}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightleftharpoons \text{CuS}(\text{s})$ is high because the solubility product is low.

- (1) (A) is false and (R) is true.
- (2) Both (A) and (R) are true but (R) is not explanation for (A).
- (3) Both (A) and (R) are false.
- (4) Both (A) and (R) are true and (R) is the explanation for (A)

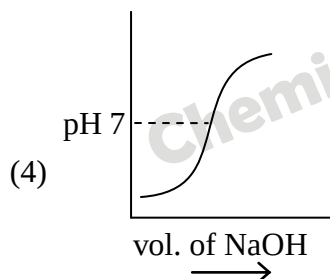
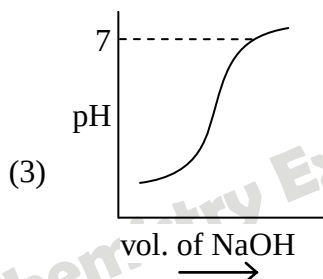
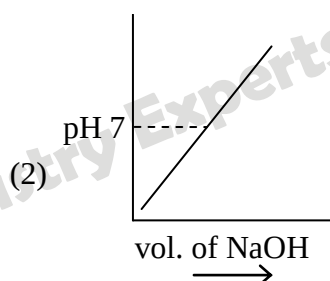
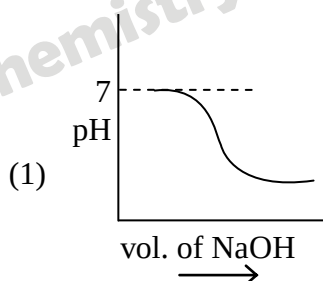
51. While titrating dilute HCl solution with aqueous NaOH, which of the following will not be required ? **[Jee-Main (Sep) 2020]**

- (1) Clamp and phenolphthalein
- (2) Burette and porcelain tile
- (3) Pipette and distilled water
- (4) Bunsen burner and measuring cylinder

52. An acidic buffer is obtained on mixing :

- (1) 100 mL of 0.1 M HCl and 200 mL of 0.1 M NaCl **[Jee-Main (Sep) 2020]**
- (2) 100 mL of 0.1 M HCl and 200 mL of 0.1 M CH_3COONa
- (3) 100 mL of 0.1 M CH_3COOH and 100 mL of 0.1 M NaOH
- (4) 100 mL of 0.1 M CH_3COOH and 200 mL of 0.1 M NaOH

53. 100 mL of 0.1 M HCl is taken in a beaker and to it 100 mL of 0.1 M NaOH is added in steps of 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH? **[Jee-Main (Sep) 2020]**



54. Arrange the following solutions in the decreasing order of pOH : [Jee-Main (Sep) 2020]

(A) 0.01 M HCl

(B) 0.01 M NaOH

(C) 0.01 M CH_3COONa

(D) 0.01 M NaCl

(1) (B) > (C) > (D) > (A)

(2) (A) > (C) > (D) > (B)

(3) (B) > (D) > (C) > (A)

(4) (A) > (D) > (C) > (B)

55. A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature. The partial pressure of CO_2 over the solution approaches a value of 30 bar when 44 g of CO_2 is dissolved in 1 kg of water at room temperature. The approximate pH of the soft drink is _____ $\times 10^{-1}$.

[Jee-Main (Sep) 2020]

(First dissociation constant of $\text{H}_2\text{CO}_3 = 4.0 \times 10^{-7}$; $\log 2 = 0.3$; density of the soft drink = 1 g mL^{-1})

Given 64.00

56. If the solubility product of AB_2 is $3.20 \times 10^{-11} \text{ M}^3$, then the solubility of AB_2 in pure water is _____ $\times 10^{-4} \text{ mol L}^{-1}$. [Assuming that neither kind of ion reacts with water]

[Jee-Main (Sep) 2020]

EXERCISE # JEE-ADVANCED

- What will be the resultant pH when 200 ml of an aqueous solution of HCl (pH = 2.0) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12.0) ? **[JEE 1998]**
- The solubility of $\text{Pb}(\text{OH})_2$ in water is $6.7 \times 10^{-6} \text{ M}$. Calculate the solubility of $\text{Pb}(\text{OH})_2$ in a buffer solution of pH = 8. **[JEE 1999]**
- A buffer solution can be prepared from a mixture of **[JEE 1999]**
 - sodium acetate and acetic acid in water
 - sodium acetate and hydrochloric acid in water
 - ammonia and ammonium chloride in water
 - ammonia and sodium hydroxide in water.
- The pH of 0.1 M solution of the following salts increases in the order **[JEE 1999]**
 - $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$
 - $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$
 - $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$
 - $\text{HCl} < \text{NaCl} < \text{NaCN} < \text{NH}_4\text{Cl}$
- The average concentration of SO_2 in the atmosphere over a city on a certain day is 10 ppm, when the average temperature is 298 K. Given that the solubility of SO_2 in water at 298 K is $1.3653 \text{ moles litre}^{-1}$ and the pK_a of H_2SO_3 is 1.92, estimate the pH of rain on that day. **[JEE 2000]**
- An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is **[JEE 2001]**
 - 40 mL
 - 20 mL
 - 10 mL
 - 4 mL
- For sparingly soluble salt ApBq , the relationship of its solubility product (L_s) with its solubility (S) is: **[JEE 2001]**
 - $L_s = S^{p+q}, p^p \cdot q^q$
 - $L_s = S^{p+q}, p^p \cdot q^p$
 - $L_s = S^{pq}, p^p \cdot q^q$
 - $L_s = S^{pq}, (p \cdot q)^{p+q}$
- 500 ml of 0.2 M aqueous solution of acetic acid is mixed with 500 mL of 0.2 M HCl at 25°C .
 - Calculate the degree of dissociation of acetic acid in the resulting solution and pH of the solution.
 - If 6 g of NaOH is added to the above solution, determine final pH. Assume there is no change in volume on mixing. K_a of acetic acid is $1.75 \times 10^{-5} \text{ M}$. **[JEE 2002]**

9. Will the pH of water be same at 4°C and 25°C ? Explain. [JEE 2003]
10. A solution which is 10^{-3} M each in Mn^{2+} , Fe^{2+} , Zn^{2+} and Hg^{2+} is treated with 10^{-16} M sulphide ion. If K_{sp} , MnS , FeS , ZnS and HgS are 10^{-15} , 10^{-23} , 10^{-20} and 10^{-54} respectively, which one will precipitate first ? [JEE 2003]
 (A) FeS (B) MnS (C) HgS (D) ZnS
11. HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is [JEE 2004]
 (A) 0.01% (B) 0.0001% (C) 0.1% (D) 0.5%
12. 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point. Given $K_a(\text{HA}) = 5 \times 10^{-6}$ and $h \ll 1$. [JEE 2004]
13. CH_3NH_2 (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is [JEE 2005]
 (A) 1.6×10^{-11} (B) 8×10^{-11} (C) 5×10^{-5} (D) 2×10^{-2}
14. The species present in solution when CO_2 is dissolved in water : [JEE 2006]
 (A) CO_2 , H_2CO_3 , HCO_3^{2-} (B) H_2CO_3 , CO_3^{2-}
 (C) CO_3^{2-} , HCO_3^- (D) CO_2 , H_2CO_3
15. Solubility product constants (K_{SP}) of salts of types MX, MX_2 and M_3X at temperature 'T' are 4.0×10^{-8} , 3.2×10^{-14} and 2.7×10^{-15} , respectively. Solubilities (mol dm^{-3}) of the salts at temperature 'T' are in the order : [JEE 2008]
 (A) $\text{MX} > \text{MX}_2 > \text{M}_3\text{X}$ (B) $\text{M}_3\text{X} > \text{MX}_2 > \text{MX}$
 (C) $\text{MX}_2 > \text{M}_3\text{X} > \text{MX}$ (D) $\text{MX} > \text{M}_3\text{X} > \text{MX}_2$
16. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{5}$ M HCl in water at 25°C. The concentration of H^+ at equivalence point is [JEE 2008]
 ($K_w = 1 \times 10^{-14}$ at 25°C)
 (A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

17. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4} . The pH of a 0.01 M solution of its sodium salt is [JEE 2009]
18. Aqueous solutions of HNO_3 , KOH , CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) – [JEE -2010]
 (A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa
 (C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa
19. In 1 L saturated solution of AgCl [$K_{\text{sp}}(\text{AgCl}) = 1.6 \times 10^{-10}$], 0.1 mol of CuCl [$K_{\text{sp}}(\text{CuCl}) = 1.0 \times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x} . The value of 'x' is. [JEE -2011]
20. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol/L) of Ag_2CrO_4 in a 0.1 M AgNO_3 solution is [JEE 2013]
 (A) 1.1×10^{-11} (B) 1.1×10^{-10} (C) 1.1×10^{-12} (D) 1.1×10^{-9}
21. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA , 1M) is $1/100^{\text{th}}$ of that of a strong acid (HX , 1M), at 25°C . The K_a of HA is [JEE 2013]
 (A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

Paragraph For Questions 22 and 23

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7°C was measured for the beaker and its contents. (Expt-1). Because the enthalpy of neutralisation of a strong acid with a strong base is a constant ($-57.0 \text{ kJ mol}^{-1}$), this experiment could be used to measure the calorimeter constant. In a second experiment (Expt-2), 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0M NaOH (under identical conditions to (Expt-1)) where a temperature rise of 5.6°C was measured.

(Consider heat capacity of all solutions as $4.2 \text{ J g}^{-1}\text{K}^{-1}$ and density of all solutions as 1.0 g mL^{-1})

22. Enthalpy of dissociation (in kJ mol^{-1}) of acetic acid obtained from the Expt-2 is [JEE 2015]
 (A) 1.0 (B) 10.0 (C) 24.5 (D) 51.4
23. The pH of the solution after Expt-2 [JEE 2015]
 (A) 2.8 (B) 4.7 (C) 5.0 (D) 7.0
24. The solubility of a salt of weak acid (**AB**) at pH 3 is $Y \times 10^{-3} \text{ mol L}^{-1}$. The value of **Y** is _____. (Given that the value of solubility product of **AB** (K_{sp}) = 2×10^{-10} and the value of ionization constant of **HB** (K_a) = 1×10^{-8}) [JEE 2018]

25. Dilution processes of different aqueous solutions, with water, are given in LIST-I. The effects of dilution of the solutions on $[H^+]$ are given in LIST-II.

(Note: Degree of dissociation (α) of weak acid and weak base is $\ll 1$; degree of hydrolysis of salt $\ll 1$; $[H^+]$ represents the concentration of H^+ ions)

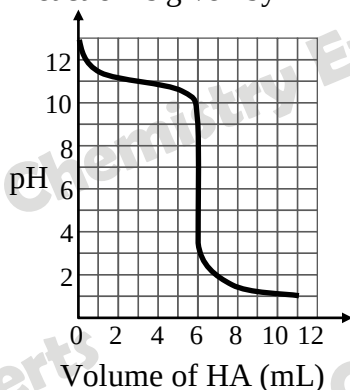
LIST-I**LIST-II****[JEE 2018]**

- | | |
|--|---|
| P. (10 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 60 mL | 1. The value of $[H^+]$ does not change on dilution |
| Q. (20 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 80 mL | 2. The value of $[H^+]$ changes to half of its initial value on dilution |
| R. (20 mL of 0.1 M HCl + 20 mL of 0.1 M ammonia solution) diluted to 80 mL | 3. The value of $[H^+]$ changes to two times of its initial value on dilution |
| S. 10 mL saturated solution of $Ni(OH)_2$ in equilibrium with excess solid $Ni(OH)_2$ is diluted to 20 mL (solid $Ni(OH)_2$ is still present after dilution). | 4. The value of $[H^+]$ changes to $\frac{1}{\sqrt{2}}$ times of its initial value on dilution |
| | 5. The value of $[H^+]$ changes to $\sqrt{2}$ times of its initial value on dilution |

Match each process given in LIST-I with one or more effect(s) in LIST-II. The correct option is

- (A) $P \rightarrow 4$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 1$ (B) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 3$
 (C) $P \rightarrow 1$; $Q \rightarrow 4$; $R \rightarrow 5$; $S \rightarrow 3$ (D) $P \rightarrow 1$; $Q \rightarrow 5$; $R \rightarrow 4$; $S \rightarrow 1$

26. A solution of 0.1 M weak base (B) is titrated with 0.1 M of a strong acid (HA). The variation of pH of the solution with the volume of HA added is shown in the figure below. What is the pK_b of the base? The neutralization reaction is given by $B + HA \rightarrow BH^+ + A^-$. **[JEE-2020]**



27. An acidified solution of 0.05 M Zn^{2+} is saturated with 0.1 M H_2S . What is the minimum molar concentration (M) of H^+ required to prevent the precipitation of ZnS ? **[JEE-2020]**

Use $K_{sp}(ZnS) = 1.25 \times 10^{-22}$ and

Overall dissociation constant of H_2S , $K_{NET} = K_1 K_2 = 1 \times 10^{-21}$

ANSWER KEY**EXERCISE # JEE-MAINS**

1. 3 2. 3 3. 4 4. 4 5. 3 6. 1 7. 1
 8. 4 9. 4 10. 2 11. 3 12. 4 13. 3 14. 4
 15. 3 16. 1 17. 3 18. 1 19. 4 20. 4 21. 1
 22. 4 23. 1 24. 1 25. 4 26. 4 27. 3 28. 2
 29. 4 30. 4 31. 3 32. 3 33. 2 34. 1 35. 3
 36. 3 37. 1 38. 1 39. 2 40. 1 41. 1 42. 2
 43. 2 44. 4 45. [BONUS] NTA Ans. (4) 46. 4 47. 2
 48. [10.60] NTA Ans. (10.6) 49. [5.23] NTA Ans. (5.22 to 5.24) 50. 2
 51. 4 52. 2 53. 4 54. 4 55. 37 56. 2.00

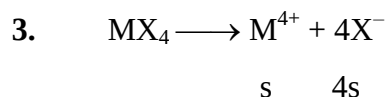
EXERCISE # JEE-ADVANCED

1. pH = 11.3010 2. $s = 1.203 \times 10^{-3} \text{M}$ 3. ABC 4. B 5. pH = 4.86
 6. A 7. A 8. a 0.0175%, b 4.757 9. No it will be > 7
 10. C 11. A 12. pH = 9 13. B 14. A 15. D 16. D
 17. 8 18. CD 19. 7 20. A 21. A 22. A 23. B
 24. 4.47 25. D 26. 3.00 27. 0.20

HINT AND SOLUTIONS
EXERCISE # JEE-MAIN

1. $K_{sp} = 4s^3 = 4 \times (1.0 \times 10^{-5})^3 = 4 \times 10^{-15}$

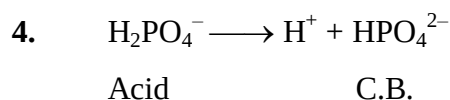
2. $K_{sp} = 4s^3$
 $= 4x^3$



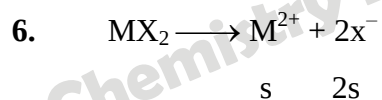
$$K_{sp} = s \times (4s)^4$$

$$= 256 s^5$$

$$s = \left(\frac{K_{sp}}{256} \right)^{1/5}$$



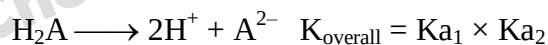
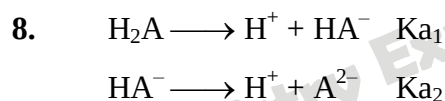
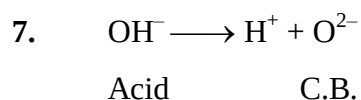
5. $pH = 5.4$
 $[H^+] = 10^{-5.4} = 10^{0.6} \times 10^{-6} = 4 \times 10^{-6}$



$$K_{sp} = 4s^3 = 4 \times 10^{-12}$$

$$s = 10^{-3} \text{ M}$$

$$[M^{2+}] = 10^{-3} \text{ M}$$



9.
$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{pK}_a = 4.5 \text{ (if acid is half ionize)}$$

$$\text{pOH} = 9.5$$

10.
$$K_{\text{sp}} = s^2 = 1.0 \times 10^{-8}$$

$$s = 0^{-4} \text{ M}$$

$$= 10^{-6} \times 283 \text{ g/L}$$

$$= 283 \times 10^{-5} \text{ g/100 mL}$$

$$= 2.83 \times 10^{-3} \text{ g/100 mL}$$

11.
$$\text{pH} = \frac{1}{2} (\text{pK}_w + \text{pK}_a - \text{pK}_b) = \frac{1}{2} (14 + 4.8 - 4.78) = \frac{1}{2} \times 14.02 = 7.01$$

12. For pptⁿ to start

$$K_{\text{sp}} = Q_{\text{ip}}$$

$$[\text{Ba}^{2+}] \times 10^{-4} = 5.1 \times 10^{-9}$$

$$[\text{Ba}^{2+}] = 5.1 \times 10^{-5} \text{ M}$$

13. For pptⁿ to start

$$K_{\text{sp}} = Q_{\text{ip}}$$

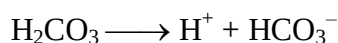
$$0.05 \times [\text{Br}^-] = 5 \times 10^{-13}$$

$$[\text{Br}^-] = 10^{-11} \text{ M}$$

$$\text{moles of Br}^- = 1 \times 10^{-11}$$

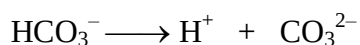
$$\text{mass of KBr} = 1 \times 10^{-11} \times 120 = 1.2 \times 10^{-9} \text{ g}$$

14.
$$K_1 \gg K_2$$



$$c - x \qquad x \qquad x - y$$

$$\approx x$$



$$x - y \qquad x + y \qquad y$$

$$\approx x \qquad \approx x$$

$$[\text{H}^+] \approx [\text{HCO}_3^-] = 1.2 \times 10^{-6} \text{ M}$$

$$[\text{CO}_3^{2-}] = K_{a2} = 4.8 \times 10^{-11} \text{ M}$$

15. $K_{sp} = Q_{ip}$

$$10^{-3} \times [\text{OH}^-]^2 = 1.1 \times 10^{-11}$$

$$[\text{OH}^-]_{\min} = 10^{-4}$$

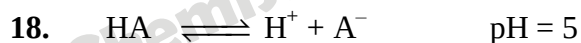
$$\text{pOH} = 4 \Rightarrow \text{pH} = 10$$

16. In reaction (ii) $\text{H}_2\text{PO}_4^{2-}$ given H^+ ion.

17. $K_{sp} = 27s^3$

$$27s^3 = 1.6 \times 10^{-30}$$

$$s = \left(\frac{1.6 \times 10^{-30}}{27} \right)^{1/3}$$



$$1 - x \quad \quad x \quad \quad x$$

$$[\text{H}^+] = 10^{-5}$$

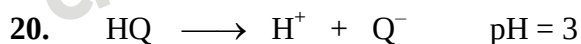
$$K_a = \frac{10^{-5} \times 10^{-5}}{1} = 10^{-10}$$

19. If $K_{sp} < Q_{ip}$

$$K_{sp} = 1.7 \times 10^{-10}$$

For option (4)

$$Q_{ip} = 1 \times 10^{-2} \times (1 \times 10^{-3})^2 = 10^{-8}$$



$$0.1 - x \quad \quad x \quad \quad x$$

$$[\text{H}^+] = 10^{-3}$$

$$K_a = \frac{10^{-3} \times 10^{-3}}{0.1} = 10^{-5}$$

21. For pptⁿ to start

$$K_{sp} = Q_{ip}$$

$$5.1 \times 10^{-9} = [\text{Ba}^{2+}] \times [\text{CO}_3^{2-}]$$

$$[\text{Ba}^{2+}] \times 10^{-4} = 5.1 \times 10^{-9}$$

$$[\text{Ba}^{2+}] = 5.1 \times 10^{-5} \text{ M}$$

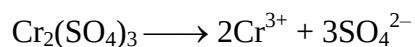
22. $[\text{OH}^-] = 5 \times 10^{-2}$

$$\text{pOH} = 2 - \log 5$$

$$\Rightarrow \text{pH} = 12 + \log 5 = 12.7$$



$$K_{\text{sp}} = 4s^3 \Rightarrow s = \left(\frac{K_{\text{sp}}}{4} \right)^{1/3}$$



$$K_{\text{sp}} = 108s^5 \Rightarrow s = \left(\frac{K_{\text{sp}}}{108} \right)^{1/5}$$

For BaSO_4 $K_{\text{sp}} = s^2 \Rightarrow s = (K_{\text{sp}})^{1/2}$

For CrCl_3 $K_{\text{sp}} = 27s^4 \Rightarrow s = \left(\frac{K_{\text{sp}}}{27} \right)^{1/4}$

24. Moles of $\text{CH}_3\text{COOH} = \frac{5}{60} = 0.083$

Moles of $\text{CH}_3\text{COONa} = \frac{7.5}{62} = 0.091$

$$\text{pH} = 4.76 + \log \frac{0.091}{0.083}$$

$$\text{pH} > 4.76$$

$$\text{Range of pH} \Rightarrow \text{pKa} \pm 1$$

25. $\text{pH}_i = 1$ $[\text{H}^+]_i = 0.1$

$$\text{pH}_f = 2$$
 $[\text{H}^+]_f = 0.01$

$$M_i V_i = M_f V_f$$

$$0.1 \times 1 = 0.01 \times V_f \Rightarrow V_f = 10 \text{ L}$$

$$\text{water should be added} = 9 \text{ L}$$

26. Refer notes.



3s 4s

$$K_{sp} = (3s)^3 \times (4s)^4 = 6912 s^7$$

$$s = \left(\frac{K_{sp}}{6912} \right)^{1/7}$$

28. $\text{pH} = \frac{1}{2} (\text{pK}_w + \text{pK}_a - \text{pK}_b)$

$$= \frac{1}{2} (14 + 3.2 - 3.4)$$

$$= \frac{1}{2} \times 13.8 = 6.9$$

29. pH range of methy orange = 3.1 – 4.4

in titration of W.A. Vs SA at end point pH should be less than 7.

30. pptⁿ to start

$$K_{sp} = Q_{ip}$$

$$[\text{Ba}^{2+}]_f [\text{SO}_4^{2-}]_f = K_{sp} \text{ BaSO}_4$$

$$\frac{50 \times 1}{500} \times \frac{M_i \times 450}{500} = 1 \times 10^{-10}$$

$$M_i = 1.1 \times 10^{-9} \text{ M}$$



$$1.2 \times 10^{-20} = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]}$$

$$1.2 \times 10^{-20} = \frac{(0.2)^2 [\text{S}^{2-}]}{0.1}$$

$$[\text{S}^{2-}] = 3 \times 10^{-20} \text{ M}$$

32. CH_3COOK is salt of WA + S.B.

So $\text{pH} > 7$

33. $K_{\text{sp}} \text{PbCl}_2 = 3.2 \times 10^{-8} = 4s^3$

$$s = 2 \times 10^{-3} \text{ M} = 0.414 \text{ g/L}$$

$$\text{Vol. of water require to dissolve 1g of PbCl}_2 = \frac{0.1}{0.414} = 0.241 \text{ L}$$

34. (i) M Mol of H^+ = 15

M Mol of OH^- = 5

$$[\text{H}^+]_f = \frac{10}{100} = 0.1 \Rightarrow \text{pH} = 1$$

35. $2\text{NH}_4\text{OH} + \text{H}_2\text{SO}_4 \longrightarrow (\text{NH}_4)_2\text{SO}_4 + \text{H}_2\text{O}$

6 2 0 -

2 0 2 -

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]} = 4.7 + \log \frac{4}{2} = 4.7 + \log 2$$

$$\text{pH} = 9.3 - \log 2 = 9$$

36. $\text{Ca}(\text{OH})_2 + \text{Na}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{NaOH}$

0.1 m mole 0.014 m mole 0 0

0.086 m mole 0 0.014 m mol 0.014 m mole

$$\text{Mass of CaSO}_4 = 0.014 \times 136 = 1.90 \text{ g}$$

$$[\text{OH}^-] = \frac{0.014}{0.2} = 0.28 \text{ M}$$

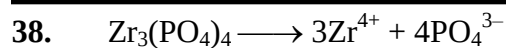
37. $\text{Ag}_2\text{CO}_3 \rightleftharpoons 2\text{Ag}^+ + \text{CO}_3^{2-}$

s_1 $2s_1 + 0.1$ s_1

$$\approx 0.1$$

$$8 \times 10^{-12} = (0.1)^2 \times s_1$$

$$s_1 = 8 \times 10^{-10} \text{ M}$$



$$K_{\text{sp}} = 6912 \text{ s}^7$$

$$s = \left(\frac{K_{\text{sp}}}{6912} \right)^{1/7}$$

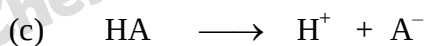
39. Initial pH > 7, at equivalence point pH = 7

40. (a) $[\text{H}^+] = \frac{400 \times 0.1 \times 2 - 400 \times 0.1}{800} = 0.05 \text{ M}$

$$\text{pH} = 1.3$$

(b) $K_{\text{w}} = [\text{H}^+][\text{OH}^-]$

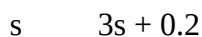
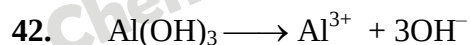
$$T \uparrow, K_{\text{w}} \uparrow$$



$$K_{\text{a}} = \frac{c\alpha^2}{1 - \alpha} \Rightarrow 10^{-5} = 10^{-5} \times \frac{\alpha^2}{1 - \alpha} \Rightarrow \alpha = \frac{1}{2}$$

(d) Le-Chatelier principle is applicable to common ion effect.

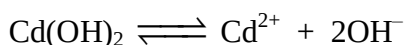
41. $\text{pH} = \frac{1}{2} (\text{p}K_{\text{w}} - \text{p}K_{\text{b}} - \log c) = \frac{1}{2} (14 - 5 - \log 2 \times 10^{-2}) = 5.35$



$$\approx 0.2$$

$$(0.2)^3 \times s = 2.4 \times 10^{-24} \Rightarrow s = 3 \times 10^{-22} \text{ M}$$

43. $K_{\text{sp}} = 4s^3 = 4 \times (1.84 \times 10^{-5})^3$



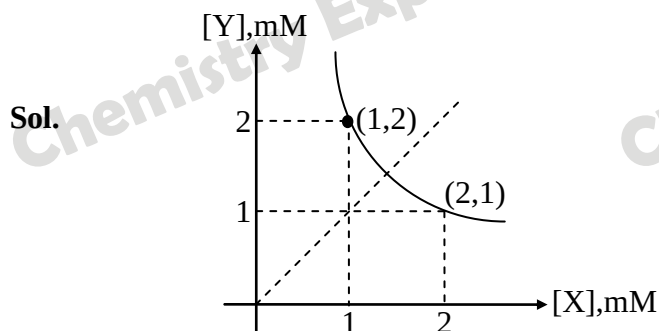
$$\approx 0.01$$

$$s \times (0.01)^2 = 4 \times (1.84 \times 10^{-5})^3$$

$$s = 2.49 \times 10^{-10} \text{ M}$$

44. (4)

45. [BONUS] NTA Ans. (4)



Since both (1, 2) and (2, 1) are present on the curve which is possible only when we consider salt to be XY.

$$\text{So, } K_{sp} = [X][Y] = 10^{-3} \times 2 \times 10^{-3} = 2 \times 10^{-6} \text{ M}^2$$

but option has $2 \times 10^{-6} \text{ M}^3$ which is wrong dimensionally so either the answer should be : XY, $2 \times 10^{-6} \text{ M}^2$ or question should be BONUS

46. (4)

Sol. Concentration of Pb^{2+} in resulting solution = $\frac{0.134}{400} \times 300 = 0.1005 \text{ M}$

Concentration of Cl^- in resulting solution = $\frac{0.4}{400} \times 100 = 0.1 \text{ M}$

$$Q = (0.1005)(0.1)^2 = 0.1005 \times 10^{-2} > K_{sp}$$

47. (2)

Sol. $K_{sp} = s(3s)^3$

$$27s^4 = 6 \times 10^{-31}$$

$$s = \left(\frac{6}{27} \times 10^{-31} \right)^{\frac{1}{4}}$$

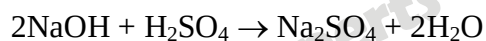
$$s = (0.222 \times 10^{-31})^{1/4}$$

$$3s = (81 \times 0.222 \times 10^{-31})^{1/4} = (18 \times 10^{-31})^{1/4}$$

48. [10.60] NTA Ans. (10.6)

Sol. Molarity of NaOH solution = $\frac{4}{40 \times 100} = .001 \text{ M}$

$$\text{Molarity of H}_2\text{SO}_4 \text{ solution} = \frac{9.8}{98 \times 100} = .001\text{M}$$



0.04 mole 0.01 mole

0.02 mole 0

$$\text{Molarity of NaOH in resulting solution} = \frac{0.02}{50} = 0.004\text{M}$$

$$\text{pH} = 14 + \log (0.004) = 10.60$$

49. [5.23] NTA Ans. (5.22 to 5.24)

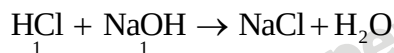
Sol. mmols of HCl in 250 ml solution = $0.1 \times 250 = 25$

$$\text{mmols of CH}_3\text{COOH in 250 ml solution} = \frac{3}{60} \times 1000 = 50$$

$$\text{mmols of HCl in 20 ml solution} = \frac{25}{500} \times 20 = 1$$

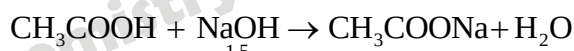
$$\text{mmols of CH}_3\text{COOH in 20 ml solution} = \frac{50}{500} \times 20 = 2$$

After adding $\frac{1}{2}$ ml, of 5M NaOH



1 1

0 0 1



0.5

0

1.5

$$\text{pH} = \text{pK}_a + \log \frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]}$$

$$= 4.75 + \log \frac{1.5}{0.5}$$

$$= 4.75 + \log 3$$

$$= 4.75 - 0.4771 = 5.23$$

50. (2)



$$K_{eq} = \frac{1}{K_{sp}}$$

If K_{sp} is low then K_{eq} will be high but K_{eq} decide only extent of reaction not the rate of reaction.

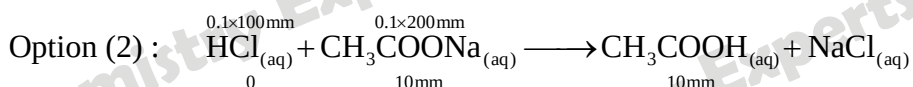
51. (4)

Sol. Titration of HCl with NaOH is acid-base titration and no busen burner and measuring cylinder is required for this.

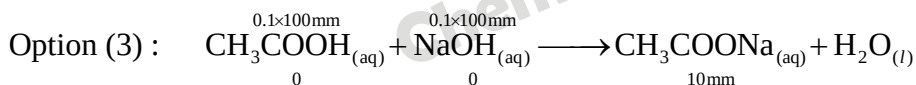
52. (2)

Sol. Acidic buffer solution contains weak acid and its salt with strong base.

Option (1) : Solution is containing strong acid and its salt with strong base.



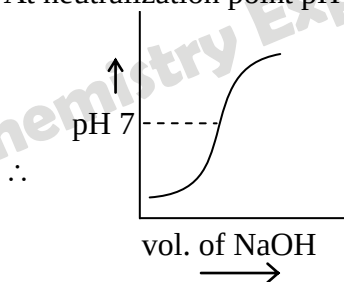
Resulting solution contains weak acid and its salt with strong base.



Resulting solution contains only salt of weak acid and strong base.

53. (4)

Sol. At neutralization point $\text{pH} = 7$ because acid and base both are strong.



54. (4)

Sol. pH of 0.01 M HCl = 2

pH of 0.01 M NaOH = 12

pH of 0.01 M CH_3COONa > 7

pH of 0.01 M NaCl = 7

order of pOH : A > D > C > B

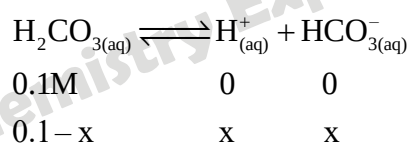
55. (37)

Sol. Using Henry's law

$$k_H \times \frac{1}{1000} = 30$$

$$k_H \times \frac{n_{\text{CO}_2}}{1000} = 3$$

$$n_{\text{CO}_2} = 0.1$$

Molarity of $\text{CO}_2 = 0.1 \text{ M}$ $\therefore \text{CO}_2$ dissolved remains in the form of H_2CO_3 

$$K_a = 4 \times 10^{-7} = \frac{x^2}{(0.1 - x)}$$

$$\therefore x^2 = 0.1 \times 4 \times 10^{-7}$$

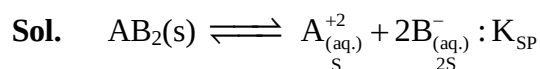
$$[\because x \ll 1]$$

$$x = 2 \times 10^{-4}$$

$$\therefore \text{pH} = -\log(2 \times 10^{-4})$$

$$= 3.7$$

56. (2.00)



$$K_{\text{SP}} = S^1 \times (2S)^2 = 4S^3$$

$$3.2 \times 10^{-11} = 4 \times S^3$$

$$S = 2 \times 10^{-4} \text{ M/L}$$

EXERCISE # JEE-ADVANCED



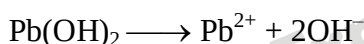
$$\begin{array}{cccc} 2\text{m mol} & 3\text{mmol} & 10 & - \\ 0 & 1\text{mmol} & 2\text{m mol} & - \end{array}$$

$$[\text{OH}^-] = \frac{1}{500} = 2 \times 10^{-3}$$

$$\text{pH} = 11 + \log 2 = 11.3$$

2. $\text{pH} = 8 \Rightarrow \text{pOH} = 6$

$$[\text{OH}^-] = 10^{-6} \text{ M}$$



$$K_{\text{sp}} = 4s^2 = 4 \times (6.7)^3 \times 10^{-18} = 1.2 \times 10^{-15}$$



$$\begin{array}{ccc} s & & 2s + 10^{-6} \end{array}$$

$$s(2s + 10^{-6}) = 1.2 \times 10^{-15}$$

$$s \times 10^{-2} = 1.2 \times 10^{-15}$$

$$s = 1.2 \times 10^{-3} \text{ M}$$

4. $\text{HCl} \Rightarrow \text{pH} < 7$

$$\text{NaCl} \Rightarrow \text{pH} = 7$$

$$\text{NaCN} \Rightarrow \text{pH} > 7$$

$$\text{NH}_4\text{Cl} \Rightarrow \text{pH} < 7$$

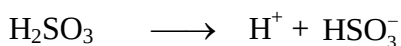
$$\text{NaCN} < \text{NaCl} > \text{NH}_4\text{Cl} > \text{HCl}$$

5. Amount of SO_2 in atmosphere = $\frac{10}{10^6} = 10 \times 10^{-6}$

$$\text{Molar conc. of } \text{SO}_2 \text{ in water} = \text{Amount of } \text{SO}_2 \times \text{Solubility of } \text{SO}_2$$

$$= 10 \times 10^{-6} \times 1.3653$$

$$= 1.3653 \times 10^{-5} \text{ M}$$



$$\begin{array}{ccc} 1.3653 \times 10^{-5} - x & x & x \end{array}$$

$$K_a = \frac{x^2}{1.3653 \times 10^{-5} - x} = 10^{-1.92}$$

$$x = 1.364 \times 10^{-5} \quad \text{pH} = 4.865$$

6. Equation of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ = Equation of NaOH

$$\frac{10}{250} \times \frac{63}{126} \times 2 = 0.1 \times V(\text{L})$$

$$V(\text{L}) = \frac{1}{25}$$

$$V(\text{mL}) = 40$$

7. $\text{A}_p\text{B}_q(\text{s}) \longrightarrow p\text{A}^{q+}(\text{aq}) + q\text{B}^{p-}(\text{aq})$



$$K_{\text{sp}} = (\text{ps})^p (\text{qs})^q$$

$$= p^p q^q (\text{s})^{p+q}$$

8. $[\text{CH}_3\text{COOH}]_{\text{f}} = 0.1 \text{ M}$

$$[\text{HCl}]_{\text{f}} = 0.1 \text{ M}$$

- (a) $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$

$$\begin{array}{ccc} 0.1-x & x & x+0.1 \end{array}$$

$$1.75 \times 10^{-5} = \frac{x(x+0.1)}{0.1-x}$$

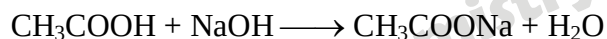
$$x = 1.75 \times 10^{-5}$$

$$0.1 \times \alpha = 1.75 \times 10^{-5}$$

$$\alpha = 1.75 \times 10^{-4}$$

$$[\text{H}^+] = 0.1 \Rightarrow \text{pH} = 1$$

- (b) Moles of $\text{NaOH} = \frac{6}{40} = 0.15 \text{ mol}$



$$n_i \quad 0.1 \quad 0.05 \quad 0 \quad -$$

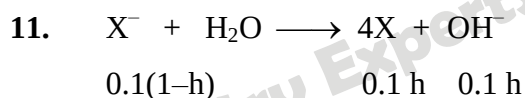
$$n_f \quad 0.05 \quad 0 \quad 0.05 \quad -$$

$\therefore$ 0.1 moles NaOH will be consumed by 0.1 mol of HCl

$$\text{pH} = \text{pK}_a = 4.75$$

9. $T \uparrow, K_w \uparrow, \text{H}^+ \uparrow, \text{pH} \downarrow$

10. Which has minimum value of K_{sp} i.e. HgS



$$K_h = \frac{K_w}{K_a(HX)} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$$

$$K_h = \frac{Ch^2}{1-h} = \frac{0.1h^2}{1-h} = 10^{-9}$$

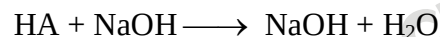
$$0.1h^2 = 10^{-9}$$

$$h^2 = 10^{-8}$$

$$h = 10^{-4}$$

$$\% h = 0.01 \%$$

12. At end point HA will completely reacts with NaOH



$$[A^-] = 0.05 \text{ M}$$

$$pH = \frac{1}{2} (pK_w + pK_a + \log C)$$

$$= \frac{1}{2} (14 + 6 - \log 5 + \log 0.05)$$

$$= \frac{1}{2} (20 - \log 5 - \log 20)$$

$$\Rightarrow pH = 9$$

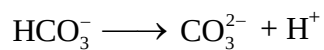
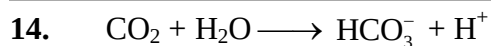


0.1 mol	0.08 mol	0
0.02 mol	0	0.05 mol

$$[OH^-] = K_b \times \frac{[CH_3NH_2]}{[CH_3NH_3^+]}$$

$$= 5 \times 10^{-4} \times \frac{0.02}{0.08} = \frac{5}{4} \times 10^{-4}$$

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{4}{5} \times 10^{-10} = 8 \times 10^{-11} \text{ M}$$



15. For MX

$$K_{\text{sp}} = s^2 = 4 \times 10^{-8}$$

$$s = 2 \times 10^{-4} \text{ M}$$

For MX_2

$$K_{\text{sp}} = 4s^3 = 32 \times 10^{-14}$$

$$s^3 = 8 \times 10^{-15}$$

$$s = 2 \times 10^{-5} \text{ M}$$

For M_3X

$$K_{\text{sp}} = 27s^4 = 2.7 \times 10^{-15}$$

$$s = 10^{-14} \text{ M}$$

$$\text{MX} > \text{M}_3\text{X} > \text{MX}_2$$



It equivalence point

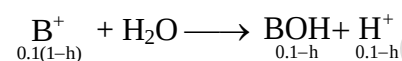
Moles of BOH = moles of HCl

$$2.5 \times \frac{2}{5} = \frac{2}{15} \times V(\text{mL})$$

$$V(\text{mL}) = 7.5 \text{ mL}$$

$$V_f = 10 \text{ mL}$$

$$[\text{BCl}] = \frac{1}{10} = 0.1 \text{ M}$$



$$K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{10^{-12}} = 10^{-2}$$

$$10^{-2} = \frac{0.1-h^2}{1-h}$$

$$h = 0.27$$

$$\Rightarrow [\text{H}^+] = 0.1 \times h = 0.027 \text{ M}$$

17.
$$\text{pH} = \frac{1}{2} (\text{pK}_w + \text{pK}_a + \log C) = \frac{1}{2} (14 + 4 + \log 0.01) = \frac{1}{2} \times 16 = 8$$

18. (A) $\text{HNO}_3 + \text{CH}_3\text{COOH}$ (S.A. + W.A.)

It can't be a buffer

(B) $\text{KOH} + \text{CH}_3\text{COONa}$ (S.B. + Salt)

It can't be a buffer

(C) $\text{CH}_3\text{COONa} + \text{HNO}_3 \longrightarrow \text{CH}_3\text{COOH} + \text{NaNO}_3$

It can form buffer if volume of CH_3COONa and HNO_3 is different.

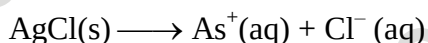
(D) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$

It is a buffer

19. For CuCl

$$K_{sp} = s^2 = 10^{-6}$$

$$s = 10^{-3}$$



$$s \qquad s + 10^{-3}$$

$$s(s + 10^{-3}) = 1.6 \times 10^{-10}$$

$$s \times 10^{-3} = 1.6 \times 10^{-10}$$

$$s = 1.6 \times 10^{-7}$$

20. For Ag_2CrO_4 $K_{sp} = 1.1 \times 10^{-12}$



$$2s + 0.1 \qquad s$$

$$\approx 0.1$$

$$(0.1)^2 \times s = 1.1 \times 10^{-12}$$

$$s = 1.1 \times 10^{-10} \text{ M}$$

21. $r \propto [\text{H}^+]^1$

$$\frac{\sqrt{K_a \times C}}{1} = \frac{1}{100}$$

$$\sqrt{K_a \times C} = 10^{-2}$$

$$K_a = 10^{-4}$$

22. Form exp.-1

$$5.7 \times 1000 = C \times 5.7$$

Form exp-2

$$q = C \times 5.6$$

$$q = 5.65 \text{ kJ}$$

$$\Delta H = -\frac{5.6}{0.1} = -56 \text{ kJ. / mol}$$

$$\Delta H_{\text{ion}} = -56 + 57 = 1 \text{ kJ/ mol}$$

23. $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

$$0.2 \text{ mol} \quad 0.1 \text{ mol} \quad 0 \quad 0$$

$$0.1 \text{ mol} \quad 0 \quad 0.1 \text{ mol} \quad 0$$

$$\text{pH} - \text{pK}_a = 4.7$$

$$\begin{aligned} 24. \quad s &= \sqrt{K_{\text{sp}} \left(\frac{H^+}{K_a} + 1 \right)} \\ &= \sqrt{20 \times 10^{-10} \left(\frac{10^{-3}}{10^{-8}} + 1 \right)} \\ &= \sqrt{2 \times 10^{-5}} = 4.47 \times 10^{-3} \text{ M} \end{aligned}$$

25. (P) $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

$$2\text{m mol} \quad 1\text{m mol} \quad 0 \quad -$$

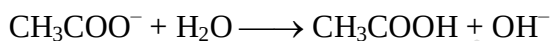
$$1\text{m mol} \quad 0 \quad 1\text{m mol} \quad -$$

If forms a buffer and pH does not change with dilution.

(Q) $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

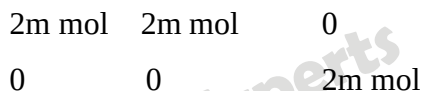
$$2\text{m mol} \quad 2\text{m mol} \quad 0 \quad -$$

$$0 \quad 0 \quad 2\text{m mol} \quad -$$

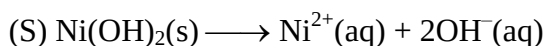


$$CH^+ = \sqrt{\frac{K_w \times K_a}{C}}$$

$$[H^+] \propto \frac{1}{\sqrt{C}}$$



$$[\text{H}^+] = \sqrt{\frac{K_w}{K_b} \times C} \Rightarrow [\text{H}^+] \propto \sqrt{C}$$



On dilution $[\text{OH}^-]$ remains same in saturated solution of $\text{Ni}(\text{OH})_2$

26. (3.00)

Sol. At 3 ml pH of solution is 11



at 3 ml best buffer action is shown

So $\boxed{\text{pOH} = \text{pK}_b = 3}$ (As basic buffer is formed)

So $\boxed{\text{P}_{\text{K}_b} = 3}$

27. (0.20)

Sol. For ZnS $K_{\text{sp}}(\text{ZnS}) = [\text{Zn}^{2+}] \times [\text{S}^{2-}] = 1.25 \times 10^{-22}$

$$= 0.05 \times [\text{S}^{2-}] = 1.25 \times 10^{-22}$$

$$[\text{S}^{2-}] = 25 \times 10^{-22}$$



$$0.1 \text{ M} \quad 25 \times 10^{-22}$$

$$K_a = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]} = 1 \times 10^{-21} = \frac{[\text{H}^+]^2 \times 25 \times 10^{-22}}{0.1} = 1 \times 10^{-21}$$

$$[\text{H}^+]^2 = \frac{1}{25}$$

$$[\text{H}^+] = \frac{1}{5}$$

$$[\text{H}^+] = 0.2 \text{ M}$$

EXERCISE # (JEE-MAIN)

1. Which one of the following aqueous solutions will exhibit highest boiling point ? [AIEEE-2004]
(1) 0.01M Na_2SO_4 (2) 0.01M KNO_3 (3) 0.015M urea (4) 0.015M glucose
2. Which of the following liquid pairs shows a positive deviation from Raoult's law ? [AIEEE-2004]
(1) Water-hydrochloric acid (2) Benzene-methanol
(3) Water-nitric acid (4) Acetone-chloroform
3. Which one of the following statement is False ? [AIEEE-2004]
(1) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction
(2) The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$ where M is the molarity of the solution
(3) The correct order of osmotic pressure for 0.01M aqueous solution of each compound is $\text{BaCl}_2 > \text{KCl} > \text{CH}_3\text{COOH} > \text{Sucrose}$
(4) Two sucrose solutions of same molality prepared in different solvent will have the same freezing point depression
4. If α is the degree of dissociation of Na_2SO_4 , the vant of Hoff's factor (i) used for calculating the molecular mass is [AIEEE-2005]
(1) $1 - \alpha$ (2) $1 + \alpha$ (3) $1 - 2\alpha$ (4) $1 + 2\alpha$
5. Benzene and toluene form nearly ideal solutions. At 20°C , the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20°C for a solution containing 78 g of benzene and 46 g of toluene in torr is - [AIEEE-2005]
(1) 25 (2) 50 (3) 53.5 (4) 37.5
6. Two solutions of a substance (non electrolyte) are mixed in the following manner. 480 ml of 1.5 M first solution + 520 mL of 1.2 M second solution. What is the molarity of the final mixture ? [AIEEE-2005]
(1) 1.50 M (2) 1.20 M (3) 2.70 M (4) 1.344 M
7. Equimolar solutions in the same solvent have [AIEEE-2006]
(1) Same freezing point but different boiling point
(2) Same boiling point but different freezing point
(3) Different boiling and different freezing point
(4) Same boiling and same freezing points

8. 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2g of water. The vapour pressure of water for this aqueous solution at $100^\circ C$ is [AIEEE-2006]
 (1) 7.60 Torr (2) 76.00 Torr (3) 752.40 Torr (4) 759.00 Torr
9. A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be [AIEEE-2007]
 (1) 350 (2) 300 (3) 700 (4) 360
10. A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60 g mol^{-1}) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 g cm^{-3} , molar mass of the substance will be [AIEEE-2007]
 (1) 90.0 g mol^{-1} (2) 115.0 g mol^{-1} (3) 105.0 g mol^{-1} (4) 210.0 g mol^{-1}
11. At $80^\circ C$, the vapour pressure of pure liquid 'A' is 520 mm Hg and that of pure liquid 'B' is 1000 mm Hg. If a mixture solution of 'A' and 'B' boils at $80^\circ C$ and 1 atm pressure, the amount of 'A' in the mixture is (1 atm = 760 mm Hg) [AIEEE-2008]
 (1) 52 mol % (2) 34 mol % (3) 48 mol % (4) 50 mol %
12. The vapour pressure of water at $20^\circ C$ is 17.5 mm Hg. If 18 g of glucose ($C_6H_{12}O_6$) is added to 178.2 g of water at $20^\circ C$, the vapour pressure of the resulting solution will be [AIEEE-2008]
 (1) 17.675 mm Hg (2) 15.750 mm Hg (3) 16.500 mm Hg (4) 17.325 mm Hg
13. Two liquids X and Y form an ideal solution. At 300K, vapour pressure of the solution containing 1 mol of X and 3 mol of Y is 550 mm Hg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. Vapour pressure (in mmHg) of X and Y in their pure states will be, respectively : [AIEEE-2009]
 (1) 400 and 600 (2) 500 and 600 (3) 200 and 300 (4) 300 and 400
14. A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution ? [AIEEE-2009]
 (1) The solution is non-ideal, showing -ve deviation from Raoult's law
 (2) n-heptane shows +ve deviation while ethanol shows -ve deviation from Raoult's law
 (3) The solution formed is an ideal solution.
 (4) The solution is non-ideal, showing +ve deviation from Raoult's law
15. If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is ($K_f = 1.86 \text{ K kg mol}^{-1}$) : [AIEEE-2010]
 (1) 0.0186 K (2) 0.0372 K (3) 0.0558 K (4) 0.0744 K

16. On mixing, heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components (heptane and octane) are 105 kPa and 45 kPa respectively. Vapour pressure of the solution obtained by mixing 25.0 of heptane and 35 g of octane will be (molar mass of heptane = 100 g mol^{-1} and of octane = 114 g mol^{-1}) : **[AIEEE-2010]**
 (1) 144.5 kPa (2) 72.0 kPa (3) 36.1 kPa (4) 96.2 kPa
17. The degree of dissociation (α) of a weak electrolyte, A_xB_y is related to van't Hoff factor (i) by the expression : **[AIEEE-2011]**
 (1) $\alpha = \frac{x+y-1}{i-1}$ (2) $\alpha = \frac{x+y+1}{i-1}$ (3) $\alpha = \frac{i-1}{(x+y-1)}$ (4) $\alpha = \frac{i-1}{(x+y+1)}$
18. Ethylene glycol is used as an antifreeze in a cold climate. Mass of ethylene glycol which should be added to 4 kg of water to prevent it from freezing at -6°C will be : **[AIEEE-2011]**
 (K_f for water = $1.86 \text{ K kg mol}^{-1}$, and molar mass of ethylene glycol = 62 g mol^{-1})
 (1) 400.00 g (2) 304.60 g (3) 804.32 g (4) 204.30 g
19. A 5% solution of cane sugar (molar mass 342) is isotonic with 1% of a solution of an unknown solute. The molar mass of unknown solute in g/mol is : **[AIEEE-2011]**
 (1) 136.2 (2) 171.2 (3) 68.4 (4) 34.2
20. The molality of a urea solution in which 0.0100g of urea, $[(\text{NH}_2)_2\text{CO}]$ is added to 0.3000 dm^3 of water at STP is : **[AIEEE-2011]**
 (1) 0.555 m (2) $5.55 \times 10^{-4} \text{ m}$ (3) 33.3 m (4) $3.33 \times 10^{-2} \text{ m}$
21. The freezing point of a 1.00 m aqueous solution of HF is found to be -1.91°C . The freezing point constant of water, K_f , is $1.86 \text{ K kg mol}^{-1}$. The percentage dissociation of HF at this concentration is **[JEE (MAIN)-2012 ONLINE]**
 (1) 2.7% (2) 30% (3) 10% (4) 5.2%
22. Liquids A and B form an ideal solution. At 30°C , the total vapour pressure of a solution containing 1 mol of A and 2 moles of B is 250 mm Hg. The total vapour pressure becomes 300 mm Hg when 1 more mol of A is added to the first solution. The vapour pressures of pure A and B at the same temperature are **[JEE (MAIN)-2012 ONLINE]**
 (1) 450, 150 mm Hg (2) 250, 300 mm Hg (3) 125, 150 mm Hg (4) 150, 450 mm Hg
23. A solution containing 0.85 g of ZnCl_2 in 125.0 g of water freezes at -0.23°C . The apparent degree of dissociation of the salt is : **[JEE (MAIN)-2012 ONLINE]**
 (k_f for water = $1.86 \text{ K kg mol}^{-1}$, atomic mass ; Zn = 65.3 and Cl = 35.5)
 (1) 1.36% (2) 2.47% (3) 73.5% (4) 7.35%
24. K_f for water is $1.86 \text{ K kg mol}^{-1}$. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must you add to get the freezing point of the solution lowered to -2.8°C ? **[AIEEE-2012]**
 (1) 27 g (2) 72 g (3) 93 g (4) 39 g
25. 12g of a nonvolatile solute dissolved in 108g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is : **[JEE (MAIN)-2013 ONLINE]**
 (1) 60 (2) 80 (3) 40 (4) 20

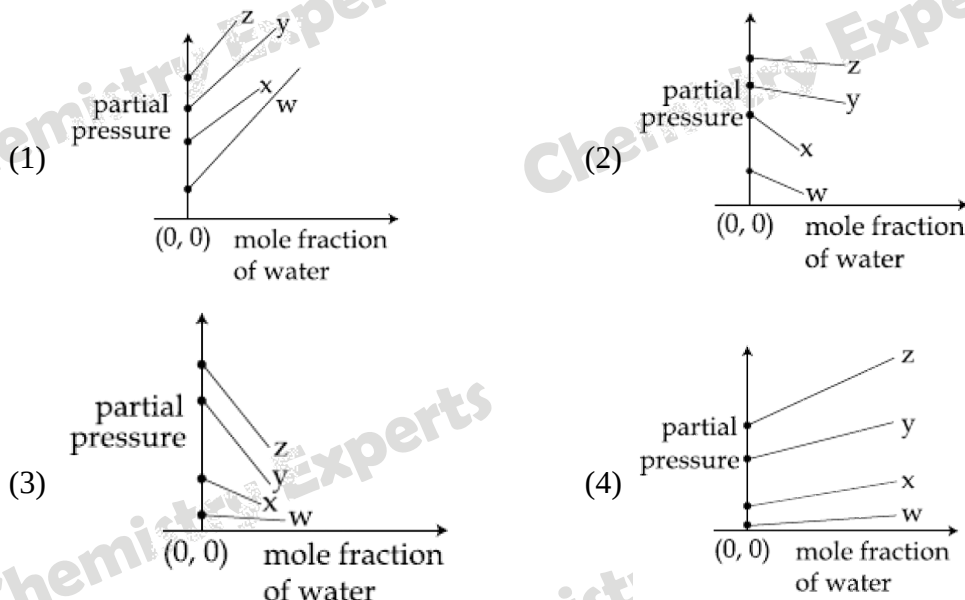
26. A molecule M associates in a given solvent according to the equation $M \longrightarrow (M)_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is : **[JEE (MAIN)-2013 ONLINE]**
 (1) 2 (2) 4 (3) 5 (4) 3
27. Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be : **[JEE (MAIN)-2013 ONLINE]**
 (1) 0.137 (2) 0.205 (3) 0.237 (4) 0.435
28. How many grams of methyl alcohol should be added to 10 litre tank of water to prevent its freezing at 268 K ? **[JEE (MAIN)-2013 ONLINE]**
 (K_f for water is $1.86 \text{ K kg mol}^{-1}$)
 (1) 899.04 g (2) 886.02 g (3) 868.06 g (4) 880.07 g
29. Consider separate solution of 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$, 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$, 0.250 M $\text{KBr}(\text{aq})$ and 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ at 25°C . Which statement is true about these solutions, assuming all salts to be strong electrolytes ? **[JEE-MAIN-2014]**
 (1) 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ has the highest osmotic pressure.
 (2) 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$ has the highest osmotic pressure.
 (3) They all have the same osmotic pressure.
 (4) 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$ has the highest osmotic pressure.
30. For an ideal Solution of two components A and B, which of the following is true ? **[JEE(MAIN)-2014 ONLINE]**
 (1) $\Delta H_{\text{mixing}} < 0$ (zero)
 (2) A – A, B – B and A – B interactions are identical
 (3) A – B interaction is stronger than A – A and B – B interactions
 (4) $\Delta H_{\text{mixing}} > 0$ (zero)
31. The observed osmotic pressure for a 0.10 M solution of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$ at 25°C is 10.8 atm. The expected and experimental (observed) values of Van't Hoff factor (i) will be respectively : **[JEE (MAIN)-2014 ONLINE]**
 ($R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$)
 (1) 3 and 5.42 (2) 5 and 3.42 (3) 4 and 4.00 (4) 5 and 4.42
32. The molarity of a solution obtained by mixing 750 mL of 0.5(M)HCl with 250 mL of 2(M)HCl will be :- **[JEE (MAIN)-2013]**
 (1) 0.875 M (2) 1.00 M (3) 1.75 M (4) 0.975 M
33. The vapour pressure of acetone at 20°C is 185 torr. When 1.2 g of non-volatile substance was dissolved in 100 g of acetone at 20°C , its vapour pressure was 183 torr. The molar mass (g mol^{-1}) of the substance is : **[JEE (MAIN)-2015]**
 (1) 128 (2) 488 (3) 32 (4) 64

34. A solution at 20°C is composed of 1.5 mol of benzene and 3.5 mol of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively : **[JEE (MAIN)-2015 ONLINE]**
 (1) 38.0 torr and 0.589 (2) 30.5 torr and 0.389
 (3) 35.8 torr and 0.280 (4) 35.0 torr and 0.480
35. Determination of the molar mass of acetic acid in benzene using freezing point depression is affected by : **[JEE (MAIN)-2015 ONLINE]**
 (1) association (2) dissociation (3) complex formation (4) partial ionization
36. 18 g glucose ($C_6H_{12}O_6$) is added to 178.2 g water. The vapour pressure of water (in torr) for this aqueous solution is : **[JEE (MAIN)-2016]**
 (1) 759.0 (2) 7.6 (3) 76.0 (4) 752.4
37. The solubility of N_2 in water at 300 K and 500 torr partial pressure is 0.01 g L^{-1} . The solubility (in g L^{-1}) at 750 torr partial pressure is : **[JEE (MAIN)-2016-ONLINE]**
 (1) 0.02 (2) 0.005 (3) 0.015 (4) 0.0075
38. An aqueous solution of a salt MX_2 at certain temperature has a van't Hoff factor of 2. The degree of dissociation for this solution of the salt is : **[JEE (MAIN)-2016-ONLINE]**
 (1) 0.50 (2) 0.80 (3) 0.67 (4) 0.33
39. The freezing point of benzene decreases by 0.45°C when 0.2 g of acetic acid is added to 20 g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be: **[JEE (MAIN)- 2017]**
 (K_f for benzene = $5.12 \text{ K kg mol}^{-1}$)
 (1) 64.6% (2) 80.4% (3) 74.6% (4) 94.6%
40. For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point ? **[JEE (MAIN)-2018]**
 (1) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ (2) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (3) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (4) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$
41. Which one of the following statements regarding Henry's law not correct ? **[JEE MAIN-2019(Jan.)]**
 (1) The value of K_H increases with function of the nature of the gas
 (2) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids.
 (3) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.
 (4) Different gases have different K_H (Henry's law constant) values at the same temperature.

42. Liquids A and B form an ideal solution in the entire composition range. At 350 K, the vapor pressures of pure A and pure B are 7×10^3 Pa and 12×10^3 Pa, respectively. The composition of the vapor in equilibrium with a solution containing 40 mole percent of A at this temperature is : **[JEE MAIN-2019(Jan.)]**
 (1) $x_A = 0.76$; $x_B = 0.24$ (2) $x_A = 0.28$; $x_B = 0.72$
 (3) $x_A = 0.37$; $x_B = 0.36$ (4) $x_A = 0.4$; $x_B = 0.6$
43. Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2 K. The relation between K_b and K_f is: **[JEE MAIN-2019(Jan.)]**
 (1) $K_b = 1.5 K_f$ (2) $K_b = K_f$ (3) $K_b = 0.5 K_f$ (4) $K_b = 2 K_f$
44. The freezing point of a diluted milk sample is found to be -0.2°C , while it should have been -0.5°C for pure milk. How much water has been added to pure milk to make the diluted sample ? **[JEE MAIN-2019(Jan.)]**
 (1) 2 cups of water to 3 cups of pure milk (2) 1 cup of water to 3 cups of pure milk
 (3) 3 cups of water to 2 cups of pure milk (4) 1 cup of water to 2 cups of pure milk
45. Molecules of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) dimerise in benzene. 'w' g of the acid dissolved in 30 g of benzene shows a depression in freezing point equal to 2K. If the percentage association of the acid to form dimer in the solution is 80, then w is : **[JEE MAIN-2019(Jan.)]**
 (Given that $K_f = 5 \text{ K kg mol}^{-1}$, Molar mass of benzoic acid = 122 g mol^{-1})
 (1) 1.5 g (2) 1.0 g (3) 2.4 g (4) 1.8 g
46. K_2HgI_4 is 40% ionised in aqueous solution. The value of its van't Hoff factor (i) is :- **[JEE MAIN-2019(Jan.)]**
 (1) 1.6 (2) 2.0 (3) 2.2 (4) 1.8
47. Freezing point of a 4% aqueous solution of X is equal to freezing point of 12% aqueous solution of Y. If molecular weight of X is A, then molecular weight of Y is : **[JEE MAIN-2019(Jan.)]**
 (1) 2A (2) A (3) 3A (4) 4A
48. A solution containing 62 g ethylene glycol in 250 g water is cooled to -10°C . If K_f for water is $1.86 \text{ K kg mol}^{-1}$, the amount of water (in g) separated as ice is : **[JEE MAIN-2019(Jan.)]**
 (1) 64 (2) 32 (3) 16 (4) 48
49. The vapour pressures of pure liquids A and B are 400 and 600 mmHg, respectively at 298 K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are : **[JEE MAIN-2019(April)]**
 (1) 450 mmHg, 0.4, 0.6 (2) 500 mmHg, 0.4, 0.6
 (3) 500 mmHg, 0.5, 0.5 (4) 450 mmHg, 0.5, 0.5

50. For the solution of the gases w, x, y and z in water at 298 K, the Henry's law constants (K_H) are 0.5, 2, 35 and 40 k bar, respectively. The correct plot for the given data is :

[JEE MAIN-2019(April)]



51. Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is :

(x_M = Mole fraction of 'M' in solution;

x_N = Mole fraction of 'N' in solution;

y_M = Mole fraction of 'M' in vapour phase;

y_N = Mole fraction of 'N' in vapour phase)

[JEE MAIN-2019(April)]

(1) $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

(2) $(x_M - y_M) < (x_N - y_N)$

(3) $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

(4) $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

52. The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01 M BaCl_2 in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol L^{-1}) in solution is:

[JEE MAIN-2019(April)]

(1) 16×10^{-4}

(2) 6×10^{-2}

(3) 4×10^{-2}

(4) 4×10^{-4}

53. Molal depression constant for a solvent is 4.0 kg mol^{-1} . The depression in the freezing point of the solvent for 0.03 mol kg^{-1} solution of K_2SO_4 is :

[JEE MAIN-2019(April)]

(Assume complete dissociation of the electrolyte)

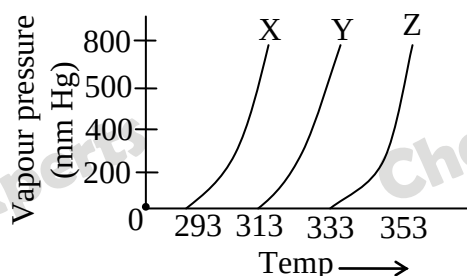
(1) 0.12 K

(2) 0.36 K

(3) 0.24 K

(4) 0.18 K

54. At room temperature, a dilute solution of urea is prepared by dissolving 0.60 g of urea in 360 g of water. If the vapour pressure of pure water at this temperature is 35 mmHg, lowering of vapour pressure will be: (molar mass of urea = 60 g mol⁻¹) **[JEE MAIN-2019(April)]**
 (1) 0.017 mmHg (2) 0.027 mmHg (3) 0.031 mmHg (4) 0.028 mmHg
55. 1g of a non-volatile non-electrolyte solute is dissolved in 100 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1: 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$, is : **[JEE MAIN-2019(April)]**
 (1) 10 : 1 (2) 5 : 1 (3) 1 : 5 (4) 1 : 0.2
56. A solution is prepared by dissolving 0.6 g of urea (molar mass = 60 g mol⁻¹) and 1.8 g of glucose (Molar mass = 180 g mol⁻¹) in 100 mL of water at 27°C. The osmotic pressure of the solution is: (R = 0.08206 L atm K⁻¹ mol⁻¹) **[JEE MAIN-2019(April)]**
 (1) 4.92 atm (2) 2.46 atm (3) 8.2 atm (4) 1.64 atm
57. At 35°C, the vapour pressure of CS₂ is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS₂ in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is: **[Jee Main (Jan. 2020)]**
 (1) CS₂ and acetone are less attracted to each other than to themselves
 (2) a mixture of 100 mL CS₂ and 100 mL acetone has a volume < 200 mL
 (3) Raoult's law is not obeyed by this system
 (4) heat must be absorbed in order to produce the solution at 35°C
58. Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non volatile solute are together sealed in a container. Over time : **[Jee Main (Jan. 2020)]**
 (1) The volume of the solution decreases and the volume of the solvent increases
 (2) The volume of the solution increases and the volume of the solvent decreases
 (3) The volume of the solution does not change and the volume of the solvent decreases
 (4) The volume of the solution and the solvent does not change
59. A graph of vapour pressure and temperature for these different liquids X, Y, and Z is shown below :



The following inference are made :

- (A) X has higher intermolecular interactions compared to Y.

[Jee Main (Jan. 2020)]

(B) X has lower intermolecular interactions compared to Y

(C) Z has lower intermolecular interactions compared to Y

The correct inference(s) is/are :

(1) (B)

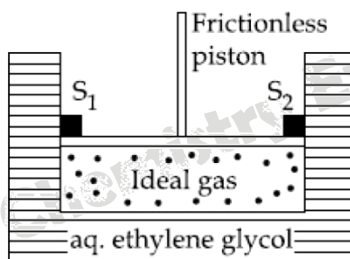
(2) (C)

(3) (A)

(4) (A) and (C)

60. How much amount of NaCl should be added [to 600g of water ($\rho = 1.00\text{g/mL}$) to decrease the freezing point of water to -0.2°C ? _____. (The freezing point depression constant for water = 2K kg mol^{-1}) **[Jee Main (Jan. 2020)]**

61. A cylinder containing an ideal gas (0.1 mol of 1.0 dm^3) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be _____.
(Given, $K_f(\text{water}) = 2.0\text{ K kg mol}^{-1}$, $R = 0.08\text{ dm}^3\text{ atm K}^{-1}\text{ mol}^{-1}$) **[Jee Main (Jan. 2020)]**



62. An open beaker of water in equilibrium with water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules :

[Jee-Main (Sep) 2020]

(1) leaves the solution increases

(2) leaves the solution decreases

(3) leaves the vapour decreases

(4) leaves the vapour increases

63. The size of a raw mango shrinks to a much smaller size when kept in a concentrated salt solution. Which one of the following processes can explain this ? **[Jee-Main (Sep) 2020]**

(1) Diffusion

(2) Osmosis

(3) Reverse osmosis

(4) Dialysis

64. Henry's constant (in kbar) for four gases α , β , γ and δ in water at 298 K is given below :

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(Density of water = 10^3 kg m^{-3} at 298 K)

[Jee-Main (Sep) 2020]

This table implies that :

(1) α has the highest solubility in water at a given pressure

(2) The pressure of a 55.5 molal solution at δ is 250 bar

(3) Solubility of γ at 308 K is lower than at 298 K

(4) The pressure of a 55.5 molal solution of γ is 1 bar

65. A set of solutions is prepared using 180 g of water as a solvent and 10 g of different non-volatile solutes A, B and C. The relative lowering of vapour pressure in the presence of these solutes are in the order [Given, molar mass of A = 100 g mol^{-1} ; B = 200 g mol^{-1} ; C = $10,000 \text{ g mol}^{-1}$] **[Jee-Main (Sep) 2020]**
- (1) $A > B > C$ (2) $A > C > B$ (3) $C > B > A$ (4) $B > C > A$
66. If 250 cm^3 of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K, the ratio of the molecular masses of A and B is $\text{_____} \times 10^{-2}$ (to the nearest integer). **[Jee-Main (Sep) 2020]**
Given 177.00
67. At 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg. At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. What is the vapour pressure in mm Hg of n-heptane in its pure state _____ ? **[Jee-Main (Sep) 2020]**
68. The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2 L of the glucose solution is $x \times 10^{-3}$ atm. x is _____. (nearest integer)
Given : 166.00 **[Jee-Main (Sep) 2020]**
69. The elevation of boiling point of 0.10 m aqueous $\text{CrCl}_3 \cdot x\text{NH}_3$ solution is two times that of 0.05 m aqueous CaCl_2 solution. The value of x is [Assume 100% ionisation of the complex and CaCl_2 , coordination number of Cr as 6, and that all NH_3 molecules are present inside the coordination sphere] **[Jee-Main (Sep) 2020]**

EXERCISE # (JEE-ADVANCED)

1. A 0.004 M solution of Na_2SO_4 is isotonic with a 0.010 M solution of glucose at same temperature. The apparent degree of dissociation of Na_2SO_4 is [JEE 2004]
(A) 25% (B) 50% (C) 75% (D) 85%
2. 1.22 g of benzoic acid is dissolved in (i) 100 g acetone (K_b for acetone = 1.7) and (ii) 100 g benzene (K_b for benzene = 2.6). The elevation in boiling points T_b is 0.17°C and 0.13°C respectively. [JEE 2004]
(a) What are the molecular weights of benzoic acid in both the solutions?
(b) What do you deduce out of it in terms of structure of benzoic acid?
3. The elevation in boiling point, when 13.44 g of freshly prepared CuCl_2 are added to one kilogram of water, is [Some useful data, K_b (H_2O) = $0.52 \text{ K kg mol}^{-1}$, mol. wt. of CuCl_2 = 134.4 gm] [JEE 2005]
(A) 0.05 (B) 0.1 (C) 0.16 (D) 0.21
4. 72.5 g of phenol is dissolved in 1 kg of a solvent (k_f = 14) which leads to dimerization of phenol and freezing point is lowered by 7 kelvin. What percent of total phenol is present in dimeric form? [JEE 2006]
5. When 20 g of naphtholic acid ($\text{C}_{11}\text{H}_8\text{O}_2$) is dissolved in 50 g of benzene (K_f = $1.72 \text{ K kg mol}^{-1}$), a freezing point depression of 2 K is observed. The van't Hoff factor (i) is [JEE 2007]
(A) 0.5 (B) 1 (C) 2 (D) 3

Paragraph for Question No. Q.6 to Q.8

Properties such as boiling point, freezing point and vapour pressure of a pure solvent change when solute molecules are added to get homogeneous solution. These are called colligative properties. Applications of colligative properties are very useful in day-to-day life. One of its examples is the use of ethylene glycol and water mixture as anti-freezing liquid in the radiator of automobiles.

A solution M is prepared by mixing ethanol and water. The mole fraction of ethanol in the mixture is 0.9.

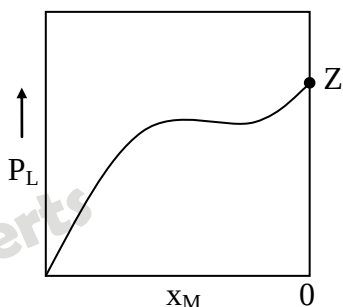
- Given :
- Freezing point depression constant of water (K_f^{water}) = $1.86 \text{ K kg mol}^{-1}$
 - Freezing point depression constant of ethanol (K_f^{ethanol}) = $2.0 \text{ K kg mol}^{-1}$
 - Boiling point elevation constant of water (K_b^{water}) = $0.52 \text{ K kg mol}^{-1}$
 - Boiling point elevation constant of ethanol (K_b^{ethanol}) = $1.2 \text{ K kg mol}^{-1}$
 - Standard freezing point of water = 273 K
 - Standard freezing point of ethanol = 155.7 K
 - Standard boiling point of water = 373 K

Standard boiling point of ethanol = 351.5 K
 Vapour pressure of pure water = 32.8 mm Hg
 Vapour pressure of pure ethanol = 40 mm Hg
 Molecular weight of water = 18 g mol^{-1}
 Molecular weight of ethanol = 46 g mol^{-1}

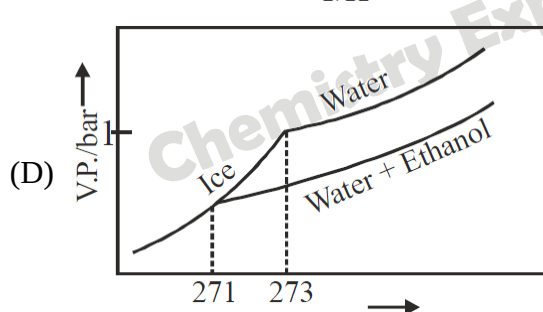
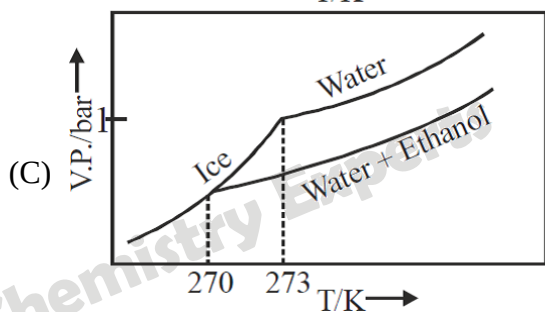
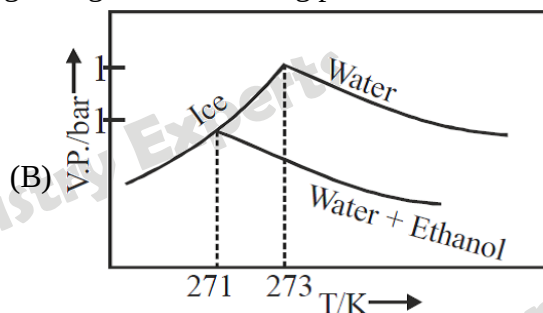
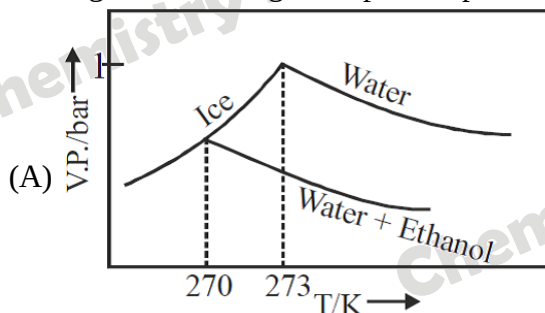
In answering the following questions, consider the solutions to be ideal dilute solutions and solutes to be non-volatile and non-dissociative

6. The freezing point of the solution M is [JEE 2008]
 (A) 268.7 K (B) 268.5 K (C) 234.2 K (D) 150.9 K
7. The vapour pressure of the solution M is [JEE 2008]
 (A) 39.3 mm Hg (B) 36.0 mm Hg (C) 29.5 mm Hg (D) 28.8 mm Hg
8. Water is added to the solution M such that the mole fraction of water in the solution becomes 0.9. The boiling point of this solution is [JEE 2008]
 (A) 380.4 K (B) 376.2 K (C) 375.5 K (D) 354.7 K
9. The Henry's law constant for the solubility of N_2 gas in water at 298 K is $1.0 \times 10^5 \text{ atm}$. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water at 298 K and 5 atm pressure is— [JEE 2009]
 (A) 4.0×10^{-4} (B) 4.0×10^{-5} (C) 5.0×10^{-4} (D) 4.0×10^{-5}
10. The freezing point (in $^{\circ}\text{C}$) of a solution containing 0.1 g of $\text{K}_3[\text{Fe}(\text{CN})_6]$ (Mol. Wt. 329) in 100 g of water ($K_f = 1.86 \text{ K kg mol}^{-1}$) is [JEE 2011]
 (A) -2.3×10^{-2} (B) -5.7×10^{-2} (C) -5.7×10 (D) -1.2×10^{-2}
11. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C . Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is: (take $K_b = 0.76 \text{ K kg mol}^{-1}$) [JEE 2011]
 (A) 724 (B) 740 (C) 736 (D) 718
12. Benzene and naphthalene form an ideal solution at room temperature. For this process, the true statement(s) is(are) [J-Adv. 2013]
 (A) ΔG is positive (B) ΔS_{system} is positive (C) $\Delta S_{\text{surroundings}} = 0$ (D) $\Delta H = 0$
13. A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g mol^{-1} , Assuming no change in volume upon dissolution, the molality of a 3.2 molar solution is [JEE-Adv. 2014]
14. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride-ammonia complex (which behaves as a strong electrolyte) is -0.0558°C , the number of chloride (s) in the coordination sphere of the complex is [JEE-Adv. 2015]
 [K_f of water = $1.86 \text{ K kg mol}^{-1}$]

15. Mixture(s) showing positive deviation from Raoult's law at 35°C is (are) [JEE-Adv. 2016]
 (A) carbon tetrachloride + methanol (B) carbon disulphide + acetone
 (C) benzene + toluene (D) phenol + aniline
16. For a solution formed by mixing liquids L and M, the vapour pressure of L plotted against the mole fraction of M in solution is shown in the following figure. Here x_L and x_M represent mole fractions of L and M, respectively, in the solutions. The correct statement(s) applicable to this system is(are) [JEE Adv. 2017]

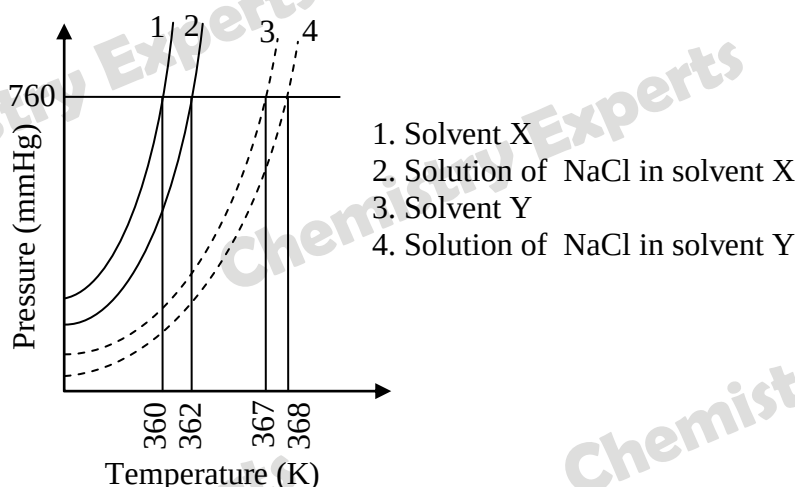


- (A) Attractive intermolecular interactions between L-L in pure liquid L and M-M in pure liquid M are stronger than those between L-M when mixed in solution.
 (B) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed when $x_L \rightarrow 0$
 (C) The point Z represents vapour pressure of pure liquid L and Raoult's law is obeyed when $x_L \rightarrow 1$
 (D) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed when $x_L = 0$ $x_L = 1$
17. Pure water freezes at 273 K and 1 bar. The addition of 34.5 g of ethanol to 500 g of water changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol⁻¹. The figures shown below represent plots of vapour pressure (V.P.) Among the following, the option representing change in the freezing point is



18. Liquids A and B form ideal solution over the entire range of composition. At temperature T, equimolar binary solution of liquids A and B has vapour pressure 45 Torr. At the same temperature, a new solution of A and B having mole fractions x_A and x_B respectively, has vapour pressure of 22.5 Torr. The value of X_A/X_B in the new solution is _____.
(given that the vapour pressure of pure liquid A is 20 Torr at temperature T) [JEE Adv. 2018]

19. The plot given below shows P – T curves (where P is the pressure and T is the temperature) for two solvents X and Y and isomolal solutions of NaCl in these solvents. NaCl completely dissociates in both the solvents. On addition of equal number of moles of a non-volatile solute S in equal amount (in kg) of these solvents, the elevation of boiling point of solvent X is three times that of solvent Y. Solute S is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent Y, the degree of dimerization in solvent X is _____. [JEE Adv. 2018]



On addition of equal number of moles of a non-volatile solute S in equal amount (in kg) of these solvents, the elevation of boiling point of solvent X is three times that of solvent Y. Solute S is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent Y, the degree of dimerization in solvent X is _____.

20. On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, its vapor pressure decreases from 650 mm Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is _____. [JEE Adv. 2019]
(Given data: Molar mass and the molal freezing point depression constant of benzene are 78 g mol⁻¹ and 5.12 K kg mol⁻¹, respectively)
21. Liquids A and B form ideal solution for all compositions of A and B at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of A have the total vapour pressure of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid B in bar? [JEE Adv. 2020]

ANSWER KEY

EXERCISE # (JEE-MAIN)

1.	(1)	2.	(2)	3.	(4)	4.	(4)	5.	(2)	6.	(4)	7.	(4)
8.	(3)	9.	(1)	10.	(4)	11.	(4)	12.	(4)	13.	(1)	14.	(4)
15.	(3)	16.	(2)	17.	(3)	18.	(3)	19.	(3)	20.	(2)	21.	(1)
22.	(1)	23.	(3)	24.	(3)	25.	(4)	26.	(1)	27.	(3)	28.	(3)
29.	(3)	30.	(2)	31.	(4)	32.	(1)	33.	(4)	34.	(1)	35.	(1)
36.	(4)	37.	(3)	38.	(1)	39.	(4)	40.	(1)	41.	(2)	42.	(2)
43.	(4)	44.	(3)	45.	(3)	46.	(4)	47.	(3)	48.	(1)	49.	(2)
50.	(3)	51.	(1)	52.	(2)	53.	(2)	54.	(1)	55.	(3)	56.	(1)

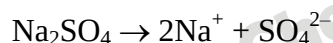
EXERCISE # (JEE-ADVANCED)

- (C)
- (a) 122, (b) It means that benzoic acid remains as it is in acetone while it dimerises in benzene
- (C)
- 35% (degree of asso = 70%)
- (A)
- (D)
- (B)
- (B)
- (A)
- (A)
- (A)
- (B,C,D)
- (8)
- (1)
- (A, B)
- (A,C)
- (C)
- (19)
- (0.05)
- (1.02)
- (0.20)

HINT AND SOLUTIONS

EXERCISE # (JEE-MAIN)

1. Elevation in boiling point is a colligative property which depends upon the number of solute particles. Greater the number of solute particles in a solution, higher the extent of elevation in boiling point.



2. Benzene-methanol shows a positive deviation from Raoult's law

3. $\Delta T_f = iK_f m$; K_f depends on nature of solvent.



$$\text{Vant Hoff factor (i)} = \frac{1-\alpha+2\alpha+\alpha}{1} = 1+2\alpha.$$

5. $P_B = P_B^0 X_B$ $P_B^0 = 75$ torr

$$X_B = \frac{78/78}{(78/78)+(46/92)} = \frac{1}{1+0.5} = \frac{1}{1.5} \quad P_B = 75 \times \frac{1}{1.5} = 50 \text{ torr.}$$

6. Total millimoles of solute = $480 \times 1.5 + 520 \times 1.2 = 720 + 624 = 1344$.

$$\text{Total volume} = 480 + 520 = 1000.$$

$$\text{Molarity of the final mixture} = \frac{1344}{1000} = 1.344 \text{ M.}$$

7. Equimolar solutions in the same solvent have same boiling and same freezing points because colligative properties depends on concentration of solution and does depend on nature of solute.

$$8. \quad \frac{P^0 - P_s}{P^0} = \frac{n}{n+N}; \quad \frac{760 - P_s}{760} = \frac{0.1}{9.9+0.1}$$

$$\therefore P_s = 752.4 \text{ torr}$$

9. According to Raoult's law

$$P = P_A + P_B = P_A^0 X_A + P_B^0 X_B$$

$$\text{or } 290 = 0.6 \times P_A^0 + 0.4 \times 200$$

$$\text{or } 290 = P_A^0 \times (0.6) + 200 \times (1 - 0.6)$$

$$\text{or } P_A^0 = 350 \text{ mm.}$$

10. Isotonic solutions have same osmotic pressure.

$$\pi_1 = C_1 RT, \pi_2 = C_2 RT$$

For isotonic solution, $\pi_1 = \pi_2$

$$\therefore C_1 = C_2.$$

$$\text{or } \frac{1.5/60}{V} = \frac{5.25/M}{V} \quad [\text{where } M = \text{molecular weight of the substance}]$$

$$\text{or } \frac{1.5}{60} = \frac{5.25}{M} \quad \text{or } M = 210.$$

11. $P_{\text{total}} = P_A^0 X_A + P_B^0 (1 - X_A) = 520 \times X_A + 1000 \times (1 - X_A) = 760$

$$520X_A + 1000 - 1000X_A = 760; \quad 480X_A = 240$$

$$\therefore X_A = 0.5 \quad \text{So, mixture contains 50\% moles of A}$$

12. $P_A = P_A^0 X_A = 17.5 \times \frac{\frac{178.2/18}{18} + \frac{18}{180}}{\frac{178.2}{18} + \frac{18}{180}} = 17.325$

13. $P_{\text{total}} = P_A^0 X_A + P_B^0 X_B = P_A^0 \times \frac{1}{4} + P_B^0 \times \frac{3}{4} = 550 \Rightarrow P_A^0 + 3P_B^0 = 550 \times 4 \dots (i)$

$$\text{Similarly, } 560 = P_A^0 \times \frac{1}{5} + P_B^0 \times \frac{4}{5} \Rightarrow P_A^0 + 4P_B^0 = 560 \times 5 \dots (ii)$$

eq. (ii) – eq. (i)

$$P_B^0 = 560 \times 5 - 550 \times 4 = 600$$

$$\text{so, } P_A^0 = 400.$$

14. The solution is non-ideal, showing +ve deviation from Raoult's Law.

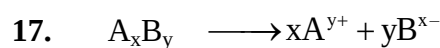
15. $\text{Na}_2\text{SO}_4(s) \xrightarrow{\text{H}_2\text{O}} 2\text{Na}^+(\text{aq.}) + \text{SO}_4^{2-}(\text{aq.})$

$$\Delta T_f = i K_f m = 3 \times 1.86 \times 0.01 = 0.0558 \text{ K.}$$

16. $P_T = X_{\text{Heptane}} P_{\text{Heptane}}^0 + X_{\text{Octane}} P_{\text{Octane}}^0$

$$= \frac{0.25}{0.557} \times 105 + \frac{0.307}{0.557} \times 45$$

$$47.127 + 24.80 = 71.92 \approx 72 \text{ kPa}$$



$$1 - \alpha \quad x\alpha \quad y\alpha$$

$$i = 1 - \alpha + x\alpha + y\alpha = 1 + \alpha(x + y - 1)$$

$$\text{So, } \alpha = \frac{i-1}{(x+y-1)}$$

18. $\Delta T_f = i \times K_f \times m ; \quad 6 = 1 \times 1.86 \times \frac{w}{62 \times 4}$

$$w = 800 \text{ g}$$

19. $\pi_1 = \pi_2$

$$i_1 C_1 = i_2 C_2$$

$$1 \times \frac{5 \times 1000}{342 \times 100} = 1 \times \frac{1 \times 1000}{M \times 100}$$

$$M = 68.4 \text{ g/mol}$$

20. $m = \frac{0.01}{60 \times 0.3} = 5.55 \times 10^{-4} \text{ m}$

21. $\Delta T_f = i \times K_f \times m ; \quad 1.91 = i \times 1.86 \times 1$

$$i = \frac{1.91}{1.86} = 1.027 \quad \therefore i = 1 + \alpha = 1.027$$

$$\therefore \alpha = 0.027, \quad \% \text{ dissociation} = 2.7 \%$$

22. Since $P_s = X_A P_A^0 + X_B P_B^0$

$$\Rightarrow 250 = \frac{1}{3} \times P_A^0 + \frac{2}{3} \times P_B^0 \dots\dots\dots (1) \quad \Rightarrow 300 = 0.5 \times P_A^0 + 0.5 \times P_B^0 \dots\dots\dots (2)$$

On solving eq. (1) and (2) we get $P_A^0 = 450 \text{ mm Hg} \quad P_B^0 = 150 \text{ mm Hg}$

23. $\Delta T_f = i \times K_f \times m ; \quad 0.23 = i \times 1.86 \times \frac{0.85 \times 1000}{136.3 \times 125}$

$$i = 2.47 \quad \therefore i = 1 + 2\alpha = 2.47$$

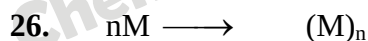
$$\therefore \alpha = 0.735, \quad \% \text{ dissociation} = 73.5 \%$$

24. $\Delta T_f = i \times K_f \times m ; \quad 2.8 = 1 \times 1.86 \times \frac{w}{62 \times 1}$

$$w = 93.3 \text{ g}$$

$$25. \quad \frac{P^0 - P_s}{P^0} = \frac{n}{n + N}; \quad 0.1 = \frac{\frac{12}{M}}{\frac{12}{M} + \frac{108}{18}}$$

$$M \approx 20$$



$$1 - \alpha \quad \frac{\alpha}{n} \Rightarrow i = 1 - \alpha + \frac{\alpha}{n} = 0.9$$

$$\text{Given ; } \frac{\frac{\alpha}{n}}{0.9} = 0.2 \quad \therefore \alpha = 0.18 n$$

$$\Rightarrow i = 1 - 0.18 n + \frac{0.18 n}{n} = 0.9 \quad : \quad n = 1.55$$

$$27. \quad P_s = X_b P_b^0 + X_t P_t^0 = 0.5 \times 119 + 0.5 \times 37 = 78$$

$$P_t = X_t P_t^0 = Y_t P_s$$

$$\Rightarrow 18.5 = Y_t \times 78 \Rightarrow Y_t = 0.237$$

$$28. \quad \Delta T_f = 273 - 268 = 5 \text{ K}$$

$$\Delta T_f = i \times K_f \times m; \quad \Rightarrow 5 = 1 \times 1.86 \times \frac{w}{32 \times 10}$$

$$w = 860.2 \text{ g}$$

$$29. \quad \text{Effective concentration of particles} = iC$$

$$C_2H_5OH(aq) = 1 \times 0.5 = 0.5 \text{ M}$$

$$Mg_3(PO_4)_2(aq) = 5 \times 0.1 = 0.5 \text{ M}$$

$$KBr(aq) = 2 \times 0.25 = 0.5 \text{ M}$$

$$KBr(aq) = 4 \times 0.125 = 0.5 \text{ M}$$

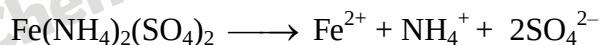
$$30. \quad \Delta H_{\text{mixing}} = 0 \text{ (zero)}$$

A – A, B – B and A – B interactions are identical

$$31. \quad \pi_1 = iCRT; \quad \Rightarrow 10.8 = i \times 0.1 \times 0.082 \times 298$$

$$\Rightarrow i = 4.42 \text{ (experimental)}$$

$$\Rightarrow i = 5 \text{ (expected)}$$



32. $M_1V_1 + M_2V_2 = M_3V_3$

$$0.5 \times 750 + 2 \times 250 = M_3 \times 1000$$

$$M_3 = 0.875 \text{ M}$$

33. $\frac{P^0 - P_s}{P_s} = \frac{n}{N} \Rightarrow \frac{185 - 183}{183} = \frac{1.2/M}{100/58}$

$$M = 64 \text{ g mol}^{-1}$$

34. $P_s = X_b P_B^0 + X_t P_t^0 = \frac{1.5}{5} \times 74.7 + \frac{3.5}{5} \times 22.3 = 22.41 + 15.687 = 38.097$

$$P_b = X_b P_b^0 = Y_b P_s$$

$$\Rightarrow 22.41 = Y_b \times 38$$

$$\Rightarrow Y_b = 0.589$$

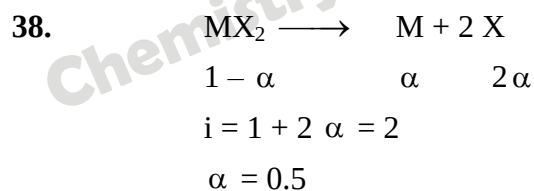
35. Acetic acid is associated in benzene due to intermolecular H – bonding.

36. $\frac{P^0 - P_s}{P^0} = \frac{n}{n + N}; \quad \frac{760 - P_s}{760} = \frac{0.1}{9.9 + 0.1}$

$$\therefore P_s = 752.4 \text{ torr}$$

37. Solubility $\propto$ Partial pressure of gas

$$\frac{0.01}{S} = \frac{500}{750}; \quad S = 0.015 \text{ g L}^{-1}$$



39. $\Delta T_f = i \times K_f \times m$

$$0.45 = i \times 5.12 \times \frac{0.2 \times 1000}{60 \times 20}$$

$$i = 0.527 = 1 - \frac{\alpha}{2}; \quad \alpha = 0.945$$

40. $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ produces least number of particles in solution so, depression in freezing will also be least and freezing point will be maximum.

41. $P_{\text{gas}} = K_H X_{\text{gas}}$

$K_H \uparrow$ Solubility $\downarrow$

$T \uparrow, K_H \uparrow$

42. $X_A = 0.4$

$P_A = X_A P_A^0 = Y_A P_s$

$0.4 \times 7 \times 10^3 = Y_A \times 10^4$

$Y_A = 0.28$

$Y_B = 0.72$

$P_A^0 = 7 \times 10^3 \text{ Pa}$

$P_B^0 = 12 \times 10^3 \text{ Pa}$

$P_s = X_A P_A^0 + X_B P_B^0$

$= 0.4 \times 7 \times 10^3 + 0.6 \times 12 \times 10^3 = 2.8 \times 10^3 + 7.2 \times 10^3 = 10^4$

43. $\therefore \Delta T_b = K_b \times m$

$K_b = \frac{2}{1} = 2$

$\therefore \Delta T_f = K_f \times m$

$K_f = \frac{2}{2} = 1$

$\therefore K_b = 2K_f$

44. ΔT_f for diluted milk = 0.2

ΔT_f for pure milk = 0.5

$\frac{(\Delta T_f)_{\text{diluted}}}{(\Delta T_f)_{\text{pure}}} = \frac{(iK_f m)_{\text{diluted}}}{(iK_f m)_{\text{pure}}}$

$\frac{0.2}{0.5} = \frac{m_1}{m_2}; \frac{m_1}{m_2} = \frac{2}{5}$

Assume that Solution is dilute so molality = molarity

$\frac{M_1}{M_2} = \frac{2m}{5m}$

So 3 cups of water are added to 2 cups of pure milk.

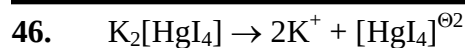
45. For dimerisation

$i = 1 + \alpha \left(\frac{1}{n} - 1 \right) = 1 + 0.8 \left(\frac{1}{2} - 1 \right) = 0.6$

$\Delta T_f = i \times K_f \times m$

$2 = 0.6 \times 5 \times \frac{W/122}{30} \times 1000$

$W = 2.44$



$$i = 1 + (n - 1) \alpha = 1 + (3 - 1) \times 0.4 = 1 + 2 \times 0.4$$

$$\text{i.e.: } i = 1.8$$

47. $(\Delta T_f)_X = (\Delta T_f)_Y$

$$K_f M_X = K_f M_Y$$

$$\frac{4}{A} = \frac{12}{M_Y}$$

$$M_Y = 3A$$

48. Let amount of water (ing) separated as ice is x

$$\Delta T_f = i \times K_f \times m$$

$$\Rightarrow 10 = 1 \times 1.86 \times \frac{62 \times 1000}{62 \times (250 - x)} \Rightarrow x = 64$$

49. Since on mixing two liquids A and B change in volume after mixing is zero, therefore binary solution is an ideal solution.

$$\therefore P_T = P_A^0 X_A + P_B^0 X_B = (400 \times 0.5) + (600 \times 0.5) = 500 \text{ Torr}$$

$$\therefore \text{Mole fraction of A in vapour phase} = \frac{P_A^0 X_A}{P_T} = \frac{400 \times 0.5}{500} = 0.4$$

50. $P_{\text{gas}} = K_H \times \text{mole fraction of gas.}$

$$\text{i.e. } P_{\text{gas}} = K_H \cdot X_{\text{gas}} \Rightarrow P_{\text{gas}} = K_H (1 - X_{H_2O})$$

$$P_{\text{gas}} = K_H - K_H \cdot X_{H_2O}$$

$$\Rightarrow \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ y & c & m & x \end{matrix}$$

For Z, K_H is maximum

51. $P_M^0 = 450 \text{ mm of Hg}$ $P_N^0 = 700 \text{ mm of Hg}$

That is 'N' is more volatile liquid as compare to 'M' so,

$$\frac{X_N}{X_M} > \frac{Y_N}{Y_M}$$

$$\text{or } \frac{X_M}{X_N} > \frac{Y_M}{Y_N}$$

52. $\pi_1 = \pi_2 \times 4$

$$i_1 C_1 RT = i_2 C_2 RT \times 4$$

$$2 \times C_1 = 3 \times 0.01 \times 4$$

$$C_1 = 6 \times 10^{-2} \text{M}$$

53. $\Delta T_f = i k_f m$

$$= 3 \times 4 \times 0.03 = 0.36 \text{ K}$$

54. $\frac{\Delta P}{P^0} = \frac{n}{n + N}$

$$\frac{\Delta P}{35} = \frac{0.01}{0.01 + 20}$$

$$\Delta P = \frac{0.01}{20.01} \times 35 = 0.017 \text{ mm}$$

55. $\Delta T_b = K_b \times m$

$$\frac{(\Delta T_b)_A}{(\Delta T_b)_B} = \frac{(K_b)_A \times m}{(K_b)_B \times m} = \frac{1}{5}$$

56. $\pi = CRT$

$$= \frac{\left(\frac{0.6}{60} + \frac{1.5}{180} \right)}{0.1} \times 0.08206 \times 300$$

$$\Rightarrow \pi = 4.92 \text{ atm.}$$

57. (2)

Sol. Since total pressure of solution is greater than vapour pressure of CS_2 and acetone which indicates solution is showing positive deviation from ideal behaviour.

58. (2)

Sol. Due to presence of solute vapour pressure of solution will be less than V.P. of pure solvent. So, vapour move from higher pressure to lower pressure.

59. (1)

Sol. order of B.P. $z > y > x$

order of V.P. $z < y < x$

order of inter molecular forces : $z > y > x$

60. [1.75 or 1.76]

NTA Ans. (1.74 to 1.76)

$$\text{Sol. } \frac{0.2}{10} = 2 \times \frac{n \times 2}{3600} \times 1000$$

$$n = 0.03$$

$$w = 0.03 \times 58.5 \text{ g}$$

$$= 1.75 \text{ or } 1.76$$

$$\text{NTA ans } 1.74 \text{ to } 1.76$$

61. 2.17 or 2.18

NTA Ans. (2.18 to 2.23)

$$\text{Sol. } \Delta T_f = 2 \times 0.5 = 1$$

$$T = 273 - 1 = 272 \text{ K}$$

$$V = \frac{0.1 \times 0.08 \times 272}{1}$$

$$= 2.176$$

$$= 2.18 \text{ L}$$

62. NTA Ans. (4)

Sol. On adding glucose (non volatile solute) vapour pressure of solution will be less than vapour pressure of pure solvent. So some vapours will condense to maintain this vapour pressure.

63. (2)

Sol. When raw mango is kept in a concentrated salt solution it shrinks due to osmosis.

64. (2)

Sol. According to Henry's Law :

$$P = K_H \cdot X_{\text{gas}}$$

$$\therefore \text{solubility} \propto \frac{1}{K_H} \text{ at a particular pressure.}$$

Option (1) : Since ' α ' Gas highest value of K_H so it has lowest solubility.

Option (2) : Since molarity of δ is 55.5

$$\text{So mole fraction of } \delta = \frac{55.5}{55.5 + 55.5}$$

$$= \frac{1}{2} = 0.5$$

$$\begin{aligned} \therefore P &= K_H \times X_{\delta} \\ &= 0.5 \times 0.5 \text{ k bar} \\ &= 250 \text{ bar.} \end{aligned}$$

Option (3) : Solubility of gas decreases with increase in temperature.

Option (4) : Since molality is 55.5

$$\text{So, mole fraction of } \gamma = \frac{55.5}{55.5 + 55.5}$$

$$\begin{aligned} \therefore P &= K_H \cdot X_\gamma \\ &= 2 \times 10^{-5} \times 0.5 \text{ k bar} \\ &= 10^{-2} \text{ bar.} \end{aligned}$$

65. (1)

Sol. Relative lowering of V.P. =

$$\left(\frac{\Delta P}{P^0} \right)_A = \frac{\frac{10}{100}}{\frac{10}{100} + \frac{180}{18}} : \left(\frac{\Delta P}{P^0} \right)_B = \frac{\frac{10}{200}}{\frac{10}{200} + \frac{180}{18}} \left(\frac{\Delta P}{P^0} \right)_C = \frac{\frac{10}{10,000}}{\frac{10}{10,000} + \frac{180}{18}} :$$

$$\left(\frac{\Delta P}{P^0} \right)_A > \left(\frac{\Delta P}{P^0} \right)_B > \left(\frac{\Delta P}{P^0} \right)_C$$

66. 177

Sol. If both solutions are isotonic

$\therefore$ Molarity of A = Molarity of B

$$\frac{0.73 / M_A}{250} \times 1000 = \frac{1.65 / M_B}{1000} \times 1000$$

$$\therefore \frac{M_A}{M_B} = \frac{0.73}{0.25 \times 1.65} = 1.77 = 177 \times 10^{-2}$$

So correct answer is 177.

67. NTA Ans. (600)

$$\text{Sol. } 550 = P_A^0 \times \frac{1}{4} + P_B^0 \times \frac{3}{4}$$

$$2200 = P_A^0 + 3P_B^0 \quad \dots(i)$$

$$2800 = P_A^0 + 4P_B^0 \quad \dots(ii)$$

$$560 = P_A^0 \times \frac{1}{5} + P_B^0 \times \frac{4}{5}$$

$$P_B^0 = 600, P_A^0 = 400$$

68. (167)

Sol. Osmotic pressure = $\pi = i \times C \times RT$

For NaCl $i = 2$ so

$$\pi_{\text{NaCl}} = i \times C_{\text{NaCl}} \times RT \quad C_{\text{NaCl}} = \text{conc. of NaCl}$$

$$0.1 = 2 \times C_{\text{NaCl}} \times RT$$

$$C_{\text{NaCl}} = \frac{0.05}{RT} \quad C_{\text{glucose}} = \text{conc. of glucose}$$

For glucose $i = 1$ so

$$\pi_{\text{Glucose}} = i \times C_{\text{glucose}} \times RT$$

$$0.2 = 1 \times C_{\text{glucose}} \times RT$$

$$C_{\text{Glucose}} = \frac{0.2}{RT} \quad \eta_{\text{NaCl}} = \text{No. of moles NaCl}$$

$$\eta_{\text{NaCl}} \text{ in 1 L} = C_{\text{NaCl}} \times V_{\text{Litre}}$$

$$= \frac{0.05}{RT} \quad \eta_{\text{glucose}} = \text{No. of moles glucose}$$

$$\eta_{\text{glucose}} \text{ in 2 L} = C_{\text{glucose}} \times V_{\text{Litre}} = \frac{0.4}{RT}$$

$$V_{\text{Total}} = 1 + 2 = 3\text{L}$$

$$\text{so Final conc. NaCl} = \frac{0.05}{3RT}$$

$$\text{Final conc. glucose} = \frac{0.4}{3RT}$$

$$\pi_{\text{Total}} = \pi_{\text{NaCl}} + \pi_{\text{glucose}}$$

$$= [0 \times C_{\text{NaCl}} + C_{\text{glucose}}] \times RT$$

$$= \left(\frac{2 \times 0.05}{3RT} + \frac{0.4}{3RT} \right) \times RT$$

$$= \frac{0.5}{3} \text{ atm} = 0.1666 \text{ atm}$$

$$= 166.6 \times 10^{-3} \text{ atm}$$

$$\Rightarrow 167.00 \times 10^{-3} \text{ atm}$$

$$\text{so } x = 167.00$$

69. (05.00)

Sol. 0.05 m aqueous CaCl_2 solution = 0.15 molal solution of non electrolyte

Since the elevation in boiling point is double so the molality of given solution of complex = 0.3 m

Hence the complex must be producing three ions in the solution, so its

formula = $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

so $x = 5$

EXERCISE # (JEE-ADVANCED)

1. $(\pi_{\text{obs}})_{\text{Na}_2\text{SO}_4} = \pi_{\text{glucose}}$

or $\frac{10}{4} = \frac{1+2\alpha}{1}$ or $10 = 4 + 8\alpha$

$\alpha = \frac{10-4}{8} = 0.75$ % of $\alpha = 75\%$

2. (i) In first case,

$$\Delta T_b = K_b \times m = K_b \times \frac{\text{Wt. of solute}}{\text{Mol. wt. of solute}} \times \frac{1000}{\text{wt. of solvent}}$$

or $0.17 = 1.7 \times \frac{1.22}{M \times 100 \times 10^{-3}}$

or **M = 122 g/mole**

Thus the benzoic acid exists as a monomer in acetone

(ii) In second case,

$$\Delta T_b = K_b \times \frac{\text{Wt. of solute}}{\text{Mol. wt. of solute}} \times \frac{1000}{\text{wt. of solvent}}$$

or $0.13 = 2.6 \times \frac{1.22}{M' \times 100 \times 10^{-3}}$ or **M' = 224**

Double molecular weight of benzoic acid (244) in acetone solution indicates that benzoic acid exists as a dimer in acetone.

3. $\Delta T_b = i K_b m$ i (vant Haff factor) of $\text{CuCl}_2 = 3$

$$\Delta T_b = 3 \times 0.52 \times \frac{13.44}{134.4 \times 1} = 0.156 = 0.16.$$

4. $\Delta T_f = i \times K_f \times \text{molality}$

$$7 = i \times 14 \times \frac{72.5}{94 \times 1}; \quad i = 0.648$$

$$i = 0.648 = 1 - \frac{\alpha}{2}; \quad \alpha = 0.704$$

5. $\Delta T_f = i \times K_f \times \text{molality} \Rightarrow 2 = i \times 1.72 \times \frac{20}{172} \times \frac{1000}{50}$

$$\Rightarrow 2 = 4i$$

$$\Rightarrow i = 1/2 = 0.5$$

$$6. \quad \Delta T_f = K_f \cdot m = 2 \times \frac{0.1}{0.9 \times 46} \times 1000 = \frac{2000}{414} = \frac{1000}{207} = 4.83$$

$$\Delta T'_f = T_f - 4.83$$

$$\Delta T'_f = 155.7 - 4.83$$

$$\Delta T'_f = 150.9 \text{ K}$$

7. Total vapour pressure

$$P = P_A^0 X_A \text{ (considering solute to be non-volatile as given in the question)}$$

$$P = 40 \times 0.9 = 36 \text{ mm Hg}$$

$$8. \quad \Delta T_b = K_b \cdot m = 0.52 \times \frac{0.1}{0.9 \times 18} \times 1000 = \frac{520}{9 \times 18} = 3.20$$

$$T_b = 373 + 3.20 \quad T_b = 376.2 \text{ K}$$

$$9. \quad P_{N_2} = K_H \times x_{N_2}$$

$$x_{N_2} = \frac{1}{10^5} \times 0.8 \times 5 = 4 \times 10^{-5} \text{ per mole}$$

In 10 mole solubility is 4×10^{-4} .

$$10. \quad \Delta T_f = i \times K_f \times m = 4 \times 1.86 \times \frac{0.1}{329 \times 0.1} = 2.3 \times 10^{-2}$$

$$\Rightarrow T_f = 0 - 2.3 \times 10^{-2} = -2.3 \times 10^{-2} \text{ } ^\circ\text{C}.$$

$$11. \quad \Delta T_b = 2^\circ\text{C}; \quad m_a = 2.5 \text{ g}$$

$$m_{\text{solvent}} = 100 \text{ g}$$

$$K_b = 0.76 \text{ K. kg. mol}^{-1}$$

$$P_{\text{solution}} = ?$$

$$\Delta T_b = K_b \times m$$

$$2 = 0.76 \times m \quad \therefore m = \frac{2}{0.76}$$

$$\frac{P^0 - P}{P} = m \times MM \times 10^{-3} \quad \therefore \frac{760 - P}{P} = \frac{2}{0.76} \times 18 \times 10^{-3}$$

$$760 - P = \frac{36}{760} P \quad \therefore 760 = \frac{796}{760} P$$

$$\therefore P = 760 \left(\frac{796}{760} \right) \text{ torr} = 725.6 \text{ torr} \approx 724 \text{ torr}$$

$$12. \quad \Delta G = - \text{ve}, \quad \Delta S_{\text{system}} = + \text{ve} \text{ Always for solution formation.}$$

$$\Delta S_{\text{surr.}} = 0$$

No heat exchange between solution and surrounding.

$$\Delta H = 0$$

For ideal solution.

13. 3.2 M = 3.2 moles of solute present in 1L of solution.

Volume of solute = volume of solution = 1L = 1000 mL

Mass of solute = Volume $\times$ density = 1000 mL $\times$ 0.4 g/mL = 400 g = 0.4 kg

$$\text{Molality} = \frac{3.2}{0.4} = 8 \text{ m}$$

14. $\Delta T_f = K_f \times i \times m$

$$0.0558 = 1.86 \times i \times 0.01$$

$$i = 3$$

Given complex behaves as a strong electrolyte

$$\alpha = 100 \%$$

$n = 3$ (number of particles given by complex)

$\therefore$ complex is $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

number of Cl^- ions in the co-ordination sphere of the complex = 1

15. (carbon tetrachloride + methanol) and (carbon disulphide + acetone) do not having any attraction forces.

16. From the graph it is clear that there is +ve deviation from Raoult law with respect to L, therefore option A is correct. When $X_L \longrightarrow 1$, then Z will have value equal to P_L^0 (Vapour pressure of pure L) so, option C is also correct.

17. $\Delta T_f = K_f \times m = 2 \times \frac{34.5 \times 2}{46} = 2 \times 1.5 = 3$

So, freezing point of ethanol + water solution = $273 - 3 = 270 \text{ K}$

18. $X_A = X_B = \frac{1}{2}$; $P_T = \frac{1}{2} \times P_A^0 + \frac{1}{2} \times P_B^0$ (Given $P_A^0 = 20 \text{ Torr}$)

$$90 = 45 \times 2 = P_A^0 + P_B^0 \dots\dots\dots(1) ; P_B^0 = 70$$

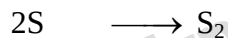
$$22.5 = X_A P_A^0 + (1 - X_A) P_B^0 \dots\dots\dots(1)$$

$$22.5 = X_A \times 20 + (1 - X_A) 70 = 70 - 50 X_A$$

$$X_A = \frac{47.5}{50} = \frac{19}{20} ; X_B = \frac{1}{20} ; \frac{X_A}{X_B} = 19$$

19. $2 = 2 \times K_{b(x)} \times m$; $1 = 2 \times K_{b(y)} \times m$; $\frac{K_{b(x)}}{K_{b(y)}} = 2$

$$\Delta T_{b(x)} = \left(1 - \frac{\alpha_1}{2}\right) K_{b(x)} m \quad ; \quad \Delta T_{b(y)} = \left(1 - \frac{\alpha_2}{2}\right) K_{b(y)} m$$



$$1 - \alpha \quad \alpha/2 \quad ; \quad i = 1 - \alpha + \alpha/2 = 1 - \alpha/2$$

$$3 = \frac{\Delta T_{b(x)}}{\Delta T_{b(y)}} = \frac{\left(1 - \frac{\alpha_1}{2}\right) K_{b(x)}}{\left(1 - \frac{0.7}{2}\right) K_{b(y)}} \quad ; \quad \left(1 - \frac{\alpha_1}{2}\right) = \frac{3 \times 0.65}{2}$$

$$\alpha = 0.05$$

$$20. \quad \frac{P^\circ - P_s}{P_s} = \frac{n}{N} = \frac{0.5}{M_{\text{solute}} \times 0.5} = \frac{650 - 640}{640}$$

$$M_{\text{solute}} = 64$$

$$\Delta T_f = i K_f m = 1 \times 5.12 \times \frac{0.5 \times 1000}{64 \times 39} = 1.025$$

21. (0.20)

Sol. Let P_A° and P_B° are vapour pressure of pure A and B liquid respectively

When $X_A = 0.25$

$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$0.3 = 0.25 P_A^\circ + 0.75 P_B^\circ \quad \text{.....(1)}$$

When $X_A = 0.5$

$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$0.4 = 0.5 P_A^\circ + 0.5 P_B^\circ \quad \text{.....(2)}$$

From equation (1) & (2)

$$P_B^\circ = 0.2 \text{ bar}$$

$$P_A^\circ = \frac{3}{5} \text{ bar}$$

EXERCISE # (JEE-MAIN)

1. Which one of the following characteristics is not correct for physical adsorption? **[AIEEE 2003]**
- (1) Adsorption on solids is reversible
 - (2) Adsorption increases with increase in temperature
 - (3) Adsorption is spontaneous
 - (4) Both enthalpy and entropy of adsorption are negative.
2. The disperse phase in colloidal iron (III) hydroxide and colloidal gold is positively and negatively charged, respectively. Which of the following statements is NOT correct? **[AIEEE 2005]**
- (1) Coagulation in both sols can be brought about by electrophoresis
 - (2) Mixing the sols has no effect
 - (3) Sodium sulphate solution causes coagulation in both sols
 - (4) Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxide sol.
3. The volume of colloidal particle V_C as compared to the volume of a solute particle in a true solution V_S could be : **[AIEEE 2005]**
- (1) ~ 1 (2) $\sim 10^{23}$ (3) $\sim 10^{-3}$ (4) $\sim 10^3$
4. In Langmuir's model of adsorption of a gas on a solid surface : **[AIEEE 2006]**
- (1) the rate of dissociation of adsorbed molecules from the surface does not depend on the surface covered
 - (2) the adsorption at a single site on the surface may involve multiple molecules at the same time
 - (3) the mass of gas striking a given area of surface is proportional to the pressure of the gas
 - (4) the mass of gas striking a given area of surface is independent of the pressure of the gas
5. Gold numbers of protective colloids A, B, C and D are 0.50, 0.01, 0.10 and 0.005, respectively. The correct order of their protective powers is **[AIEEE 2008]**
- (1) $C < B < D < A$ (2) $A < C < B < D$ (3) $B < D < A < C$ (4) $D < A < C < B$

6. In context with the industrial preparation of hydrogen from water gas ($\text{CO} + \text{H}_2$), which of the following is the correct statement? **[AIEEE 2008]**

- (1) CO is removed by absorption in aqueous Cu_2Cl_2 Solution
- (2) H_2 is removed through occlusion with Pd
- (3) CO is oxidized to CO_2 with steam in the presence of a catalyst followed by absorption of CO_2 in alkali
- (4) CO and H_2 are fractionally separated using differences in their densities

7. Which of the following statements is incorrect regarding physisorptions ?

[AIEEE 2009]

- (1) More easily liquefiable gases are adsorbed readily.
- (2) Under high pressure it results into multi molecular layer on adsorbent surface.
- (3) Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive.
- (4) It occurs because of van der Waal's forces.

8. According to Freundlich adsorption isotherm which of the following is correct?

[AIEEE 2012]

- (1) $\frac{x}{m} \propto p^0$
- (2) $\frac{x}{m} \propto p^1$
- (3) $\frac{x}{m} \propto p^{1/n}$
- (4) All the above are correct for different ranges of pressure

9. The coagulating power of electrolytes having ions Na^+ , Al^{3+} and Ba^{2+} for arsenic sulphide sol increases in the order :

[JEE(Mains) 2013]

- | | |
|---|---|
| (1) $\text{Al}^{3+} < \text{Ba}^{2+} < \text{Na}^+$ | (2) $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$ |
| (3) $\text{Ba}^{2+} < \text{Na}^+ < \text{Al}^{3+}$ | (4) $\text{Al}^{3+} < \text{Na}^+ < \text{Ba}^{2+}$ |

10. 3 gram of activated charcoal was added to 50 mL of acetic acid solution (0.06N) in a flask. After an hour it was filtered and the strength of the filtrate was found to be 0.042 N. The amount of acetic acid adsorbed (per gram of charcoal) is :

[JEE(Main)-2015]

- | | | | |
|-----------|-----------|-----------|-----------|
| (1) 18 mg | (2) 36 mg | (3) 42 mg | (4) 54 mg |
|-----------|-----------|-----------|-----------|

11. Match the catalysts to the correct processes :

[JEE(Main)-2015]

Catalyst

Process

(A) TiCl_3

(i) Wacker process

(B) PdCl_2

(ii) Ziegler - Natta polymerization

(C) CuCl_2

(iii) Contact process

(D) V_2O_5

(iv) Deacon's process

(1) (A) - (iii), (B) - (ii), (C) - (iv), (D) - (i) (2) (A) - (ii), (B) - (i), (C) - (iv), (D) - (iii)

(3) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i) (4) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)

12. The following statements relate to the adsorption of gases on a solid surface. Identify the incorrect statement among them : [JEE(Main)-2015]

(1) On adsorption decrease in surface energy appears as heat

(2) Enthalpy of adsorption is negative

(3) On adsorption, the residual forces on the surface are increased

(4) Entropy of adsorption is negative

13. Under ambient conditions, which among the following surfactants will form micelles in aqueous solution at lowest molar concentration ? [JEE(Main)-2015]

(1) $\text{CH}_3-(\text{CH}_2)_8-\text{COO}^- \text{Na}^+$

(2) $\text{CH}_3(\text{CH}_2)_{11} \overset{\oplus}{\text{N}}(\text{CH}_3)_3 \text{Br}^-$

(3) $\text{CH}_3-(\text{CH}_2)_{13}-\text{OSO}_3^- \text{Na}^+$

(4) $\text{CH}_3(\text{CH}_2)_{15} \overset{\oplus}{\text{N}}(\text{CH}_3)_3 \text{Br}^-$

14. For a linear plot of $\log(x/m)$ versus $\log p$ in a Freundlich adsorption isotherm, which of the following statements is correct? (k and n are constants) [JEE(Main)-2016]

(1) $1/n$ appears as the intercept

(2) Only $1/n$ appears as the slope.

(3) $\log(1/n)$ appears as the intercept.

(4) Both k and $1/n$ appear in the slope term.

15. The Tyndall effect is observed only when following conditions are satisfied: [JEE-Main 2017]

(a) The diameter of the dispersed particles is much smaller than the wavelength of the light used.

(b) The diameter of the dispersed particle is not much smaller than the wavelength of the light used.

(c) The refractive indices of the dispersed phase and dispersion medium are almost similar in magnitude.

(d) The refractive indices of the dispersed phase and dispersion medium differ greatly in magnitude.

(1) (a) and (d)

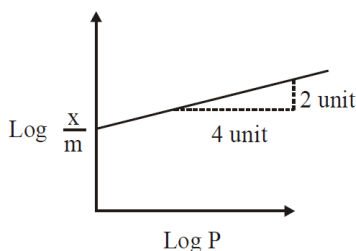
(2) (b) and (d)

(3) (a) and (c)

(4) (b) and (c)

16. Adsorption of a gas follows Freundlich adsorption isotherm. In the given plot, x is the mass of the gas adsorbed on mass m of the adsorbent at pressure p . $\frac{x}{m}$ is proportional to

[JEE-Main 2019]

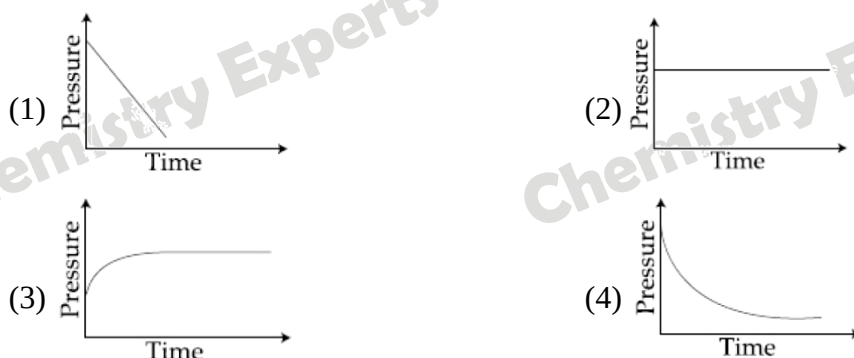


- (1) $P^{\frac{1}{4}}$ (2) P^2 (3) P (4) $P^{\frac{1}{2}}$
17. For coagulation of arsenious sulphide sol, which one of the following salt solution will be most effective ? [JEE-Main 2019]
 (1) Na_3PO_4 (2) AlCl_3 (3) BaCl_2 (4) NaCl
18. Which of the following is not an example of heterogeneous catalytic reaction ? [JEE-Main 2019]
 (1) Ostwald's process (2) Combustion of coal
 (3) Hydrogenation of vegetable oils (4) Haber's process
19. Haemoglobin and gold sol are examples of : [JEE-Main 2019]
 (1) negatively and positively charged sols, respectively
 (2) positively and negatively charged sols, respectively
 (3) negatively charged sols
 (4) positively charged sols
20. An example of solid sol is : [JEE-Main 2019]
 (1) Hair cream (2) Gem stone (3) Paint (4) Butter
21. Among the colloids cheese (C), milk (M) and smoke (S), the correct combination of the dispersed phase and dispersion medium, respectively is :- [JEE-Main 2019]
 (1) C : solid in liquid; M : liquid in liquid; S : gas in solid
 (2) C : liquid in solid; M : liquid in liquid; S : solid in gas
 (3) C : solid in liquid; M : solid in liquid; S : solid in gas
 (4) C : liquid in solid; M : liquid in solid; S : solid in gas

22. Given [JEE-Main 2019]
- | | | | | |
|------------------------|----------------|-----------------|-----------------|-----------------|
| Gas | H ₂ | CH ₄ | CO ₂ | SO ₂ |
| Critical Temperature/K | 33 | 190 | 304 | 630 |
- On the basis of data given above, predict which of the following gases shows least adsorption on a definite amount of charcoal ? [JEE-Main 2019]
- (1) CO₂ (2) H₂ (3) SO₂ (4) CH₄
23. The hardness of a water sample (in terms of equivalents of CaCO₃) containing 10⁻³ M CaSO₄ is: (molar mass of CaSO₄ = 136 g mol⁻¹) [JEE-Main 2019]
- (1) 50 ppm (2) 90 ppm (3) 10 ppm (4) 100 ppm
24. Among the following, the false statement is : [JEE-Main 2019]
- (1) Latex is a colloidal solution of rubber particles which are positively charged
 (2) Lyophilic sol can be coagulated by adding an electrolyte.
 (3) It is possible to cause artificial rain by throwing electrified sand carrying charge opposite to the one on clouds from an aeroplane.
 (4) Tyndall effect can be used to distinguish between a colloidal solution and a true solution.
25. Adsorption of a gas follows Freundlich adsorption isotherm. x is the mass of the gas adsorbed on mass m of the adsorbent. The plot of $\log \frac{x}{m}$ versus $\log p$ is shown in the given graph. $\frac{x}{m}$ is proportional to : [JEE-Main 2019]
-
- (1) $p^{\frac{2}{3}}$ (2) $p^{\frac{3}{2}}$ (3) p^2 (4) p^3
26. 100 mL of a water sample contains 0.81 g of calcium bicarbonate and 0.73 g of magnesium bicarbonate. The hardness of this water sample expressed in terms of equivalents of CaCO₃ is : (molar mass of calcium bicarbonate is 162 g mol⁻¹ and magnesium bicarbonate is 146 g mol⁻¹) [JEE-Main 2019]
- (1) 10,000 ppm (2) 100 ppm (3) 5,000 ppm (4) 1000 ppm
27. 0.27 g of a long chain fatty acid was dissolved in 100 cm³ of hexane. 10 mL of this solution was added dropwise to the surface of water in a round watch glass. Hexane evaporates and a monolayer is formed. The distance from edge to centre of the watch glass is 10 cm. What is the height of the monolayer ? [Density of fatty acid = 0.9 g cm⁻³; $\pi=3$] [JEE-Main 2019]
- (1) 10⁻² m (2) 10⁻⁴ m (3) 10⁻⁸ m (4) 10⁻⁶ m

28. The aerosol is a kind of colloid which: **[JEE-Main 2019]**
(1) gas is dispersed in liquid (2) solid is dispersed in gas
(3) liquid is dispersed in water (4) gas is dispersed in solid
29. 10 mL of 1mM surfactant solution forms a monolayer covering 0.24 cm^2 on a polar substrate. If the polar head is approximated as cube, what is its edge length? **[JEE-Main 2019]**
(1) 0.1 nm (2) 2.0 pm (3) 1.0 pm (4) 2.0 nm
30. A gas undergoes physical adsorption on a surface and follows the given Freundlich adsorption isotherm equation $\frac{x}{m} = kp^{0.5}$ **[JEE-Main 2019]**
Adsorption of the gas increases with :
(1) Decrease in p and decrease in T (2) Decrease in p and increase in T
(3) Increase in p and decrease in T (4) Increase in p and increase in T
31. The correct option among the following is : **[JEE-Main 2019]**
(1) Brownian motion in colloidal solution is faster if the viscosity of the solution is very high.
(2) Colloidal medicines are more effective because they have small surface area.
(3) Addition of alum to water makes it unfit for drinking.
(4) Colloidal particles in lyophobic sols can be precipitated by electrophoresis.
32. Peptization is a: **[JEE-Main 2019]**
(1) process of converting precipitate into colloidal solution
(2) process of converting a colloidal solution into precipitate
(3) process of bringing colloidal molecule into solution
(4) process of converting soluble particles to form colloidal solution
33. Among the following, the INCORRECT statement about colloids is: **[JEE-Main 2019]**
(1) The osmotic pressure of a colloidal solution is of higher order than the true solution at the same concentration.
(2) The range of diameters of colloidal particles is between 1 and 1000 nm.
(3) They can scatter light.
(4) They are larger than small molecules and have high molar mass.

34. A mixture of gases O_2 , H_2 and CO are taken in a dosed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is : **[Jee Main (Jan. 2020)]**



35. The flocculation value of HCl for arsenic sulphide sol. is 30 m mol L^{-1} . If H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams, of H_2SO_4 in 250 ml required for the above purpose is _____.

(molecular mass of $H_2SO_4 = 98 \text{ g/mol}$)

[Jee Main (Jan. 2020)]

36. The hardness of a water sample containing 10^{-3} M $MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is _____.

(molar mass of $MgSO_4$ is 120.37 g/mol)

[Jee Main (Jan. 2020)]

37. Which of the following is used for the preparation of colloids ? **[Jee Main (Jan. 2020)]**

- (1) Bredig's Arc Method (2) Ostwald Process
(3) Mond Process (4) Van Arkel Method

38. Amongst the following statements regarding adsorption, those that are valid are :

(a) ΔH becomes less negative as adsorption proceeds. **[Jee-Main (Sep) 2020]**

(b) On a given adsorbent, ammonia is adsorbed more than nitrogen gas.

(c) On adsorption, the residual force acting along the surface of the adsorbent increases.

(d) With increase in temperature, the equilibrium concentration of adsorbate increases.

- (1) (a) and (b) (2) (c) and (d) (3) (d) and (a) (4) (b) and (c)

39. Tyndall effect is observed when: **[Jee-Main (Sep) 2020]**

(1) The refractive index of dispersed phase is greater than that of the dispersion medium.

(2) The diameter of dispersed particles is much smaller than the wavelength of light used.

(3) The diameter of dispersed particles is similar to the wavelength of light used.

(4) The diameter of dispersed particles is much larger than the wavelength of light used.

40. Match the following :

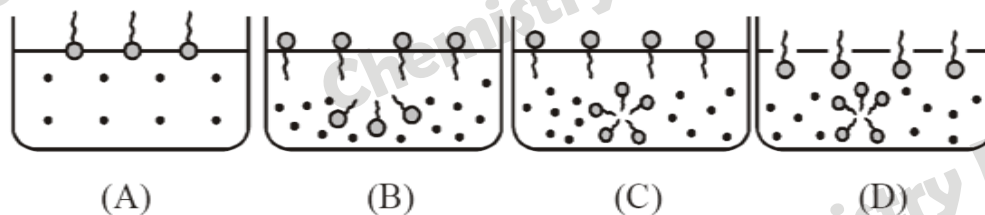
- | | |
|---------------|----------------|
| (i) Foam | (a) smoke |
| (ii) Gel | (b) cell fluid |
| (iii) Aerosol | (c) jellies |
| (iv) emulsion | (d) rubber |
| | (e) froth |
| | (f) milk |

[Jee-Main (Sep) 2020]

- | | |
|--|--|
| (1) (i)-(d); (ii)-(b); (iii)-(e); (iv)-(f) | (2) (i)-(b); (ii)-(c); (iii)-(e); (iv)-(d) |
| (3) (i)-(d); (ii)-(b); (iii)-(a); (iv)-(e) | (4) (i)-(e); (ii)-(c); (iii)-(a); (iv)-(f) |

41. Identify the correct molecular picture showing what happens at the critical micellar concentration (CMC) of an aqueous solution of a surfactant (○ polar head ; ~ non-polar tail ; ● water).

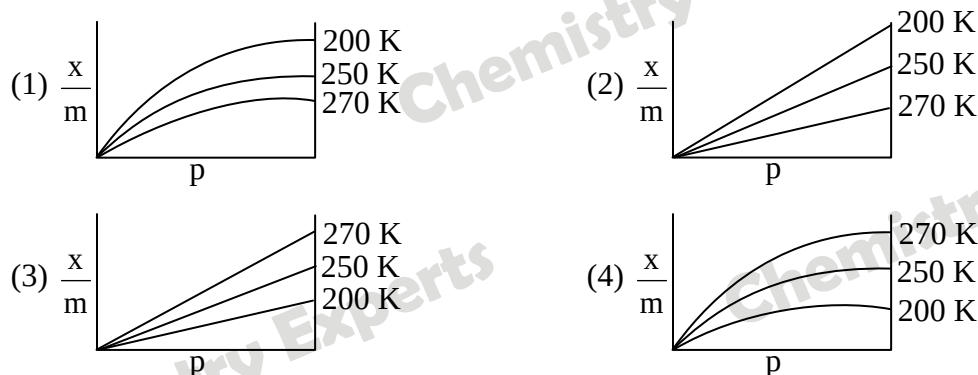
[Jee-Main (Sep) 2020]



- | | | | |
|---------|---------|---------|---------|
| (1) (B) | (2) (C) | (3) (A) | (4) (D) |
|---------|---------|---------|---------|

42. Adsorption of a gas follows Freundlich adsorption isotherm. If x is the mass of the gas adsorbed on mass m of the adsorbent, the correct plot of $\frac{x}{m}$ versus p is :

[Jee-Main (Sep) 2020]

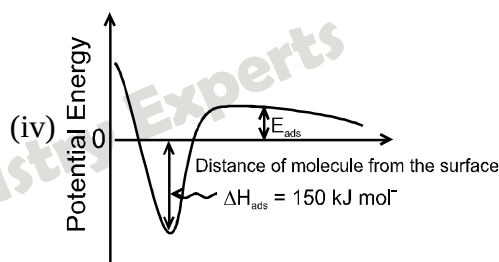
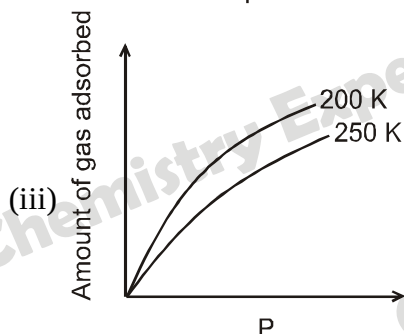
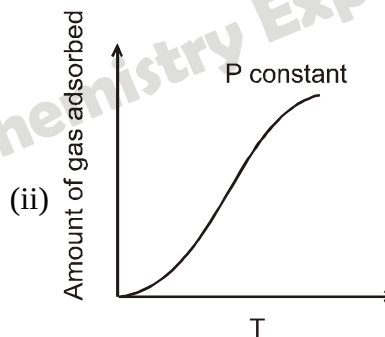
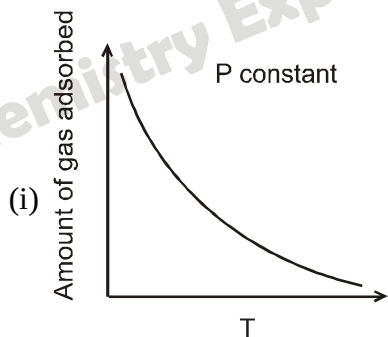


43. Kraft temperature is the temperature : **[Jee-Main (Sep) 2020]**
- (1) above which the aqueous solution of detergents starts boiling.
 - (2) below which the aqueous solution of detergents starts freezing.
 - (3) above which the formation of micelles takes place.
 - (4) below which the formation of micelles takes place.
44. For Freundlich adsorption isotherm, a plot of $\log (x/m)$ (y-axis) and $\log p$ (x-axis) gives a straight line. The intercept and slope for the line is 0.4771 and 2, respectively. The mass of gas, adsorbed per gram of adsorbent if the initial pressure is 0.04 atm, is _____ $\times 10^{-4}g$.
($\log 3 = 0.4771$) **[Jee-Main (Sep) 2020]**
45. The mass of gas adsorbed, x , per unit mass of adsorbate, m , was measured at various pressures, p . A graph between $\log \frac{x}{m}$ and $\log p$ gives a straight line with slope equal to 2 and the intercept equal to 0.4771. The value of $\frac{x}{m}$ at a pressure of 4 atm is : (Given $\log 3 = 0.4771$) **[Jee-Main (Sep) 2020]**

EXERCISE # JEE-ADVANCED

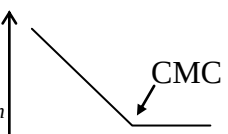
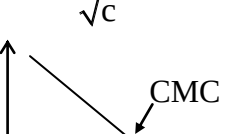
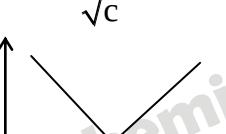
- Rate of physisorption increases with [JEE 2003]
 (A) decrease in temperature (B) increase in temperature
 (C) decrease in pressure (D) decrease in surface area
- Adsorption of gases on solid surface is generally exothermic because [JEE 2004]
 (A) enthalpy is positive (B) entropy decreases
 (C) entropy increases (D) free energy increases
- Lyophilic sols are [JEE 2005]
 (A) Irreversible sols (B) They are prepared from inorganic compound
 (C) Coagulated by adding electrolytes (D) Self-stabilizing
- Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient condition is : [JEE 2008]
 (A) $\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3\text{Br}^-$ (B) $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-\text{Na}^+$
 (C) $\text{CH}_3(\text{CH}_2)_6\text{COO}^-\text{Na}^+$ (D) $\text{CH}_3(\text{CH}_2)_{11}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
- Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 sol is : [JEE 2009]
 (A) Na_2SO_4 (B) CaCl_2 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) NH_4Cl
- Silver (atomic weight = 108 g mol^{-1}) has a density of 10.5 g cm^{-3} . The number of silver atoms on a surface of area 10^{-12} m^2 can be expressed in scientific notation as $y \times 10^x$. The value of x is: [JEE 2010]
- The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) [JEE 2011]
 (A) Adsorption is always exothermic
 (B) Physisorption may transform into chemisorption at high temperature
 (C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
 (D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation.
- Choose the correct reason(s) for the stability of the **lyophobic** colloidal particles. [JEE 2012]
 (A) Preferential adsorption of ions on their surface from the solution.
 (B) Preferential adsorption of solvent on their surface from the solution.
 (C) Attraction between different particles having opposite charges on their surface.
 (D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles.

9. The given graph / data I, II, III and IV represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice (s) about I, II, III and IV is (are) correct. [JEE 2012]



- (A) I is physisorption and II is chemisorption
 (B) I is physisorption and III is chemisorption
 (C) IV is chemisorption and II is chemisorption
 (D) IV is chemisorption and III is chemisorptions
10. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25° C. For this process, the correct statement is [JEE(Advanced)-2013]
 (A) The adsorption requires activation at 25°C.
 (B) The adsorption is accompanied by a decreases in enthalpy.
 (C) The adsorption increases with increase of temperature.
 (D) The adsorption is irreversible.
11. When O₂ is adsorbed on a metallic surface, electron transfer occurs from the metal to O₂. The **TRUE** statement(s) regarding this adsorption is(are) [JEE(Advanced)-2015]
 (A) O₂ is physisorbed
 (B) heat is released
 (C) occupancy of π^* of O₂ is increased
 (D) bond length of O₂ is increased

-
- The figure contains three separate graphs, labeled I, II, and III, each plotting surface tension on the y-axis against concentration on the x-axis.
- Graph I:** The curve starts at a high surface tension value at zero concentration and decreases as concentration increases, eventually leveling off to a constant value.
 - Graph II:** The curve starts at a low surface tension value at zero concentration and increases linearly as concentration increases.
 - Graph III:** The curve starts at a high surface tension value at zero concentration, drops sharply to a low constant value at a very low concentration, and then remains constant for all higher concentrations.

- (A)  (B) 
- (C)  (D) 

ANSWER KEY**EXERCISE # JEE-MAIN**

1.	2	2.	2	3.	4	4.	3	5.	2
6.	3	7.	3	8.	4	9.	2	10.	1
11.	2	12.	3	13.	4	14.	2	15.	2
16.	4	17.	2	18.	2	19.	2	20.	2
21.	2	22.	2	23.	4	24.	1	25.	1
26.	1	27.	4	28.	2	29.	2	30.	3
31.	2	32.	1	33.	1	34.	4		
35.	[0.37] NTA Ans. 0.36 to 0.38			36.	100	37.	1	38.	1
39.	3	40.	4	41.	4	42.	1	43.	3
44.	48.00	45.	(48)	NTA Ans. (6)					

EXERCISE # JEE-ADVANCED

1.	A	2.	B	3.	D	4.	A	5.	C
6.	7	7.	ABD	8.	AD	9.	AC	10.	B
11.	BCD	12.	D	13.	AB	14.	C		

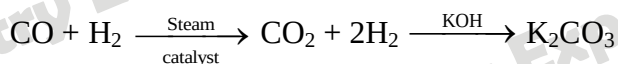
Hint and Solutions

EXERCISE # JEE-MAIN

1. Since the physical adsorption process is exothermic, the physical adsorption occurs readily at low temperature and decreases with increasing temperature. (Le Chatelier's principle).
2. On mixing, they will coagulate each other being +ve and -ve charged.
3. For true solution the diameter range is 1 to 10 Å and for colloidal solution diameter range is 10 to 10,000 Å.

$$\frac{V_c}{V_s} = \frac{(4/3)\pi r_c^3}{(4/3)\pi r_s^3} = \left(\frac{r_c}{r_s}\right)^3 \quad \text{Ratio of diameters} = (10/1)^3 = 10^3 \quad V_c / V_s = 10^3.$$

5. The adsorption of a gas is directly proportional to the pressure of gas.
6. CO is oxidised to CO₂ with steam in the presence of a catalyst followed by absorption of CO₂ in alkali.



7. Since adsorption is exothermic process so ΔH of adsorption is always negative.
8. $\frac{x}{m} \propto P^{1/n}$ where $n \geq 1$
9. According to Hardy Schulze rule, greater the charge on cation, greater is its coagulating power for negatively charged solution. So, order of coagulating power : $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$.

10. Initial mmoles of CH₃COOH = 0.06×50
 Final mmoles of CH₃COOH = 0.042×50
 Hence, mass of CH₃COOH adsorbed per gram of charcoal

$$= \frac{(0.06 - 0.042) \times 50 \times 10^{-3} \times 60 \times 10^3}{3} = 18 \text{ mg}$$

11. TiCl₃ → Ziegler-Natta polymerisation
 PdCl₂ → Wacker process
 CuCl₂ → Deacon's process
 V₂O₅ → Contact process

12. Adsorption takes place due to the presence of residual forces on the surface. After adsorption, these are decreased.
13. Longer hydrophobic chain, lesser CMC
14. According to the Freundlich adsorption isotherm

$$\frac{x}{m} = kP^{1/n}$$

$$\log \frac{x}{m} = \log K + \frac{x}{m} \log P$$

34. (4)

35. [0.37] NTA Ans. 0.36 to 0.38

Sol. Moles of H^+ needed for flocculation of 1L solution = 30 mmole

$$\text{Moles of } H^+ \text{ needed for 250 ml solution} = \frac{30}{4} = 7.5 \text{ mmole}$$

$$\text{Moles of } H_2SO_4 \text{ needed} = \frac{7.5}{2} \text{ mmole}$$

$$\begin{aligned} \text{Weight of } H_2SO_4 &= \frac{7.5}{2} \times 98 \times 10^{-3} \text{ gm} \\ &= 0.3675 \text{ gm} \end{aligned}$$

36. (100)

Sol. $10^{-3} \text{ M } MgSO_4$

Equivalent to $10^{-3} \text{ M } CaCO_3$

$$\Rightarrow \text{cons. of } CaCO_3 \text{ in ppm} = \frac{10^{-3} \times 100}{1000} \times 10^6 = 100 \text{ ppm}$$

37. (1)

Sol. Bredig's arc method is used for preparation of colloids.

38. (1)

Sol. (a) as adsorption proceeds, extent of adsorption decreases

So ΔH becomes less negative

(b) $(T_C) NH_3 > (T_C) N_2$

So adsorption of NH_3 is more than N_2

(c) as adsorption forces along adsorbent surface decreases.

(d) On increasing temperature extent of adsorption decreases.

39. (3)

Sol. Tyndall effect is observed when the diameter of dispersed particles is similar to the wavelength of light used.

40. (4)

Sol. Foam - Froth

Gel $\rightarrow$ Jellies

Aerosol $\rightarrow$ Smoke

Sol $\rightarrow$ Cell fluids

Solid sol $\rightarrow$ rubber

41. (4)

Sol. Polar ends interacts with water where as non polar ends moves away from water.

42. (1)

Sol. $\frac{x}{m} = k P^{1/n}$

$$\frac{x}{m} \downarrow \text{ as } T \uparrow$$

So correct curve is (1)

43. (3)

Sol. (i) Micelles formation take place only above a particular temperature called as kraft temperature (T_k)

(ii) Concentration above which micelle formation become appreciable is critical micells concentration.

44. (48.00)

Sol. $\frac{x}{m} = KP^{1/n}$

$$\log\left(\frac{x}{m}\right) = \frac{1}{n} \log P + \log K$$

$$\text{slope } \frac{1}{n} = 2$$

$$\text{intercept} = \log K = 0.4771$$

$$K = 3$$

mass of gas adsorbed per gm of adsorbent – $\frac{x}{m}$

$$\frac{x}{m} = 3 \times (0.04)^2 = 48 \times 10^{-4}$$

45. (48) NTA Ans. (6)

Sol. $\frac{x}{m} = KP^{1/n}$

$$\log\left(\frac{x}{m}\right) = \log K + \frac{1}{n} \log P$$

In the curve of $\log\left(\frac{x}{m}\right)$ VS $\log P$

$$\text{Slope} = \frac{1}{n} = 2$$

$$\text{Intercept} = \log K = 0.4771$$

$$K = 3$$

at $P = 4 \text{ atm}$

$$\frac{x}{m} = KP^{1/n}$$

$$= 3 \times (4)^2 = 48$$

EXERCISE # JEE-ADVANCED

- Physical adsorption is exothermic process so its rate decreases with increase in temperature.
- In adsorption there is bond formation between the gases and solid surface which decreases the entropy so to make it spontaneous the enthalpy change must be negative.
- In lyophilic sols the dispersed phase has great affinity (attraction) towards dispersion medium. So they are self stabilizing.
- Longer the hydrophobic part of the molecule easy will be the formation of micelle. (Longest hydrocarbon chain)
- Most effective coagulating agent for Sb_2S_3 is $\text{Al}_2(\text{SO}_4)_3$ because of high charge.

6. Volume of one mole of silver atoms = $\frac{108}{10.5} \text{ cm}^3/\text{mole}$

$$\text{volume of one silver atom} = \frac{108}{10.5} \times \frac{1}{6.022 \times 10^{23}} \text{ cm}^3$$

$$\text{so, } \frac{4}{3} \pi R^3 = \frac{108}{10.5} \times \frac{1}{6.022 \times 10^{23}} = 1.708 \times 10^{-23} \quad [\text{neglecting the void space}]$$

$$R^3 = 0.407 \times 10^{-23} \text{ cm}^3$$

$$R^3 = 0.407 \times 10^{-29} \text{ m}^3$$

Area of each silver atom

$$\pi R^2 = \pi \times (0.407 \times 10^{-29} \text{ m}^3)^{2/3}$$

$$\text{so, number of silver atoms in given area.} = \frac{10^{-12}}{(0.407 \times 10^{-29} \text{ m}^3)^{2/3}} = \frac{10^8}{(\pi \times 2)} = 1.6 \times 10^7 = y \times 10^x$$

$$x = 7$$

- (A) $\Delta H = -ve$ for adsorption

(B) fact

(D) chemical bonds are stronger than vander waal's forces so chemical adsorption is more exothermic.
- (A) due to preferential adsorption of common ions

(C) due to repulsion not due to attraction

(D) The layer of oppositely charged particles around any colloidal particles will decrease the potential energy of system as a whole.

9. In physisorption on increasing temperature at constant pressure, adsorption decreases while in chemical adsorption on increasing temperature due to requirement of activation energy adsorption will increase at same pressure. So, I is physisorption while II is chemisorption. III is physical adsorption as on increasing temperature, extent of adsorption is decreasing. IV is representing enthalpy change (which is high) during chemical adsorption (due to bond formation) So, is valid for chemical adsorption. So, answer is (A) and (C)
10. As the adsorption of methylene blue over activated charcoal is physisorption (Reference : NCERT), it is accompanied by decrease in enthalpy.
11. (A) As electron transfer is involved, so chemisorption is taking place.
(B) Adsorption is exothermic process.
(C) The π^*_{2p} orbitals of O_2 accommodate the incoming electron as they are partially filled.
(D) As electron enters antibonding MO, bond length increases.
So, answer is (B,C,D)